SECTION 00 31 32

Geotechnical Data Report

for

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Prepared by

Geotechnical Branch

Engineering Division

Jacksonville District Corps of Engineers

July 2025

SECTION 00 31 32

GEOTECHNICAL DATA REPORT

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GEOTECHNICAL DATA REPORT

1 SCOPE

The information provided in this section encompasses the geotechnical field investigations available for this project. The investigations consist of borings. The associated boring logs and laboratory data are presented in paragraphs 4.5 and 4.6, respectively. A character of materials paragraph is included to provide a comprehensive description of the materials utilizing both recent and historical knowledge of the project area. Also included in this section are definitions of terms and boring log notes, which provide additional explanation of the boring logs and drilling techniques.

Items discussed in the character of materials paragraph may not appear explicitly on the boring logs. Based on historic knowledge of the project area, the character of materials paragraph includes items that supplement the data documented by the boring logs. When reviewing the boring logs, use all of the data on the logs, including the materials description, legend, and blow counts. When evaluating the subsurface conditions, use all of the data, including the character of materials paragraph and boring logs.

During the solicitation period, any questions that pertain to the information provided in this section should be addressed to the contract specialist identified in Block 9 of SF1442. After contract award, questions should be addressed to the appointed ACO/COR, which will coordinate with Geosystems Branch.

2 CHARACTER OF MATERIALS

2.1 THE LOCAL GEOLOGY

The shallow sediments on the upland adjacent to Kings Bay Submarine Base are typified as fine to medium-grained sands interbedded with silty and/or clayey fine sands that extend to a depth of at least 50 feet below ground level.

Kings Bay Naval Submarine Base is situated at the edge of a salt marsh. The present marshlands surrounding Kings Bay developed over the past several thousand years as sea level slowly rose to reoccupy the area behind the existing barrier islands. Marshland sediments anticipated to be part of the shoaled maintenance dredging areas consist primarily of current-borne sands, silts, and clays, in addition to a significant quantity of organic material contributed by indigenous plants and animals.

2.2 MATERIALS ENCOUNTERED

Special information regarding materials is provided in the Special Instructions paragraph of the Dredging Specifications, Section 35 20 23; in situ rock is not required to be dredged. Due to the age of borings, the material recovered are no longer available for inspection. Prospective offerors are cautioned to use great care in the evaluation of provided data.

2.2.1 Channel O&M, Ranges E through I, Stations 31+800 through 48+175

The materials to be removed from the maintenance areas in the federal channel accumulated as a result of shoaling since the previous dredging and dredged materials descriptions are based on current and historical records. Depending on the location or depth, these materials will consist primarily of fine to medium-grained quartz sand, silt, and clay, with various amounts of gravel sized shell and/or limestone fragments.

2.2.2 Special Feature Areas, to Include Magnetic Silencing Facility – North, Magnetic Silencing Facility – South, and Site Six – South

The materials to be removed by maintenance dredging from the special feature areas outside the federal channel are described by VB-KBIC13-03 (MSF – South), VB-KB15-49 (Site Six – South), and VB-KB15-61 (MSF – North); the material encountered accumulated as a result of shoaling since the previous dredging in 2025 and dredged materials descriptions are based on current and historical records. Depending on the location and/or depth, these materials will primarily consist of silty sand, silt, and clay. Gravel sized shell and/or limestone fragments may also be present.

3 DEFINITIONS

Definitions not explicitly indicated in the sections below are typical industry standard definitions from their respective ASTMs.

3.1 DEFINITIONS OF BASIC TERMS

Carbonate – Soil component that reacts with HCl of an indeterminate origin (shell, rock, etc.).
Fill – Material that has been placed by man, described with all soil characteristics.
Layer – Rock or soil with a thickness of 6 inches or less.
Lens – A geologic deposit of variable thickness, which disappears laterally in all directions and cannot be correlated to adjacent borings.
Rock – A naturally occurring substance composed of one or more minerals bound together. This geologic term includes a range of engineering properties: strength, hardness, permeability, weathering, and discontinuity. These properties are noted or can be inferred from the boring logs as blow counts, penetration rate, RQD, hardness, etc.
Shell – Material composed of predominantly (>75%) coarse-grained sand to gravel-sized whole or broken shell.

3.2 CURRENT LOGGING TERMS

Definition of terms used on boring logs dated since 2000:

Banded - 0.02' (6 mm) to 0.1' (3.0 cm).
Boulder-Sized – Particles greater than 12 inches in diameter.
Cavity - Voids greater than the diameter of the core.
Coarse Grained - Grain diameter greater than 0.079" (2 mm) for sedimentary rocks or 0.197" (5 mm) for igneous or metamorphic rocks.
Coarse Grained Sand-Sized – Less than 10 percent of fine and medium grained sand sizes are present.
Coarse Gravel-Sized – Particles greater than 3/4 of an inch but less than 3 inches in diameter.
Cobble-Sized – Particles greater than 3 inches but less than 12 inches in diameter.
Decomposed – Applicable to saprolitic rock; rock is essentially reduced to a soil with a relic rock texture; can be molded or crumbled by hand.
Dipping (Dip) - 20 to 45 degrees.
Discontinue – Particles were present within the unit above but are no longer present.
Fine Grained - Grain diameter between 0.004" (0.1 mm) and 0.016" (0.4 mm) for sedimentary rocks or 0.039" (1 mm) for igneous or metamorphic rocks.
Fine Grained Sand-Sized - Less than 10 percent of medium and coarse grained sand sizes are present.
Fine Gravel-Sized – Particles greater than No. 4 sieve but less than 3/4 of an inch in diameter.
Flat (Dip) - 0 to 20 degrees.
Fossiliferous – Greater than 40 percent fossils.
Hard (Hardness) - Difficult to scratch with a knife (cannot be pitted with a geology hammer, but can be chipped with moderate blows of the hammer).
Highly Fractured - Spacing 0.3' (9.1 cm) to 1' (30.5 cm).

Highly Weathered - Entire section is discolored; alteration is greater than 50%; some areas of slightly weathered rock are present; some minerals are leached away; retains only a fraction of its original strength (wet strength usually lower than dry strength).
Intact – Spacing greater than 6' (1.8 m).
Intensely Fractured - Spacing less than 0.3' (9.1 cm).
Massive – Over 3' (0.9 m) thick.
Medium-Bedded - 0.3' (9.1 cm) to 1' (30.5 cm) thick.
Medium Grained - Grain diameters between 0.016" (0.4 mm) to 0.079" (2 mm) for sedimentary rocks or 0.039" (1 mm) to 0.197" (5 mm) for igneous or metamorphic rocks.
Medium Grained Sand-Sized - Less than 10 percent of fine and coarse grained sand sizes are present.
Moderately Fractured - Spacing 1' (30.5 cm) to 3' (0.9 m).
Moderately Hard (Hardness) - Can be scratched easily with a knife, but cannot be scratched by a fingernail (can be pitted with moderate blows of a geology hammer).
Moderately Open (Fracture Aperture) - 0.020" (0.5 mm) to 0.098" (2.5 mm).
Moderately Weathered - Discoloration is evident; surface is pitted and altered, with alterations penetrating well below rock surfaces; 10% to 50% of the rock is altered; strength is noticeably less than unweathered rock.
Non-fossiliferous – No observed fossils.
Open (Fracture Aperture) - 0.098" (2.5 mm) to 0.394" (10 mm).
Pitted - Voids 0.039" (1mm) to 0.236" (6mm) in diameter.
Porous - Voids less than 0.039" (1mm) in diameter.
Slightly Fractured - Spacing 3' (0.9 m) to 6' (1.8 m).
Slightly Weathered - Superficial discoloration, alteration and/or discoloration along discontinuities; less than 10% of the rock volume is altered; strength is essentially unaffected.
Soft (Hardness) - Can be scratched with a fingernail (cannot be crumbled between fingers, but can be easily pitted with light blows of a geology hammer).
Solid - Absence of voids.
Sparsely Fossiliferous – Less than 40 percent fossils.
Steeply Dipping (Dip) - 45 to 90 degrees.
Thick-Bedded – 1' (30.5 cm) to 3' (0.9 m) thick.
Thin-Bedded - 0.1' (3.0 cm) to 0.3' (9.1 cm) thick.
Thin Parting - Paper thin to 0.002' (0.6 mm).
Tight (Fracture Aperture) - 0.004" (0.1 mm) to 0.020" (0.5 mm).
Unweathered - No evidence of any mechanical or chemical alteration.
Very Fine Grained - Grain diameter less than 0.004" (0.1 mm); individual grains or crystals are too small to be seen with the naked eye.
Very Hard (Hardness) - Cannot be scratched with a knife (chips can be broken off only with heavy blows of a geology hammer).
Very Soft (Hardness) - Can be deformed by hand (has a rock-like character, but can be easily broken by hand).
Very Tight (Fracture Aperture) - less than 0.004" (0.1 mm).
Very Wide (Fracture Aperture) - 0.394" (10 mm) to 0.984" (25 mm).
Vuggy - Voids 0.236" (6mm) to the diameter of the core.

3.3 PREVIOUS LOGGING TERMS

Definition of terms used on boring logs dated before 2000:

Dense – Equivalent to SPT N-value of 30 to 50.
Incompetent – Rock that disintegrates while coring; weak.
Indurated – Rock or soil hardened or consolidated by pressure or cementation. Very difficult to break by hand.
Poorly-Indurated – See semi-indurated.
Seam – Rock or soil with average thickness of 2 to 3 inches.

Semi-Indurated – Rock or soil with a lesser degree of hardening or consolidation by pressure or cementation. Crumbles with little effort by hand.

3.4 TESTING AND PROCEDURE METHODS

Test/Procedure	Method
Abrasion Resistance of Large-Size Coarse Aggregate, Los Angeles Machine	ASTM C535
Abrasion Resistance of Small-Size Coarse Aggregate, Los Angeles Machine	ASTM C131
Air Content by the Pressure Method	ASTM C231
Air Content by the Volumetric Method	ASTM C173
Air-Entraining Admixtures for Concrete	ASTM C233
Atterberg Limits, wet preparation method, test method a multipoint	ASTM D4318
Bulk Specific Gravity of Bituminous Mixtures	ASTM D2726
Capping Cylindrical Concrete Specimens	ASTM C617
Carbonate Content	ASTM D4373
Compaction: 4" Mold, Modified Proctor	ASTM D1557
Compaction: 4" Mold, Standard Proctor	ASTM D698
Compaction: 6" Mold, Modified Proctor	ASTM D1557
Compaction: 6" Mold, Standard Proctor	ASTM D698
Compressive Strength of Cast in Place Concrete Cylinders	ASTM C873
Compressive Strength of Cylindrical Concrete Specimens (Set of 4)	ASTM C39
Compressive Strength of Lightweight Concrete	ASTM C495
Consolidation, each load of rebound increment with complete time	ASTM D2435
Density and Water Content of Soil by Nuclear Methods (Minimum 4)	ASTM D2922/D3017
Density of Bituminous Concrete in Place by Nuclear Methods (Minimum 4)	ASTM D2950
Density of Soil in Place by the Sand-Cone Method	ASTM D1556
Direct Shear	ASTM D3080
Drilled Cores and Sawed Beams of Concrete	ASTM C42
Extraction of Bituminous Paving Mixtures	ASTM D2172
Freezing and Thawing (up to 55 cycles)	ASTM D5312
Grain size sieve Analysis	ASTM D422
Hydrometer Analysis	ASTM D422
Making and Curing Concrete Test Specimens in the Field (Set of 6)	ASTM C31
Making and Curing Concrete Test Specimens in the Lab (Set of 4)	ASTM C192
Marshall Resistance to Plastic Flow (Set of 3)	ASTM D1559
Material Passing No. 200 Sieve	ASTM D1140
Modulus of Elasticity	ASTM D3148
Moisture Content	ASTM D2216
Munsell Color	Munsell Soil Color Charts
Organic Content, Test Method C	ASTM D2974
Organic Impurities in Fine Aggregate	ASTM C40
Permeability: Constant Head Permeability	ASTM D2434
Permeability: Falling Head Permeability	ASTM D5084
Petrographic Examination	ASTM C295

Test/Procedure	Method
Sampling Aggregate (Minimum 4 hours)	ASTM D75
Sampling Freshly Mixed Concrete	ASTM C172
Sedimentation Rate (USACE Method)	No ASTM
Sieve Analysis of Aggregate	ASTM C136
Slump of Hydraulic-Cement Concrete	ASTM C143
Soil Classification (Boring logs)	ASTM D 2488
Specific Gravity and Absorption of Fine Aggregate	ASTM C128
Specific Gravity for Rock	ASTM C127
Specific Gravity for Soil	ASTM D854
SPLITTING TENSILE STRENGTH	ASTM D3967
Splitting Tensile Strength of Concrete Cylinders	ASTM C496
Subsample Preparation ASTM D4220, Group B	ASTM D4220,
Sulfate Soundness	ASTM C88
Total Evaporable Moisture Content	ASTM C566
Triaxial Compression Test for Rock with Strain Gages or LVDTs (per confining pressure)	ASTM D2664
Triaxial compression: Consolidated Undrained (CU) Triaxial Compression Test R Test with Pore Pressure Measurements (per confining pressure)	ASTM D4767
Triaxial compression: Unconsolidated Undrained (UU) Triaxial Compression Test Q Test with Pore Pressure Measurements (per confining pressure)	ASTM D2850
Unbonded Caps for Compressive Strength	ASTM C1231
Unconfined Compressive Strength for Rock Greater than 9,000 psi with Strain	ASTM D2938
Unconfined Compressive Strength for Rock up to 9,000 psi with Strain	ASTM D2938
Unconfined Compressive Strength for Soil	ASTM D2166
Unit Weight and Absorption	ASTM C127
Unit Weight of Aggregate	ASTM C29 -
Visual Percent Shell	No ASTM
Wetting and Drying	ASTM D5313

4 GEOTECHNICAL DATA

4.1 SUMMARY OF FIELD INVESTIGATIONS

The table below summarizes the field investigations data set available for this project.

Table 1. Available Vibracore Data

Boring Designation	State Plane, GA-East, NAD83		Project Location
	X	Y	
VB-KBIC13-03	868545	283068	Range E
VB-KBIC13-04	867721	286573	
VB-KB15-49	867091	286328	
VB-KB15-61	867715	285069	
VB-KB15-63	868076	285705	

Boring Designation	State Plane, GA-East, NAD83		Project Location
	X	Y	
VB-KB15-W4	868917	284424	Range E
VB-KBIC16-4	867840	286302	
VB-KBIC13-02	866321	287795	Range F
VB-KB15-47	866360	287768	
VB-KB15-62	866613	288591	
VB-KBIC16-3	866624	288586	
VB-KB15-W5	867008	288001	Range F Widener
VB-KBIC13-01	865566	288492	Range G
VB-KBIC16-2	865550	288532	
VB-KBIC19-11	862905	289633	
VB-KBIC19-12	863187	290080	
VB-KBIC19-13	863382	290464	
VB-KBIC19-14	864687	288597	
VB-KBIC19-15	864966	288993	
VB-KBIC19-16	865121	289357	
VB-KBIC19-06	859361	293021	Range H
VB-KBIC19-07	859567	293203	
VB-KBIC19-08	861500	290850	
VB-KBIC19-09	861744	291009	
VB-KBIC19-10	862087	291303	
VB-KBIC16-5	859193	295286	Range I / Mooring Area
VB-KBIC16-6	859353	294247	
VB-KBIC19-03	857974	294647	
VB-KBIC19-04	858631	295121	
VB-KBIC19-05	859354	295605	
* Coordinates presented correspond to the project coordinate system and datum			

4.2 SUMMARY OF INDEX TESTING DATA

The table below summarizes the index testing available for this project.

Table 2: Index Testing Available for this Project

Boring Designation	Sample Designation	USCS	LL	PL	PI	Org %	Visual Shell %	w_n (%)	G_S	Munsell Color
VB-KBIC13-01	1	OH					0			2.5Y 3/2
	2	CH					0			2.5Y 3/2
VB-KBIC16-2	1	MH					0	30.3		2.5Y 3/1
	2	GP-GM					0			2.5Y 4/3

Boring Designation	Sample Designation	USCS	LL	PL	PI	Org %	Visual Shell %	w_n (%)	G_S	Munsell Color
VB-KBIC16-3	1	SM					0			2.5Y 4/1
	2	SM					0			2.5Y 3/1
VB-KBIC16-4	1	SP					0.1			2.5Y 7/2
	2	SP-SM					0.1			2.5Y 5/1
	3	SP					1			2.5Y 5/1
	4	GP					0			2.5Y 4/1
VB-KBIC16-5	1	SP					4.1			2.5Y 7/2
	2	SP					0.9			2.5Y 7/2
VB-KBIC16-6	1	SP-SM					0.1			2.5Y 4/1
	2	SM					0			2.5Y 4/1
	3	SM					0			2.5Y 6/2
VB-KBIC19-3	1	SM	75	41	34	7		199.0	2.5	10Y 2.5/1
	2	MH	62	32	30	3.7		173.0	2.85	10Y 2.5/1
	3	SM	NP	NP	NP	5.7		163.0	2.69	10Y 2.5/1
VB-KBIC19-4	1	SM	80	45	35	6.0		116.9	2.53	N 2.5/
	2	SM	93	47	46	2.9		103.2	2.72	N 2.5/
	3	SM	92	55	37	2.8		68.6	2.70	2.5Y 3/1
	4	MH	94	52	42	7.5		180.8	2.78	5Y 2.5/2
VB-KBIC19-5	1	SP-SM	NP	NP	NP	1.5		48.9	2.59	5Y 6/1
	2	SP-SM	NP	NP	NP	1.0		25.9	2.60	2.5Y 4/1
VB-KBIC19-6	1	SP	NP	NP	NP	0.6		26.1	2.59	2.5YR 4/1
	2	SM	NP	NP	NP	3.5		64.7	2.86	2.5YR 3/1
	3	SP-SM	NP	NP	NP	2.4		70.9	2.68	10Y 2.5/1
	4	SM	NP	NP	NP	2.2		40.4	2.80	5Y 3/1
VB-KBIC19-7	1	SM	NP	NP	NP	2.5		31.8	2.86	10Y 4/1
	2	MH	67	33	34	4.9		99.7	2.56	5Y 3/1
	3	SM	39	25	14	3.9		94.1	2.74	5Y 3/1
VB-KBIC19-8	1	MH	87	58	29	11.2		280.5	2.47	10Y 2.5/1
	2	MH	90	56	34	9.8		287.2	2.83	10Y 2.5/1
	3	MH	97	55	42	12.0		259.3	2.84	10Y 2.5/1
VB-KBIC19-9	1	MH	84	49	35	8.4		273.9	2.54	5Y 3/1
	2	MH	85	49	36	9.5		268.7	2.78	5Y 3/1
	3	MH	78	47	31	9.4		263.3	2.75	5Y 3/1
VB-KBIC19-10	1	MH	105	55	50	8.7		266.9	2.49	5Y 3/1
	2	MH	81	49	32	8.1		238.5	2.80	5Y 3/1
	3	MH	101	50	51	10.8		246.8	2.74	5Y 3/1
VB-KBIC19-11	1	MH	108	58	50	12.3		327.9	2.51	5G 2.5/1
	2	MH	110	59	51	14.8		296.7	2.65	5G 2.5/1
	3	MH	112	66	46	12.7		253.5	2.76	5G 2.5/1

Boring Designation	Sample Designation	USCS	LL	PL	PI	Org %	Visual Shell %	w_n (%)	G_S	Munsell Color
VB-KBIC19-12	1	MH	102	55	47	13.8		260.4	2.47	5G 2.5/1
	2	MH	101	54	47	13.0		264.7	2.77	5G 2.5/1
	3	MH	79	39	40	8.0		207.8	2.77	5G 2.5/1
VB-KBIC19-13	1	MH	97	49	48	5.8		150.7	2.53	10Y 2.5/1
	2	MH	97	52	45	11.0		274.1	2.56	10Y 2.5/1
	3	SM	93	46	47	8.6		158.7	2.67	5Y 3/1
VB-KBIC19-14	1	MH	98	46	52	7.6		231.4	2.54	5Y 2.5/1
	2	MH	105	55	50	10.9		241.7	2.66	5Y 2.5/1
	3	MH	107	52	55	13.1		242.1	2.64	5Y 3/2
VB-KBIC19-15	1	MH	81	37	44	9.3		250.4	2.51	10Y 3/1
	2	MH	104	57	47	9.2		234.0	2.74	10Y 3/1
VB-KBIC19-16	1	SC-SM	28	21	7	0.9		57.5	2.58	10Y 3/1
	2	GM	22	19	3	0.8		34.8	2.70	5Y 4/1
	3	SM	NP	NP	NP	0.5		28.3	2.67	2.5Y 7/1
USCS: Unified Soil Classification System; LL: Liquid Limit; PL: Plastic Limit; PI: Plastic Index; Org %: Organic Content; CO_3%: Carbonate Content; w_n: Moisture Content; G_s: Specific Gravity										

4.3 SUMMARY OF ADDITIONAL LABORATORY TESTING DATA

Sedimentation rate curves were obtained for this data set. Testing results are summarized in the table below. Test data sheets are included in paragraph 4.7.

50 grams of moist specimen is placed into a 100-ml beaker, the tube is filled to the 100-cm mark with water and mixed thoroughly. The sediment height in the tube is recorded in intervals for 24 hours as it is settling in the tube.

Table 3. Available Sedimentation Rate Data

Boring Number	Sample #	USCS	Sedimentation Rate	
			Remaining Settling Height (cm)	Remaining Concentration (g/L)
VB-KBIC16-02	1	MH	26.2	381.68
VB-KBIC16-03	2	SM	19.2	520.83
VB-KBIC19-3	1	SM	6.5	769
	2	MH	5.0	1000
	3	SM	5.0	1000
VB-KBIC19-4	1	SM	5.0	1000
	2	SM	4.0	1250
	3	SM	4.0	1250
	4	MH	5.5	909

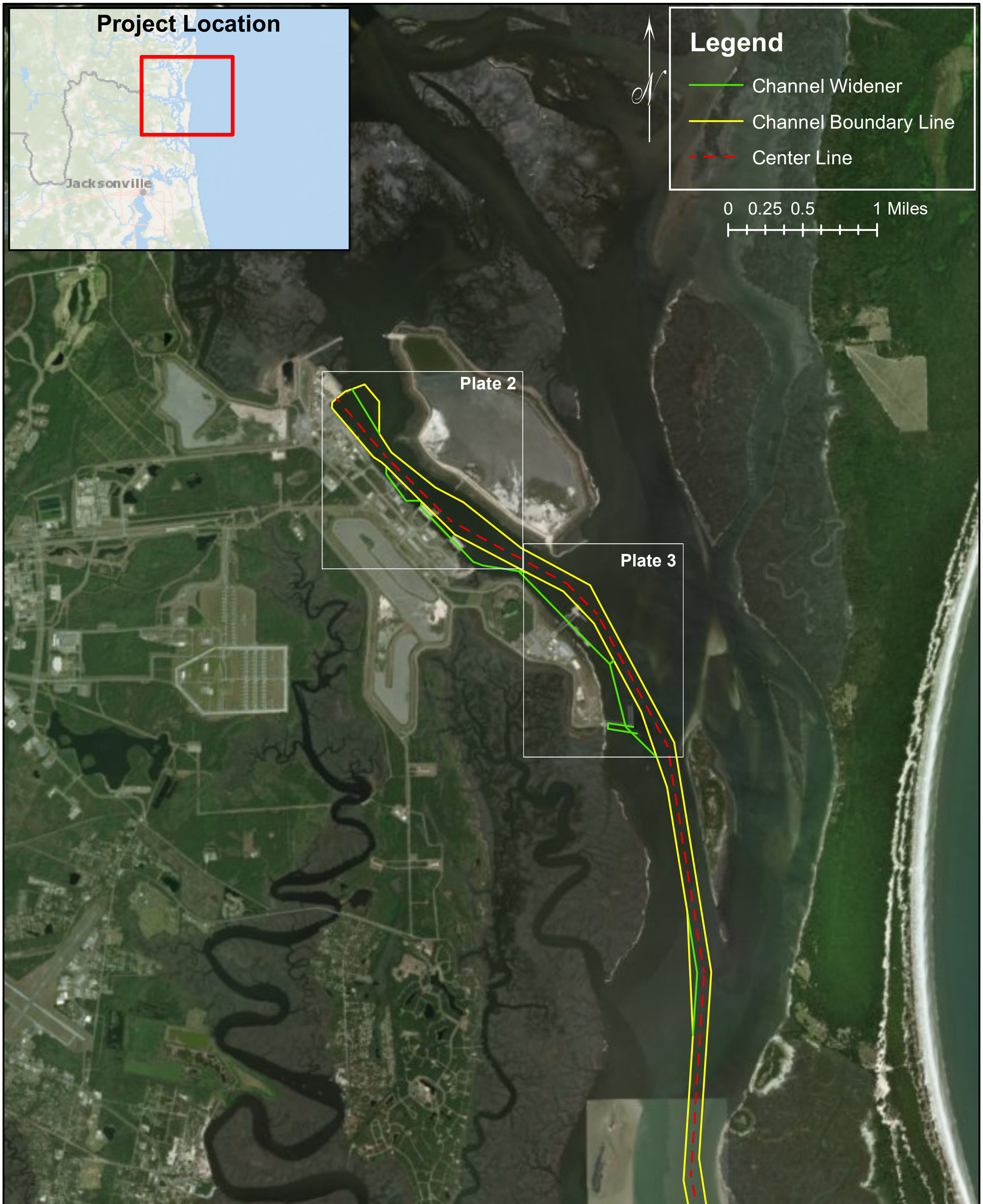
Boring Number	Sample #	USCS	Sedimentation Rate	
			Remaining Settling Height (cm)	Remaining Concentration (g/L)
VB-KBIC19-5	1	SP-SM	2.5	2000
	2	SP-SM	2.2	2273
VB-KBIC19-6	1	SP	2.7	1852
	2	SM	4.0	1250
	3	SP-SM	3.5	1429
	4	SM	4.0	1250
VB-KBIC19-7	1	SM	4.0	1250
	2	MH	5.0	1000
	3	SM	5.0	1000
VB-KBIC19-8	1	MH	4.0	1250
	2	MH	6.5	769
	3	MH	6.5	769
VB-KBIC19-9	1	MH	4.0	1250
	2	MH	6.0	833
	3	MH	7.0	714
VB-KBIC19-10	1	MH	5.5	909
	2	MH	6.5	769
	3	MH	6.5	769
VB-KBIC19-11	1	MH	5.5	909
	2	MH	6.0	833
	3	MH	8.5	588
VB-KBIC19-12	1	MH	5.5	909
	2	MH	5.5	909
	3	MH	6.5	769
VB-KBIC19-13	1	MH	6.0	833
	2	MH	15.0	333
	3	SM	5.5	909
VB-KBIC19-14	1	MH	5.5	909
	2	MH	7.0	714
	3	MH	6.5	769
VB-KBIC19-15	1	MH	5.0	1000
	2	MH	5.5	909
VB-KBIC19-16	1	SC-SM	3.5	1429
	2	GM	6.0	833
	3	SM	3.0	1667

4.4 BORING LOG NOTES

All borings were sampled using an Alpine style, pneumatic vertical hammer, vibracore drilling apparatus having a continuous 20-foot by 3 or 3 1/2-inch core barrel with clear plastic core liner until refusal was encountered. Refusal is defined as no additional penetration of the vibracore with two minutes of additional effort applied to penetrate through the resistant layer at maximum vibration.

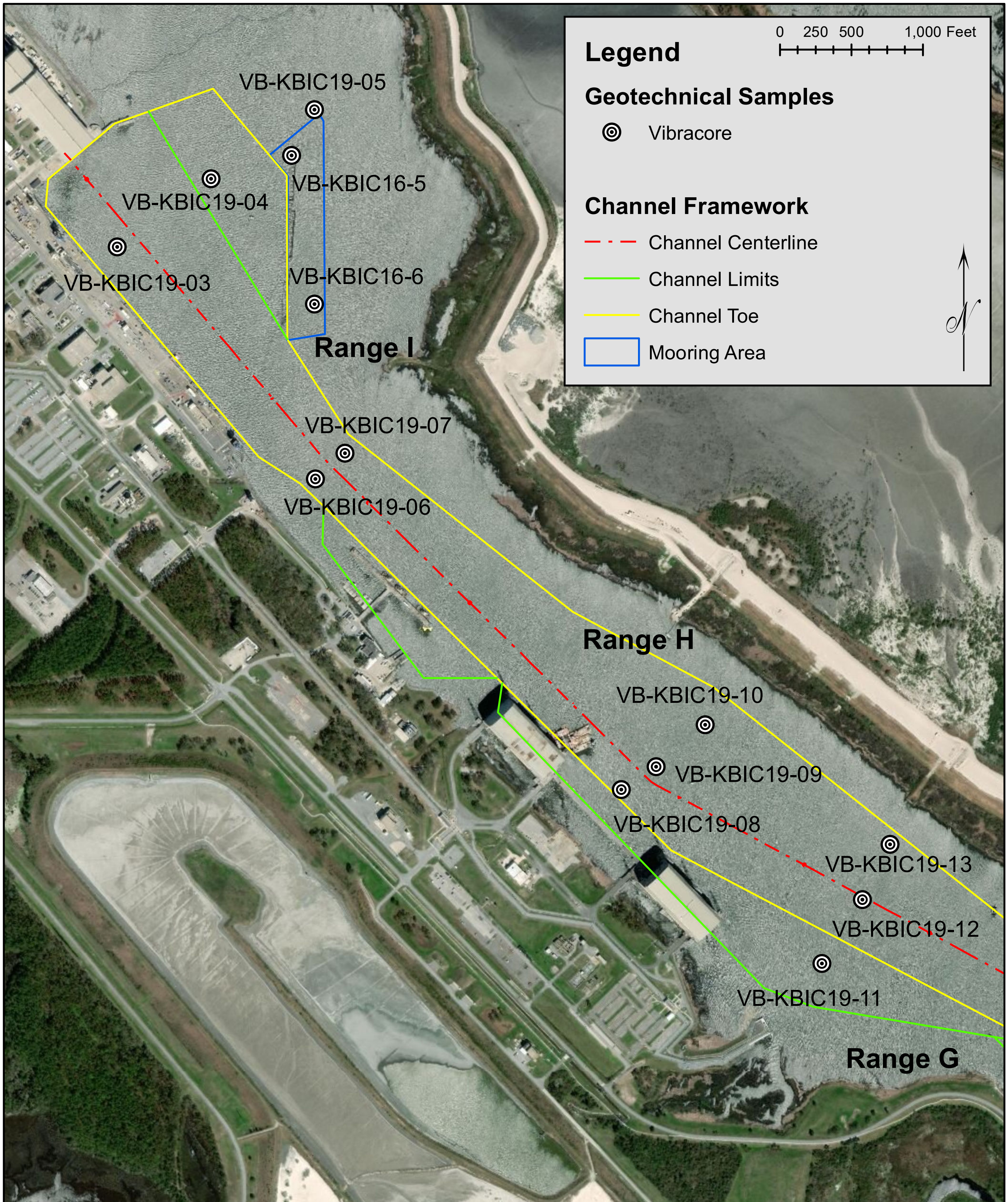
4.5 INSPECTION OF RECOVERED MATERIALS

The Material recovered from borings are not available for inspection.

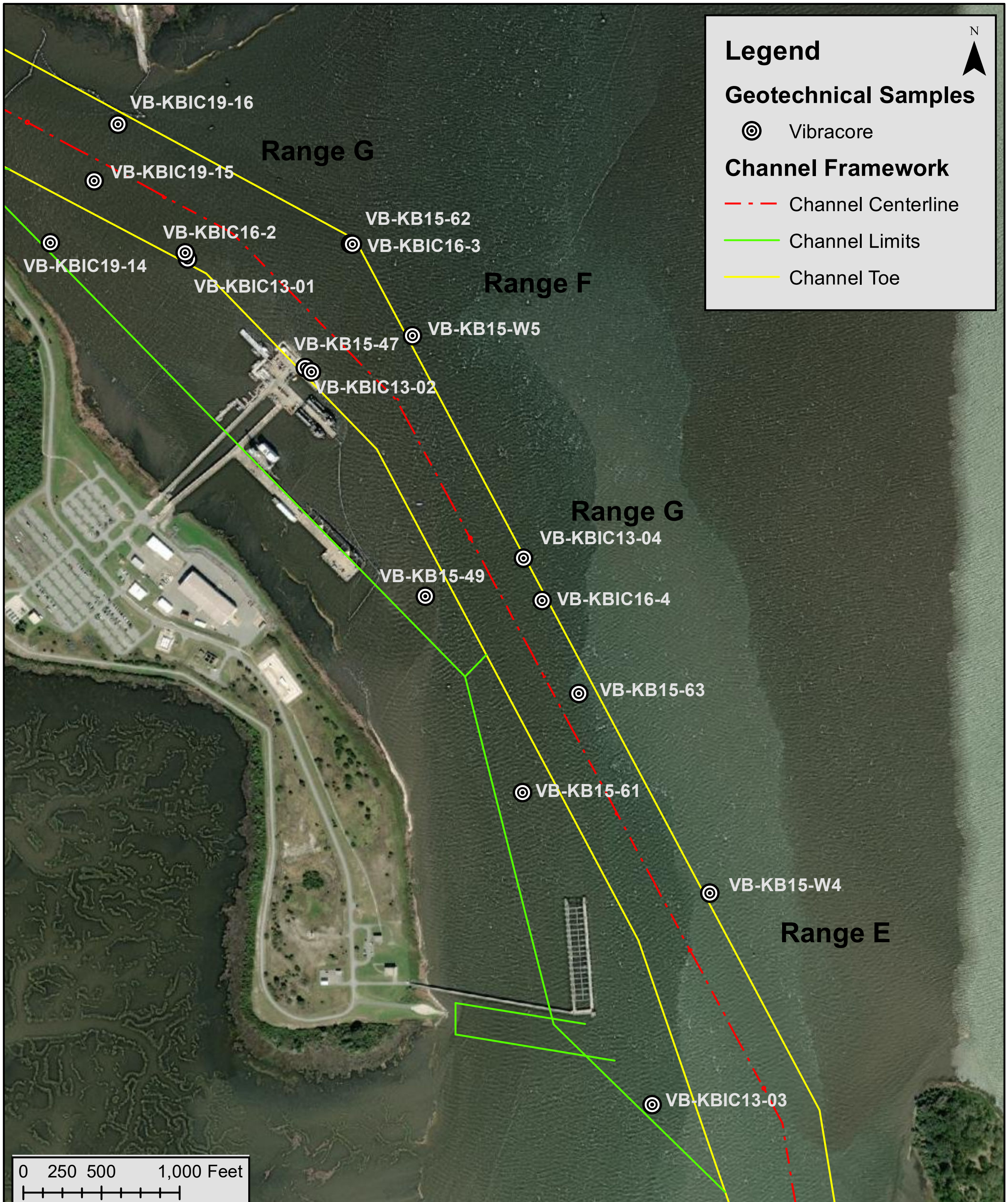


US Army Corps of Engineers®
Jacksonville District

Geotechnical Drawings	Dsn by: JLC	DEPARTMENT OF THE ARMY	Plate No.
	Dwn by: JLC	JACKSONVILLE DISTRICT, CORPS OF ENGINEERS JACKSONVILLE, FLORIDA	1
King's Bay Inner Channel O&M Geotechnical Data Report	Ckd by: BJS		
	Dated: July 2025		



US Army Corps of Engineers® Jacksonville District	Geotechnical Drawings	Dsn by: JLC	DEPARTMENT OF THE ARMY JACKSONVILLE DISTRICT, CORPS OF ENGINEERS JACKSONVILLE, FLORIDA	Plate No. 2
		Dwn by: JLC		
	Naval Submarine Base Kings Bay Maintenance Dredging	Ckd by: BJS		
		Dated: July 2025		



US Army Corps of Engineers® Jacksonville District	Geotechnical Drawings Naval Submarine Base Kings Bay Maintenance Dredging	Dsn by: JLC Dwn by: JLC Ckd by: BJS Dated: July 2025	DEPARTMENT OF THE ARMY JACKSONVILLE DISTRICT, CORPS OF ENGINEERS JACKSONVILLE, FLORIDA	Plate No. 3

4.6 BORING LOGS

Applicable boring logs are presented on the following pages.

The borings presented in this section are available historic borings within the project area. Most of the materials within the project template presented on the boring logs have been removed by previous dredging events with subsequent shoaling of similar materials. The current elevations of the materials at these locations are shown on the contract drawings.

While the Government's borings are representative of subsurface conditions at their respective locations and vertical reaches, local variations in the characteristics of the subsurface materials of this region are to be expected. Accordingly, prospective offerors shall form their own conclusions from the examination of the recovered materials prior to submission of their offer.

Boring Designation VB-KBIC13-01

DRILLING LOG	DIVISION: Jacksonville District	INSTALLATION:	SHEET 1 OF 1 SHEETS

1. PROJECT: Kings Bay 103 Evaluation, Contract #: W912EP-09-D-0013-0016	9. SIZE AND TYPE OF BIT: 3.0 In.
	10. COORDINATE SYSTEM/DATUM: Georgia State Plane East; HORIZONTAL: NAD 1983; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KBIC13-01; LOCATION COORDINATES: X = 865,566 Y = 288,492	11. MANUFACTURER'S DESIGNATION OF DRILL: ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: ; CONTRACTOR FILE NO.:	12. TOTAL SAMPLES: DISTURBED 2; UNDISTURBED (UD):
4. NAME OF DRILLER: P. McClellan	13. TOTAL NUMBER CORE BOXES:
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL: ; BEARING:	14. ELEVATION GROUND WATER: 37.1 Ft.
	15. DATE BORING: STARTED 04-30-13 10:30; COMPLETED 04-30-13 10:45
6. THICKNESS OF OVERBURDEN: 0.0 Ft.	16. ELEVATION TOP OF BORING: -33.6 Ft.
7. DEPTH DRILLED INTO ROCK: 0.0 Ft.	17. TOTAL RECOVERY FOR BORING: 12.25 Ft.
8. TOTAL DEPTH OF BORING: 18.0 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: W. Sexton

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS Depths and elevations based on measured values	% REC.	BOX OR SAMPLE	REMARKS
-33.6	0.0		CLAY, organic, little silt, trace fine-grained quartz sand, very soft, very dark grayish brown (2.5Y-3/2), (OH).		1	Sample #1, Depth = 0.8' Mean (mm): 0.01, Phi Sorting: 2.15 Shell: 0%, Fines (230): 94.88% (OH)
-37.2	3.6		CLAY, inorganic, little silt, trace fine-grained quartz sand, soft, very dark grayish brown (2.5Y-3/2), (CH).		2	Sample #2, Depth = 4.4' Mean (mm): 0.01, Phi Sorting: 1.97 Shell: 0%, Fines (230): 97.67% (CH)
-41.6	8.0		CLAY, little silt, soft, very dark grayish brown (2.5Y-3/2), (CL).			
-45.3	11.7					
-45.5	11.9		Silty fine to medium SAND, trace bioclastic, subround, light olive brown (2.5Y-5/3), (SM).			
-45.9	12.3		CLAY, soft, very dark grayish brown (2.5Y-3/2), (CL).			
			End of Boring			

Depth scale: 0, 5, 10, 15, 20

FLORIDA DEP ROSS KINGS BAY_FY13.GPJ FL DEP ROSS.GDT 8/12/14

SAJ FORM 1836
JUN 02

Boring Designation VB-KBIC13-02

DRILLING LOG	DIVISION Jacksonville District	INSTALLATION	SHEET 1 OF 1 SHEETS

1. PROJECT: Kings Bay 103 Evaluation, Contract #: W912EP-09-D-0013-0016	9. SIZE AND TYPE OF BIT: 3.0 In.
	10. COORDINATE SYSTEM/DATUM: Georgia State Plane East; HORIZONTAL: NAD 1983; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KBIC13-02; LOCATION COORDINATES: X = 866,321 Y = 287,795	11. MANUFACTURER'S DESIGNATION OF DRILL: ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: ; CONTRACTOR FILE NO.:	12. TOTAL SAMPLES: DISTURBED 2; UNDISTURBED (UD):
4. NAME OF DRILLER: P. McClellan	13. TOTAL NUMBER CORE BOXES:
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL: ; BEARING:	14. ELEVATION GROUND WATER: 42.3 Ft.
	15. DATE BORING: STARTED 04-30-13 11:30; COMPLETED 04-30-13 11:45
6. THICKNESS OF OVERBURDEN: 0.0 Ft.	16. ELEVATION TOP OF BORING: -37.4 Ft.
7. DEPTH DRILLED INTO ROCK: 0.0 Ft.	17. TOTAL RECOVERY FOR BORING: 12.33 Ft.
8. TOTAL DEPTH OF BORING: 14.0 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: W. Sexton

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS Depths and elevations based on measured values	% REC.	BOX OR SAMPLE	REMARKS
-37.4	0.0					
-40.4	3.0		CLAY, inorganic, some fine-grained quartz sand, little silt, trace shell, very soft, dark gray (2.5Y-4/1), (CL).		1	Sample #1, Depth = 0.8' Mean (mm): 0.05, Phi Sorting: 2.21 Shell: 4%, Fines (230): 53.33% (CL)
-47.2	9.8		CLAY, inorganic, little fine-grained quartz sand, little silt, soft, very dark grayish brown (2.5Y-3/2), (CH).		2	Sample #2, Depth = 3.6' Mean (mm): 0.03, Phi Sorting: 2.36 Shell: 2%, Fines (230): 77.10% (CL)
-48.4	11.0		CLAY, little coarse sand, trace fine sand/bioclastic, 10.2-10.6' = medium quartz sand, dark greenish gray (10G-4/1), (CL).			
-48.9	11.5		Silty medium quartz SAND, few coarse sand, trace bioclastic, subround, dark greenish gray (10G-4/1), (SM).			
-49.7	12.3		Medium quartz SAND, few silt, trace bioclastic, poorly graded, subround, pale olive (5Y-6/3), (SP).			
			End of Boring			

0
5
10
15

FLORIDA DEP ROSS KINGS BAY_FY13.GPJ FL DEP ROSS.GDT 8/12/14

Boring Designation VB-KBIC13-03

DRILLING LOG	DIVISION: Jacksonville District	INSTALLATION	SHEET 1 OF 1 SHEETS

1. PROJECT: Kings Bay 103 Evaluation, Contract #: W912EP-09-D-0013-0016	9. SIZE AND TYPE OF BIT: 3.0 In.
	10. COORDINATE SYSTEM/DATUM: Georgia State Plane East; HORIZONTAL: NAD 1983; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KBIC13-03; LOCATION COORDINATES: X = 868,545 Y = 283,068	11. MANUFACTURER'S DESIGNATION OF DRILL: ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY; CONTRACTOR FILE NO.	12. TOTAL SAMPLES: DISTURBED 2; UNDISTURBED (UD)
4. NAME OF DRILLER: P. McClellan	13. TOTAL NUMBER CORE BOXES
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL; BEARING	14. ELEVATION GROUND WATER: 37.6 Ft.
	15. DATE BORING: STARTED 04-30-13 12:10; COMPLETED 04-30-13 12:25
6. THICKNESS OF OVERBURDEN: 0.0 Ft.	16. ELEVATION TOP OF BORING: -31.9 Ft.
7. DEPTH DRILLED INTO ROCK: 0.0 Ft.	17. TOTAL RECOVERY FOR BORING: 13 Ft.
8. TOTAL DEPTH OF BORING: 16.0 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: W. Sexton

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS Depths and elevations based on measured values	% REC.	BOX OR SAMPLE	REMARKS
-31.9	0.0					
-33.7	1.8		Clayey fine quartz SAND, few silt, soft, very dark grayish brown (2.5Y-3/2), (SC).		1	Sample #1, Depth = 0.6' Mean (mm): 0.09, Phi Sorting: 1.32 Shell: 1%, Fines (230): 25.54% (SC)
-35.1	3.2		CLAY, inorganic, few fine sand, trace organics, silty fine sand lenses (SM) present, soft, dark olive brown (2.5Y-3/3), (CH).			
-44.7	12.8		CLAY, organic, little silt, trace fine-grained quartz sand, silty fine sand lenses (SM) present, soft, very dark grayish brown (10YR-3/2), (OH).		2	Sample #2, Depth = 3.9' Mean (mm): 0.01, Phi Sorting: 1.96 Shell: 0%, Fines (230): 95.01% (OH)
-44.9	13.0		Silty fine SAND, medium stiff, dark greenish gray (10G-4/1), (SM).			
			End of Boring			

Depth scale: 0, 5, 10, 15, 20

FLORIDA DEP ROSS KINGS BAY_FY13.GPJ FL DEP ROSS.GDT 8/12/14

SAJ FORM 1836
JUN 02

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Boring Designation VB-KBIC13-04

DRILLING LOG	**DIVISION** Jacksonville District	**INSTALLATION**	**SHEET 1 OF 1 SHEETS**
1. PROJECT Kings Bay 103 Evaluation, Contract #: W912EP-09-D-0013-0016		**9. SIZE AND TYPE OF BIT** 3.0 In.	
		10. COORDINATE SYSTEM/DATUM Georgia State Plane East	**HORIZONTAL** NAD 1983; **VERTICAL** MLLW
2. BORING DESIGNATION VB-KBIC13-04	**LOCATION COORDINATES** X = 867,721 Y = 286,573	**11. MANUFACTURER'S DESIGNATION OF DRILL**	☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY	**CONTRACTOR FILE NO.**	**12. TOTAL SAMPLES**	**DISTURBED** 3; **UNDISTURBED (UD)**
4. NAME OF DRILLER P. McClellan		**13. TOTAL NUMBER CORE BOXES**	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED	**DEG. FROM VERTICAL**; **BEARING**	**14. ELEVATION GROUND WATER** 42.1 Ft.	
		15. DATE BORING	**STARTED** 04-30-13 09:30; **COMPLETED** 04-30-13 09:45
6. THICKNESS OF OVERBURDEN 0.0 Ft.		**16. ELEVATION TOP OF BORING** -40.3 Ft.	
7. DEPTH DRILLED INTO ROCK 0.0 Ft.		**17. TOTAL RECOVERY FOR BORING** 6.75 Ft.	
8. TOTAL DEPTH OF BORING 6.8 Ft.		**18. SIGNATURE AND TITLE OF INSPECTOR** W. Sexton	

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS Depths and elevations based on measured values	% REC.	BOX OR SAMPLE	REMARKS
-40.3	0.0					
			Fine SAND, trace silt/heavy minerals/bioclastic, clay rip-ups/flaser beds present, poorly graded, subround, 4.2-4.3' = Clayey fine SAND (SC), pale yellow (2.5Y-7/3), (SP).		1	Sample #1, Depth = 0.6' Mean (mm): 0.18, Phi Sorting: 0.42 Shell: 5%, Fines (230): 1.37% (SP)
					2	Sample #2, Depth = 4.6' Mean (mm): 0.12, Phi Sorting: 1.24 Shell: 1%, Fines (230): 20.61% (SC)
					3	Sample #3, Depth = 5.4' Mean (mm): 0.16, Phi Sorting: 0.35 Shell: 2%, Fines (230): 1.15% (SP)
-46.6	6.3		Medium quartz SAND, few coarse sand, trace bioclastic, well graded, subround, pale yellow (2.5Y-8/2), (SW).			
-46.8	6.5		Cemented clastic (silt/sand) material, dark gray (2.5Y-4/1).			
-47.1	6.8		End of Boring			

0 5 10 15

FLORIDA DEP ROSS KINGS BAY_FY13.GPJ FL DEP ROSS.GDT 8/12/14

SAJ FORM 1836
JUN 02

Boring Designation VB-KB15-47

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 1 SHEETS

1. PROJECT: Kings Bay O&M, Inner and Outer Channel	9. SIZE AND TYPE OF BIT: See Remarks
2. BORING DESIGNATION: VB-KB15-47; LOCATION COORDINATES: X = 867,037 Y = 287,164	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
3. DRILLING AGENCY: Corps of Engineers - CESAJ; CONTRACTOR FILE NO.: 6738-15-5412	11. MANUFACTURER'S DESIGNATION OF DRILL: Vibracore; ☐ AUTO HAMMER ☐ MANUAL HAMMER
4. NAME OF DRILLER: Lester Gaughf	12. TOTAL SAMPLES: DISTURBED 0; UNDISTURBED (UD) 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL: ; BEARING:	13. TOTAL NUMBER CORE BOXES: 0
6. THICKNESS OF OVERBURDEN: N/A	14. ELEVATION GROUND WATER:
7. DEPTH DRILLED INTO ROCK: N/A	15. DATE BORING: STARTED 11-24-14; COMPLETED 11-24-14
8. TOTAL DEPTH OF BORING: 10.0 Ft.	16. ELEVATION TOP OF BORING: -45.0 Ft.
	17. TOTAL RECOVERY FOR BORING: 66 %
	18. SIGNATURE AND TITLE OF INSPECTOR: Stephanie A. Setser, P.E., Geotechnical Engineer

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-45.0	0.0						-45.0		
-48.0	3.0		SAND, silty, mostly fine-grained sand-sized quartz, little silt, weak reaction with HCl, intermittent silt seams, 5Y 4/2 olive gray (SM)	66			Vibracore		
-51.6	6.6		SILT, inorganic-L, few fine-grained sand-sized quartz, trace limestone, strong reaction with HCl, (possible residual soil), 10Y 6/1 greenish gray (ML) At El. -48.9 Ft., few quartz, 10G 3/1 very dark greenish gray At El. -50.0 Ft., 10Y 6/2 light grayish olive						
-55.0	10.0	NO RECOVERY					-55.0		

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

SAJ FORM 1836
JUN 02

Boring Designation VB-KB15-49

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 1 SHEETS

1. PROJECT: Kings Bay O&M, Inner and Outer Channel	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KB15-49; LOCATION COORDINATES: X = 866,204 Y = 285,719	11. MANUFACTURER'S DESIGNATION OF DRILL: Vibracore; ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Corps of Engineers - CESAJ; CONTRACTOR FILE NO.: 6738-15-5412	12. TOTAL SAMPLES: DISTURBED 0; UNDISTURBED (UD) 0
4. NAME OF DRILLER: Lester Gaughf	13. TOTAL NUMBER CORE BOXES: 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL: ; BEARING:	14. ELEVATION GROUND WATER:
	15. DATE BORING: STARTED 11-24-14; COMPLETED 11-24-14
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: -48.7 Ft.
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 90 %
8. TOTAL DEPTH OF BORING: 10.0 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: Stephanie A. Setser, P.E., Geotechnical Engineer

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-48.7	0.0						-48.7		
-52.0	3.3		SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, trace limestone, trace shell, weak reaction with HCl, 5Y 4/2 olive gray (SP-SM) At El. -51.0 Ft., few coarse gravel-sized limestone	90			Vibracore		
-57.7	9.0		SILT, inorganic-H, little fine-grained sand-sized quartz, trace limestone, strong reaction with HCl, cemented seams, possible residual soil, 10Y 6/1 greenish gray (MH) At El. -56.9 Ft., little fine-grained sand-sized quartz						
-58.7	10.0	NR					-58.7		

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

Scale: 0, 5, 10, 15

SAJ FORM 1836
JUN 02

Boring Designation VB-KB15-61

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 1 SHEETS

1. PROJECT: Kings Bay O&M, Inner and Outer Channel	9. SIZE AND TYPE OF BIT: See Remarks
2. BORING DESIGNATION: VB-KB15-61; LOCATION COORDINATES: X = 867,730 Y = 285,079	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
3. DRILLING AGENCY: Corps of Engineers - CESAJ; CONTRACTOR FILE NO.: 6738-15-5412	11. MANUFACTURER'S DESIGNATION OF DRILL: Vibracore; ☐ AUTO HAMMER ☐ MANUAL HAMMER
4. NAME OF DRILLER: Lester Gaughf	12. TOTAL SAMPLES: DISTURBED 0; UNDISTURBED (UD) 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL: ; BEARING:	13. TOTAL NUMBER CORE BOXES: 0
6. THICKNESS OF OVERBURDEN: N/A	14. ELEVATION GROUND WATER:
7. DEPTH DRILLED INTO ROCK: N/A	15. DATE BORING: STARTED 11-24-14; COMPLETED 11-24-14
8. TOTAL DEPTH OF BORING: 10.0 Ft.	16. ELEVATION TOP OF BORING: -45.7 Ft.
	17. TOTAL RECOVERY FOR BORING: 79 %
	18. SIGNATURE AND TITLE OF INSPECTOR: Stephanie A. Setser, P.E., Geotechnical Engineer

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-45.7	0.0						-45.7		
-47.7	2.0		SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, few silt, weak reaction with HCl, 5Y 4/1 dark gray (SP-SM) From El. -46.7 to -47.7 Ft., intermittent silt seams	79			Vibracore		
-53.6	7.9		SILT, inorganic-H, some fine to medium-grained sand-sized limestone, few fine-grained sand-sized quartz, weak reaction with HCl, N 5/ gray (MH) At El. -48.8 Ft., few coarse gravel-sized limestone At El. -50.4 Ft., trace limestone At El. -53.2 Ft., few fine gravel-sized limestone						
-55.7	10.0	NR					-55.7		

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

0 5 10 15

SAJ FORM 1836
JUN 02

Boring Designation VB-KB15-62

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 1 SHEETS

1. PROJECT: Kings Bay O&M Inner and Outer Channel	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KB15-62; LOCATION COORDINATES: X = 866,956 Y = 287,936	11. MANUFACTURER'S DESIGNATION OF DRILL: Vibracore; ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Corps of Engineers - CESAJ; CONTRACTOR FILE NO.: 6738-15-5412	12. TOTAL SAMPLES: DISTURBED 0; UNDISTURBED (UD) 0
4. NAME OF DRILLER: Lester Gaughf	13. TOTAL NUMBER CORE BOXES: 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL; BEARING	14. ELEVATION GROUND WATER
	15. DATE BORING: STARTED 11-24-14; COMPLETED 11-24-14
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: -43.8 Ft.
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 100 %
8. TOTAL DEPTH OF BORING: 10.0 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: Stephanie A. Setser, P.E., Geotechnical Engineer

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-43.8	0.0						-43.8		
-44.0	0.2		SILT, inorganic-H, few fine-grained sand-sized quartz, strong reaction with HCl, 10Y 3/1 very dark greenish gray (MH)						
-44.6	0.8		SAND, silty, mostly fine to medium-grained sand-sized quartz, some silt, trace limestone, strong reaction with HCl, 10Y 3/1 very dark greenish gray (SM) At El. -44.1 Ft., 1" diameter limestone At El. -44.4 Ft., 1" diameter limestone						
			SILT, inorganic-L, some fine-grained sand-sized quartz, trace limestone, strong reaction with HCl, 10Y 3/1 very dark greenish gray (ML)	100			Vibracore		
			At El. -50.7 Ft., little fine-grained sand-sized quartz, trace limestone, cemented seams, possible residual soil,, 5Y 4/2 olive gray						
-53.8	10.0						-53.8		

Scale: 0, 5, 10, 15

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

Boring Designation VB-KB15-63

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 1 SHEETS

1. PROJECT: Kings Bay O&M, Inner and Outer Channel	9. SIZE AND TYPE OF BIT: See Remarks
2. BORING DESIGNATION: VB-KB15-63 / LOCATION COORDINATES: X = 868,084 Y = 285,718	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.) / HORIZONTAL: NAD83 / VERTICAL: MLLW
3. DRILLING AGENCY: Corps of Engineers - CESAJ / CONTRACTOR FILE NO.: 6738-15-5412	11. MANUFACTURER'S DESIGNATION OF DRILL: Vibracore / ☐ AUTO HAMMER ☐ MANUAL HAMMER
4. NAME OF DRILLER: Lester Gaughf	12. TOTAL SAMPLES: DISTURBED 0 / UNDISTURBED (UD) 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED / DEG. FROM VERTICAL: / BEARING:	13. TOTAL NUMBER CORE BOXES: 0
6. THICKNESS OF OVERBURDEN: N/A	14. ELEVATION GROUND WATER:
7. DEPTH DRILLED INTO ROCK: N/A	15. DATE BORING: STARTED 11-24-14 / COMPLETED 11-24-14
8. TOTAL DEPTH OF BORING: 10.0 Ft.	16. ELEVATION TOP OF BORING: -43.6 Ft.
	17. TOTAL RECOVERY FOR BORING: 100 %
	18. SIGNATURE AND TITLE OF INSPECTOR: Stephanie A. Setser, P.E., Geotechnical Engineer

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-43.6	0.0						-43.6		
			SAND, poorly-graded with clay, mostly fine-grained sand-sized quartz, few clay, no reaction with HCl, 5Y 4/1 dark gray (SP-SC)	100			Vibracore		
			At El. -48.8 Ft., little sand to gravel-sized limestone, strong reaction with HCl, N 4/ dark gray						
-49.9	6.3		SILT, inorganic-L, little sand to gravel-sized limestone, few fine-grained sand-sized quartz, strong reaction with HCl, possible residual soil, N 4/ dark gray (ML)						
-53.6	10.0						-53.6		

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

SAJ FORM 1836
JUN 02

Boring Designation VB-KB15-W4

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 2 SHEETS

1. PROJECT: Kings Bay O&M Entrance Channel	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KB15-W4; LOCATION COORDINATES: X = 869,447 Y = 283,341	11. MANUFACTURER'S DESIGNATION OF DRILL: Vibracore; ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Corps of Engineers - CESAJ; CONTRACTOR FILE NO.: 6738-15-5412	12. TOTAL SAMPLES: DISTURBED 5; UNDISTURBED (UD) 0
4. NAME OF DRILLER: Lester Gaughf	13. TOTAL NUMBER CORE BOXES: 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL; BEARING	14. ELEVATION GROUND WATER
	15. DATE BORING: STARTED 11-23-14; COMPLETED 11-23-14
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: -43.7 Ft.
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 100 %
8. TOTAL DEPTH OF BORING: 20.0 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: Stephanie A. Setser, P.E., Geotechnical Engineer

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-43.7	0.0						-43.7		
			SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, no reaction with HCl, 5Y 7/2 light gray (SP-SM)	290	1		Vibracore		
			From El. -49.9 to -50.6 Ft., weak reaction with HCl, intermittent silt seams				-50.6		
			At El. -50.6 Ft., no reaction with HCl, 5Y 5/2 olive gray		2				
-51.6	7.9		SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace silt, trace shell, no reaction with HCl, 5Y 7/2 light gray (SP)		3		-51.6		
			At El. -52.6 Ft., mostly fine to medium-grained sand-sized quartz, few medium-grained sand-sized shell						
-54.0	10.3		SAND, silty, mostly fine to medium-grained sand-sized quartz, some silt, strong reaction with HCl, possible residual soil, 10Y 7/1 light greenish gray (SM)		4		-54.0		
			From El. -54.8 to -55.1 Ft., silt seams						
			From El. -56.3 to -56.6 Ft., silt seams						

0
5
10
15

SAJ FORM 1836
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(Continued)

Boring Designation VB-KB15-W4

DRILLING LOG (Cont. Sheet)	INSTALLATION: Jacksonville District		SHEET 2 OF 2 SHEETS
PROJECT: Kings Bay O&M	COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.)	HORIZONTAL: NAD83	VERTICAL: MLLW
LOCATION COORDINATES: X = 869,447 Y = 283,341	ELEVATION TOP OF BORING: -43.7 Ft.		

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
			From El. -59.2 to -61.6 Ft., occasional cemented seams		4				
-61.6	17.9						-61.6		
			SILT, inorganic-L, few fine-grained sand-sized quartz, strong reaction with HCl, cemented seams, possible residual soil, 10Y 7/1 light greenish gray (ML)		5				
-63.7	20.0						-63.7		

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

3. Laboratory Testing Results

SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION
1	0.0/2.3	SP-SM*

*Lab visual classification based on gradation curve

4. Additional Laboratory Testing

1 Percent Visual Shell

Abbreviations:

15
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SAJ FORM 1836-A
JUN 02

Boring Designation VB-KB15-W5

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 2 SHEETS

1. PROJECT: Kings Bay O&M Inner and Outer Channel	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KB15-W5; LOCATION COORDINATES: X = 866,993 Y = 287,987	11. MANUFACTURER'S DESIGNATION OF DRILL: Vibracore; ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Corps of Engineers - CESAJ; CONTRACTOR FILE NO.: 6738-15-5412	12. TOTAL SAMPLES: DISTURBED 0; UNDISTURBED (UD) 0
4. NAME OF DRILLER: Lester Gaughf	13. TOTAL NUMBER CORE BOXES: 0
	14. ELEVATION GROUND WATER
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL; BEARING	15. DATE BORING: STARTED 11-23-14; COMPLETED 11-23-14
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: -45.6 Ft.
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 100 %
8. TOTAL DEPTH OF BORING: 20.0 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: Stephanie A. Setser, P.E., Geotechnical Engineer

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-45.6	0.0						-45.6		
			SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, few silt, weak reaction with HCl, 5Y 6/2 light olive gray (SP-SM)						
			From El. -47.7 to -47.9 Ft., silty sand seams						
			From El. -52.8 to -53.4 Ft., 1" diameter sandy silt nodules	100			Vibracore		
			From El. -55.6 to -56.3 Ft., intermittent silt seams						

0
5
10
15

SAJ FORM 1836
JUN 02

(Continued)

Boring Designation VB-KB15-W5

DRILLING LOG (Cont. Sheet)	INSTALLATION: Jacksonville District		SHEET 2 OF 2 SHEETS
PROJECT: Kings Bay O&M	COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.)	HORIZONTAL: NAD83	VERTICAL: MLLW
LOCATION COORDINATES: X = 866,993 Y = 287,987	ELEVATION TOP OF BORING: -45.6 Ft.		

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
			From El. -60.6 to -60.9 Ft., intermittent silt seams						
-61.9	16.3		From El. -61.7 to -61.9 Ft., silt seam						
			SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, few fine-grained sand-sized shell, weak reaction with HCl, 5Y 6/2 light olive gray (SP)	100			Vibracore		
			From El. -62.3 to -62.6 Ft., intermittent silt seams						
-63.8	18.2		At El. -62.9 Ft., little fine to medium-grained sand-sized shell, strong reaction with HCl						
		Mod. Weathered	LIMESTONE, hard, moderately weathered, medium grained, 5Y 6/2 light olive gray						
-65.6	20.0						-65.6		

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

Depth scale: 15, 20, 25, 30, 35

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JUN 02

Boring Designation VB-KBIC16-2

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 2 SHEETS

1. PROJECT: King's Bay Inner Channel	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KBIC16-2; LOCATION COORDINATES: X = 865,550 Y = 288,532	11. MANUFACTURER'S DESIGNATION OF DRILL: Snell Vibracore; ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Corps of Engineers - CESAW; CONTRACTOR FILE NO.	12. TOTAL SAMPLES: DISTURBED 2; UNDISTURBED (UD) 0
4. NAME OF DRILLER: Snell	13. TOTAL NUMBER CORE BOXES: 2
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL; BEARING	14. ELEVATION GROUND WATER
	15. DATE BORING: STARTED 07-10-16; COMPLETED 07-10-16
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: -40.2 Ft.
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 76 %
8. TOTAL DEPTH OF BORING: 9.6 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: ,

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-40.2	0.0		SILT, inorganic-H, few fine-grained sand-sized quartz, weak reaction with HCl, 2.5Y 3/1 very dark gray (MH)						
					1		-44.2		
-49.1	8.9		GRAVEL, poorly-graded with silt, mostly sand to gravel-sized limestone, few silt, trace silt, strong reaction with HCl, 2.5Y 4/3 olive brown (GP-GM)		2		-49.5 -49.8		
-49.8	9.6						Abbreviations:		

Depth scale: 0, 5, 10, 15

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

3. Laboratory Testing Results

SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION
1	4.0/4.2	MH*

SAJ FORM 1836
JUN 02

(Continued)

Boring Designation VB-KBIC16-2

DRILLING LOG (Cont. Sheet)	INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS
PROJECT King's Bay Inner Channel	COORDINATE SYSTEM/DATUM State Plane, GAE (U.S. Ft.)	HORIZONTAL NAD83	VERTICAL MLLW	
LOCATION COORDINATES X = 865,550 Y = 288,532	ELEVATION TOP OF BORING -40.2 Ft.			

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
	15		2 9.3/9.5 GP-GM* *Lab visual classification based on gradation curve SAMPLE DEPTH / LABORATORY SOIL TESTING / RESULT / UNIT 4.0 Water Content 30 % 4.0 Unit Weight / pcf						
	20								
	25								
	30								
	35								

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JUN 02

Boring Designation VB-KBIC16-3

DRILLING LOG | DIVISION: South Atlantic | INSTALLATION: Jacksonville District | SHEET 1 OF 2 SHEETS

1. PROJECT: King's Bay Inner Channel	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KBIC16-3; LOCATION COORDINATES: X = 866,624 Y = 288,586	11. MANUFACTURER'S DESIGNATION OF DRILL: Snell Vibracore; ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Corps of Engineers - CESAW; CONTRACTOR FILE NO.:	12. TOTAL SAMPLES: DISTURBED 2; UNDISTURBED (UD) 0
4. NAME OF DRILLER: Snell	13. TOTAL NUMBER CORE BOXES: 2
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL:; BEARING:	14. ELEVATION GROUND WATER:
	15. DATE BORING: STARTED 07-10-16; COMPLETED 07-10-16
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: -39.2 Ft.
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 82 %
8. TOTAL DEPTH OF BORING: 7.4 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: ,

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-39.2	0.0								
			SAND, silty, mostly fine-grained sand-sized quartz, little silt, trace organic matter, weak reaction with HCl, 2.5Y 4/1 dark gray (SM)		1		-39.6		
-40.0	0.8		SAND, silty, mostly fine-grained sand-sized quartz, little silt, weak reaction with HCl, intermittent slightly silty sand seams 1.7'-4.0', 2.5Y 4/1 dark gray (SM)						
-42.3	3.1		SAND, silty, mostly fine-grained sand-sized quartz, little silt, trace organic matter, weak reaction with HCl, 2.5Y 3/1 very dark gray (SM)		2		-42.3		
-42.8	3.6		SAND, silty, mostly fine-grained sand-sized quartz, little silt, weak reaction with HCl, intermittent slightly silty sand seams 4.0'-8.3', 2.5Y 4/1 dark gray (SM)						
-46.6	7.4						-46.6		

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

3. Laboratory Testing Results

SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION
1	0.4/0.6	SM*
2	3.1/3.3	SM*

*Lab visual classification based on gradation curve

SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT	UNIT
3.1	Water Content	9	%

Abbreviations:

0 5 10 15

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JUN 02

(Continued)

Boring Designation VB-KBIC16-3

DRILLING LOG (Cont. Sheet)	INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS
PROJECT King's Bay Inner Channel	COORDINATE SYSTEM/DATUM State Plane, GAE (U.S. Ft.)	HORIZONTAL NAD83	VERTICAL MLLW	
LOCATION COORDINATES X = 866,624 Y = 288,586	ELEVATION TOP OF BORING -39.2 Ft.			

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
	15		3.1 Unit Weight / pcf						
	20								
	25								
	30								
	35								

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Boring Designation VB-KBIC16-4

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 2 SHEETS

1. PROJECT: King's Bay Inner Channel	9. SIZE AND TYPE OF BIT: See Remarks
2. BORING DESIGNATION: VB-KBIC16-4; LOCATION COORDINATES: X = 867,840 Y = 286,302	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
3. DRILLING AGENCY: Corps of Engineers - CESAW; CONTRACTOR FILE NO.:	11. MANUFACTURER'S DESIGNATION OF DRILL: Snell Vibracore; ☐ AUTO HAMMER ☐ MANUAL HAMMER
4. NAME OF DRILLER: Snell	12. TOTAL SAMPLES: DISTURBED 4; UNDISTURBED (UD) 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL: ; BEARING:	13. TOTAL NUMBER CORE BOXES: 1
6. THICKNESS OF OVERBURDEN: N/A	14. ELEVATION GROUND WATER:
7. DEPTH DRILLED INTO ROCK: N/A	15. DATE BORING: STARTED 07-10-16; COMPLETED 07-10-16
8. TOTAL DEPTH OF BORING: 8.5 Ft.	16. ELEVATION TOP OF BORING: -39.6 Ft.
	17. TOTAL RECOVERY FOR BORING: 100 %
	18. SIGNATURE AND TITLE OF INSPECTOR: ,

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-39.6	0.0								
			SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace silt, no reaction with HCl, 2.5Y 7/2 light gray (SP)		1		-41.4		
-43.9	4.3		SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, few silt, weak reaction with HCl, 2.5Y 5/1 gray (SP-SM)		2		-44.1		
-45.6	6.0		SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, few medium-grained sand-sized limestone, trace silt, trace shell, weak reaction with HCl, SAND seam 6.7'-7.0', 2.5Y 5/1 gray (SP)		3		-46.1		
-46.6	7.0								
-47.4	7.8		SAND, silty, mostly fine-grained sand-sized quartz, little silt, weak reaction with HCl, 2.5Y 4/1 dark gray (SM)		4		-47.6		
-48.1	8.5		GRAVEL, poorly-graded, mostly sand to gravel-sized limestone, trace silt, trace quartz, strong reaction with HCl, 2.5Y 4/1 dark gray (GP)				-48.1 Abbreviations:		

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

3. Laboratory Testing Results

SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION
1	1.8/2.0	SP*
2	4.5/4.7	SP-SM*
3	6.5/6.7	SP*

0 5 10 15

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(Continued)

Boring Designation VB-KBIC16-4

DRILLING LOG (Cont. Sheet)	INSTALLATION Jacksonville District		SHEET 2 OF 2 SHEETS
PROJECT King's Bay Inner Channel	COORDINATE SYSTEM/DATUM State Plane, GAE (U.S. Ft.)	HORIZONTAL NAD83	VERTICAL MLLW
LOCATION COORDINATES X = 867,840 Y = 286,302	ELEVATION TOP OF BORING -39.6 Ft.		

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
	15		4 8.0/8.2 GP* *Lab visual classification based on gradation curve						
	20								
	25								
	30								
	35								

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Boring Designation VB-KBIC16-5

DRILLING LOG	**DIVISION** South Atlantic	**INSTALLATION** Jacksonville District	**SHEET** 1 **OF** 2 **SHEETS**

1. PROJECT King's Bay Inner Channel	**9. SIZE AND TYPE OF BIT** See Remarks
	10. COORDINATE SYSTEM/DATUM State Plane, GAE (U.S. Ft.) — **HORIZONTAL** NAD83 — **VERTICAL** MLLW
2. BORING DESIGNATION VB-KBIC16-5 — **LOCATION COORDINATES** X = 859,193 Y = 295,286	**11. MANUFACTURER'S DESIGNATION OF DRILL** Snell Vibracore — ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY Corps of Engineers - CESAW — **CONTRACTOR FILE NO.**	**12. TOTAL SAMPLES** — **DISTURBED** 2 — **UNDISTURBED (UD)** 0
4. NAME OF DRILLER Snell	**13. TOTAL NUMBER CORE BOXES** 1
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED — **DEG. FROM VERTICAL** — **BEARING**	**14. ELEVATION GROUND WATER**
	15. DATE BORING — **STARTED** 07-10-16 — **COMPLETED** 07-10-16
6. THICKNESS OF OVERBURDEN N/A	**16. ELEVATION TOP OF BORING** -5.4 Ft.
7. DEPTH DRILLED INTO ROCK N/A	**17. TOTAL RECOVERY FOR BORING** 50 %
8. TOTAL DEPTH OF BORING 10.0 Ft.	**18. SIGNATURE AND TITLE OF INSPECTOR** ,

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-5.4	0.0		SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, trace fine to medium-grained sand-sized shell, trace silt, strong reaction with HCl, 2.5Y 7/2 light gray (SP)						
					1		-8.4		
-12.2	6.8		SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, trace fine to medium-grained sand-sized shell, trace silt, strong reaction with HCl, Thin silt seams (Length:1"-2") @ 7.3' and 8.1', 2.5Y 7/2 light gray (SP)						
					2		-13.2		
-15.4	10.0						-15.4		

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

3. Laboratory Testing Results

SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION
1	3.0/3.2	SP*

Abbreviations:

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JUN 02

(Continued)

Boring Designation VB-KBIC16-5

DRILLING LOG (Cont. Sheet)	**INSTALLATION** Jacksonville District		**SHEET** 2 **OF** 2 **SHEETS**
PROJECT King's Bay Inner Channel	**COORDINATE SYSTEM/DATUM** State Plane, GAE (U.S. Ft.)	**HORIZONTAL** NAD83	**VERTICAL** MLLW
LOCATION COORDINATES X = 859,193 Y = 295,286	**ELEVATION TOP OF BORING** -5.4 Ft.		

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
	15		2 7.8/8.0 SP*						
			*Lab visual classification based on gradation curve						
	20								
	25								
	30								
	35								

Boring Designation VB-KBIC16-6

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 2 SHEETS

1. PROJECT: King's Bay Inner Channel	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KBIC16-6; LOCATION COORDINATES: X = 859,353 Y = 294,247	11. MANUFACTURER'S DESIGNATION OF DRILL: Snell Vibracore; ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Corps of Engineers - CESAW; CONTRACTOR FILE NO.:	12. TOTAL SAMPLES: DISTURBED 3; UNDISTURBED (UD) 0
4. NAME OF DRILLER: Snell	13. TOTAL NUMBER CORE BOXES: 1
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL: ; BEARING:	14. ELEVATION GROUND WATER:
	15. DATE BORING: STARTED 07-10-16; COMPLETED 07-10-16
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: -3.8 Ft.
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 70 %
8. TOTAL DEPTH OF BORING: 13.9 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: ,

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-3.8	0.0								
-5.0	1.2		SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, no reaction with HCl, 2.5Y 5/1 gray (SP-SM)		1		-4.9		
			SAND, silty, mostly fine to medium-grained sand-sized quartz, little silt, trace organic matter, weak reaction with HCl, Intermittent thin silt seams 1.2'-8.7' 11.3'-11.6', 2.5Y 4/1 dark gray (SM)		2		-7.8		
			From El. -11.3 to -16.5 Ft., mostly fine-grained sand-sized quartz, little silt, trace organic matter, no reaction with HCl, 2.5Y 6/2 light brownish gray		3		-11.7		
-17.7	13.9						-17.7		
			NOTES:				Abbreviations:		

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JUN 02

(Continued)

Boring Designation VB-KBIC16-6

DRILLING LOG (Cont. Sheet)	INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS
PROJECT King's Bay Inner Channel	COORDINATE SYSTEM/DATUM State Plane, GAE (U.S. Ft.)	HORIZONTAL NAD83	VERTICAL MLLW	
LOCATION COORDINATES X = 859,353 Y = 294,247	ELEVATION TOP OF BORING -3.8 Ft.			

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
			1. USACE Jacksonville is the custodian for these original files.						
			2. Soils are field visually classified in accordance with the Unified Soils Classification System.						
			3. Laboratory Testing Results						

SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION
1	1.1/1.3	SP-SM*
2	4.0/4.2	SM*
3	7.9/8.1	SM*

*Lab visual classification based on gradation curve

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SAJ FORM 1836-A
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Boring Designation VB-KBIC19-3

DRILLING LOG

DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 1 SHEETS	
1. PROJECT: Kings Bay Inner Channel Site Investigation	9. SIZE AND TYPE OF BIT: See Remarks		
	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.)	HORIZONTAL: NAD83	VERTICAL: MLLW
2. BORING DESIGNATION: VB-KBIC19-3	LOCATION COORDINATES: X = 857,974 Y = 294,647	11. MANUFACTURER'S DESIGNATION OF DRILL	☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Athena Technologies, Inc.	CONTRACTOR FILE NO.	12. TOTAL SAMPLES: DISTURBED 3	UNDISTURBED (UD) 0
4. NAME OF DRILLER: Adam Freeze		13. TOTAL NUMBER CORE BOXES: 0	
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED	DEG. FROM VERTICAL / BEARING	14. ELEVATION GROUND WATER: N/A	
		15. DATE BORING: STARTED 06-06-19	COMPLETED 06-06-19
6. THICKNESS OF OVERBURDEN: N/A		16. ELEVATION TOP OF BORING: -47.3 Ft.	
7. DEPTH DRILLED INTO ROCK: N/A		17. TOTAL RECOVERY FOR BORING: 87 %	
8. TOTAL DEPTH OF BORING: 5.5 Ft.		18. SIGNATURE AND TITLE OF INSPECTOR: Lauren Wargo, Civil Engineering Technician	

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-47.3	0.0						-47.3		
			SAND, silty, some fine-grained sand-sized quartz, some silt, trace organic matter, trace shell, 10Y 2.5/1 greenish black (SM)	100	n/a		Vibracore -49.0		
				100	n/a		Vibracore -50.1		
-51.9	4.6			100	n/a		Vibracore -52.1		
-52.1	4.8		SAND, poorly-graded with silt, mostly fine to coarse-grained sand-sized quartz, few silt, trace organic matter, trace shell, 5Y 5/1 gray (SP-SM)						
-52.8	5.5	NR		0			Vibracore -52.8		

BORING TERMINATED IN REFUSAL

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

SAJ FORM 1836
JUN 02

Boring Designation VB-KBIC19-4

DRILLING LOG	DIVISION South Atlantic	INSTALLATION Jacksonville District	SHEET 1 OF 1 SHEETS

1. PROJECT: Kings Bay Inner Channel Site Investigation	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KBIC19-4; LOCATION COORDINATES: X = 858,631 Y = 295,121	11. MANUFACTURER'S DESIGNATION OF DRILL: ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Athena Technologies, Inc.; CONTRACTOR FILE NO.:	12. TOTAL SAMPLES: DISTURBED 4; UNDISTURBED (UD) 0
4. NAME OF DRILLER: Adam Freeze	13. TOTAL NUMBER CORE BOXES: 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL: ; BEARING:	14. ELEVATION GROUND WATER: N/A
	15. DATE BORING: STARTED 06-06-19; COMPLETED 06-06-19
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: -47.0 Ft.
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 86 %
8. TOTAL DEPTH OF BORING: 7.3 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: Lauren Wargo, Civil Engineering Technician

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-47.0	0.0						-47.0		
			SAND, silty, mostly fine-grained sand-sized quartz, little silt, few organic matter, trace shell, 2.5YR 3/1 dark reddish gray (SM) From El. -47.8 to -50.0 Ft., N 2.5/ black	100	n/a		Vibracore -48.3		
				100	n/a		-48.7 Vibracore		
-49.2	2.2			100	n/a		Vibracore -49.5		
			SILT, inorganic-H, some fine to coarse-grained sand-sized quartz, some silt, few organic matter, trace shell, 5Y 2.5/2 black (MH)	100	n/a		Vibracore		
-53.0	6.0								
-53.3	6.3		SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace silt, 5Y 5/1 gray (SP)				-53.3		
-54.3	7.3	NR		0			Vibracore -54.3		
			BORING TERMINATED IN REFUSAL						

NOTES:

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SAJ FORM 1836
JUN 02

Boring Designation VB-KBIC19-5

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 1 SHEETS

1. PROJECT: Kings Bay Inner Channel Site Investigation	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.) — HORIZONTAL: NAD83 — VERTICAL: MLLW
2. BORING DESIGNATION: VB-KBIC19-5 — LOCATION COORDINATES: X = 859,354 Y = 295,605	11. MANUFACTURER'S DESIGNATION OF DRILL: ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Athena Technologies, Inc. — CONTRACTOR FILE NO.:	12. TOTAL SAMPLES — DISTURBED: 2 — UNDISTURBED (UD): 0
4. NAME OF DRILLER: Adam Freeze	13. TOTAL NUMBER CORE BOXES: 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED — DEG. FROM VERTICAL: — BEARING:	14. ELEVATION GROUND WATER: N/A
	15. DATE BORING — STARTED: 06-06-19 — COMPLETED: 06-06-19
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: N/A
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 81 %
8. TOTAL DEPTH OF BORING: 7.3 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: Lauren Wargo, Civil Engineering Technician

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
	0.0								
	1.9		SAND, poorly-graded with silt, mostly fine to coarse-grained sand-sized quartz, few silt, trace organic matter, trace shell, 5Y 6/1 gray (SP-SM)	100	n/a		Vibracore		
	2.4		SAND, poorly-graded, mostly fine to coarse-grained sand-sized quartz, trace silt, trace organic matter, trace shell, 5Y 5/1 gray (SP)						
	5.8		SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, trace organic matter, trace shell, 2.5Y 4/1 dark gray (SP-SM)	100	n/a		Vibracore		
	5.9								
	7.3	NR	SILT, inorganic-H, mostly silt, few fine-grained sand-sized quartz, trace organic matter, trace shell, 5Y 3/1 very dark gray (MH)	0			Vibracore		
			BORING TERMINATED IN REFUSAL						

NOTES:

1. USACE Jacksonville is the custodian for these original files.

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SAJ FORM 1836
JUN 02

Boring Designation VB-KBIC19-6

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 1 SHEETS
1. PROJECT: Kings Bay Inner Channel Site Investigation		9. SIZE AND TYPE OF BIT: See Remarks	
		10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.)	HORIZONTAL: NAD83; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KBIC19-6	LOCATION COORDINATES: X = 859,361 Y = 293,021	11. MANUFACTURER'S DESIGNATION OF DRILL	☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Athena Technologies, Inc.	CONTRACTOR FILE NO.	12. TOTAL SAMPLES	DISTURBED: 4; UNDISTURBED (UD): 0
4. NAME OF DRILLER: Adam Freeze		13. TOTAL NUMBER CORE BOXES: 0	
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED	DEG. FROM VERTICAL; BEARING	14. ELEVATION GROUND WATER: N/A	
		15. DATE BORING	STARTED: 06-06-19; COMPLETED: 06-06-19
6. THICKNESS OF OVERBURDEN: N/A		16. ELEVATION TOP OF BORING: -47.0 Ft.	
7. DEPTH DRILLED INTO ROCK: N/A		17. TOTAL RECOVERY FOR BORING: 93 %	
8. TOTAL DEPTH OF BORING: 5.5 Ft.		18. SIGNATURE AND TITLE OF INSPECTOR: Lauren Wargo, Civil Engineering Technician	

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-47.0	0.0						-47.0		
-48.8	1.8		SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace silt, trace organic matter, trace shell, 2.5YR 4/1 dark reddish gray (SP)	100	n/a		Vibracore -48.8		
-49.3	2.3		SAND, silty, mostly fine-grained sand-sized quartz, little silt, trace organic matter, trace shell, 2.5YR 3/1 dark reddish gray (SM)	100	n/a		Vibracore -49.5		
-50.6	3.6		SAND, poorly-graded with silt, mostly fine to coarse-grained sand-sized quartz, little shell, few silt, trace organic matter, 10Y 2.5/1 greenish black (SP-SM)	100	n/a		Vibracore -50.8		
-52.1	5.1		SAND, silty, mostly fine-grained sand-sized quartz, little silt, trace organic matter, trace shell, 5Y 3/1 very dark gray (SM)	100	n/a		Vibracore -52.1		
-52.5	5.5	NO REC		0			-52.5 Vibracore		
			BORING TERMINATED IN REFUSAL NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System.						

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SAJ FORM 1836
JUN 02

Boring Designation VB-KBIC19-7

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 1 SHEETS

1. PROJECT: Kings Bay Inner Channel Site Investigation	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KBIC19-7; LOCATION COORDINATES: X = 859,567 Y = 293,203	11. MANUFACTURER'S DESIGNATION OF DRILL: ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Athena Technologies, Inc.; CONTRACTOR FILE NO.:	12. TOTAL SAMPLES: DISTURBED 3; UNDISTURBED (UD) 0
4. NAME OF DRILLER: Adam Freeze	13. TOTAL NUMBER CORE BOXES: 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL:; BEARING:	14. ELEVATION GROUND WATER: N/A
	15. DATE BORING: STARTED 06-05-19; COMPLETED 06-05-19
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: -48.3 Ft.
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 78 %
8. TOTAL DEPTH OF BORING: 8.0 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: Lauren Wargo, Civil Engineering Technician

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-48.3	0.0						-48.3		
-48.5	0.2		SILT, inorganic-H, mostly silt, little fine-grained sand-sized quartz, trace organic matter, trace shell, 5Y 3/1 very dark gray (MH)	100	n/a		Vibracore		
-49.1	0.8		SAND, silty, mostly fine-grained sand-sized quartz, little silt, trace organic matter, trace shell, 10Y 4/1 dark greenish gray (SM)				-50.1		
-50.6	2.3		SILT, inorganic-H, some fine-grained sand-sized quartz, some silt, trace organic matter, trace shell, 5Y 3/1 very dark gray (MH)	100	n/a		Vibracore -50.8		
			SAND, silty, mostly fine-grained sand-sized quartz, little silt, trace organic matter, trace shell, 5Y 3/1 very dark gray (SM)	100	n/a		Vibracore		
-54.5	6.2	NR		0			-54.5 Vibracore		
-56.3	8.0		BORING TERMINATED IN REFUSAL				-56.3		

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

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SAJ FORM 1836
JUN 02

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Boring Designation VB-KBIC19-8

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 1 SHEETS

1. PROJECT: Kings Bay Inner Channel Site Investigation	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KBIC19-8; LOCATION COORDINATES: X = 861,500 Y = 290,850	11. MANUFACTURER'S DESIGNATION OF DRILL: ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Athena Technologies, Inc.; CONTRACTOR FILE NO.:	12. TOTAL SAMPLES: DISTURBED 3; UNDISTURBED (UD) 0
4. NAME OF DRILLER: Adam Freeze	13. TOTAL NUMBER CORE BOXES: 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL: ; BEARING:	14. ELEVATION GROUND WATER: N/A
	15. DATE BORING: STARTED 06-05-19; COMPLETED 06-05-19
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: -42.1 Ft.
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 48 %
8. TOTAL DEPTH OF BORING: 11.0 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: Lauren Wargo, Civil Engineering Technician

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-42.1	0.0						-42.1		
			SILT, inorganic-H, mostly silt, few organic matter, trace fine-grained sand-sized quartz, trace shell, 10Y 2.5/1 greenish black (MH)	100	n/a		Vibracore -43.6		
			From El. -43.6 to -47.4 Ft., little fine to coarse-grained sand-sized quartz	100	n/a		Vibracore -45.1		
				100	n/a		Vibracore		
-47.4	5.3						-47.4		
		NO RECOVERY		0			Vibracore		
-53.1	11.0						-53.1		
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System.						

SAJ FORM 1836
JUN 02

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Boring Designation VB-KBIC19-9

DRILLING LOG	**DIVISION** South Atlantic	**INSTALLATION** Jacksonville District	**SHEET** 1 **OF** 1 **SHEETS**
1. PROJECT Kings Bay Inner Channel Site Investigation		**9. SIZE AND TYPE OF BIT** See Remarks	
		10. COORDINATE SYSTEM/DATUM State Plane, GAE (U.S. Ft.)	**HORIZONTAL** NAD83 / **VERTICAL** MLLW
2. BORING DESIGNATION VB-KBIC19-9	**LOCATION COORDINATES** X = 861,744 Y = 291,009	**11. MANUFACTURER'S DESIGNATION OF DRILL**	☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY Athena Technologies, Inc.	**CONTRACTOR FILE NO.**	**12. TOTAL SAMPLES**	**DISTURBED** 3 / **UNDISTURBED (UD)** 0
4. NAME OF DRILLER Adam Freeze		**13. TOTAL NUMBER CORE BOXES** 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED	**DEG. FROM VERTICAL** / **BEARING**	**14. ELEVATION GROUND WATER** N/A	
		15. DATE BORING	**STARTED** 06-06-19 / **COMPLETED** 06-06-19
6. THICKNESS OF OVERBURDEN N/A		**16. ELEVATION TOP OF BORING** -44.4 Ft.	
7. DEPTH DRILLED INTO ROCK N/A		**17. TOTAL RECOVERY FOR BORING** 70 %	
8. TOTAL DEPTH OF BORING 8.3 Ft.		**18. SIGNATURE AND TITLE OF INSPECTOR** Lauren Wargo, Civil Engineering Technician	

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-44.4	0.0						-44.4		
			SILT, inorganic-H, mostly silt, few fine-grained sand-sized quartz, few organic matter, trace shell, 10Y 3/1 very dark greenish gray (MH)	100	n/a		Vibracore -45.9		
			From El. -45.9 to -47.4 Ft., some silt	100	n/a		Vibracore -47.4		
			From El. -47.4 to -50.2 Ft., mostly silt, some fine to coarse-grained sand-sized quartz	100	n/a		Vibracore		
-50.2	5.8						-50.2		
		NR		0			Vibracore		
-52.7	8.3						-52.7		
			BORING TERMINATED IN REFUSAL						
			NOTES:						
			1. USACE Jacksonville is the custodian for these original files.						
			2. Soils are field visually classified in accordance with the Unified Soils Classification System.						

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SAJ FORM 1836
JUN 02

Boring Designation VB-KBIC19-10

DRILLING LOG	**DIVISION** South Atlantic	**INSTALLATION** Jacksonville District	**SHEET** 1 **OF** 1 **SHEETS**

1. PROJECT Kings Bay Inner Channel Site Investigation	**9. SIZE AND TYPE OF BIT** See Remarks
	10. COORDINATE SYSTEM/DATUM State Plane, GAE (U.S. Ft.) — **HORIZONTAL** NAD83 — **VERTICAL** MLLW
2. BORING DESIGNATION VB-KBIC19-10 — **LOCATION COORDINATES** X = 862,087 Y = 291,303	**11. MANUFACTURER'S DESIGNATION OF DRILL** ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY Athena Technologies, Inc. — **CONTRACTOR FILE NO.**	**12. TOTAL SAMPLES** — **DISTURBED** 3 — **UNDISTURBED (UD)** 0
4. NAME OF DRILLER Adam Freeze	**13. TOTAL NUMBER CORE BOXES** 0
	14. ELEVATION GROUND WATER N/A
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED — **DEG. FROM VERTICAL** — **BEARING**	**15. DATE BORING** — **STARTED** 06-05-19 — **COMPLETED** 06-05-19
6. THICKNESS OF OVERBURDEN N/A	**16. ELEVATION TOP OF BORING** -43.4 Ft.
7. DEPTH DRILLED INTO ROCK N/A	**17. TOTAL RECOVERY FOR BORING** 51 %
8. TOTAL DEPTH OF BORING 11.0 Ft.	**18. SIGNATURE AND TITLE OF INSPECTOR** Lauren Wargo, Civil Engineering Technician

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-43.4	0.0						-43.4		
			SILT, inorganic-H, mostly silt, little fine to coarse-grained sand-sized quartz, few organic matter, trace shell, 2.5Y 3/1 very dark gray (MH)	100	n/a		Vibracore -44.9		
			From El. -44.9 to -46.4 Ft., some fine to coarse-grained sand-sized quartz, some silt	100	n/a		Vibracore -46.4		
			From El. -46.4 to -49.0 Ft., mostly silt	100	n/a		Vibracore		
-49.0	5.6						-49.0		
		NO RECOVERY		0			Vibracore		
-54.4	11.0						-54.4		

NOTES:

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2. Soils are field visually classified in accordance with the Unified Soils Classification System.

SAJ FORM 1836
JUN 02

Boring Designation VB-KBIC19-11

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 1 SHEETS

1. PROJECT: Kings Bay Inner Channel Site Investigation	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KBIC19-11; LOCATION COORDINATES: X = 862,905 Y = 289,633	11. MANUFACTURER'S DESIGNATION OF DRILL: ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Athena Technologies, Inc.; CONTRACTOR FILE NO.:	12. TOTAL SAMPLES: DISTURBED 3; UNDISTURBED (UD) 0
4. NAME OF DRILLER: Adam Freeze	13. TOTAL NUMBER CORE BOXES: 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL: ; BEARING:	14. ELEVATION GROUND WATER: N/A
	15. DATE BORING: STARTED 06-04-19; COMPLETED 06-04-19
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: -38.6 Ft.
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 52 %
8. TOTAL DEPTH OF BORING: 11.0 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: Lauren Wargo, Civil Engineering Technician

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-38.6	0.0						-38.6		
			SILT, inorganic-H, some silt, little fine to coarse-grained sand-sized quartz, few organic matter, trace shell, 5GY 2.5/1 greenish black (MH)	100	n/a		Vibracore -40.1		
			From El. -40.1 to -44.3 Ft., some fine to coarse-grained sand-sized quartz, little organic matter	100	n/a		Vibracore -41.6		
				100	n/a		Vibracore		
-44.3	5.7						-44.3		
		NO RECOVERY		0			Vibracore		
-49.6	11.0						-49.6		

NOTES:

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2. Soils are field visually classified in accordance with the Unified Soils Classification System.

SAJ FORM 1836
JUN 02

Boring Designation VB-KBIC19-12

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 1 SHEETS

1. PROJECT: Kings Bay Inner Channel Site Investigation	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KBIC19-12; LOCATION COORDINATES: X = 863,187 Y = 290,080	11. MANUFACTURER'S DESIGNATION OF DRILL: ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Athena Technologies, Inc.; CONTRACTOR FILE NO.:	12. TOTAL SAMPLES: DISTURBED 3; UNDISTURBED (UD) 0
4. NAME OF DRILLER: Adam Freeze	13. TOTAL NUMBER CORE BOXES: 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL:; BEARING:	14. ELEVATION GROUND WATER: N/A
	15. DATE BORING: STARTED 06-05-19; COMPLETED 06-05-19
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: -48.1 Ft.
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 51 %
8. TOTAL DEPTH OF BORING: 8.5 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: Lauren Wargo, Civil Engineering Technician

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-48.1	0.0						-48.1		
			SILT, inorganic-H, some fine to coarse-grained sand-sized quartz, some silt, few organic matter, trace shell, 5G 2.5/1 greenish black (MH)	100	n/a		Vibracore		
							-49.6		
			From El. -49.6 to -50.6 Ft., few fine-grained sand-sized quartz	100	n/a		Vibracore		
							-50.6		
			From El. -50.6 to -51.3 Ft., some fine to coarse-grained sand-sized quartz	100	n/a		Vibracore		
			From El. -51.3 to -52.4 Ft., 5Y 4/1 dark gray						
-52.4	4.3						-52.4		
		NO RECOVERY		0			Vibracore		
-56.6	8.5						-56.6		
			BORING TERMINATED IN REFUSAL						

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

SAJ FORM 1836
JUN 02

Boring Designation VB-KBIC19-13

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 1 SHEETS

1. PROJECT: Kings Bay Inner Channel Site Investigation	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KBIC19-13; LOCATION COORDINATES: X = 863,382 Y = 290,464	11. MANUFACTURER'S DESIGNATION OF DRILL: ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Athena Technologies, Inc.; CONTRACTOR FILE NO.:	12. TOTAL SAMPLES: DISTURBED 3; UNDISTURBED (UD) 0
4. NAME OF DRILLER: Adam Freeze	13. TOTAL NUMBER CORE BOXES: 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL: ; BEARING:	14. ELEVATION GROUND WATER: N/A
	15. DATE BORING: STARTED 06-04-19; COMPLETED 06-04-19
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: -48.3 Ft.
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 61 %
8. TOTAL DEPTH OF BORING: 11.0 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: Lauren Wargo, Civil Engineering Technician

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-48.3	0.0		SILT, inorganic-H, some fine-grained sand-sized quartz, little silt, few organic matter, trace shell, 10Y 2.5/1 greenish black (MH)	100	n/a		-48.3 Vibracore -49.8		
			From El. -49.8 to -51.3 Ft., some silt	100	n/a		Vibracore -51.3		
-51.3	3.0		SAND, silty, mostly fine to coarse-grained sand-sized quartz, little silt, few organic matter, trace shell, 5Y 3/1 very dark gray (SM)	100	n/a		Vibracore		
-54.4	6.1		SAND, poorly-graded with silt, mostly fine to coarse-grained sand-sized quartz, little silt, few organic matter, trace shell, 2.5Y 5/1 gray (SP-SM)				-55.0		
-55.0	6.7	NO RECOVERY		0			Vibracore		
-59.3	11.0						-59.3		

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

SAJ FORM 1836
JUN 02

Boring Designation VB-KBIC19-14

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 1 SHEETS

1. PROJECT: Kings Bay Inner Channel Site Investigation	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KBIC19-14; LOCATION COORDINATES: X = 864,687 Y = 288,597	11. MANUFACTURER'S DESIGNATION OF DRILL: ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Athena Technologies, Inc.; CONTRACTOR FILE NO.:	12. TOTAL SAMPLES: DISTURBED 3; UNDISTURBED (UD) 0
4. NAME OF DRILLER: Adam Freeze	13. TOTAL NUMBER CORE BOXES: 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL: ; BEARING:	14. ELEVATION GROUND WATER: N/A
	15. DATE BORING: STARTED 06-07-19; COMPLETED 06-07-19
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: -22.5 Ft.
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 65 %
8. TOTAL DEPTH OF BORING: 11.0 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: Lauren Wargo, Civil Engineering Technician

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-22.5	0.0						-22.5		
			SILT, inorganic-H, mostly silt, little fine to coarse-grained sand-sized quartz, few organic matter, trace shell, 5Y 2.5/1 black (MH)	100	n/a		Vibracore		
							-24.0		
			From El. -24.0 to -25.5 Ft., some silt, 5Y 2.5/1 black	100	n/a		Vibracore		
							-25.5		
			From El. -25.5 to -29.5 Ft., 5Y 3/2 dark olive gray	100	n/a		Vibracore		
-29.7	7.2		From El. -29.5 to -29.7 Ft., 10Y 3/1 very dark greenish gray				-29.7		
		NO RECOVERY		0			Vibracore		
-33.5	11.0						-33.5		
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System.						

0
5
10
15

SAJ FORM 1836
JUN 02

Boring Designation VB-KBIC19-15

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 1 SHEETS

1. PROJECT: Kings Bay Inner Channel Site Investigation	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, GAE (U.S. Ft.); HORIZONTAL: NAD83; VERTICAL: MLLW
2. BORING DESIGNATION: VB-KBIC19-15; LOCATION COORDINATES: X = 864,966 Y = 288,993	11. MANUFACTURER'S DESIGNATION OF DRILL: ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Athena Technologies, Inc.; CONTRACTOR FILE NO.:	12. TOTAL SAMPLES: DISTURBED 2; UNDISTURBED (UD) 0
4. NAME OF DRILLER: Adam Freeze	13. TOTAL NUMBER CORE BOXES: 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED; DEG. FROM VERTICAL: ; BEARING:	14. ELEVATION GROUND WATER: N/A
	15. DATE BORING: STARTED 06-07-19; COMPLETED 06-07-19
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: -43.4 Ft.
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 76 %
8. TOTAL DEPTH OF BORING: 9.7 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: Lauren Wargo, Civil Engineering Technician

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-43.4	0.0						-43.4		
			SILT, inorganic-H, some fine to coarse-grained sand-sized quartz, some silt, few organic matter, trace shell, 10Y 3/1 very dark greenish gray (MH)	100	n/a		Vibracore -44.9		
			From El. -49.7 to -50.6 Ft., 5Y 4/1 dark gray	100	n/a		Vibracore		
-50.8	7.4		From El. -50.6 to -50.8 Ft., 5GY 4/1 dark greenish gray				-50.8		
		NR		0			Vibracore		
-53.1	9.7		BORING TERMINATED IN REFUSAL				-53.1		

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

SAJ FORM 1836
JUN 02

Boring Designation VB-KBIC19-16

DRILLING LOG	**DIVISION** South Atlantic	**INSTALLATION** Jacksonville District	**SHEET** 1 **OF** 1 **SHEETS**

1. PROJECT Kings Bay Inner Channel Site Investigation	**9. SIZE AND TYPE OF BIT** See Remarks
	10. COORDINATE SYSTEM/DATUM State Plane, GAE (U.S. Ft.) · **HORIZONTAL** NAD83 · **VERTICAL** MLLW
2. BORING DESIGNATION VB-KBIC19-16 · **LOCATION COORDINATES** X = 865,121 Y = 289,357	**11. MANUFACTURER'S DESIGNATION OF DRILL** ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY Athena Technologies, Inc. · **CONTRACTOR FILE NO.**	**12. TOTAL SAMPLES** · **DISTURBED** 3 · **UNDISTURBED (UD)** 0
4. NAME OF DRILLER Adam Freeze	**13. TOTAL NUMBER CORE BOXES** 0
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED · **DEG. FROM VERTICAL** · **BEARING**	**14. ELEVATION GROUND WATER** N/A
	15. DATE BORING · **STARTED** 06-07-19 · **COMPLETED** 06-07-19
6. THICKNESS OF OVERBURDEN N/A	**16. ELEVATION TOP OF BORING** -48.8 Ft.
7. DEPTH DRILLED INTO ROCK N/A	**17. TOTAL RECOVERY FOR BORING** 82 %
8. TOTAL DEPTH OF BORING 3.8 Ft.	**18. SIGNATURE AND TITLE OF INSPECTOR** Lauren Wargo, Civil Engineering Technician

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-48.8	0.0						-48.8		
-49.7	0.9		SAND, silty, clayey, mostly fine to coarse-grained sand-sized quartz, little silt, little shell, trace organic matter, 10Y 3/1 very dark greenish gray (SC-SM)	100	n/a		Vibracore -50.0		
-51.1	2.3		SHELL, mostly shell, little fine to coarse-grained sand-sized quartz, little silt, trace organic matter, 5Y 4/1 dark gray	100	n/a		Vibracore -51.3		
-51.9	3.1		SAND, silty, mostly fine to coarse-grained sand-sized quartz, little silt, trace organic matter, trace shell, 2.5Y 7/1 light gray (SM)	100	n/a		Vibracore -51.9		
-52.6	3.8	NR		0			Vibracore -52.6		

BORING TERMINATED IN REFUSAL

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

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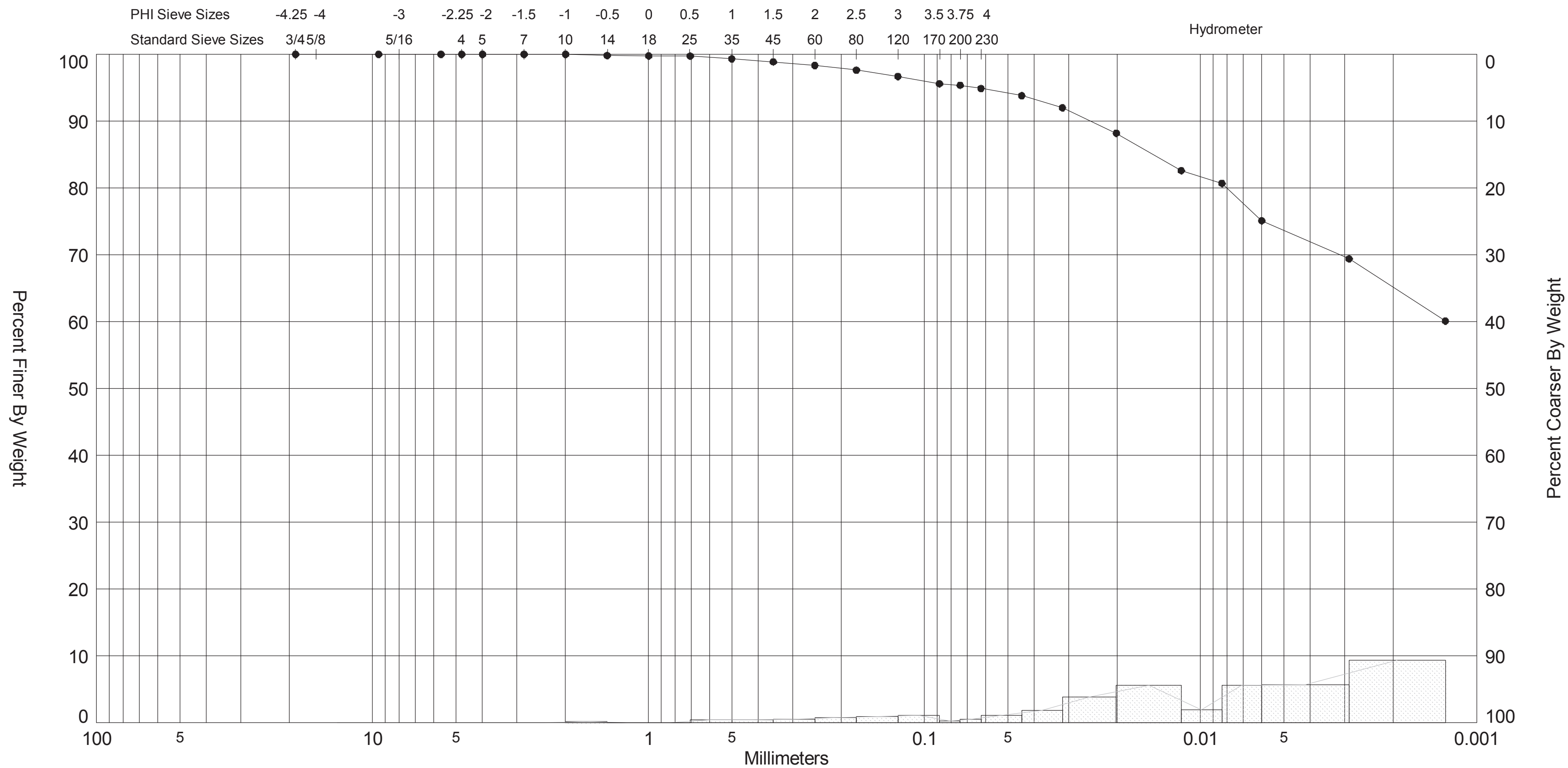
4.7 LABORATORY TESTING DATA

Applicable laboratory testing data are presented on the following pages.



Gravel		Sand			Silt and Clay
Coarse	Fine	Coarse	Medium	Fine	

Sample	Symbol	Elev. (ft)	USCS	% Fines	% Organics	% Carbonates	Median	Mean	Skew	Kurt	Sort
VB-KBIC13-01 #1	──●──	-34.4	OH	#200 - 95.37 #230 - 94.88				6.53	-0.84	3.29	2.15

Comments:

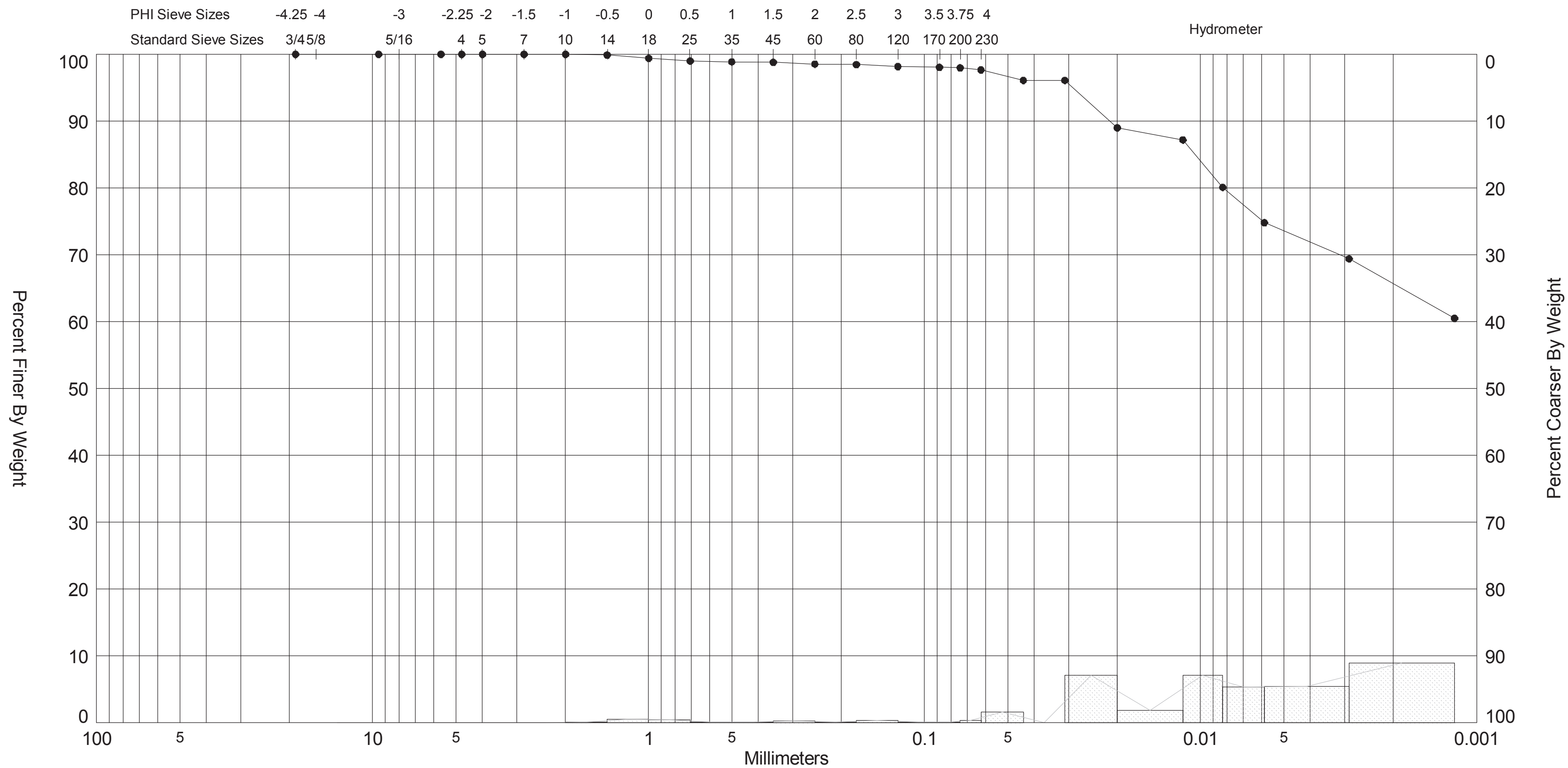
Depths and elevations based on measured values

Sample Information	
Project Name:	Kings Bay 103 Evaluation
Analysis Date:	06-10-13
Analyzed By:	CRM Jr.
Easting (X, ft):	865,566
Northing (Y, ft):	288,492
Horizontal System:	NAD 1983
Vertical System:	MLLW

US Army Corps of Engineers
701 San Marco Blvd
Jacksonville, Florida 32207
phone (904)396-7121

SIEVE ANALYSIS KINGS BAY.GPJ FL DEP ROSS.GDT 8/12/14

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



Gravel		Sand			Silt and Clay
Coarse	Fine	Coarse	Medium	Fine	

Sample	Symbol	Elev. (ft)	USCS	% Fines	% Organics	% Carbonates	Median	Mean	Skew	Kurt	Sort
VB-KBIC13-01 #2	—●—	-38.0	CH	#200 - 97.97 #230 - 97.67				6.77	-1.25	5.39	1.97

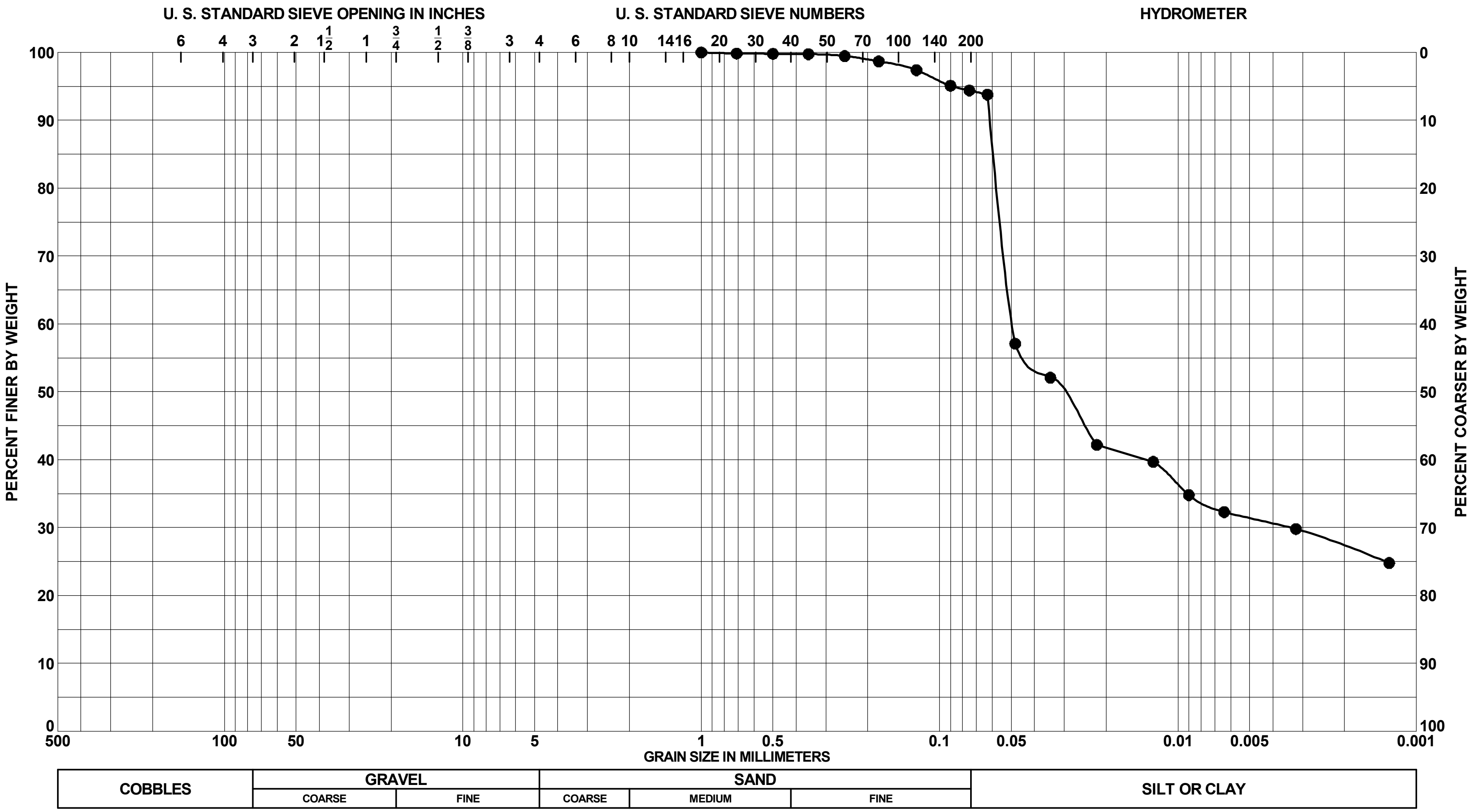
Comments:

Depths and elevations based on measured values

Sample Information	
Project Name:	Kings Bay 103 Evaluation
Analysis Date:	06-10-13
Analyzed By:	CRM Jr.
Easting (X, ft):	865,566
Northing (Y, ft):	288,492
Horizontal System:	NAD 1983
Vertical System:	MLLW

US Army Corps of Engineers
701 San Marco Blvd
Jacksonville, Florida 32207
phone (904)396-7121

00320-1



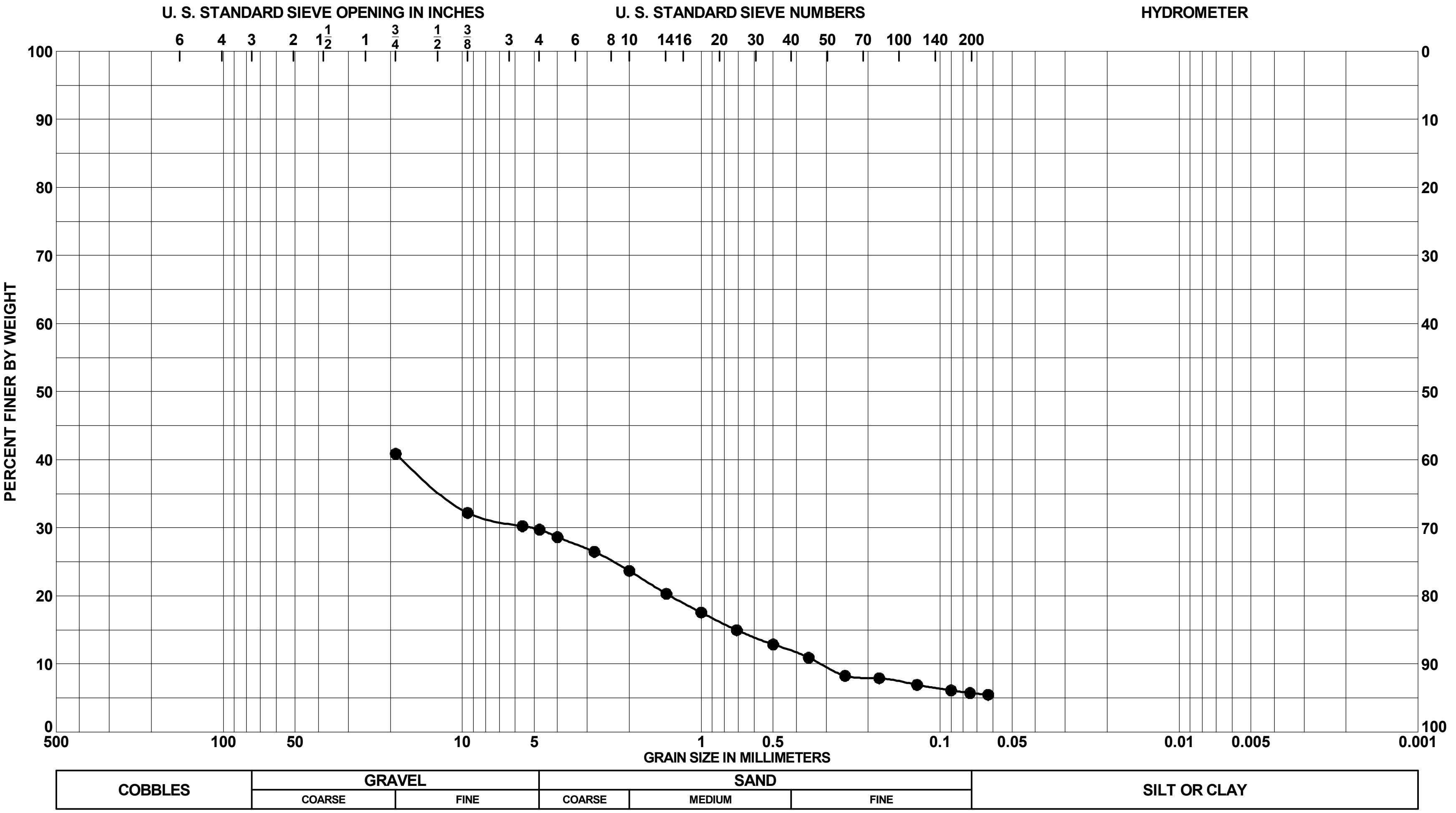
Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	%CO_3	G_s	Org %	w_n	LL	PL	PI
● 1	4.0 to 4.2 Ft.	SILT, inorganic-H, few fine-grained sand-sized quartz, very dark gray (MH)	2.5Y 3/1	0.0				30.3			

PROJECT	King's Bay Inner Channel
BORING NO.	VB-KBIC16-2
BORING ELEV.	Not Determined
DATE	9/14/2016

GRADATION CURVES

SAJ FORM 2087
JUN 02

00320-2



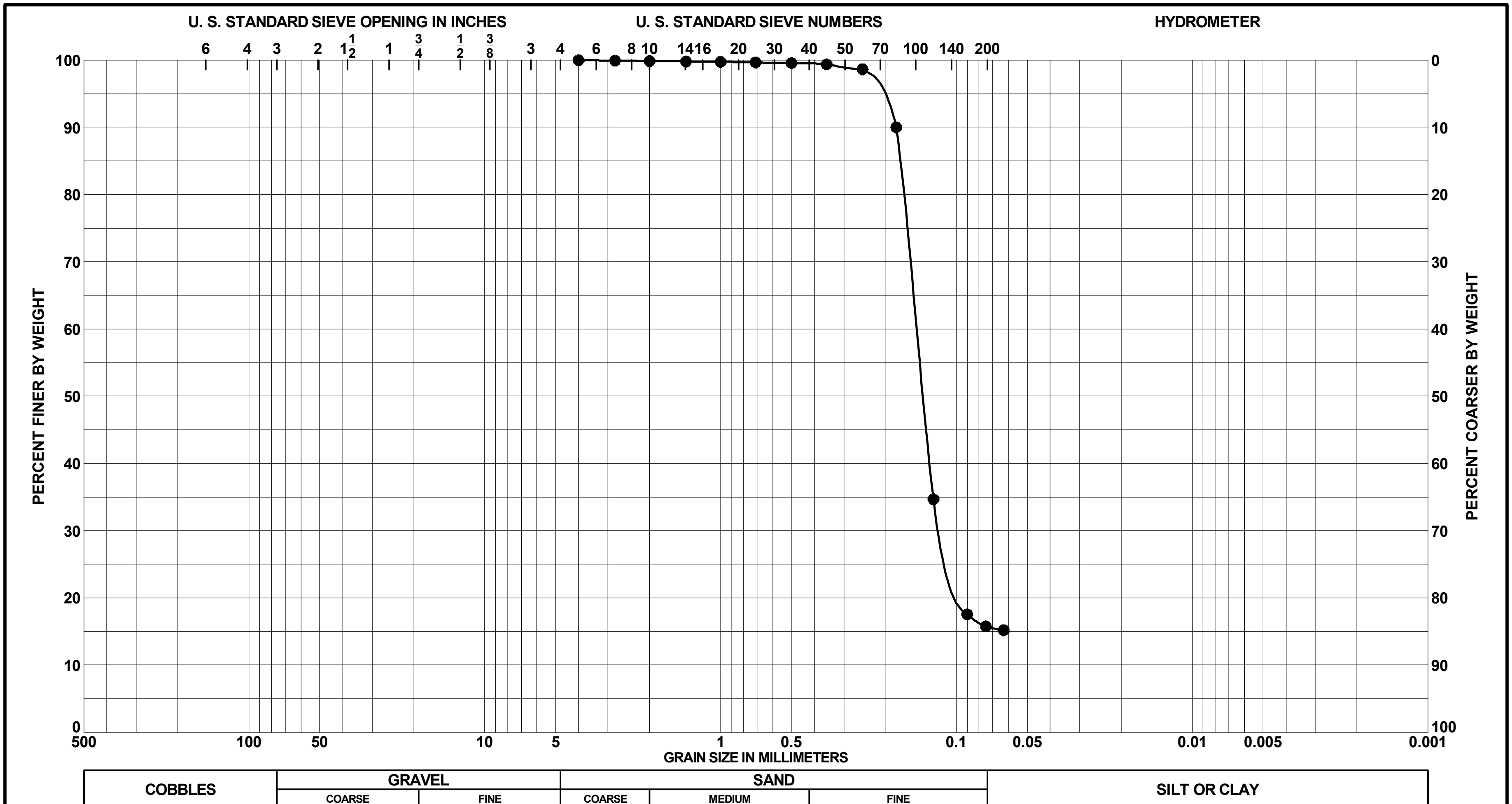
Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	$\%CO_3$	G_s	Org %	w_n	LL	PL	PI
● 2	9.3 to 9.5 Ft.	LIMESTONE, poorly-graded with silt, mostly sand to gravel-sized limestone, few silt, olive brown (GP-GM)	2.5Y 4/3	0.0							

GRADATION CURVES

PROJECT	King's Bay Inner Channel
BORING NO.	VB-KBIC16-2
BORING ELEV.	Not Determined
DATE	9/14/2016

SAJ FORM 2087
JUN 02

00320-3

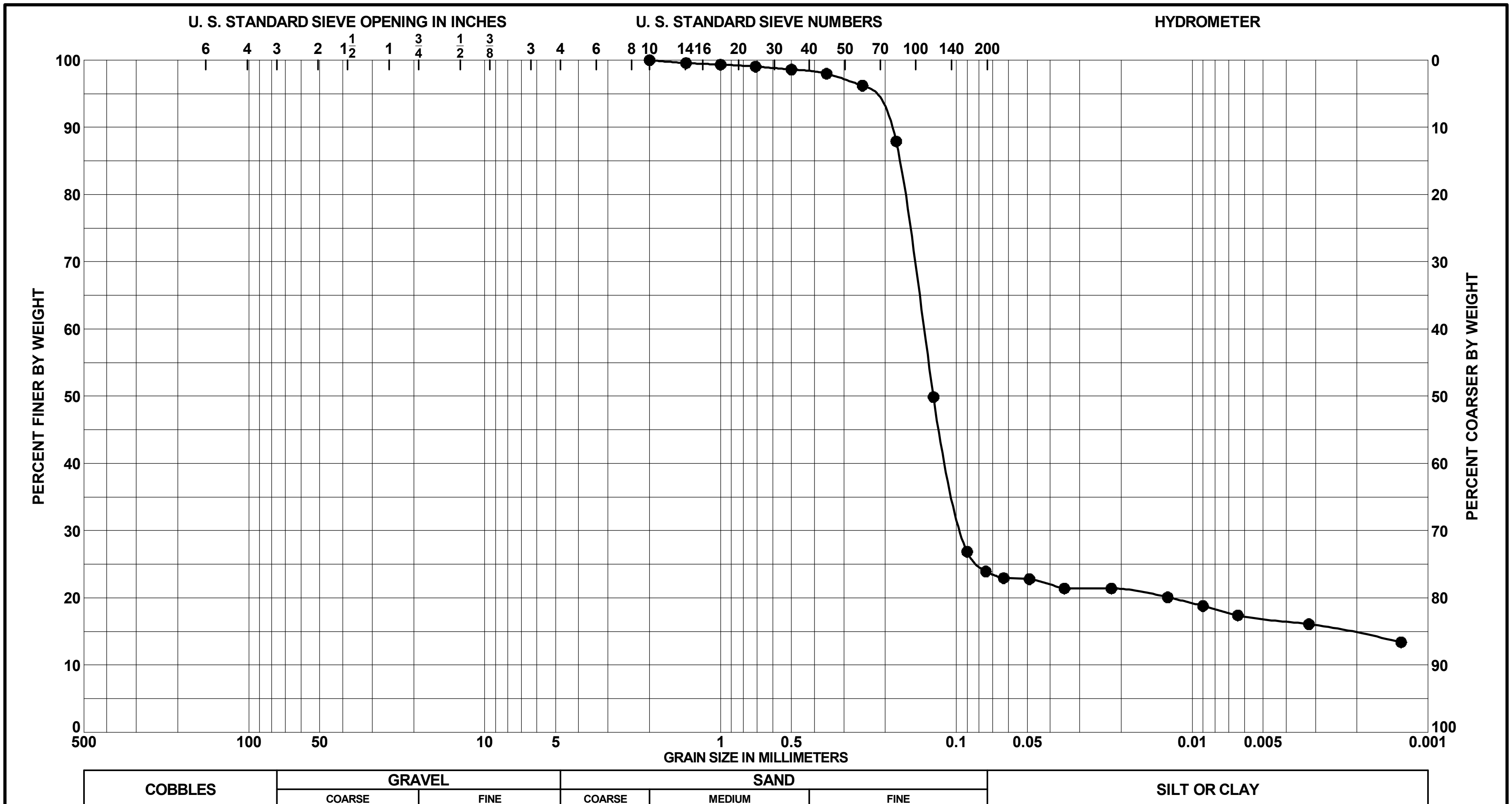


Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	%CO_3	G_s	Org %	w_n	LL	PL	PI
● 1	0.4 to 0.6 Ft.	SAND, silty, mostly fine-grained sand-sized quartz, little silt, trace organics, dark gray (SM)	2.5Y 4/1	0.0							

PROJECT	King's Bay Inner Channel
BORING NO.	VB-KBIC16-3
BORING ELEV.	Not Determined
DATE	9/14/2016

GRADATION CURVES

SAJ FORM 2087
JUN 02



00320-4

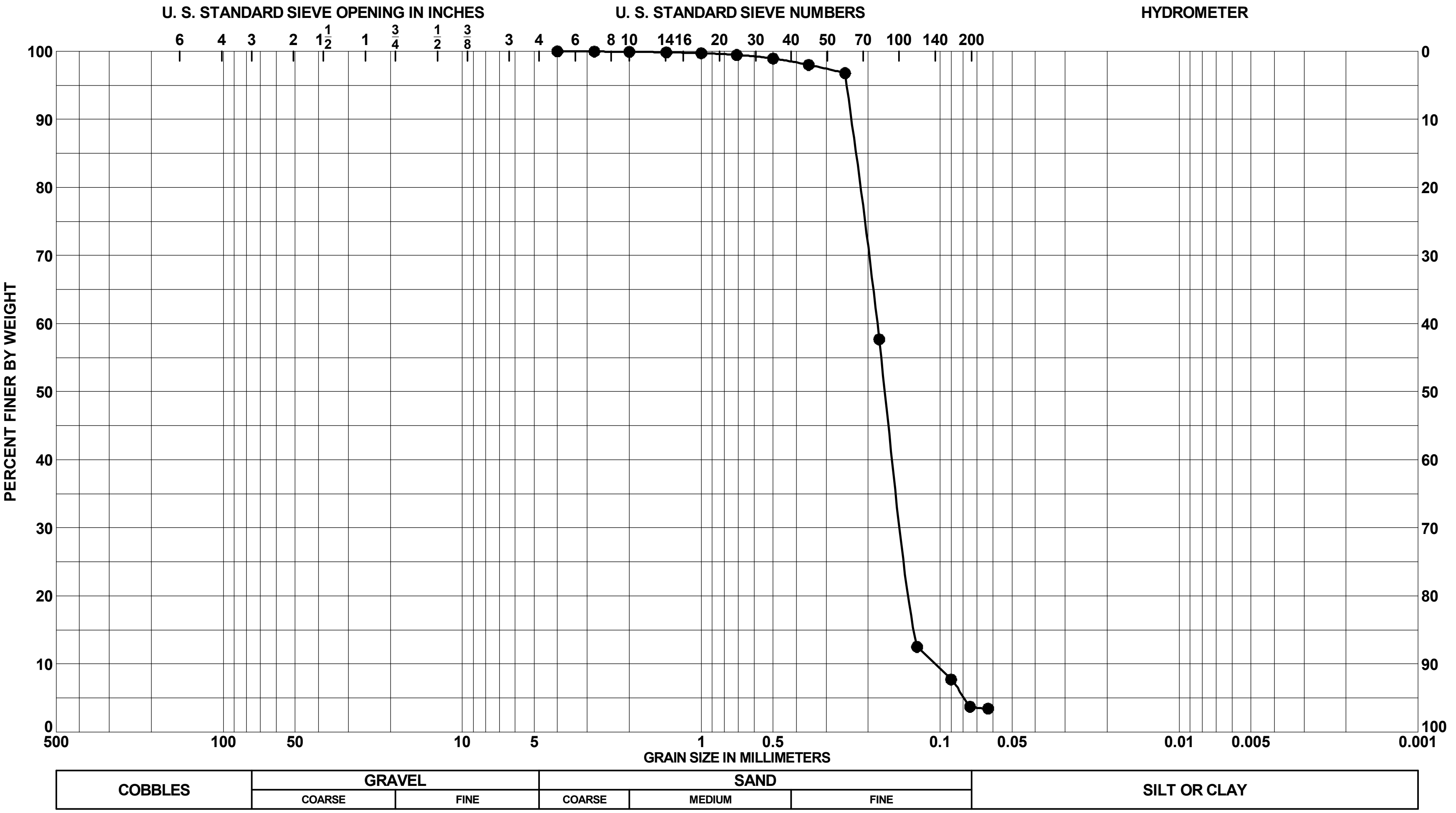
Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	%CO_3	G_s	Org %	w_n	LL	PL	PI
● 2	3.1 to 3.3 Ft.	SAND, silty, mostly fine-grained sand-sized quartz, little silt, trace organics, very dark gray (SM)	2.5Y 3/1	0.0				9.5			

PROJECT	King's Bay Inner Channel
BORING NO.	VB-KBIC16-3
BORING ELEV.	Not Determined
DATE	9/14/2016

GRADATION CURVES

SAJ FORM 2087
JUN 02

00320-5



Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	$\%CO_3$	G_s	Org %	w_n	LL	PL	PI
● 1	1.8 to 2.0 Ft.	SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace silt, light gray (SP)	2.5Y 7/2	0.1							

PROJECT King's Bay Inner Channel

BORING NO. VB-KBIC16-4

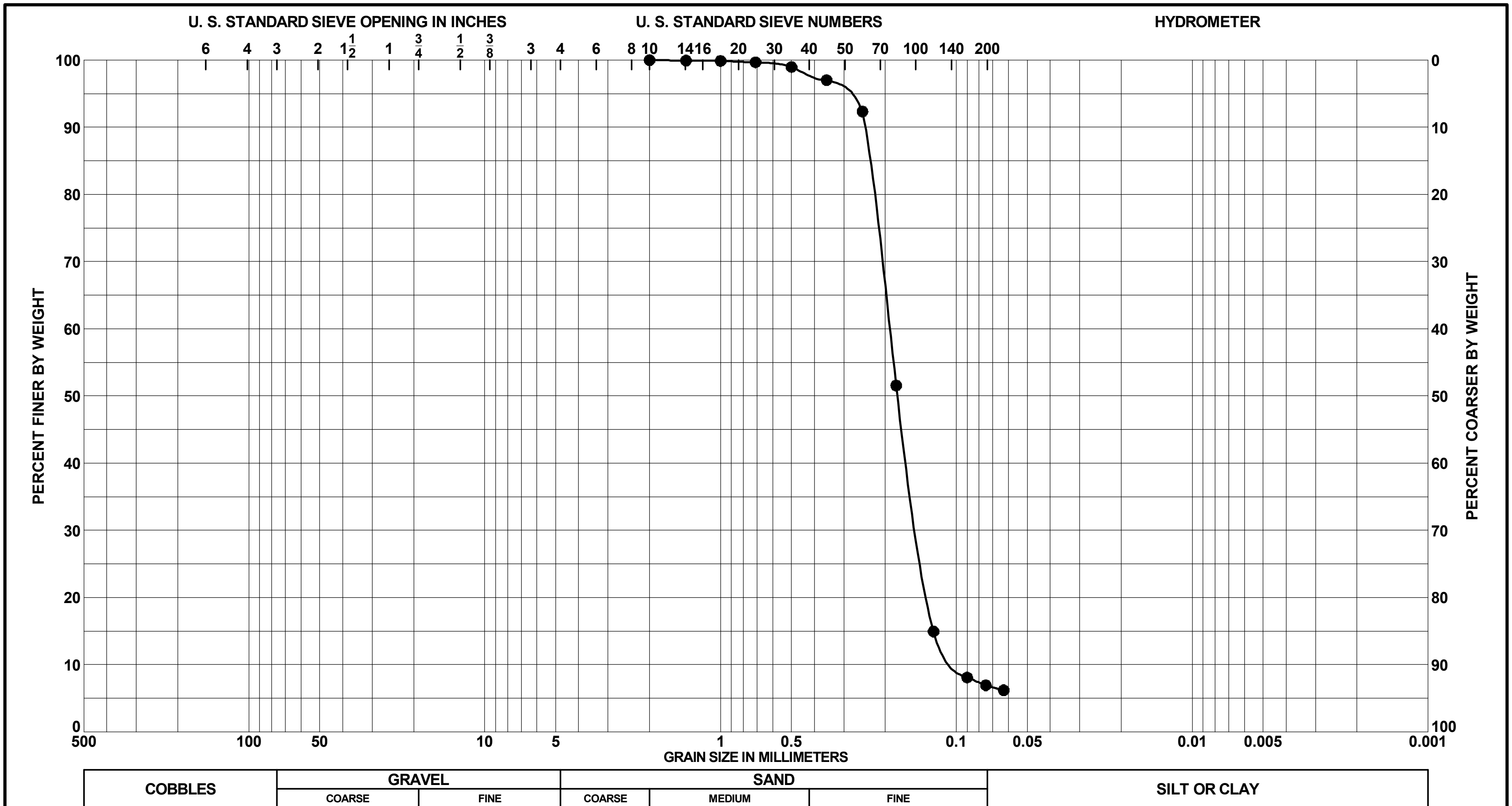
BORING ELEV. Not Determined

DATE 9/14/2016

GRADATION CURVES

SAJ FORM 2087
JUN 02

00320-6



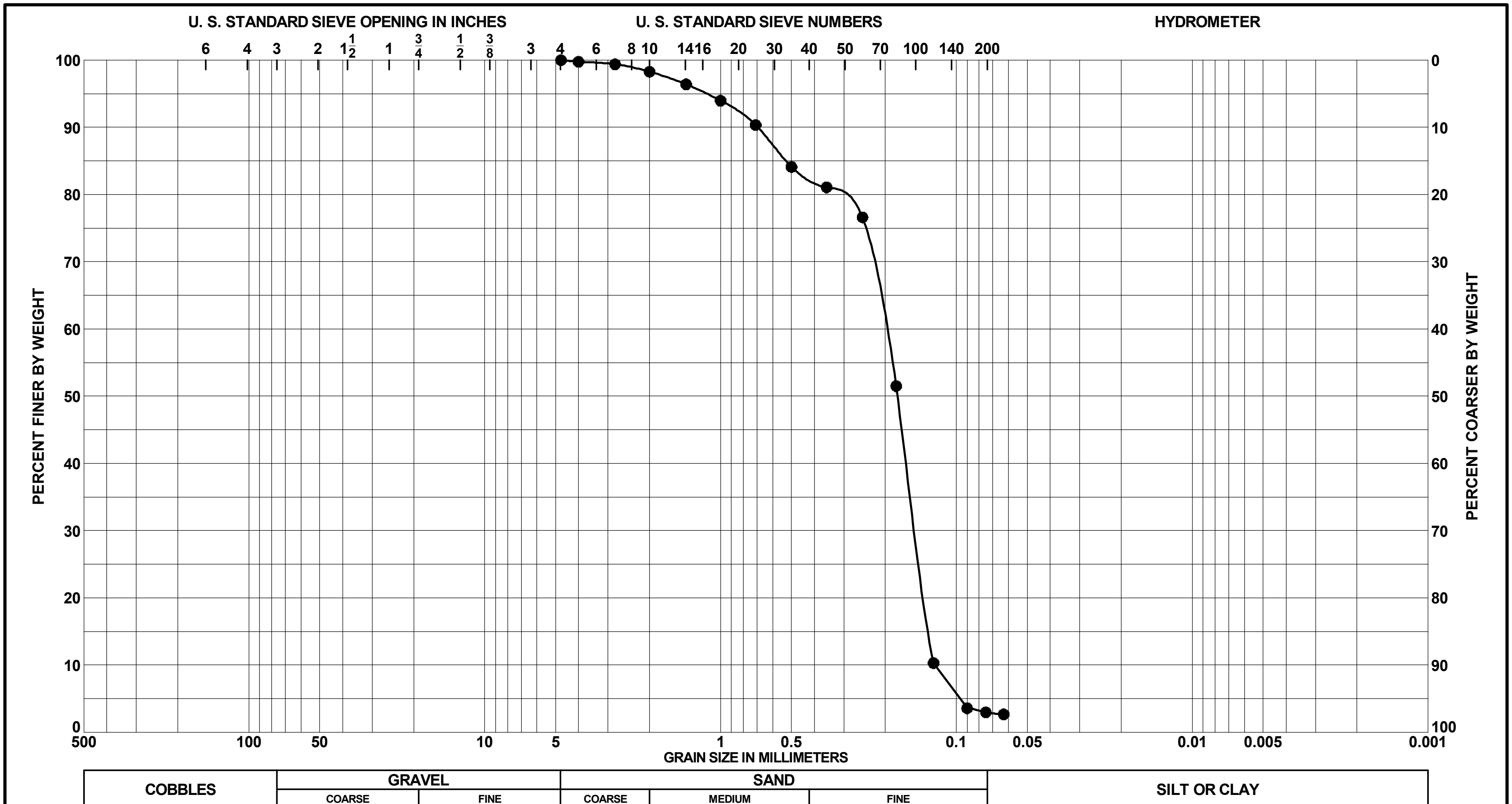
Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	$\%CO_3$	G_s	Org %	w_n	LL	PL	PI
● 2	4.5 to 4.7 Ft.	SAND, poorly graded with silt, mostly fine-grained sand-sized quartz, few silt, gray (SP-SM)	2.5Y 5/1	0.1							

PROJECT	King's Bay Inner Channel
BORING NO.	VB-KBIC16-4
BORING ELEV.	Not Determined
DATE	9/14/2016

GRADATION CURVES

SAJ FORM 2087
JUN 02

00320-7



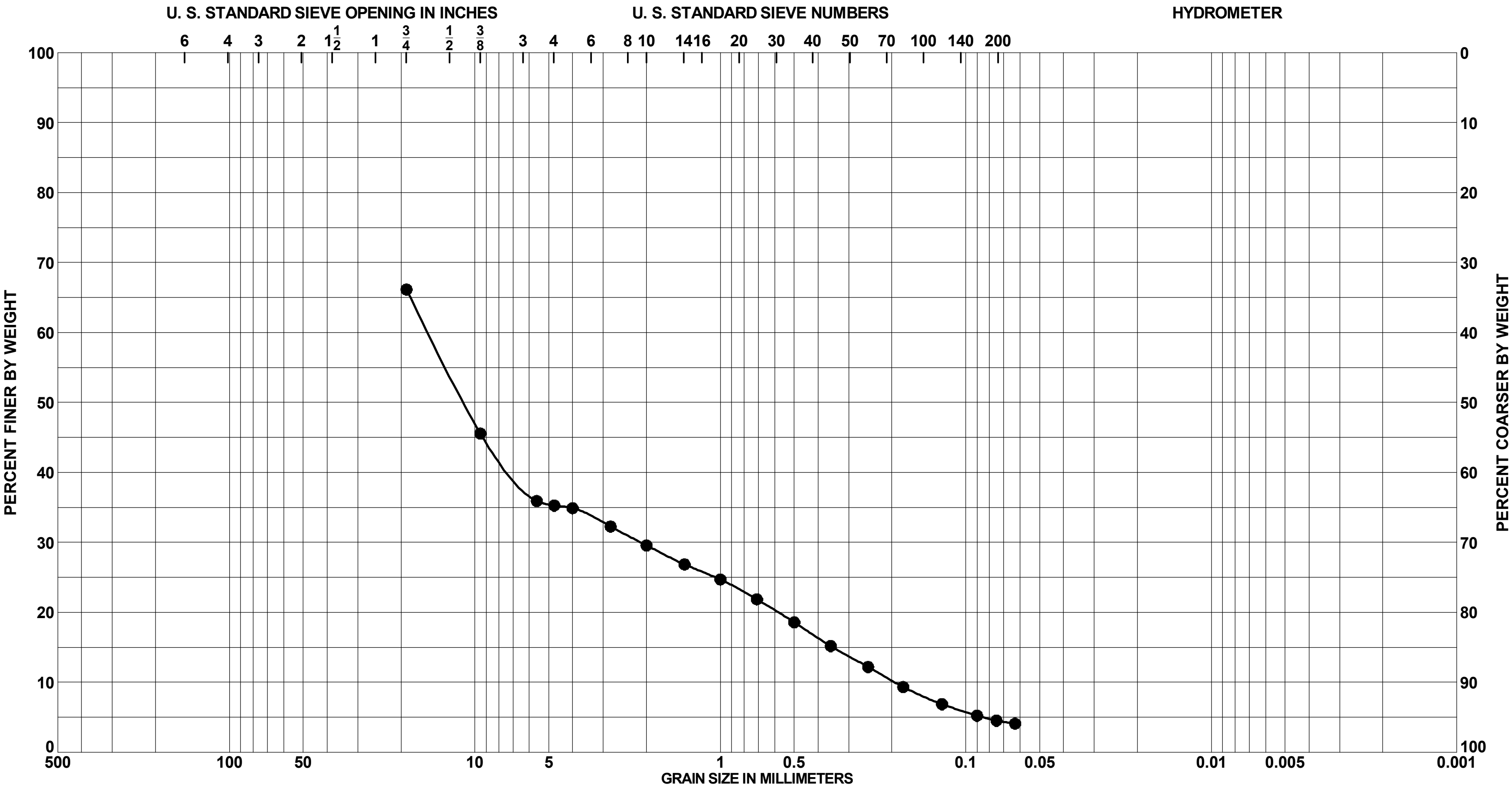
Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	%CO_3	G_s	Org %	w_n	LL	PL	PI
● 3	6.5 to 6.7 Ft.	SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, few medium-grained sand-sized limestone, trace silt, trace shell, gray (SP)	2.5Y 5/1	1.0							

PROJECT	King's Bay Inner Channel
BORING NO.	VB-KBIC16-4
BORING ELEV.	Not Determined
DATE	9/14/2016

GRADATION CURVES

SAJ FORM 2087
JUN 02

00320-8



COBBLES	GRAVEL		SAND			SILT OR CLAY
	COARSE	FINE	COARSE	MEDIUM	FINE	

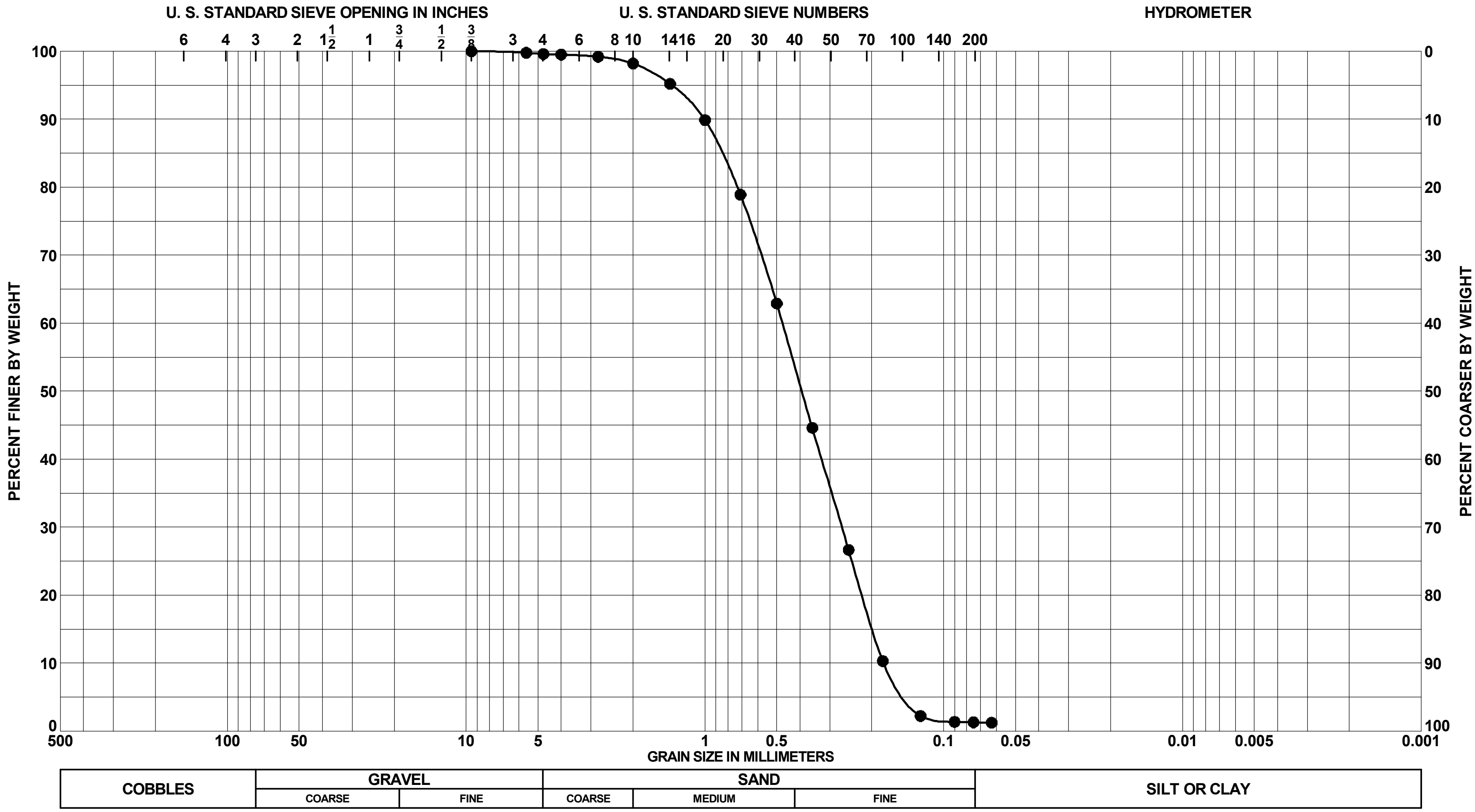
Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	%CO_3	G_s	Org %	w_n	LL	PL	PI
● 4	8.0 to 8.2 Ft.	LIMESTONE, poorly-graded, mostly sand to gravel-sized limestone, trace silt, trace quartz, dark gray (GP)	2.5Y 4/1	0.0							

PROJECT	King's Bay Inner Channel
BORING NO.	VB-KBIC16-4
BORING ELEV.	Not Determined
DATE	9/14/2016

GRADATION CURVES

SAJ FORM 2087
JUN 02

00320-9



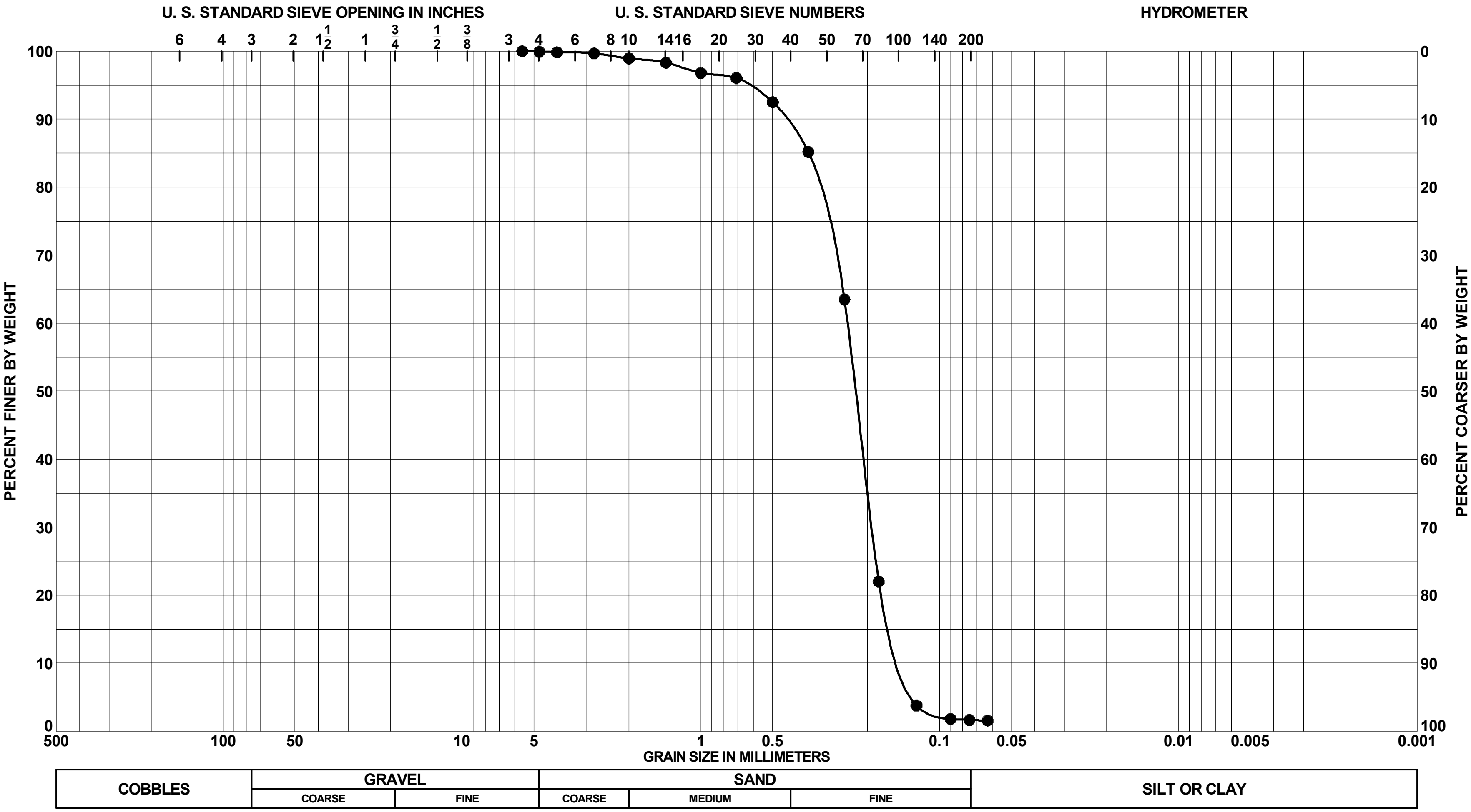
Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	%CO_3	G_s	Org %	w_n	LL	PL	PI
● 1	3.0 to 3.2 Ft.	SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, trace shell, trace silt, light gray (SP)	2.5Y 7/2	4.1							

PROJECT	King's Bay Inner Channel
BORING NO.	VB-KBIC16-5
BORING ELEV.	Not Determined
DATE	9/14/2016

GRADATION CURVES

SAJ FORM 2087
JUN 02

00320-10



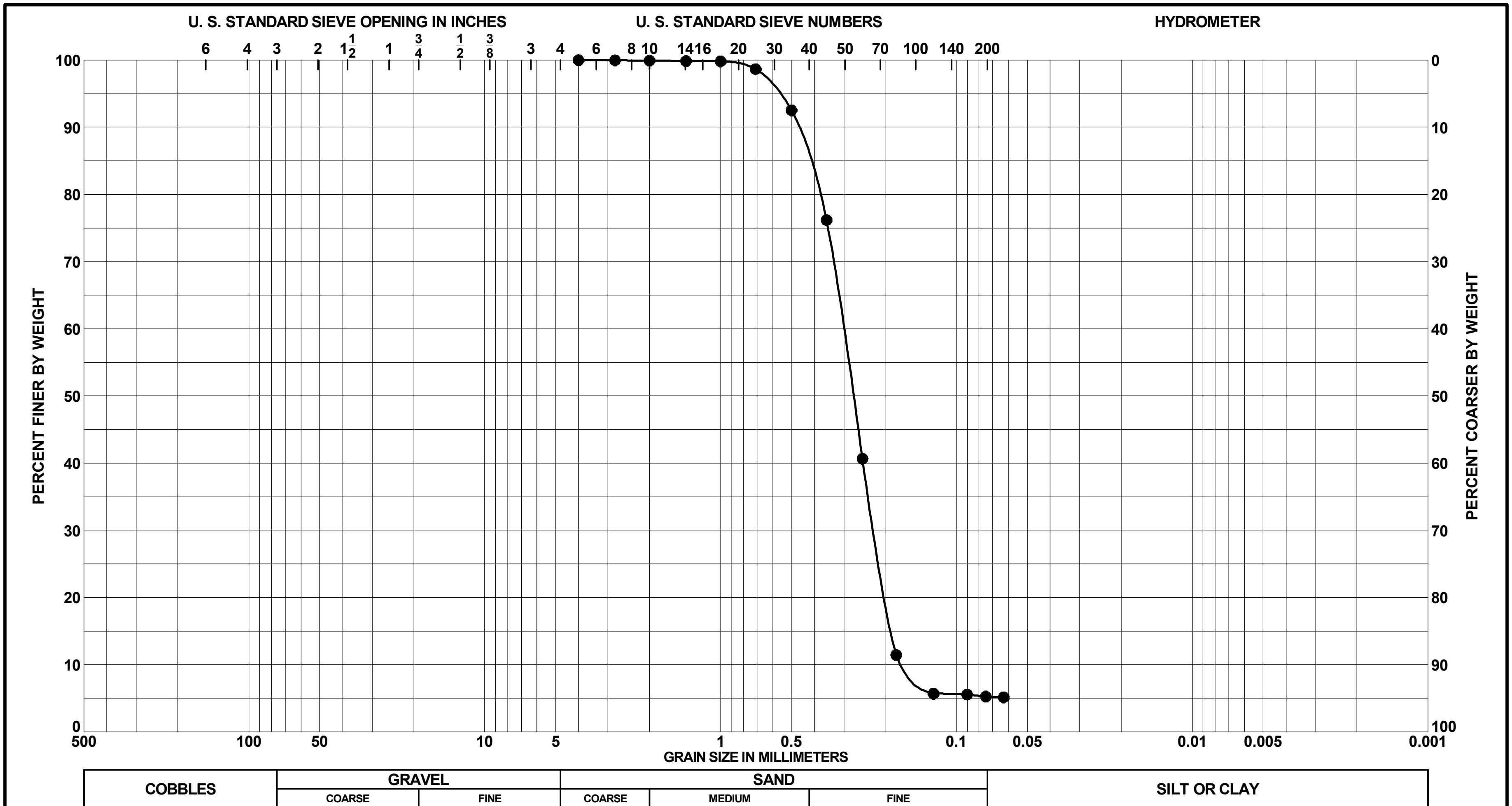
Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	%CO_3	G_s	Org %	w_n	LL	PL	PI
● 2	7.8 to 8.0 Ft.	SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, trace silt, light gray (SP)	2.5Y 7/2	0.9							

PROJECT	King's Bay Inner Channel
BORING NO.	VB-KBIC16-5
BORING ELEV.	Not Determined
DATE	9/14/2016

GRADATION CURVES

SAJ FORM 2087
JUN 02

00320-11



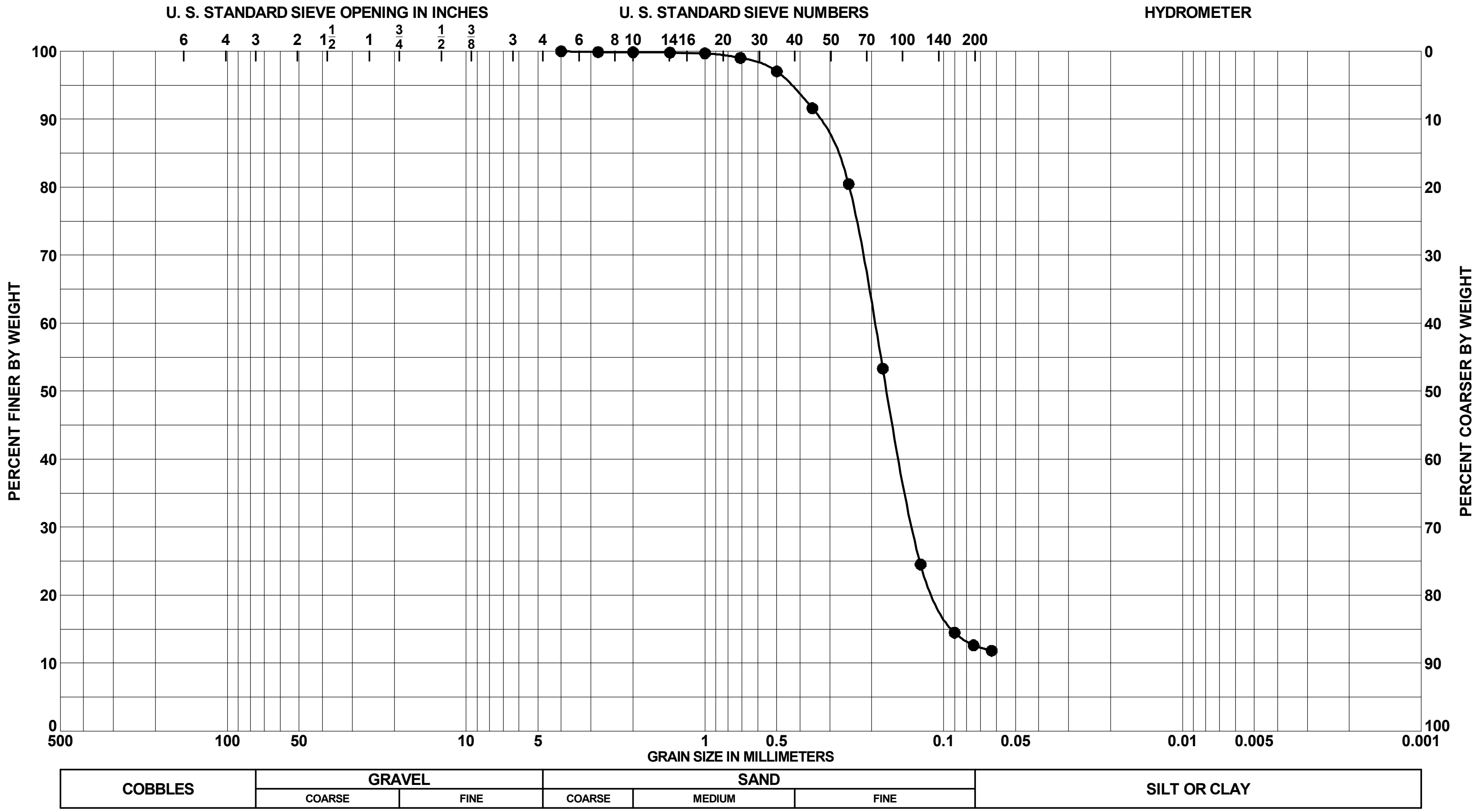
Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	%CO_3	G_s	Org %	w_n	LL	PL	PI
● 1	1.1 to 1.3 Ft.	SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, dark gray (SP-SM)	2.5Y 4/1	0.1							

GRADATION CURVES

PROJECT	King's Bay Inner Channel
BORING NO.	VB-KBIC16-6
BORING ELEV.	Not Determined
DATE	9/14/2016

SAJ FORM 2087
JUN 02

00320-12



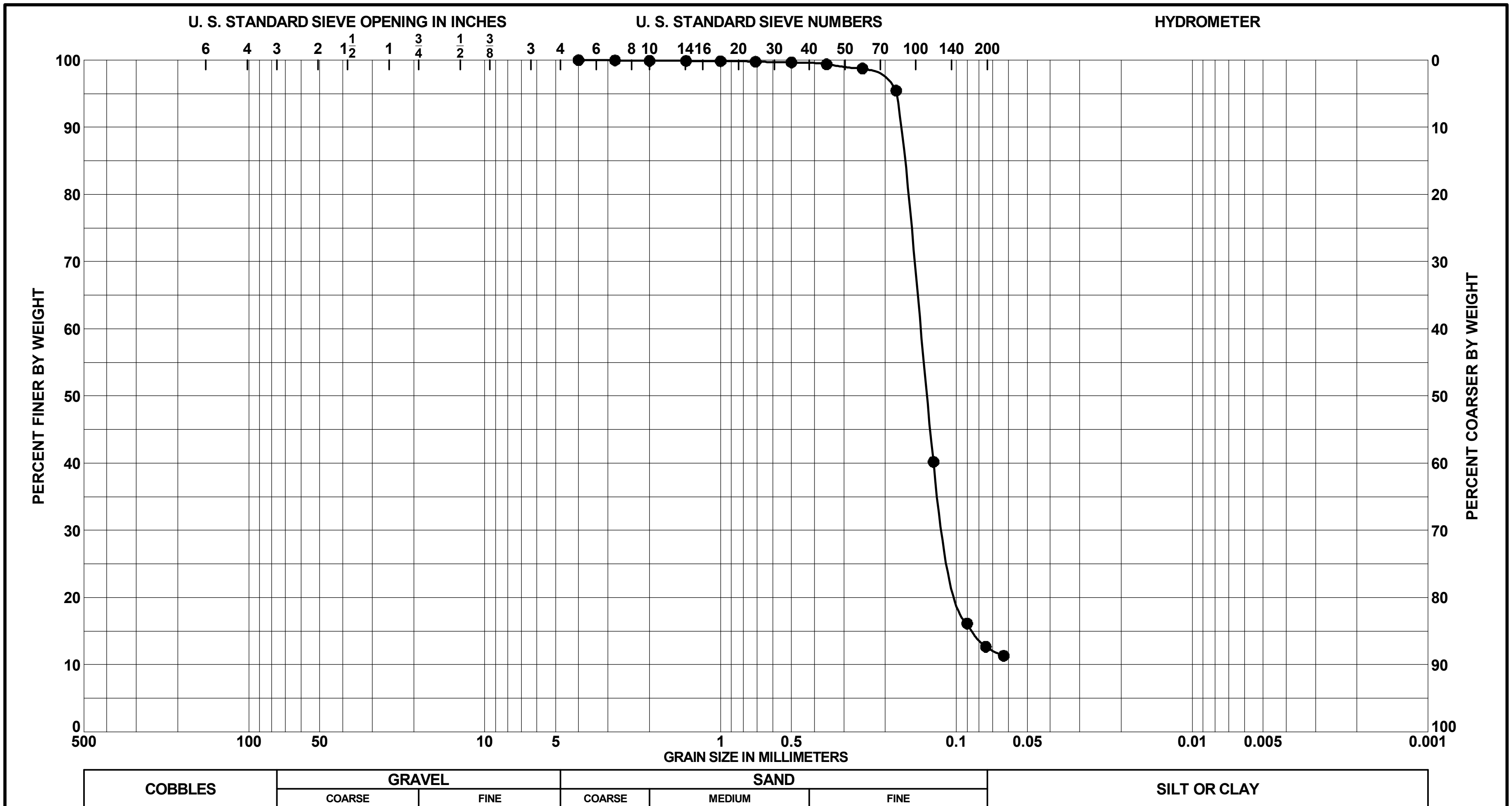
Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	$\%CO_3$	G_s	Org %	w_n	LL	PL	PI
● 2	4.0 to 4.2 Ft.	SAND, silty, mostly fine to medium-grained sand-sized quartz, little silt, trace organics, dark gray (SM)	2.5Y 4/1	0.0							

GRADATION CURVES

PROJECT	King's Bay Inner Channel
BORING NO.	VB-KBIC16-6
BORING ELEV.	Not Determined
DATE	9/14/2016

SAJ FORM 2087
JUN 02

00320-13

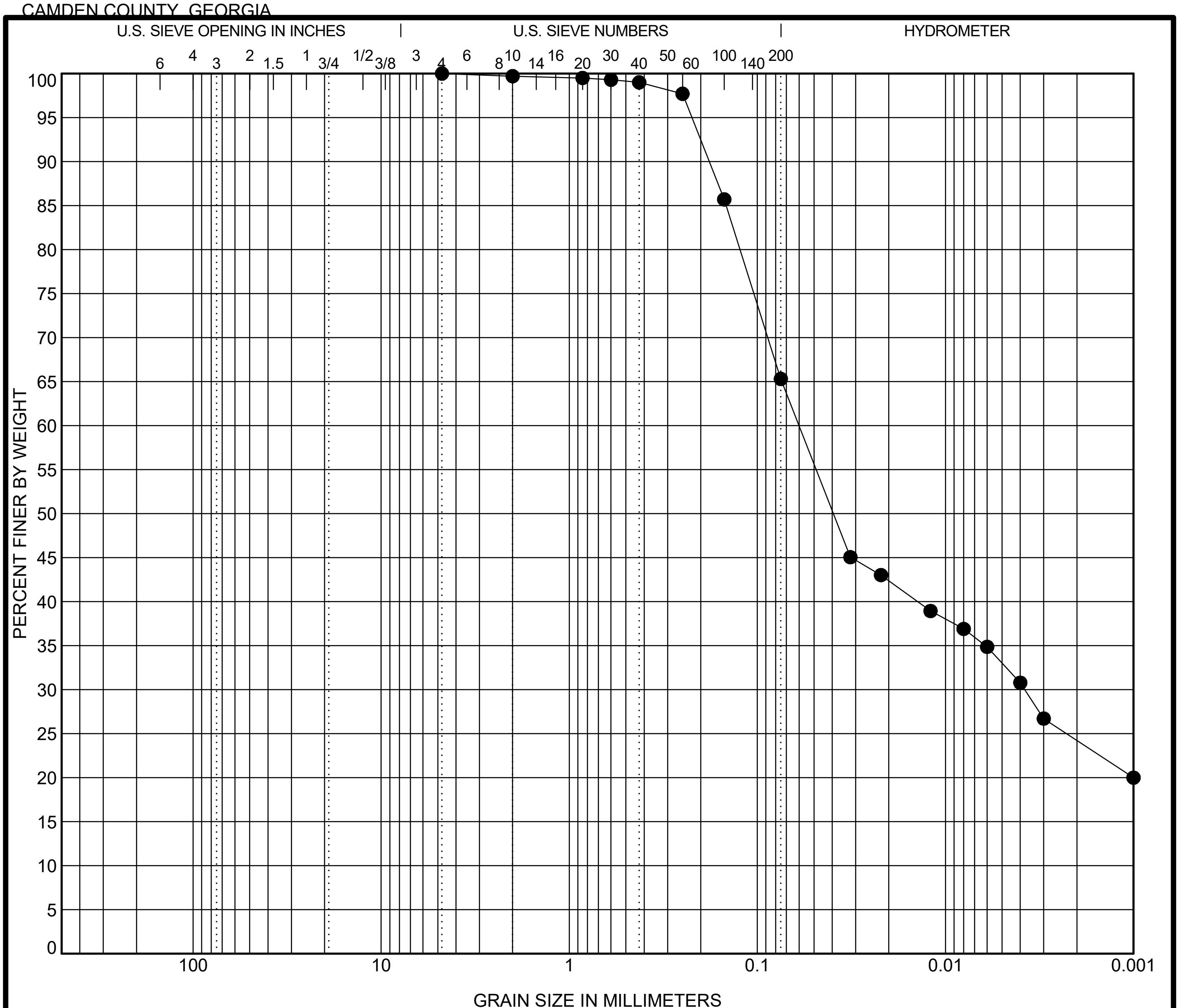


Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	$\%CO_3$	G_s	Org %	w_n	LL	PL	PI
● 3	7.9 to 8.1 Ft.	SAND, silty, mostly fine-grained sand-sized quartz, little silt, trace organics, light brownish gray (SM)	2.5Y 6/2	0.0							

PROJECT	King's Bay Inner Channel
BORING NO.	VB-KBIC16-6
BORING ELEV.	Not Determined
DATE	9/14/2016

GRADATION CURVES

SAJ FORM 2087
JUN 02



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-3	0.2 ft	**SAND, silty, trace organic matter, 10Y 2.5/1 greenish black (SM) [Estimated shell content < 1.0%]**	**75**	**41**	**34**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.06**	**0.004**		**0.0**	**34.7**	**41.1**	**24.2**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	65.3	85.7	97.7	99.0	99.3	99.5	99.7	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/19/19	SRH	MDE

Schnabel ENGINEERING

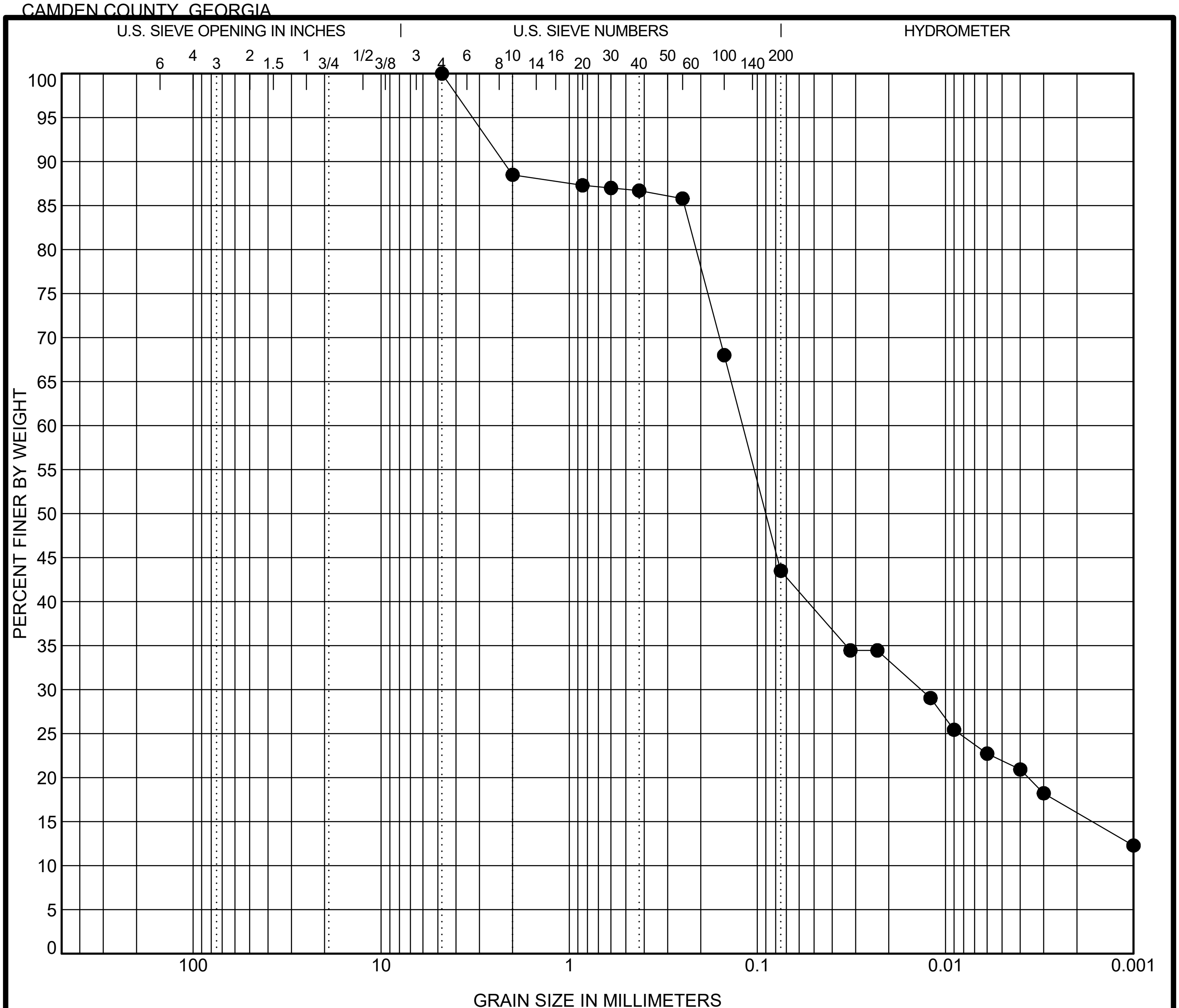
GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00

SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 7/16/19

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-3	1.7 ft	**SILT, trace organic matter, wet, 10Y 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]**	**62**	**32**	**30**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.12**	**0.013**		**0.0**	**56.5**	**27.5**	**16.0**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	43.5	68.0	85.8	86.7	87.0	87.3	88.5	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/19/19	SRH	MDE

Schnabel ENGINEERING

GRADATION CURVE

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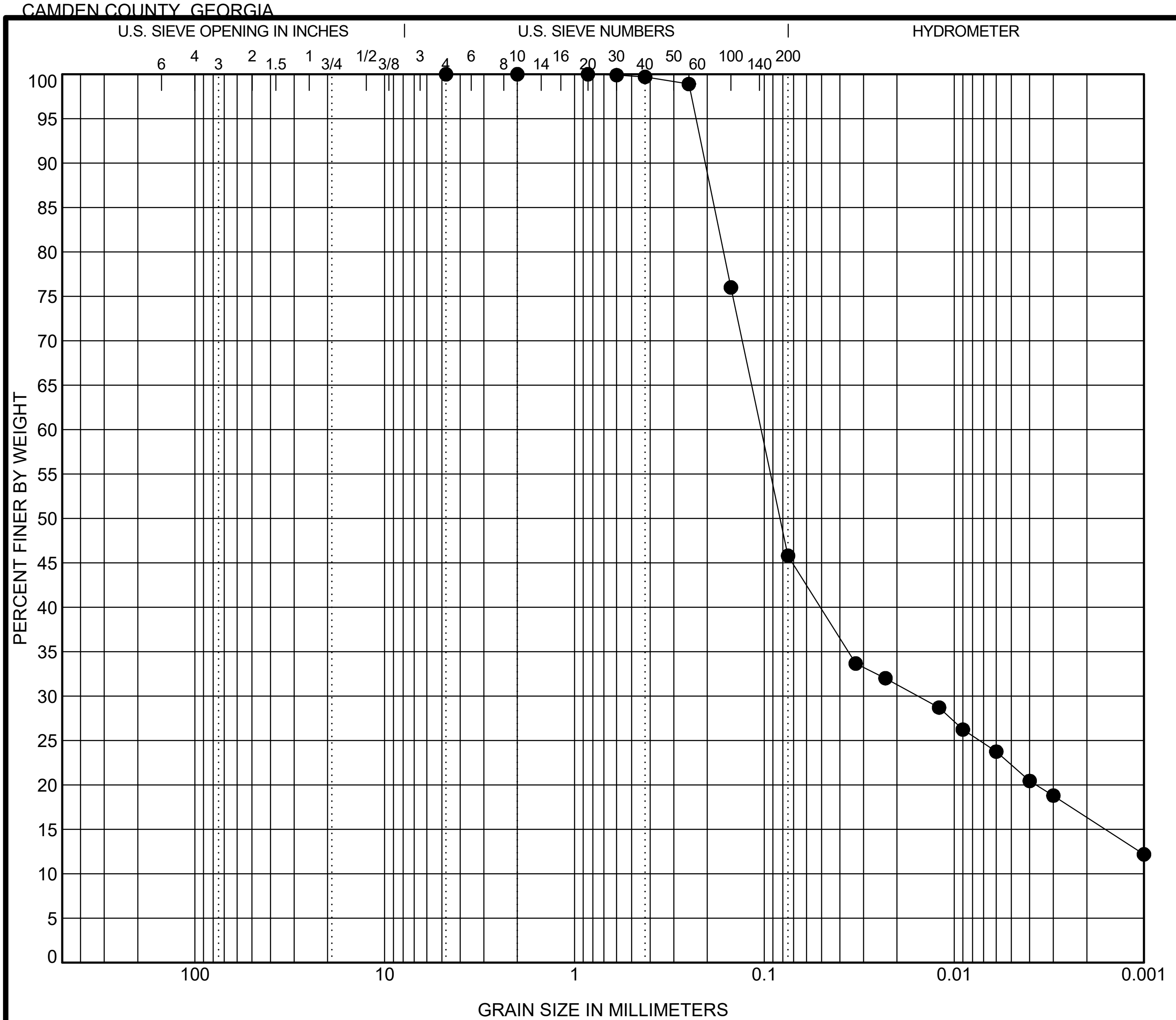
NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-3	2.8 ft	**SAND, silty, 10Y 2.5/1 greenish black (SM) [Estimated shell content < 1.0%]**	**NP**	**NP**	**NP**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**0.85**	**0.104**	**0.015**		**0.0**	**54.2**	**29.4**	**16.4**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	45.8	76.0	98.9	99.7	99.9	100.0	100.0	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/25/19	SRH	EC

Schnabel ENGINEERING

GRADATION CURVE

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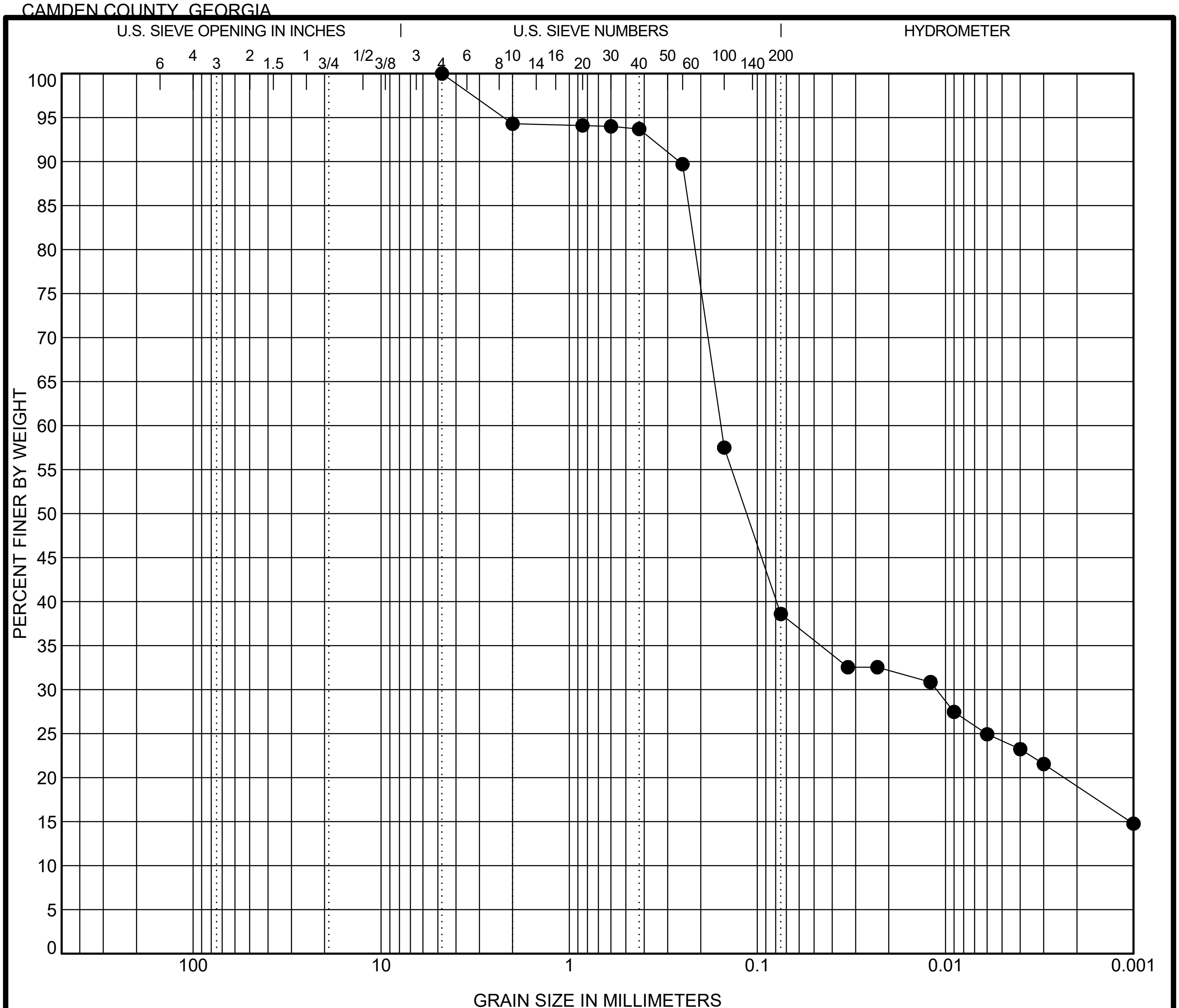
SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 7/16/19

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

Specimen		Sample Description	LL	PL	PI	Test Method
● VB-KBIC19-4	0.8 ft	**SAND, silty, N 2.5/ black (SM) [Estimated shell content < 1.0%]**	**80**	**45**	**35**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.156**	**0.011**		**0.0**	**61.4**	**19.6**	**19.0**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	38.6	57.5	89.7	93.7	94.0	94.1	94.3	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/21/19	SRH	EC

Schnabel ENGINEERING

GRADATION CURVE

Project: P617 Transit Program
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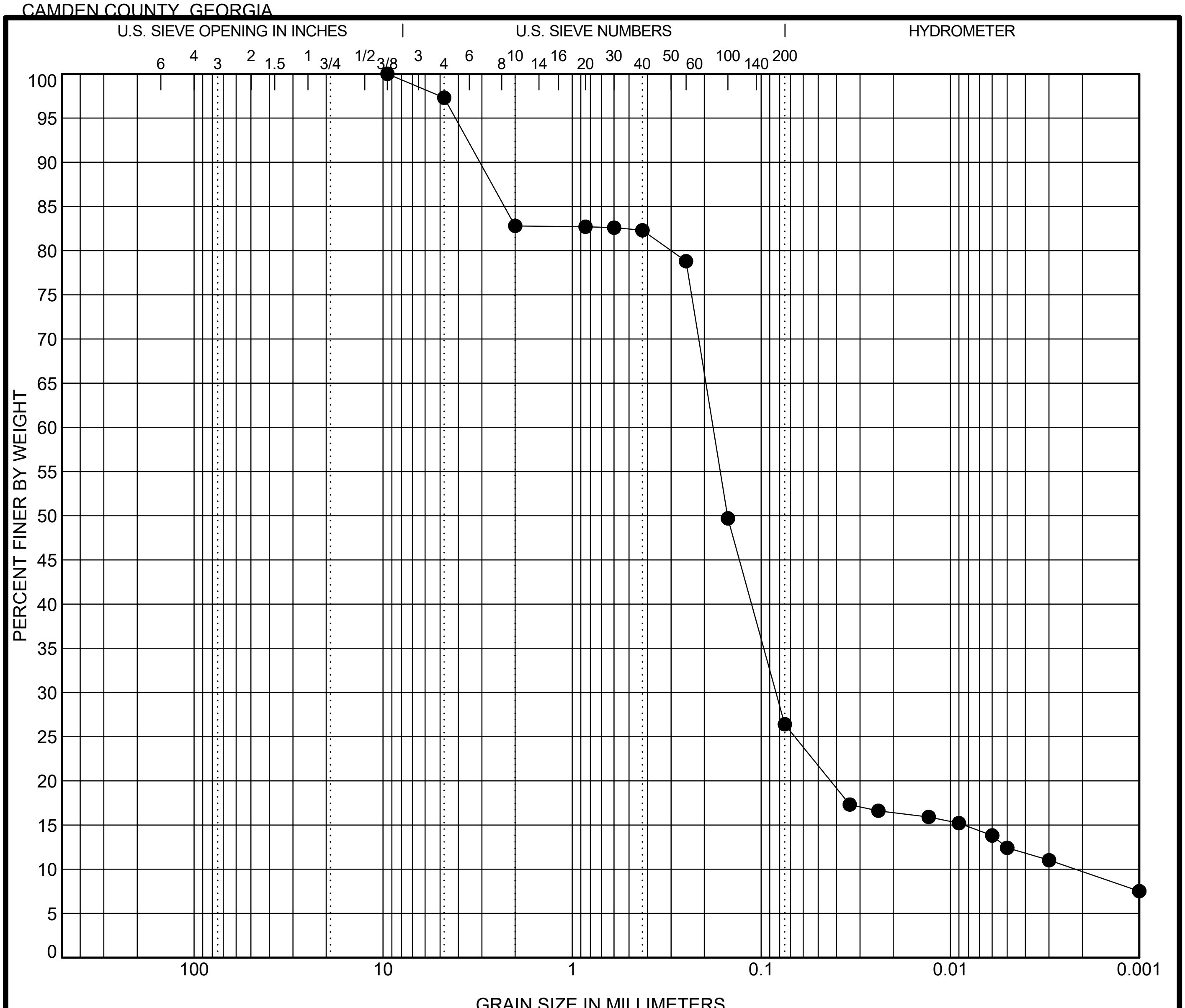
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00

SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 7/16/19



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-4	1.3 ft	**SAND, silty, N 2.5/ black (SM) [Estimated shell content <3.0%]**	**93**	**47**	**46**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**9.5**	**0.18**	**0.083**	**0.002**	**2.7**	**70.9**	**16.7**	**9.7**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4	3/8"
% Finer	26.4	49.7	78.8	82.3	82.6	82.7	82.8	97.3	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/20/19	SRH	EC

Schnabel ENGINEERING

GRADATION CURVE

Project: P617 Transit Program
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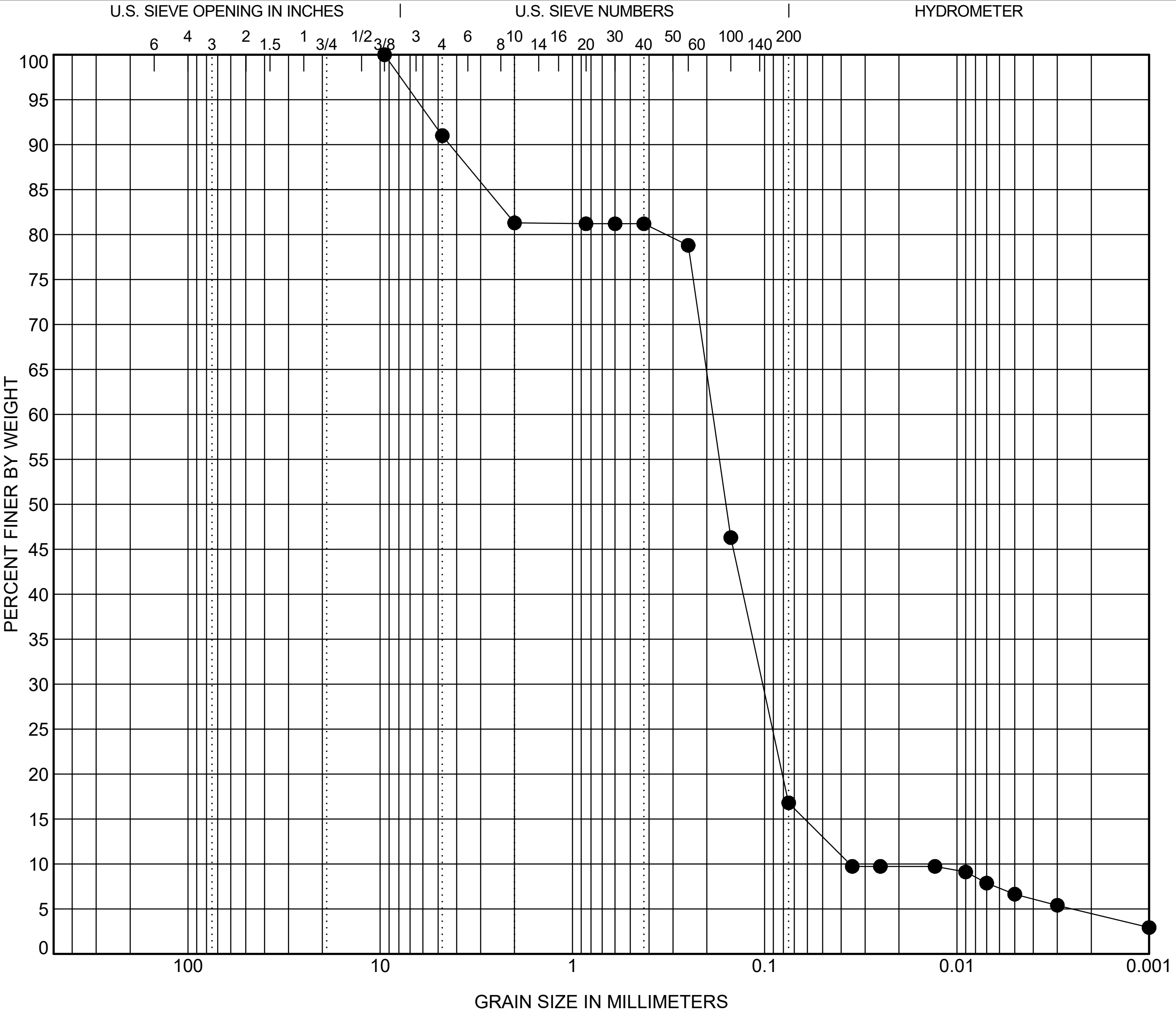
SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 7/16/19



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-4	1.7 ft	**SAND, silty, little sand, 2.5Y 3/1 very dark gray (SM) [Estimated shell content <9.0%]**	**92**	**55**	**37**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**9.5**	**0.186**	**0.102**	**0.036**	**9.0**	**74.2**	**12.3**	**4.5**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4	3/8"
% Finer	16.8	46.3	78.8	81.2	81.2	81.2	81.3	91.0	100.0

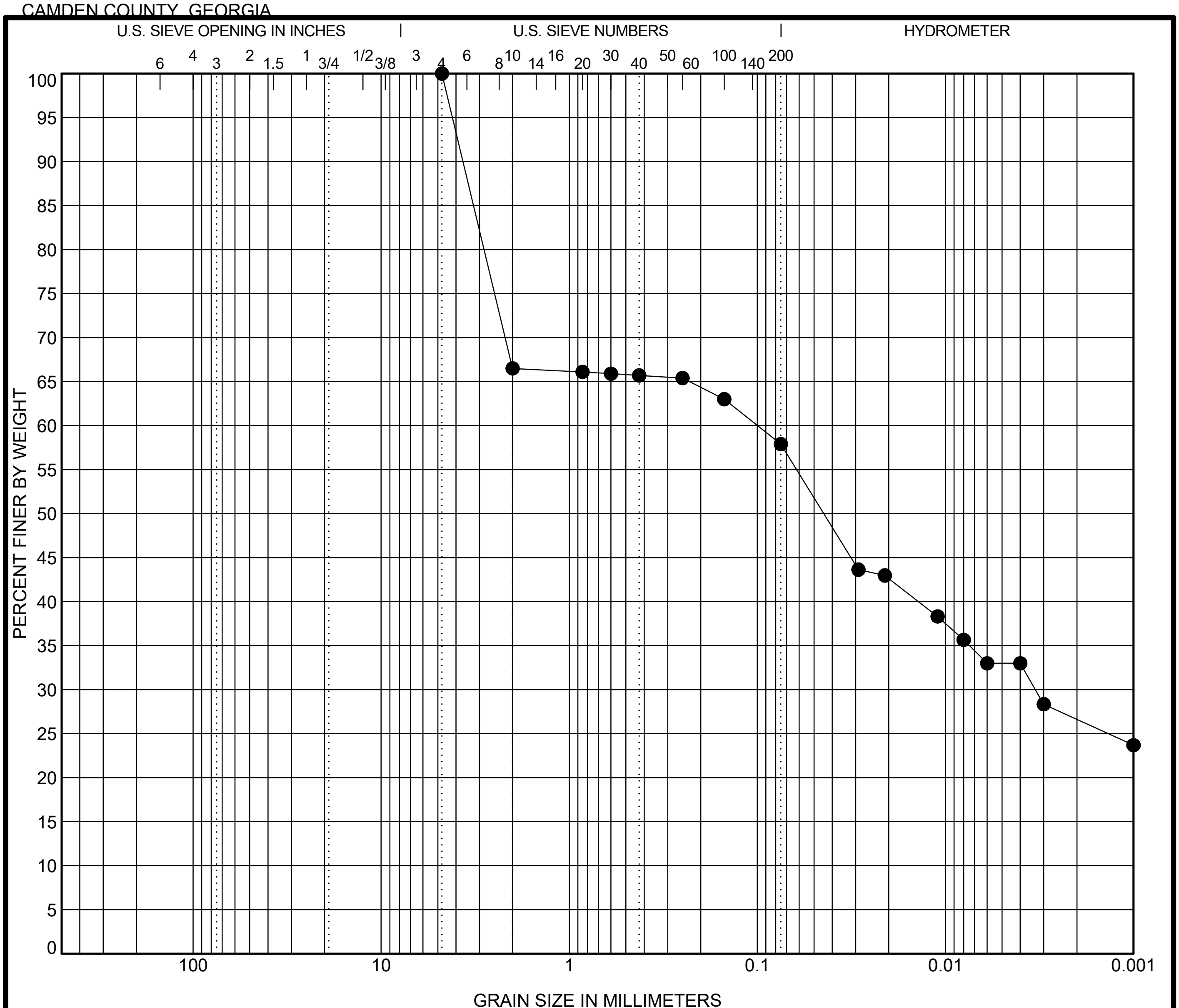
Tested By	Tested Date	Reviewed by	Calc by
CAB	6/20/19	SRH	EC

Schnabel ENGINEERING

GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-4	2.5 ft	**SILT, 5Y 2.5/2 black (MH) [Estimated shell content < 1.0%]**	**94**	**52**	**42**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.1**	**0.003**		**0.0**	**42.1**	**31.3**	**26.6**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	57.9	63.0	65.4	65.7	65.9	66.1	66.5	100.0

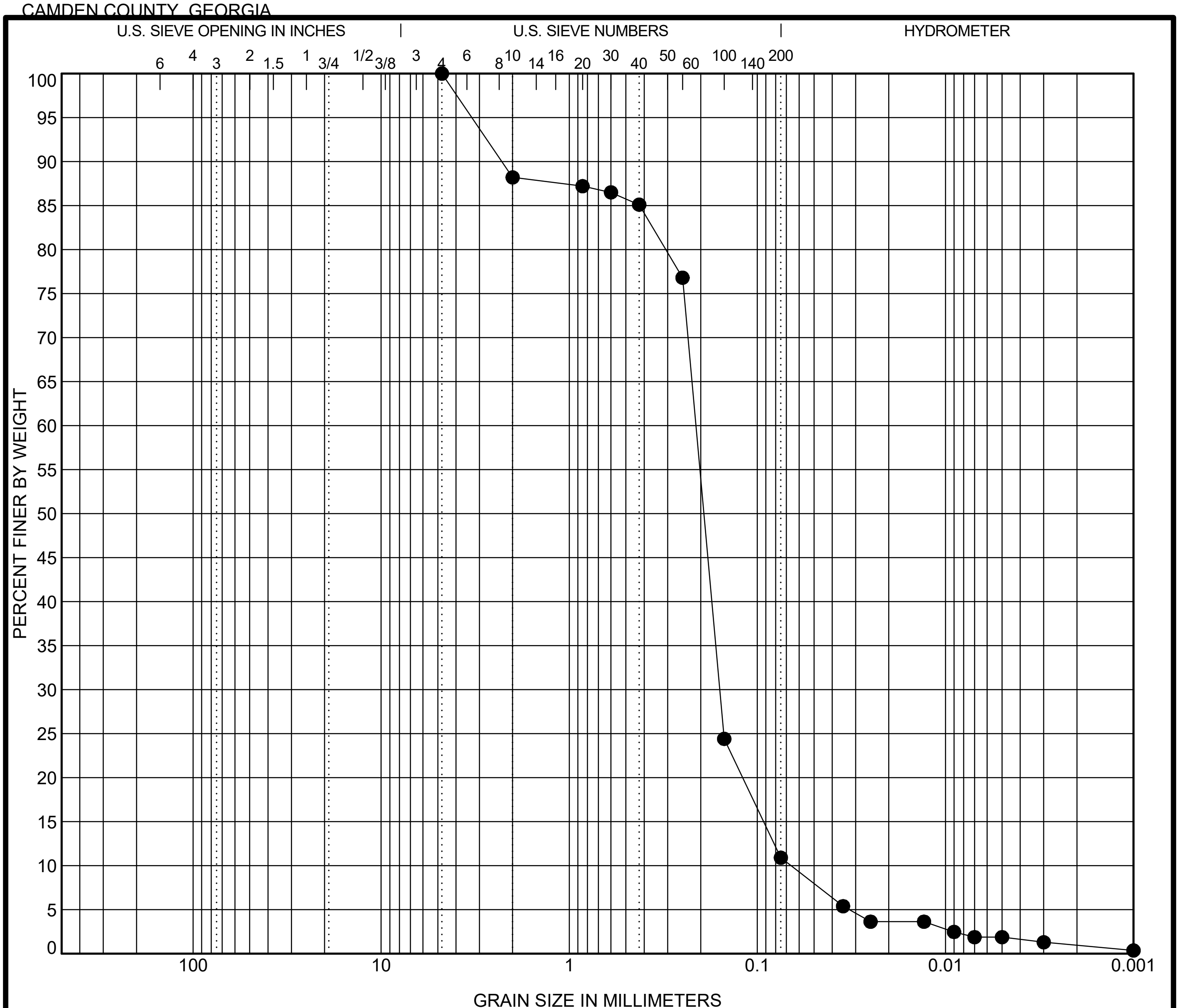
Tested By	Tested Date	Reviewed by	Calc by
CAB	6/24/19	SRH	EC

Schnabel ENGINEERING

GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-5	0.8 ft	**SAND, poorly-graded with silt, some shell fragments, 5Y 6/1 gray (SP-SM) [Estimated shell content < 1.0%]**	**NP**	**NP**	**NP**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.212**	**0.158**	**0.066**	**0.0**	**89.1**	**10.0**	**0.9**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	10.9	24.4	76.8	85.1	86.5	87.2	88.2	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/19/19	SRH	MDE

Schnabel ENGINEERING

GRADATION CURVE

Project: P617 Transit Program
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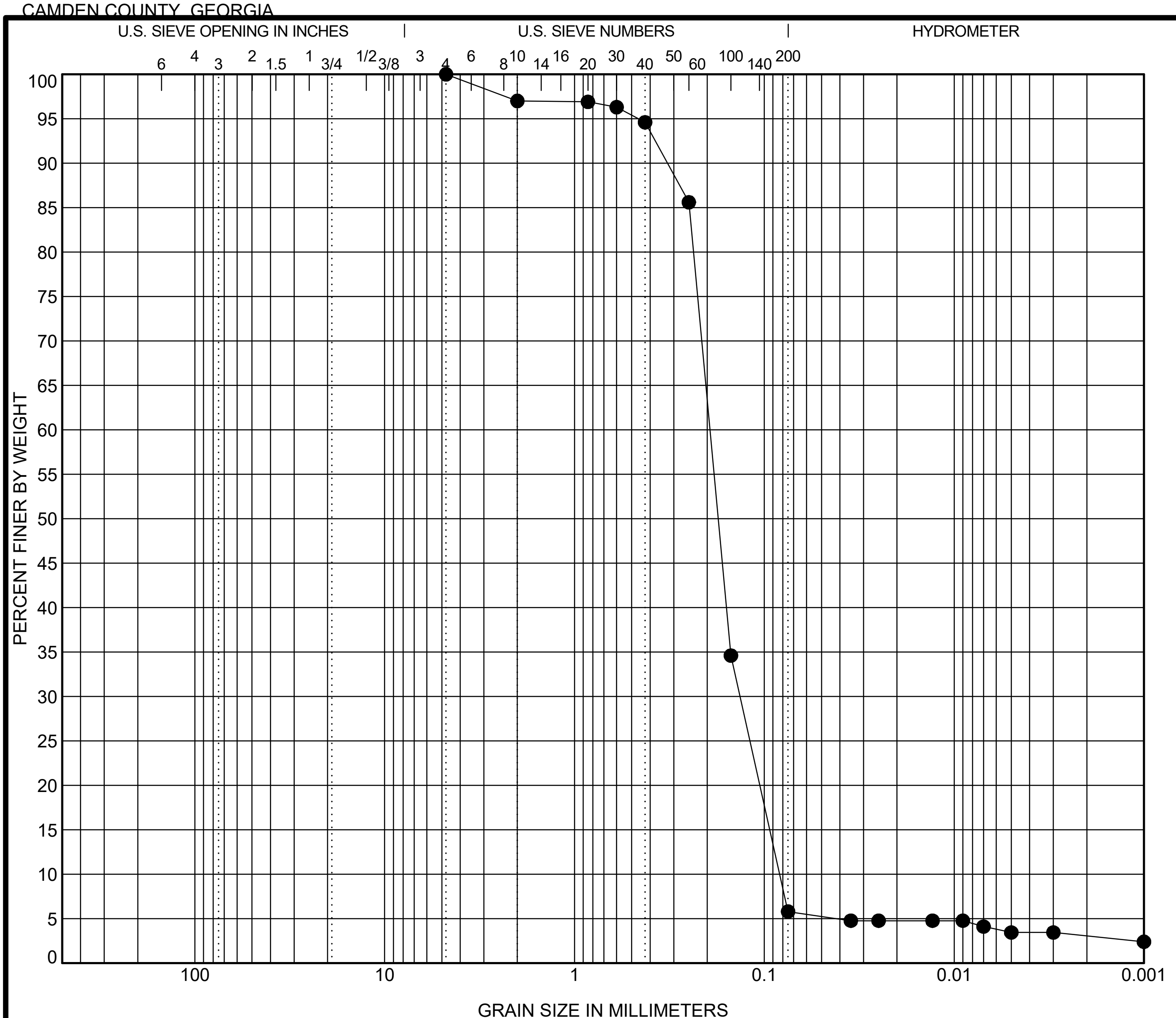
Contract: 19C19018.00

SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 7/16/19

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-5	2.5 ft	**SAND, poorly-graded with silt, 2.5Y 4/1 dark gray (SP-SM) [Estimated shell content < 1.0%]**	**NP**	**NP**	**NP**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.193**	**0.134**	**0.083**	**0.0**	**94.2**	**2.7**	**3.1**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	05.8	34.6	85.6	94.6	96.3	96.9	97.0	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/19/19	SRH	MDE

Schnabel ENGINEERING

GRADATION CURVE

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Kings Bay Inner Channel, Camden Co., GA

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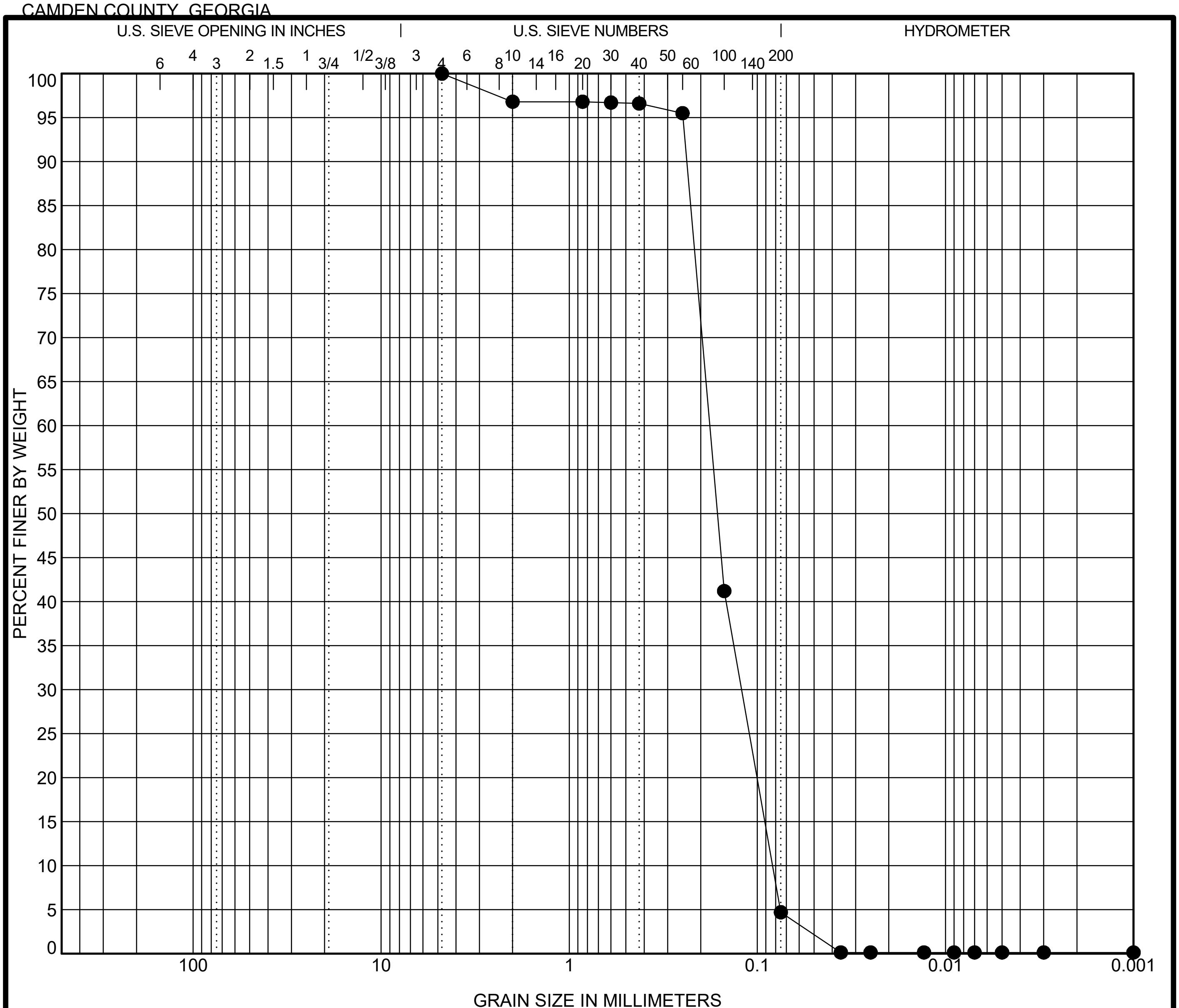
NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-6	0.2 ft	**SAND, poorly-graded, some fine to medium-grained sand-sized silt, 2.5YR 4/1 dark reddish gray (SP) [Estimated shell content < 1.0%]**	**NP**	**NP**	**NP**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.179**	**0.121**	**0.083**	**0.0**	**95.3**	**4.6**	**0.1**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	04.7	41.2	95.5	96.6	96.7	96.8	96.8	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/19/19	SRH	MDE

Schnabel ENGINEERING

GRADATION CURVE

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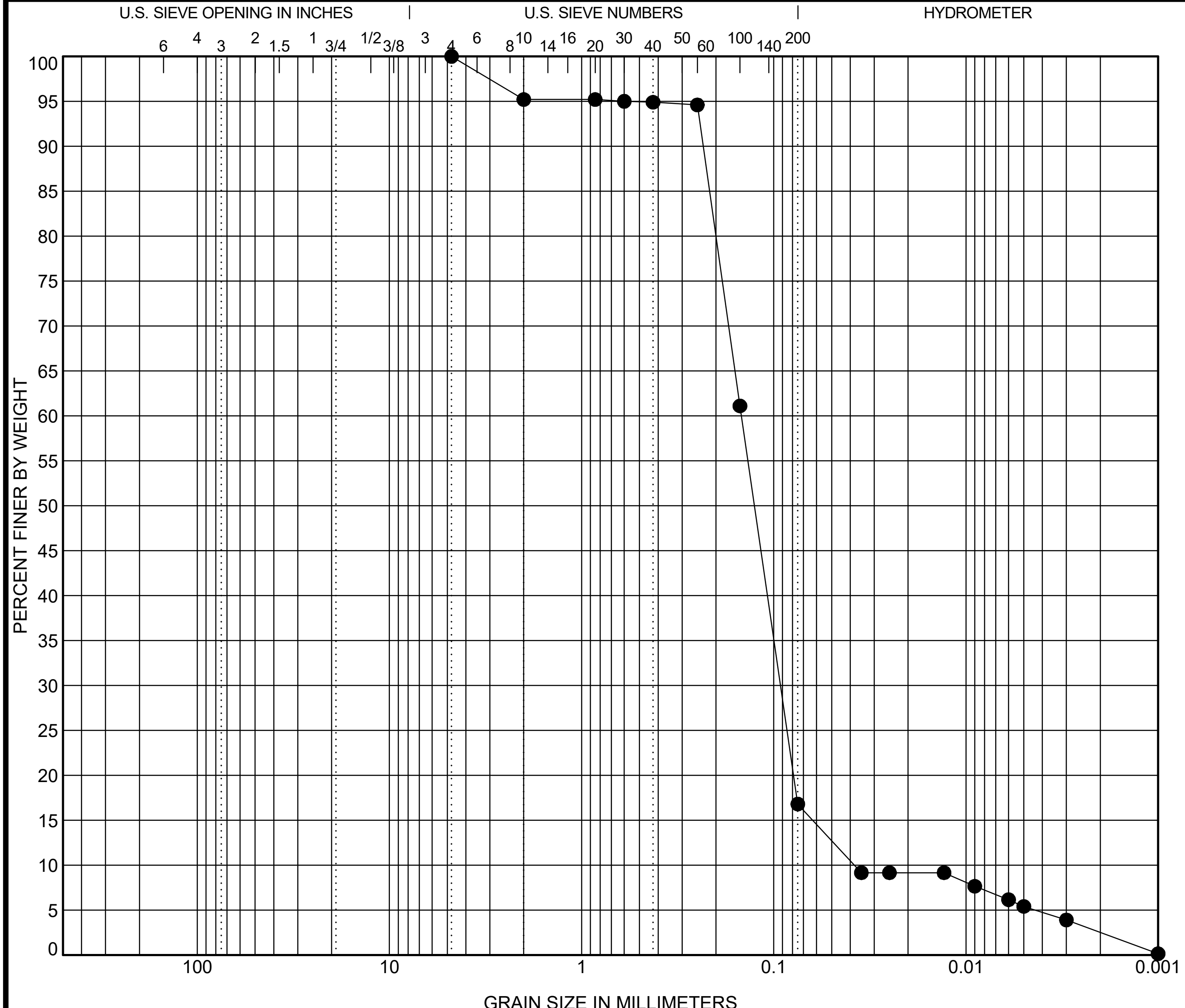
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00

SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 7/16/19



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-6	1.8 ft	**SAND, silty, 2.5YR 3/1 dark reddish gray (SM) [Estimated shell content < 1.0%]**	**NP**	**NP**	**NP**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.147**	**0.092**	**0.038**	**0.0**	**83.2**	**14.3**	**2.5**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	16.8	61.1	94.6	94.9	95.0	95.2	95.2	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/19/19	SRH	MDE

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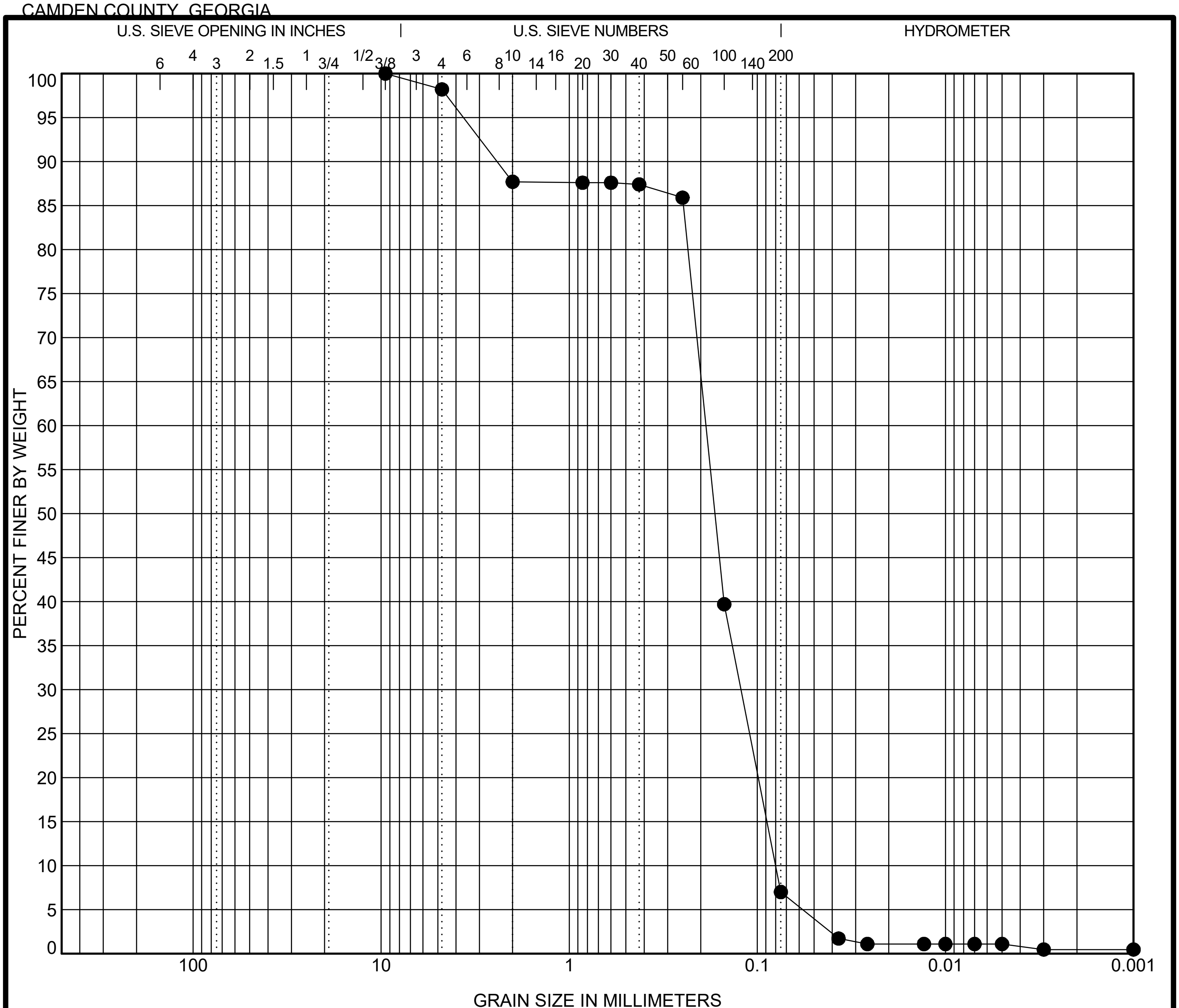
Contract: 19C19018.00

SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 7/16/19

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-6	2.5 ft	**SAND, poorly-graded with silt, some sand, 10Y 2.5/1 greenish black (SP-SM) [Estimated shell content <21.0%]**	**NP**	**NP**	**NP**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**9.5**	**0.188**	**0.122**	**0.08**	**1.8**	**91.2**	**6.5**	**0.5**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4	3/8"
% Finer	07.0	39.7	85.9	87.4	87.6	87.6	87.7	98.2	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/20/19	SRH	EC

Schnabel ENGINEERING

GRADATION CURVE

Project: P617 Transit Program
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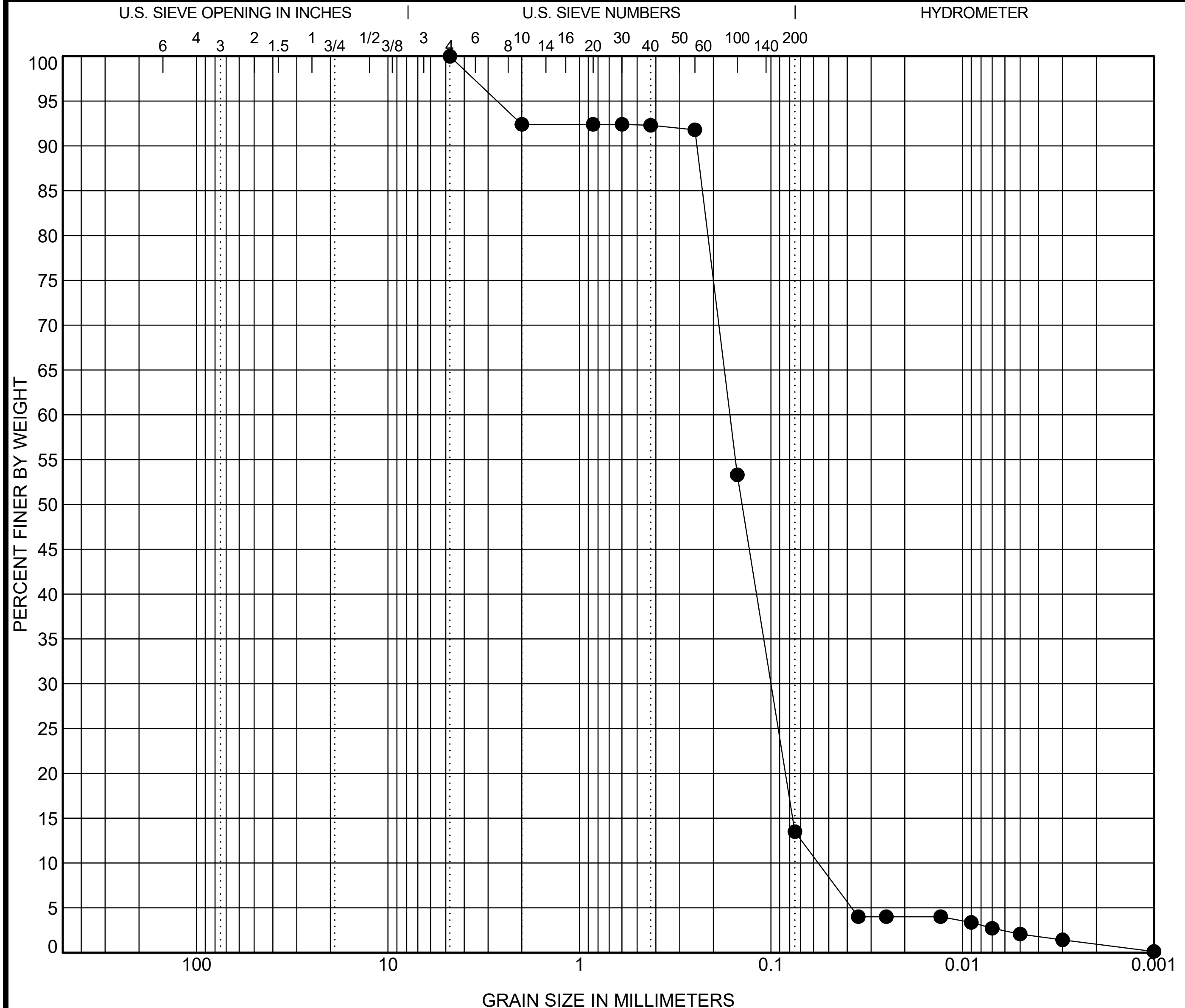


COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-6	3.8 ft	**SAND, silty, 5Y 3/1 very dark gray (SM) [Estimated shell content < 1.0%]**	**NP**	**NP**	**NP**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.164**	**0.1**	**0.057**	**0.0**	**86.5**	**12.6**	**0.9**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	13.5	53.3	91.8	92.3	92.4	92.4	92.4	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/19/19	SRH	MDE

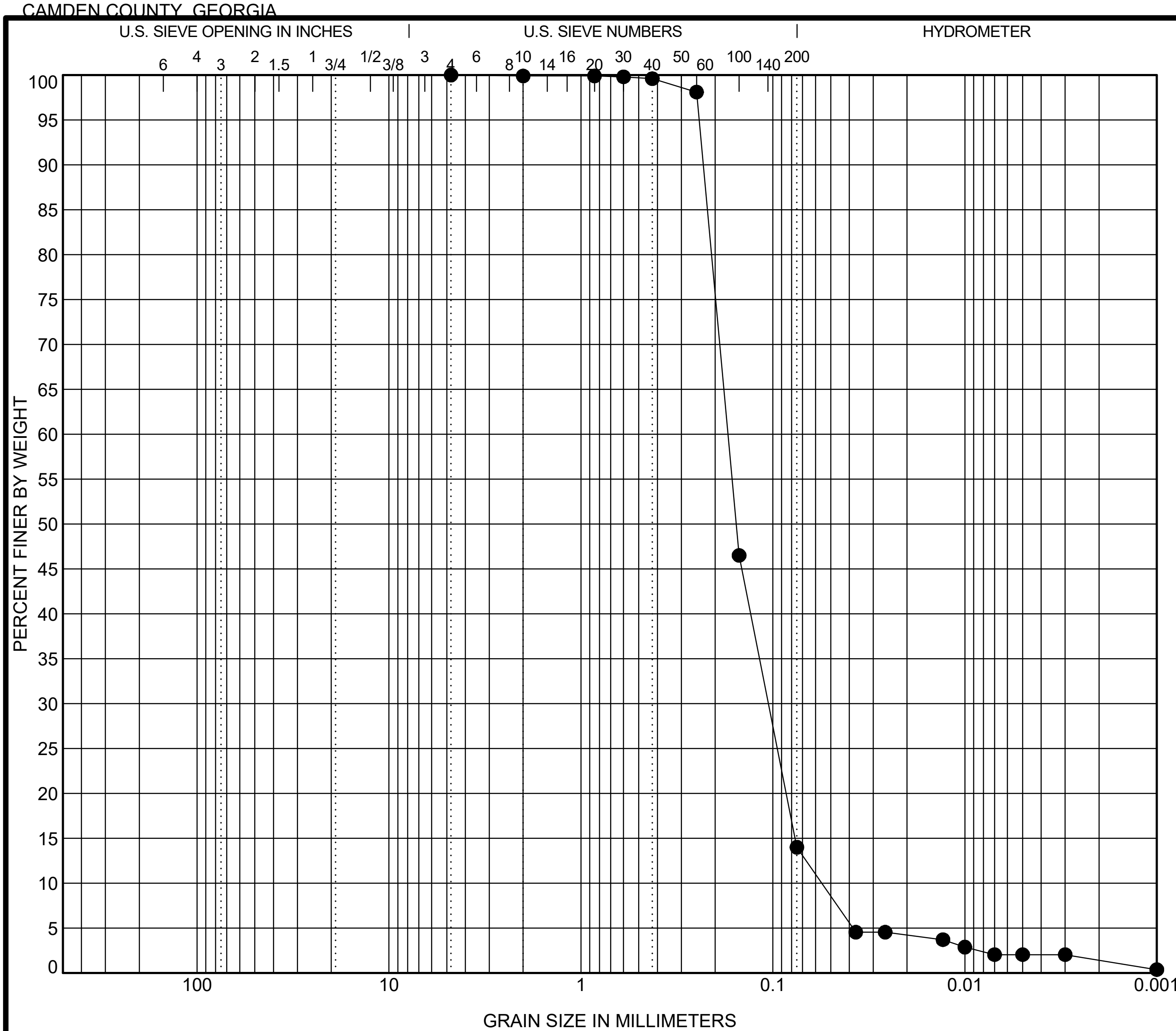
Schnabel ENGINEERING

GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-7	0.2 ft	**SAND, silty, 10Y 4/1 dark greenish gray (SM) [Estimated shell content < 1.0%]**	**NP**	**NP**	**NP**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.171**	**0.106**	**0.056**	**0.0**	**86.0**	**12.6**	**1.4**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	14.0	46.5	98.1	99.6	99.8	99.9	99.9	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/21/19	SRH	EC

Schnabel
ENGINEERING

GRADATION CURVE

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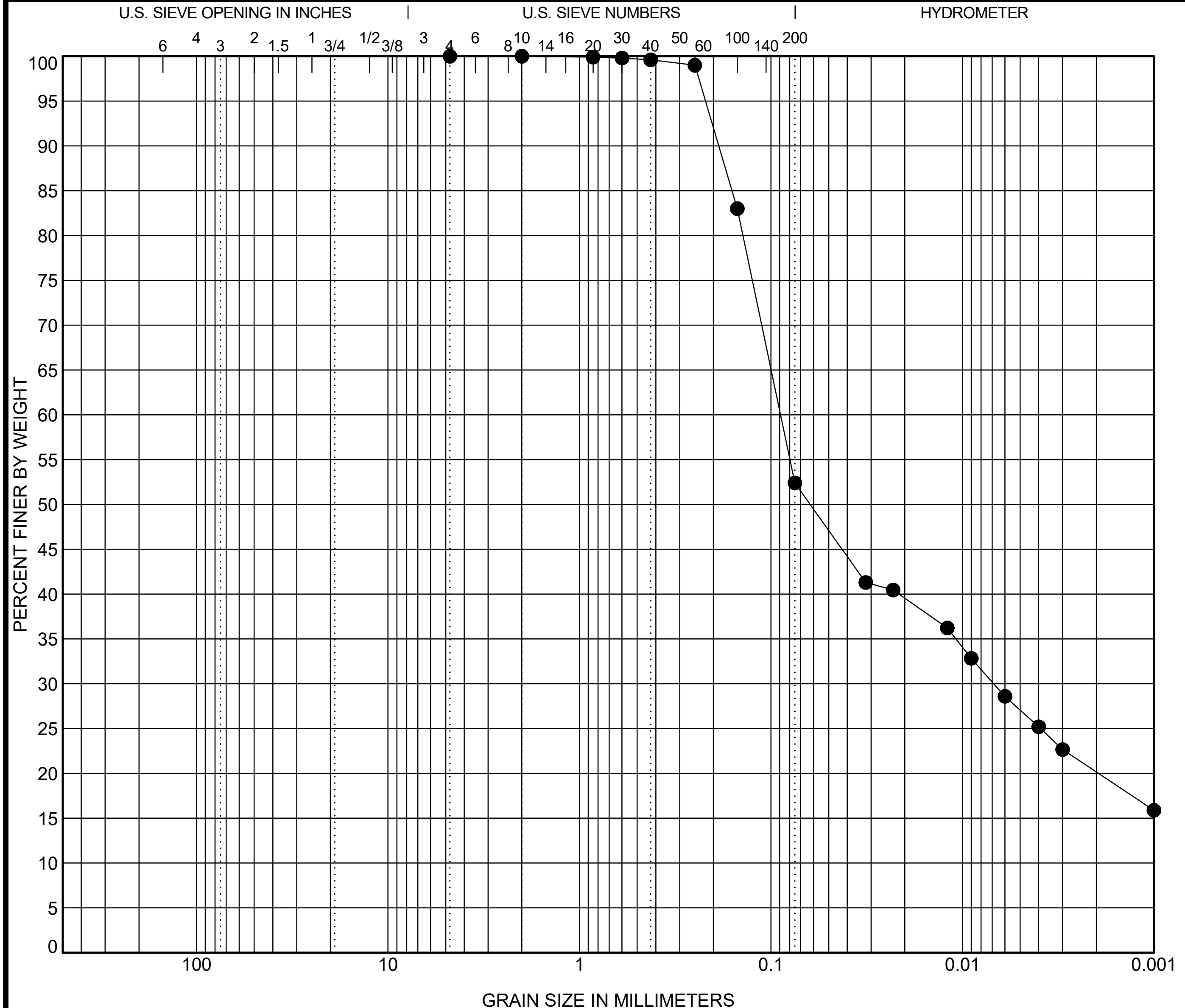
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00

SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-7	1.8 ft	**SILT, 5Y 3/1 very dark gray (MH) [Estimated shell content < 1.0%]**	**67**	**33**	**34**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**2**	**0.089**	**0.007**		**0.0**	**47.6**	**32.2**	**20.2**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	52.4	83.0	99.0	99.6	99.8	99.9	100.0	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/25/19	SRH	EC

Schnabel ENGINEERING

GRADATION CURVE

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Kings Bay Inner Channel, Camden Co., GA

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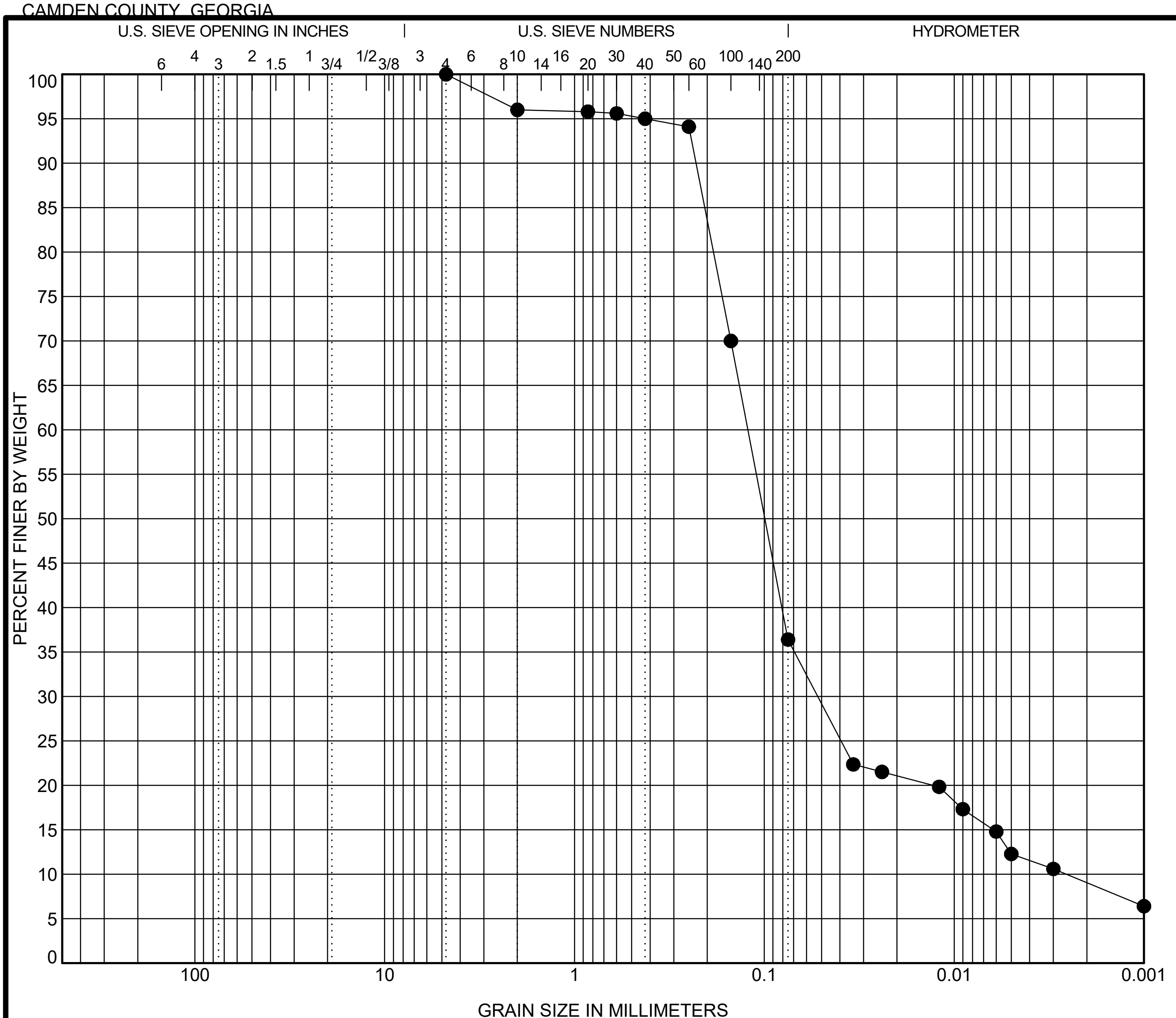
NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-7	2.5 ft	**SAND, silty, 5Y 3/1 very dark gray (SM) [Estimated shell content < 1.0%]**	**39**	**25**	**14**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.122**	**0.052**	**0.003**	**0.0**	**63.6**	**27.4**	**9.0**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	36.4	70.0	94.1	95.0	95.6	95.8	96.0	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/24/19	SRH	EC

Schnabel ENGINEERING

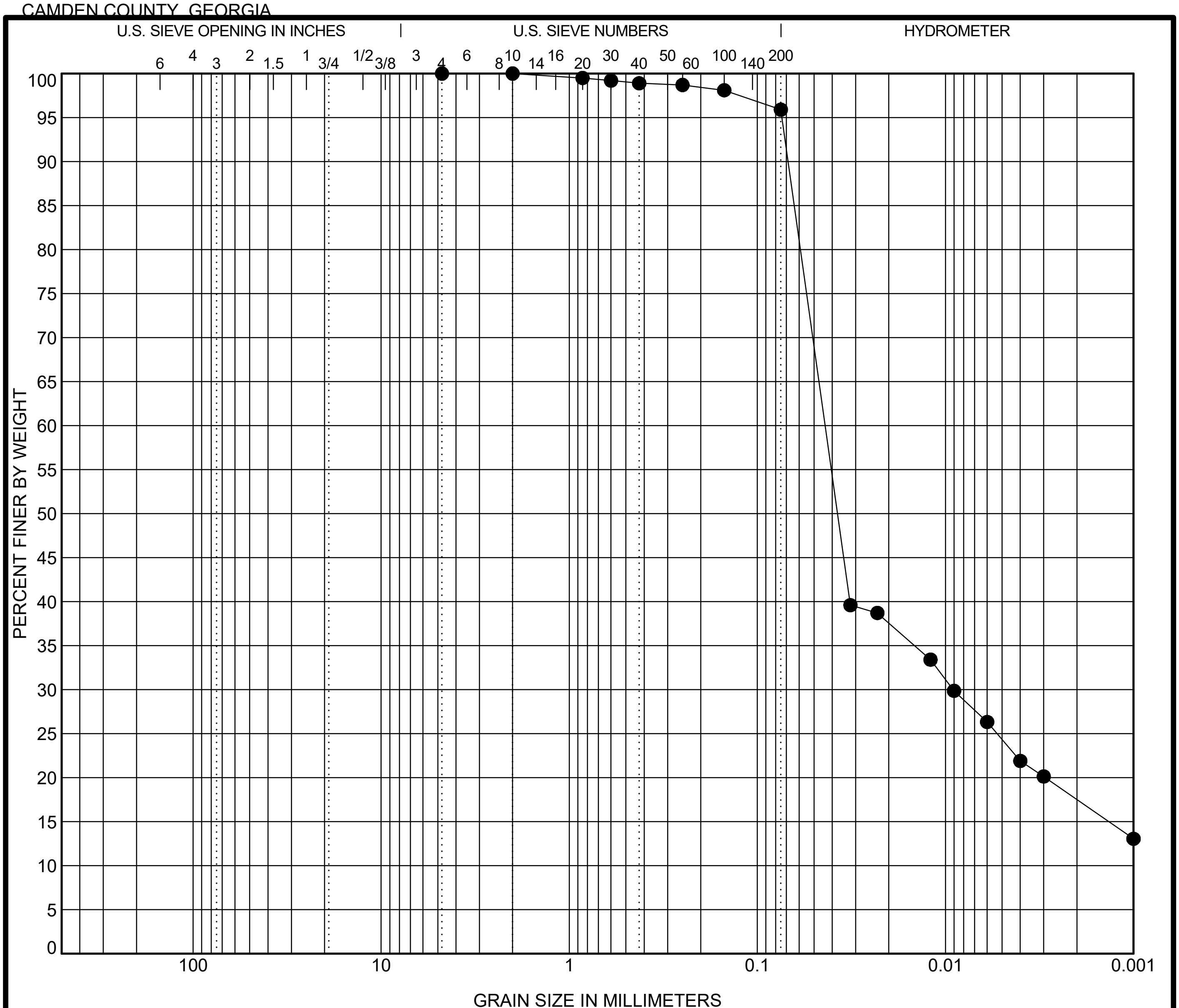
GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00

SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-8	0.5 ft	**SILT, 10Y 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]**	**87**	**58**	**29**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**2**	**0.044**	**0.009**		**0.0**	**4.1**	**78.4**	**17.5**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	95.9	98.1	98.7	98.9	99.2	99.5	100.0	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/25/19	SRH	EC

Schnabel ENGINEERING

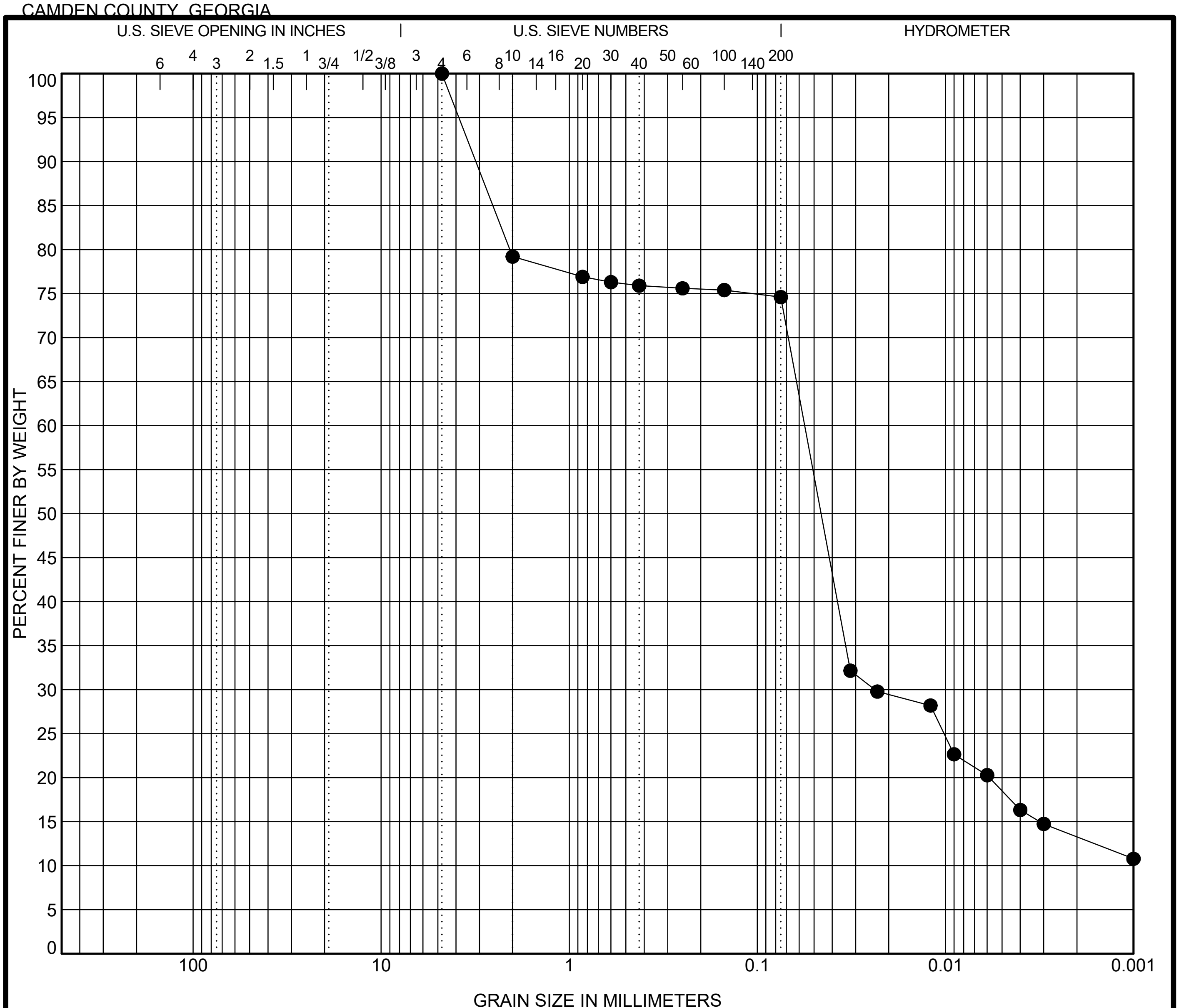
GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00

SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-8	1.5 ft	**SILT, 10Y 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]**	**90**	**56**	**34**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.056**	**0.024**		**0.0**	**25.4**	**61.3**	**13.3**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	74.6	75.4	75.6	75.9	76.3	76.9	79.2	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/24/19	SRH	EC

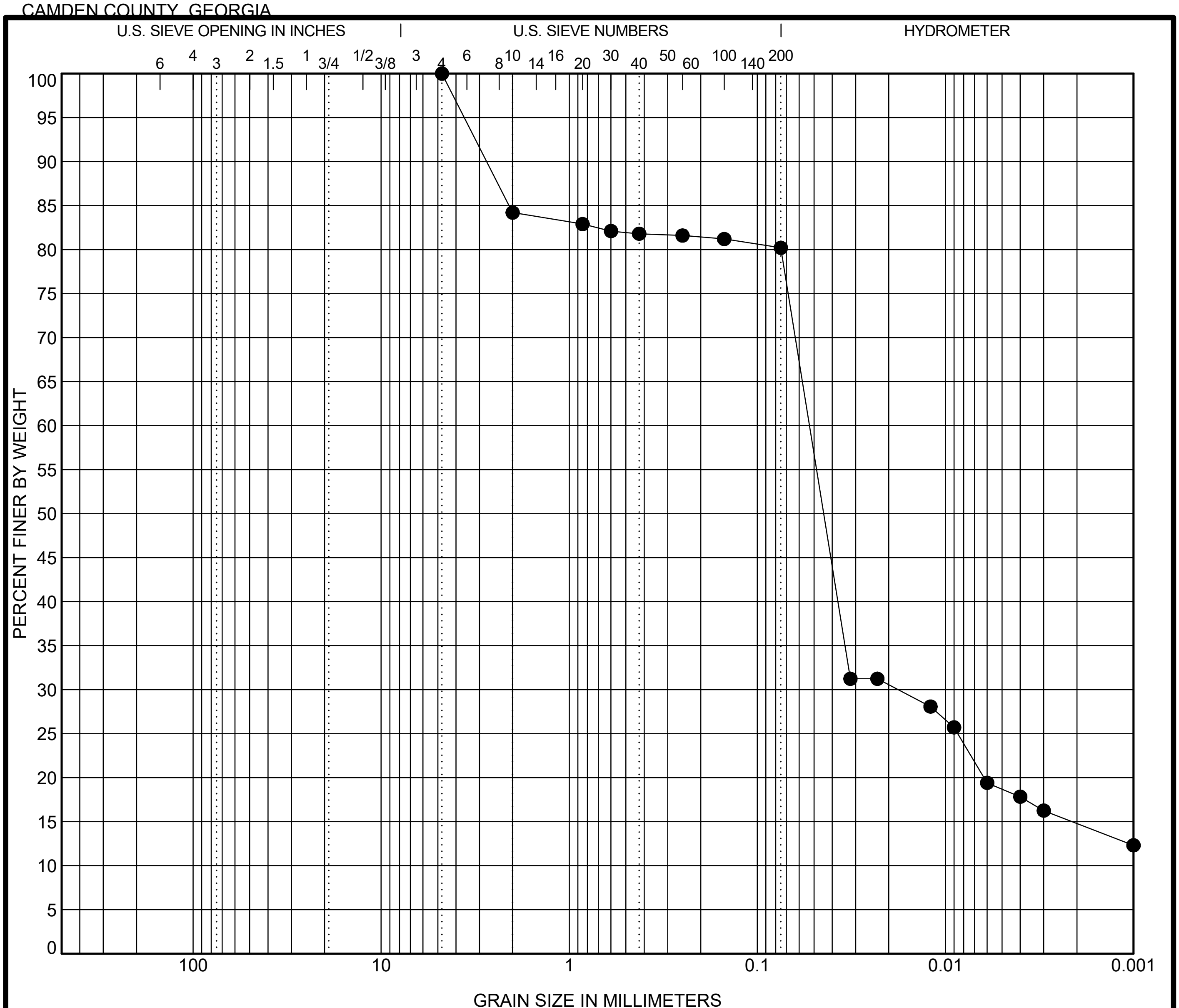
Schnabel ENGINEERING

GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-8	3.0 ft	**SILT, 10Y 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]**	**97**	**55**	**42**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.053**	**0.018**		**0.0**	**19.8**	**65.4**	**14.8**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	80.2	81.2	81.6	81.8	82.1	82.9	84.2	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/24/19	SRH	EC

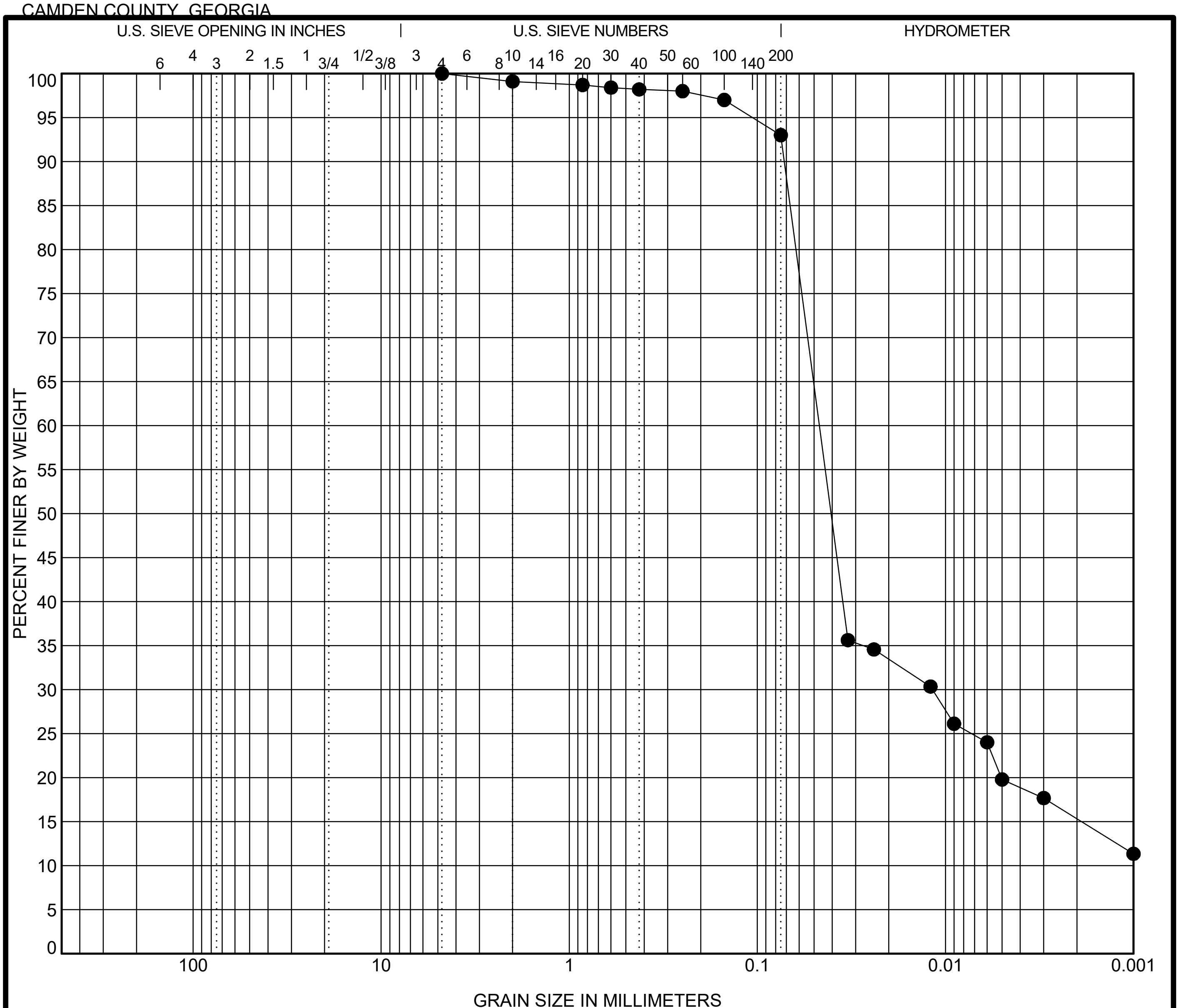
Schnabel ENGINEERING

GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00

SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-9	0.5 ft	**SILT, 10Y 3/1 very dark greenish gray (MH) [Estimated shell content < 1.0%]**	**84**	**49**	**35**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.047**	**0.012**		**0.0**	**7.0**	**77.7**	**15.3**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	93.0	97.0	98.0	98.2	98.4	98.7	99.1	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/25/19	SRH	EC

Schnabel ENGINEERING

GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

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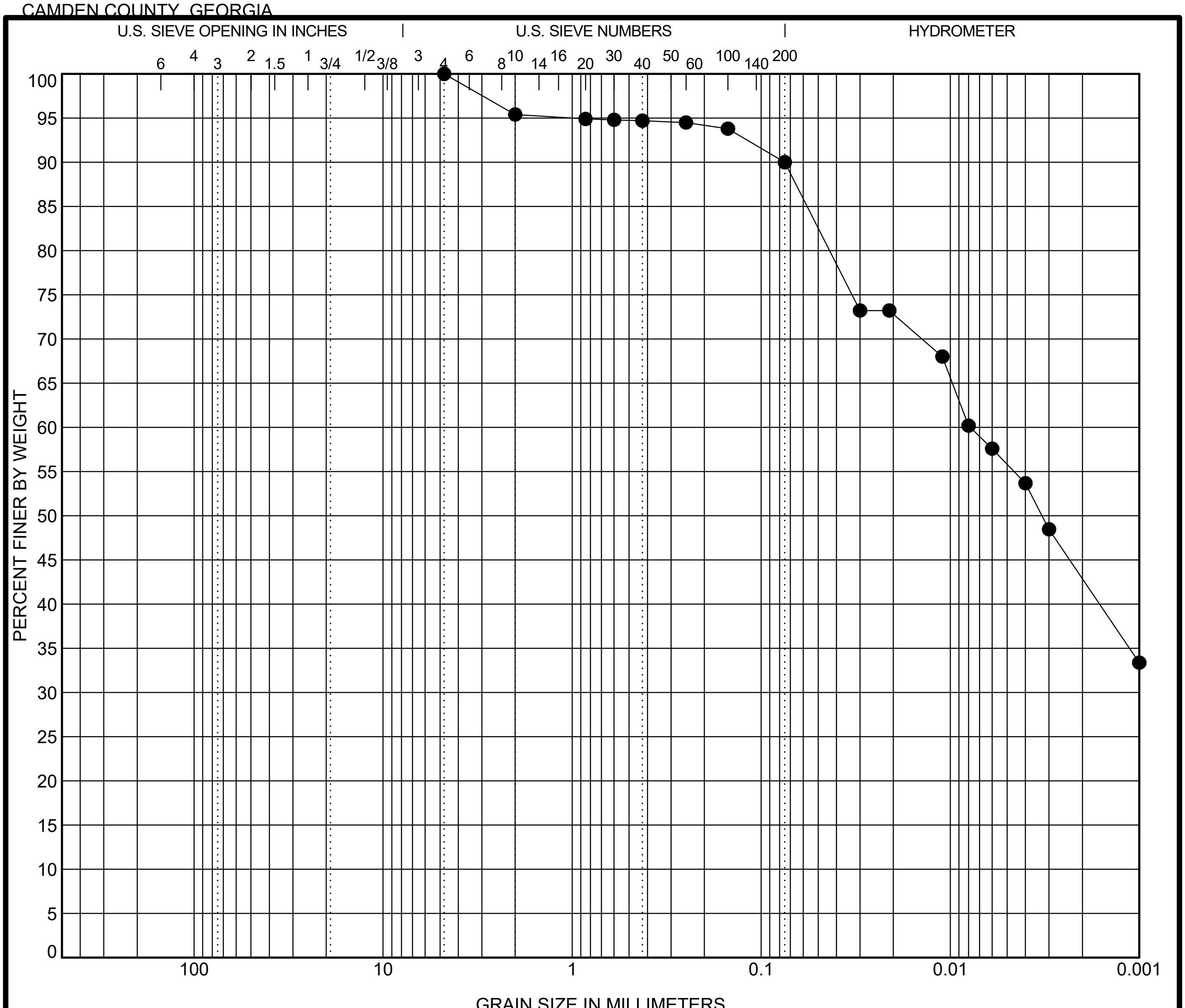

COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-9	1.5 ft	**SILT, 10Y 3/1 very dark greenish gray (MH) [Estimated shell content < 1.0%]**	**85**	**49**	**36**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.008**			**0.0**	**10.0**	**47.1**	**42.9**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	90.0	93.8	94.5	94.7	94.8	94.9	95.4	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/19/19	SRH	MDE

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GRADATION CURVE

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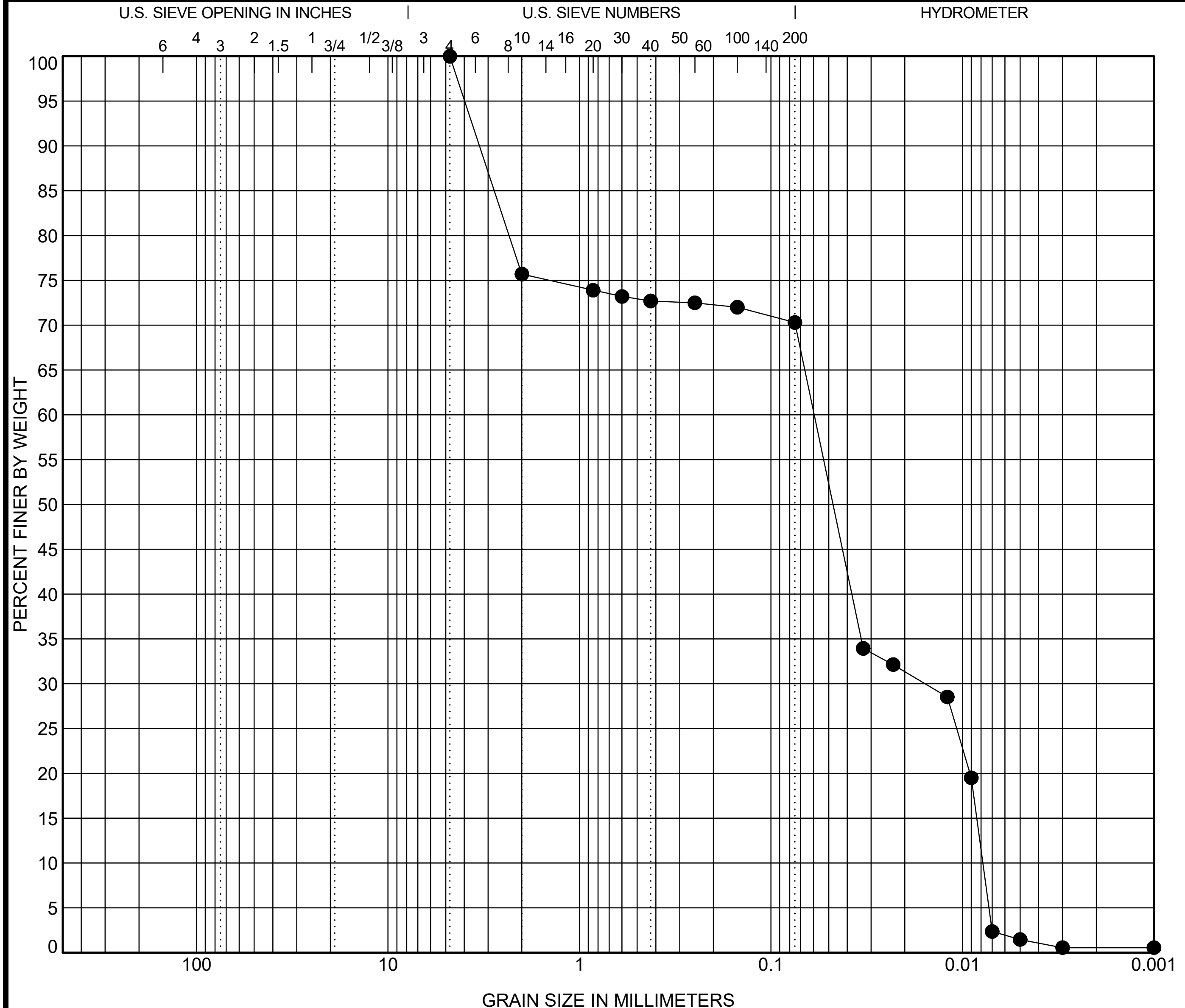
SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-9	3.0 ft	**SILT, 10Y 3/1 very dark greenish gray (MH) [Estimated shell content < 1.0%]**	**78**	**47**	**31**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.059**	**0.016**	**0.008**	**0.0**	**29.7**	**69.7**	**0.6**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	70.3	72.0	72.5	72.7	73.2	73.9	75.7	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/24/19	SRH	EC

Schnabel ENGINEERING

GRADATION CURVE

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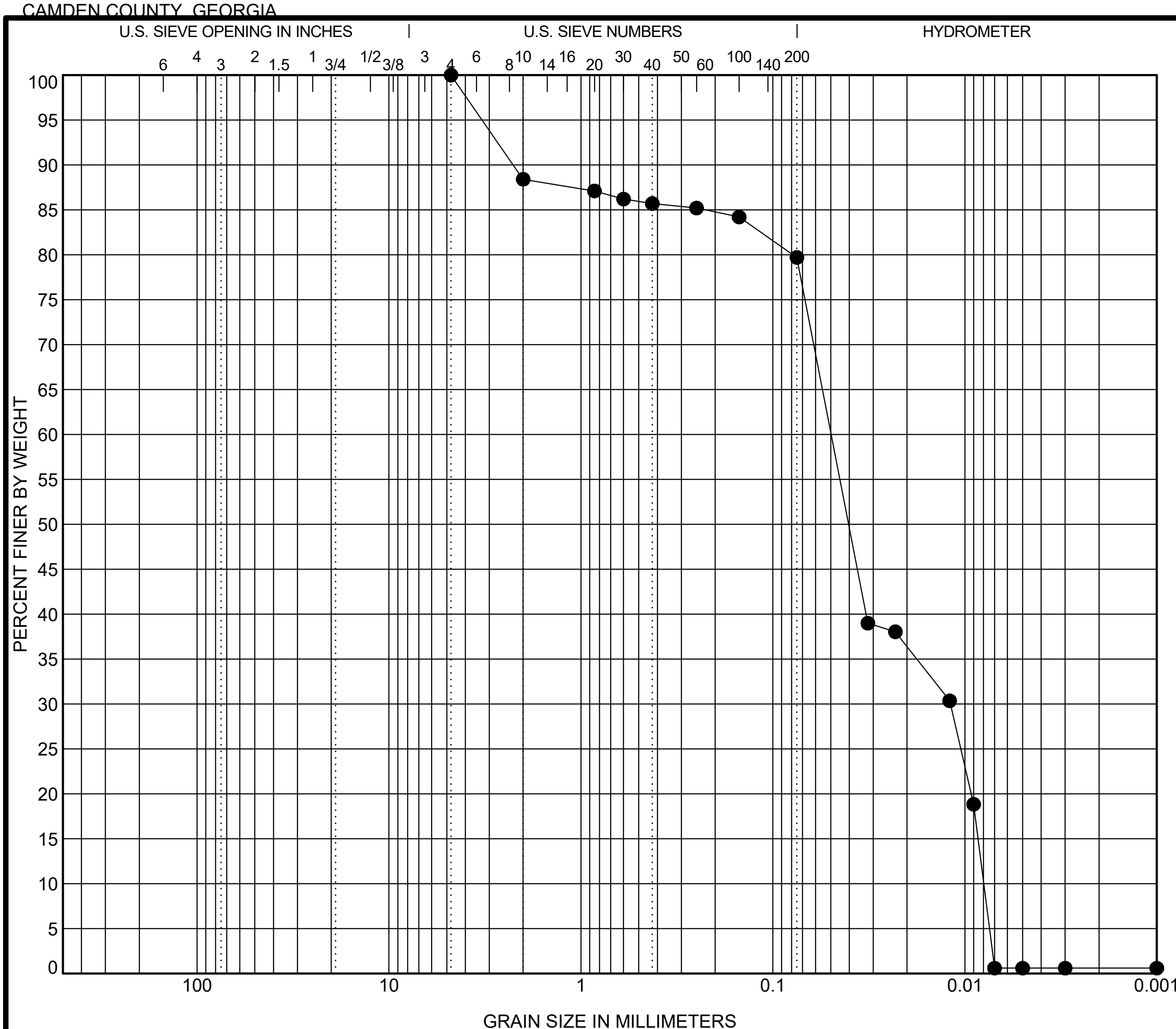
NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

Specimen		Sample Description	LL	PL	PI	Test Method
● VB-KBIC19-10	0.5 ft	**SILT, 2.5Y 3/1 very dark gray (MH) [Estimated shell content < 1.0%]**	**105**	**55**	**50**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.05**	**0.012**	**0.008**	**0.0**	**20.3**	**79.1**	**0.6**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	79.7	84.2	85.2	85.7	86.2	87.1	88.4	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/24/19	SRH	EC

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GRADATION CURVE

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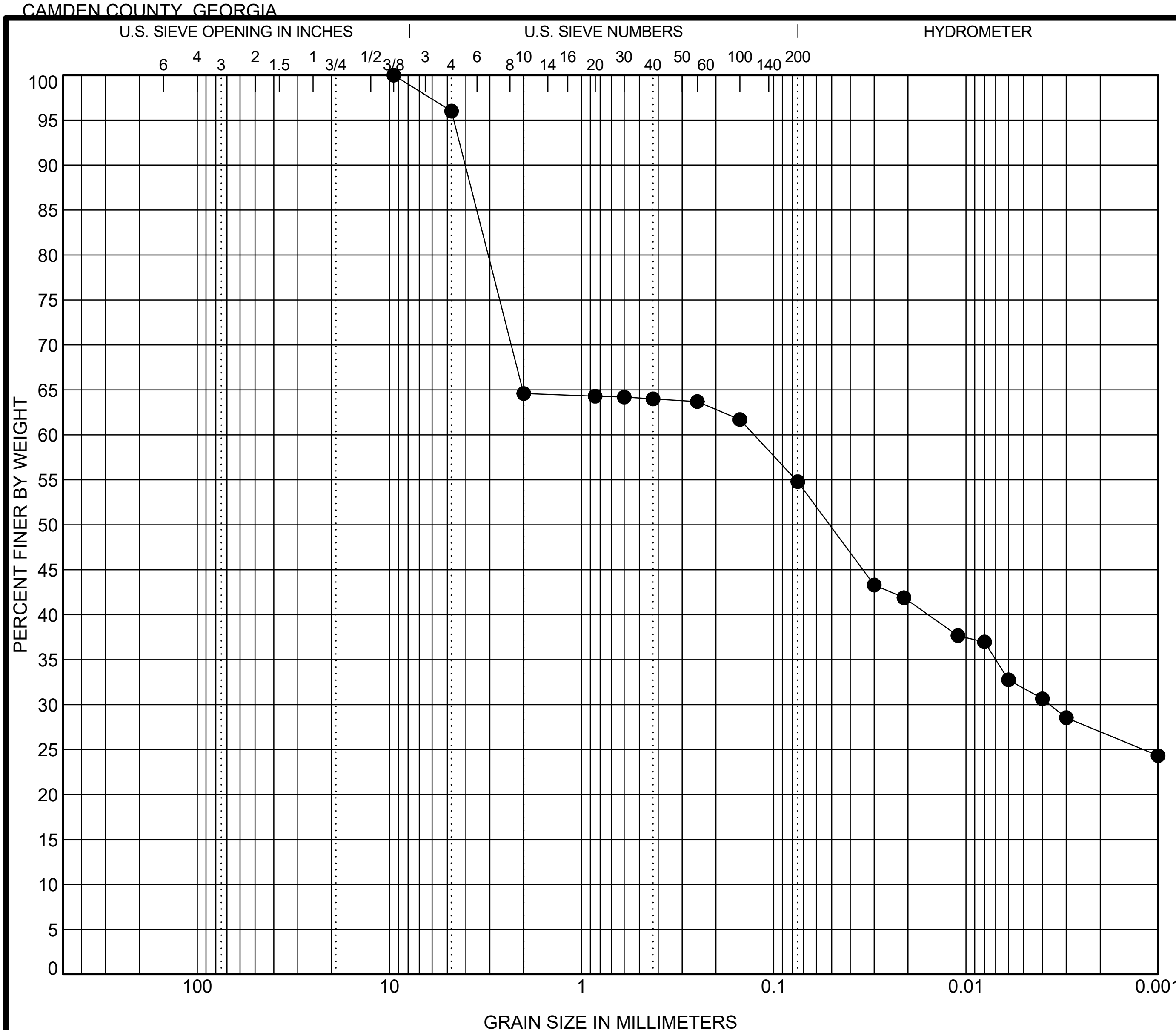
Contract: 19C19018.00



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-10	1.5 ft	**SILT, 2.5Y 3/1 very dark gray (MH) [Estimated shell content < 4.0%]**	**81**	**49**	**32**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**9.5**	**0.126**	**0.004**		**4.0**	**41.2**	**27.8**	**27.0**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4	3/8"
% Finer	54.8	61.7	63.7	64.0	64.2	64.3	64.6	96.0	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/24/19	SRH	EC

Schnabel ENGINEERING

GRADATION CURVE

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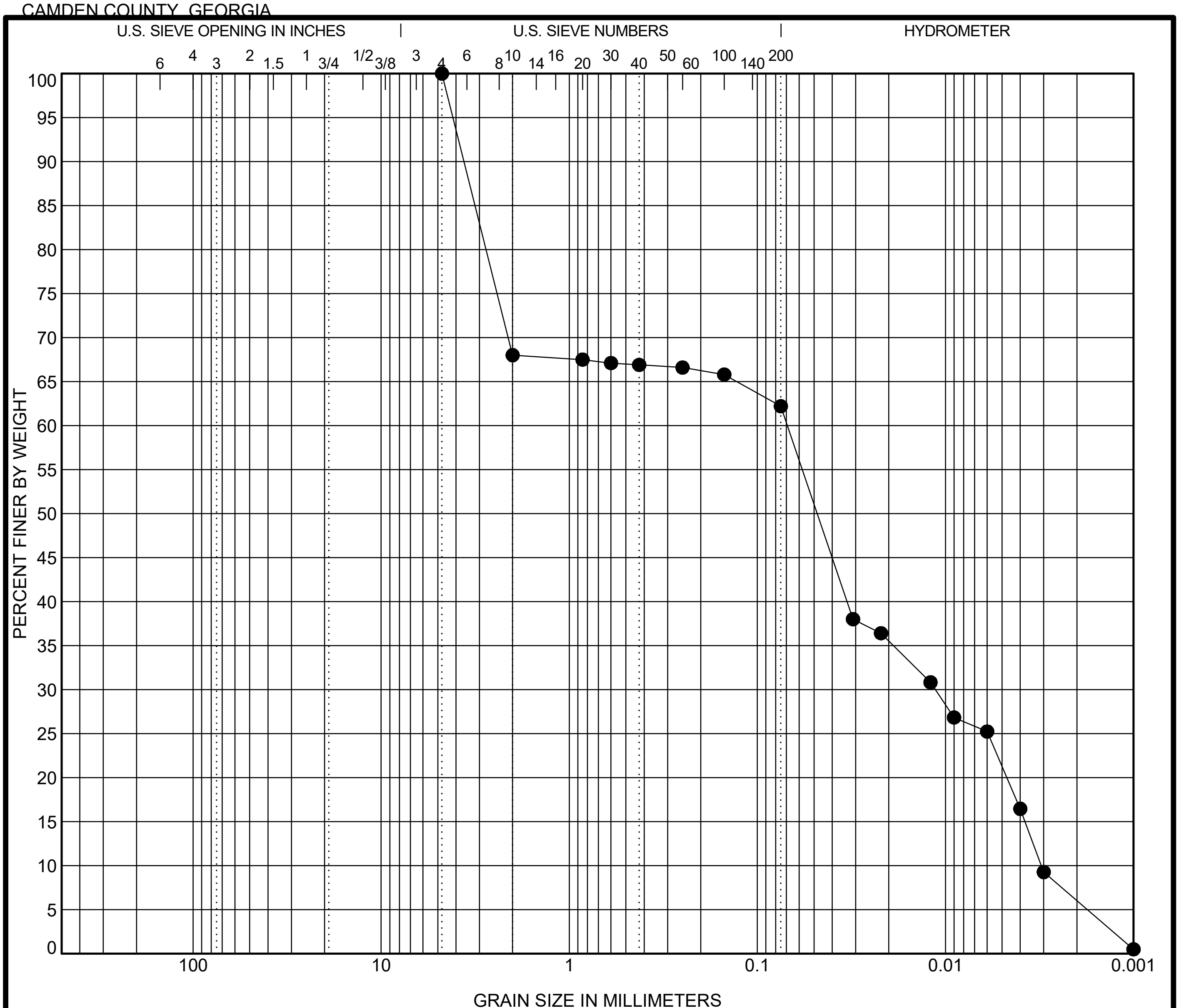
NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-10	3.0 ft	**SILT, 2.5Y 3/1 very dark gray (MH) [Estimated shell content < 1.0%]**	**101**	**50**	**51**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.069**	**0.011**	**0.003**	**0.0**	**37.8**	**56.2**	**6.0**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	62.2	65.8	66.6	66.9	67.1	67.5	68.0	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/24/19	SRH	EC

Schnabel ENGINEERING

GRADATION CURVE

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Kings Bay Inner Channel, Camden Co., GA

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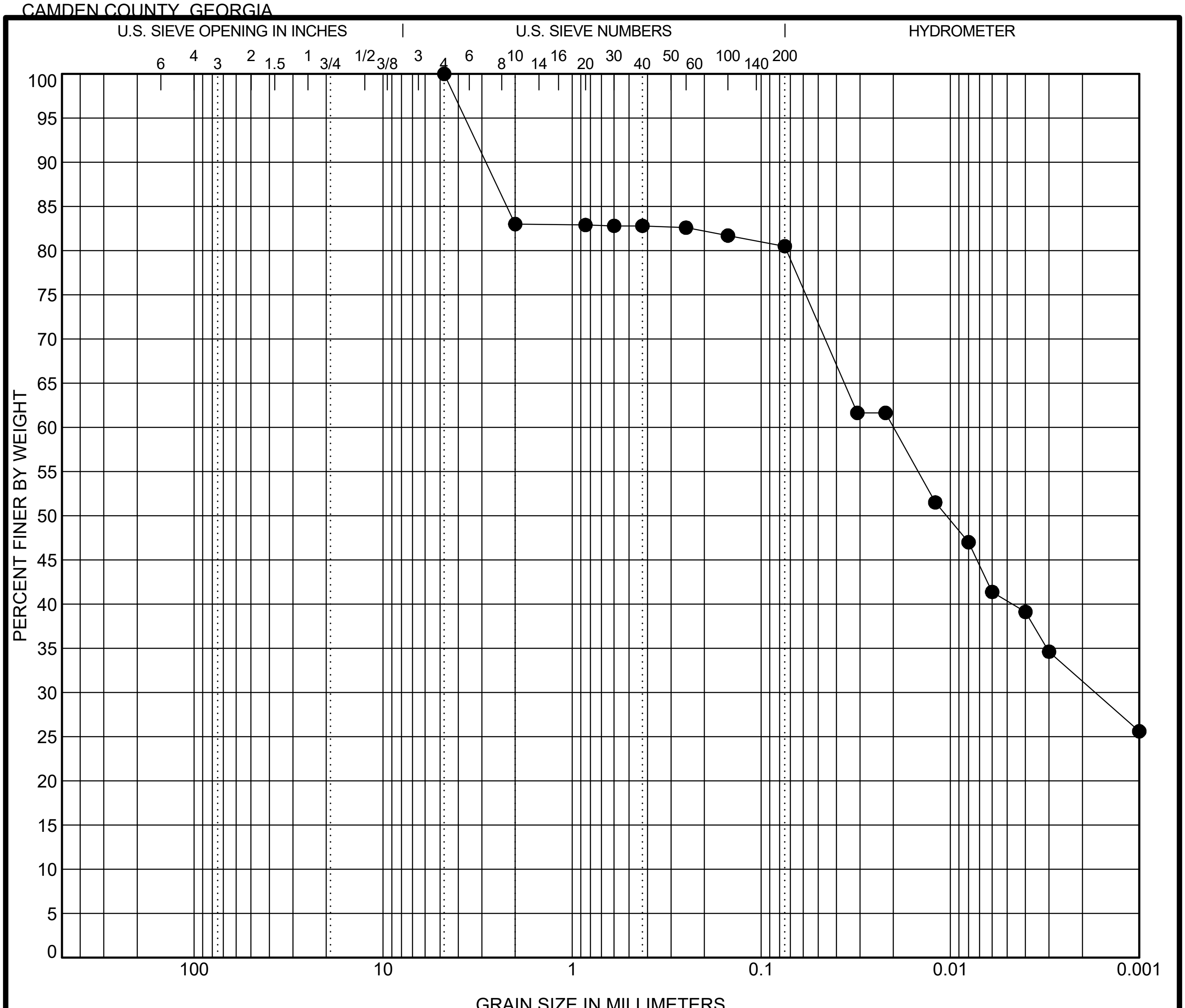
NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-11	0.5 ft	**SILT, 5GY 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]**	**108**	**58**	**50**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.02**	**0.002**		**0.0**	**19.5**	**49.2**	**31.3**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	80.5	81.7	82.6	82.8	82.8	82.9	83.0	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/20/19	SRH	EC

Schnabel ENGINEERING

GRADATION CURVE

Project: P617 Transit Program
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Kings Bay Inner Channel, Camden Co., GA

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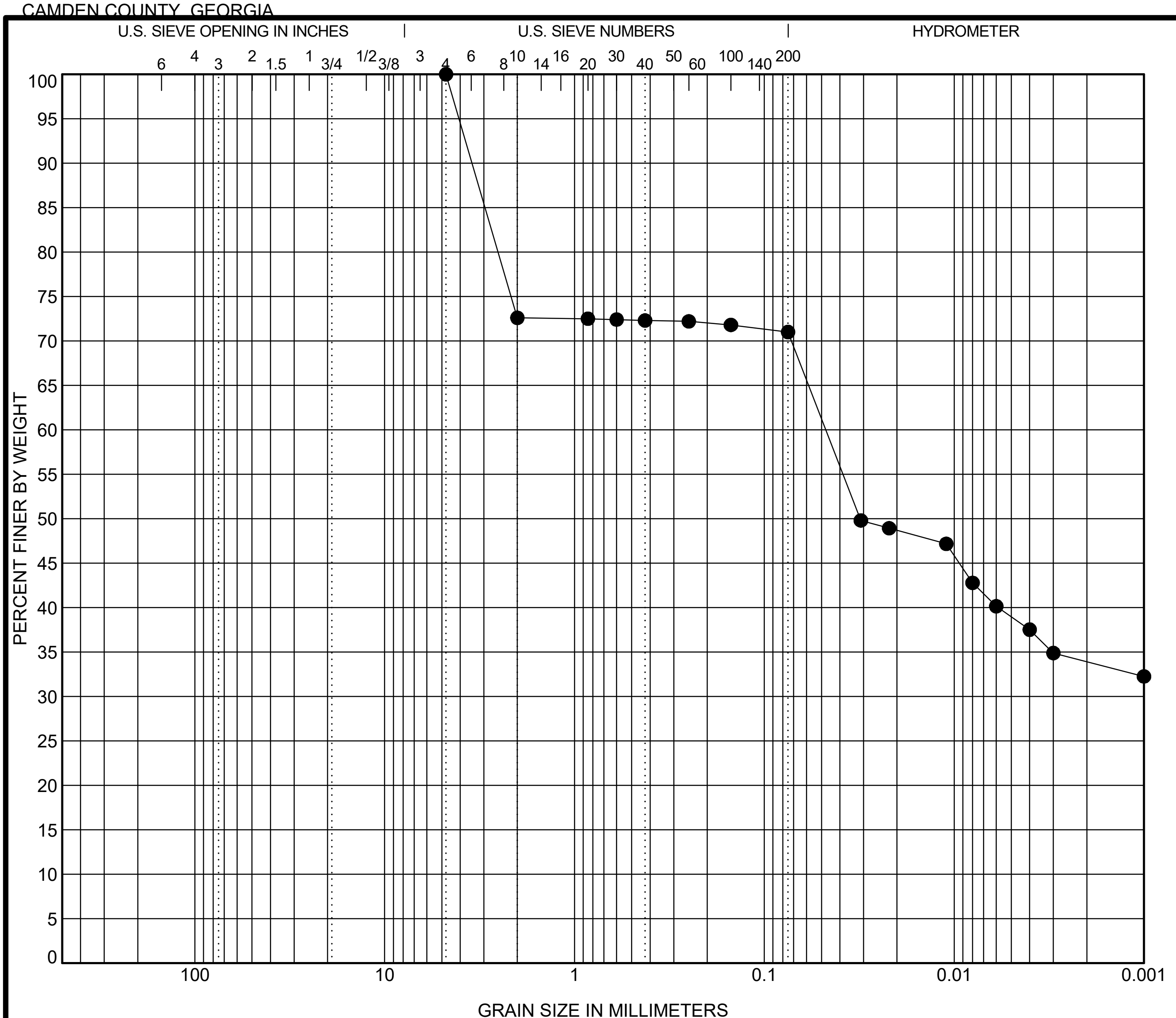

COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-11	1.5 ft	**SILT, 5GY 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]**	**110**	**59**	**51**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.047**			**0.0**	**29.0**	**37.1**	**33.9**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	71.0	71.8	72.2	72.3	72.4	72.5	72.6	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/20/19	SRH	EC

Schnabel ENGINEERING

GRADATION CURVE

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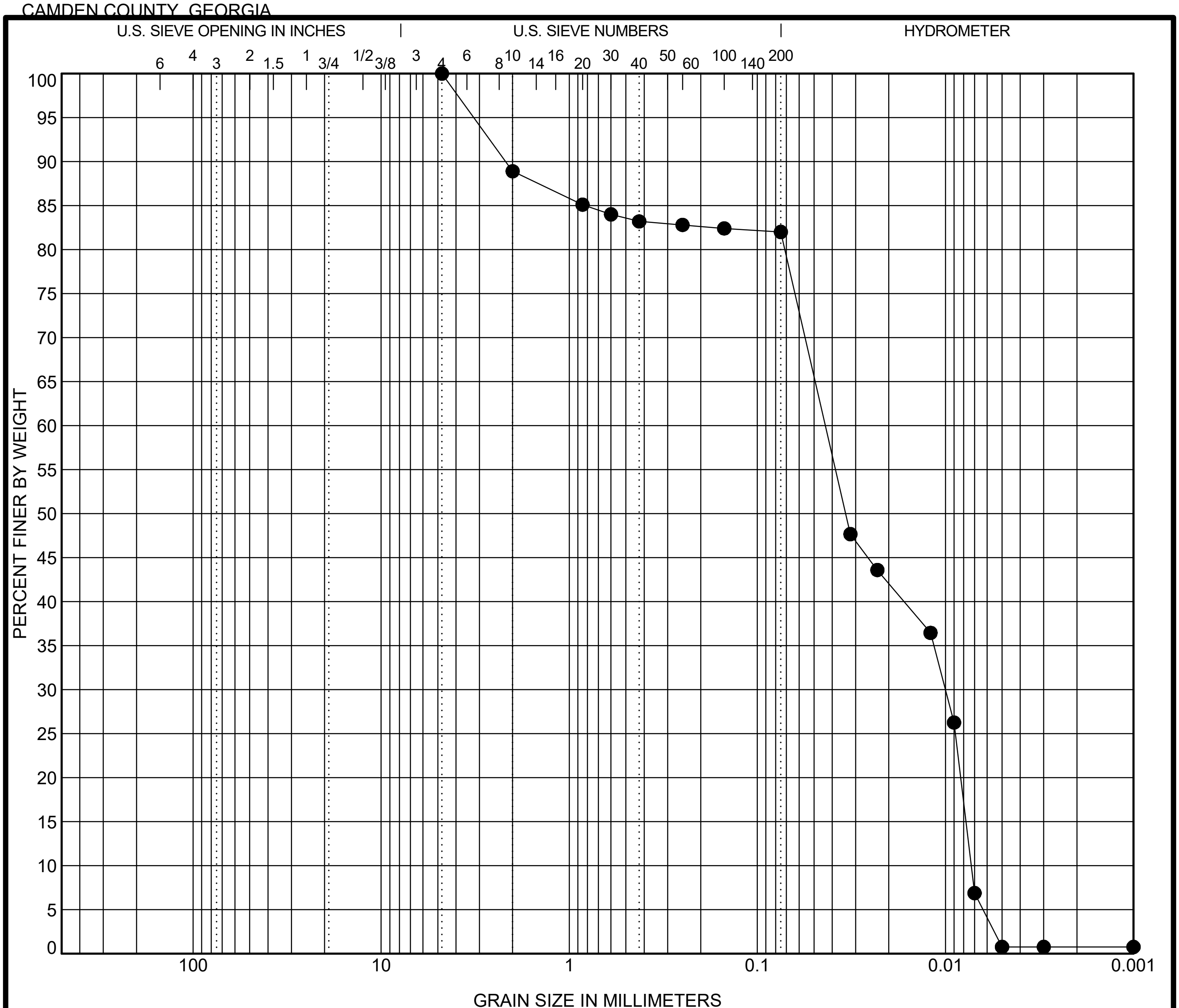
SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-11	3.0 ft	**SILT, 5GY 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]**	**112**	**66**	**46**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.043**	**0.01**	**0.007**	**0.0**	**18.0**	**81.2**	**0.8**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	82.0	82.4	82.8	83.2	84.0	85.1	88.9	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/25/19	SRH	EC

Schnabel ENGINEERING

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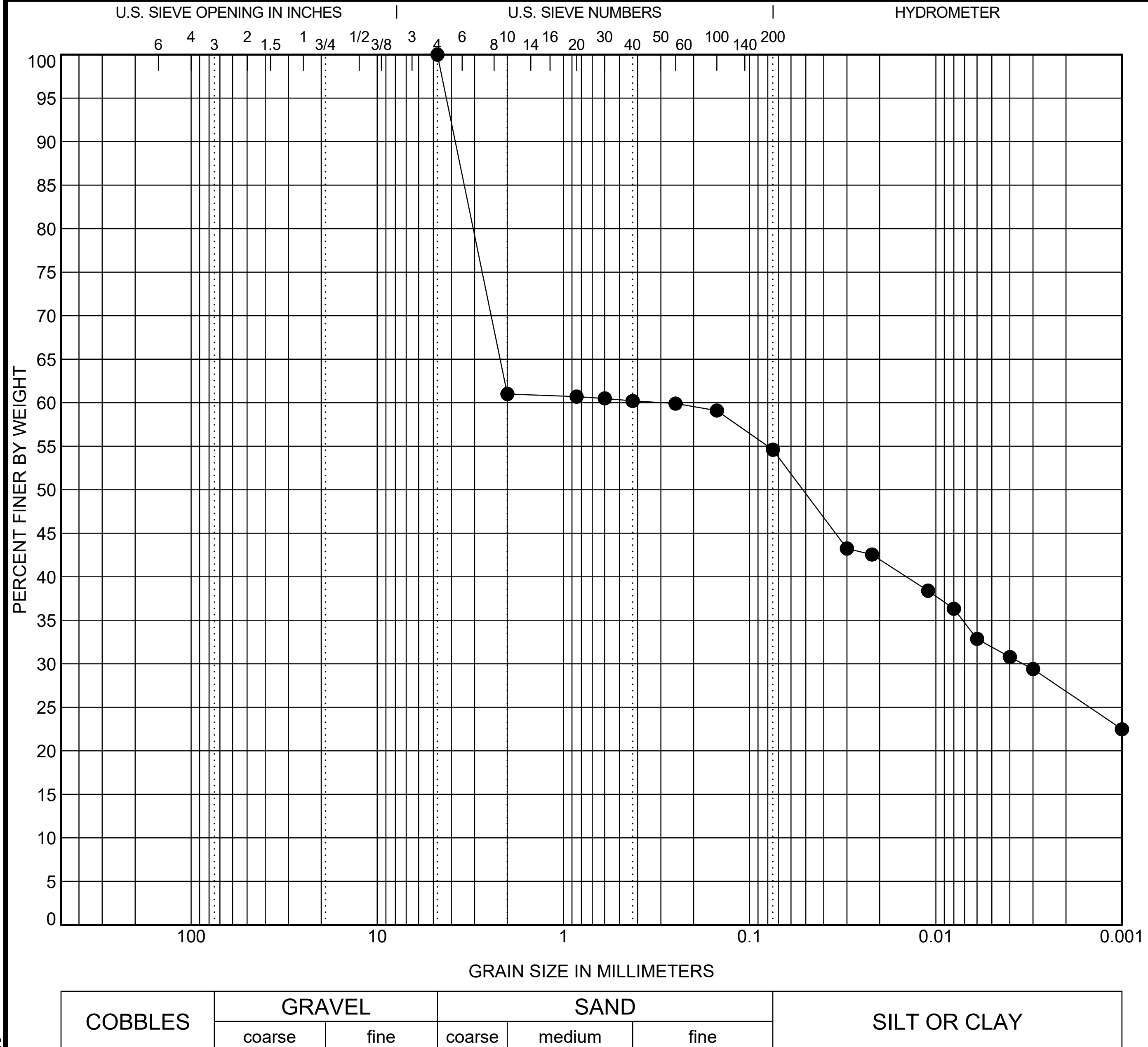
NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-12	0.5 ft	**SILT, 5G 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]**	**102**	**55**	**47**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.298**	**0.003**		**0.0**	**45.4**	**27.8**	**26.8**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	54.6	59.1	59.9	60.2	60.5	60.7	61.0	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/21/19	SRH	EC

Schnabel ENGINEERING

GRADATION CURVE

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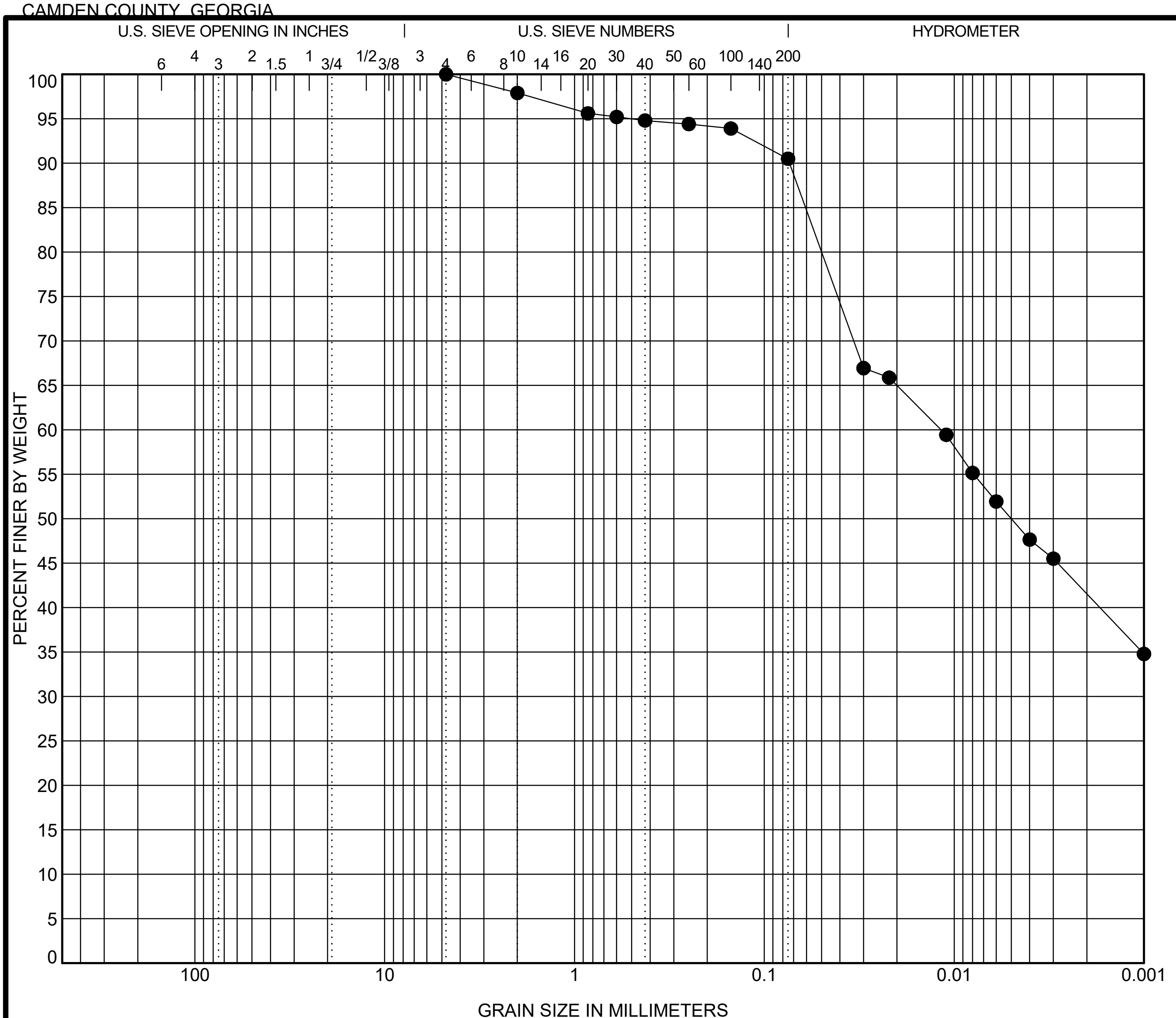
NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-12	1.5 ft	**SILT, 5G 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]**	**101**	**54**	**47**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.012**			**0.0**	**9.5**	**49.0**	**41.5**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	90.5	93.9	94.4	94.8	95.2	95.6	97.9	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/21/19	SRH	EC

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GRADATION CURVE

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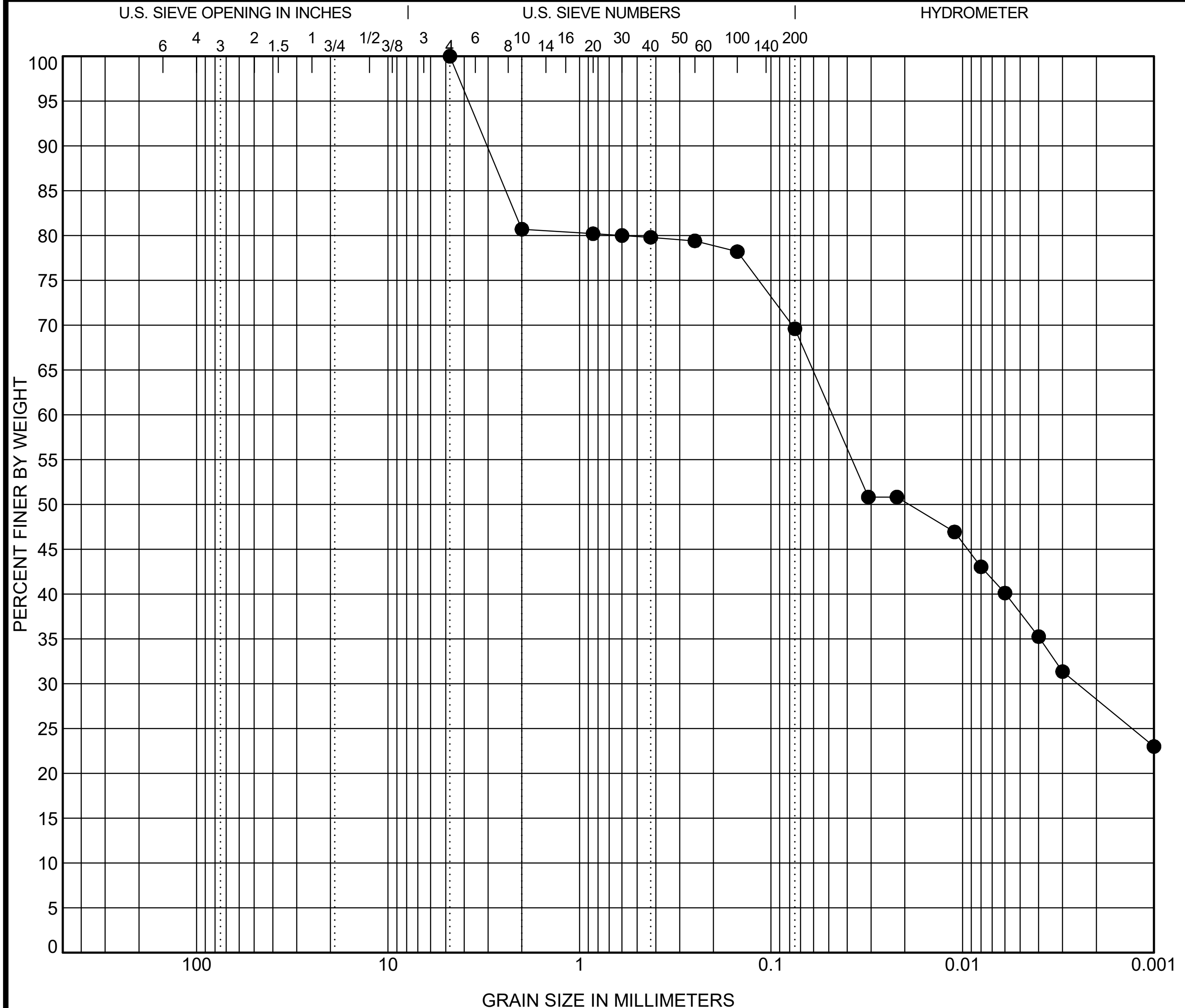

COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-12	2.5 ft	**SILT, 5G 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]**	**79**	**39**	**40**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.048**	**0.003**		**0.0**	**30.4**	**41.3**	**28.3**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	69.6	78.2	79.4	79.8	80.0	80.2	80.7	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/19/19	SRH	EC

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ENGINEERING

GRADATION CURVE

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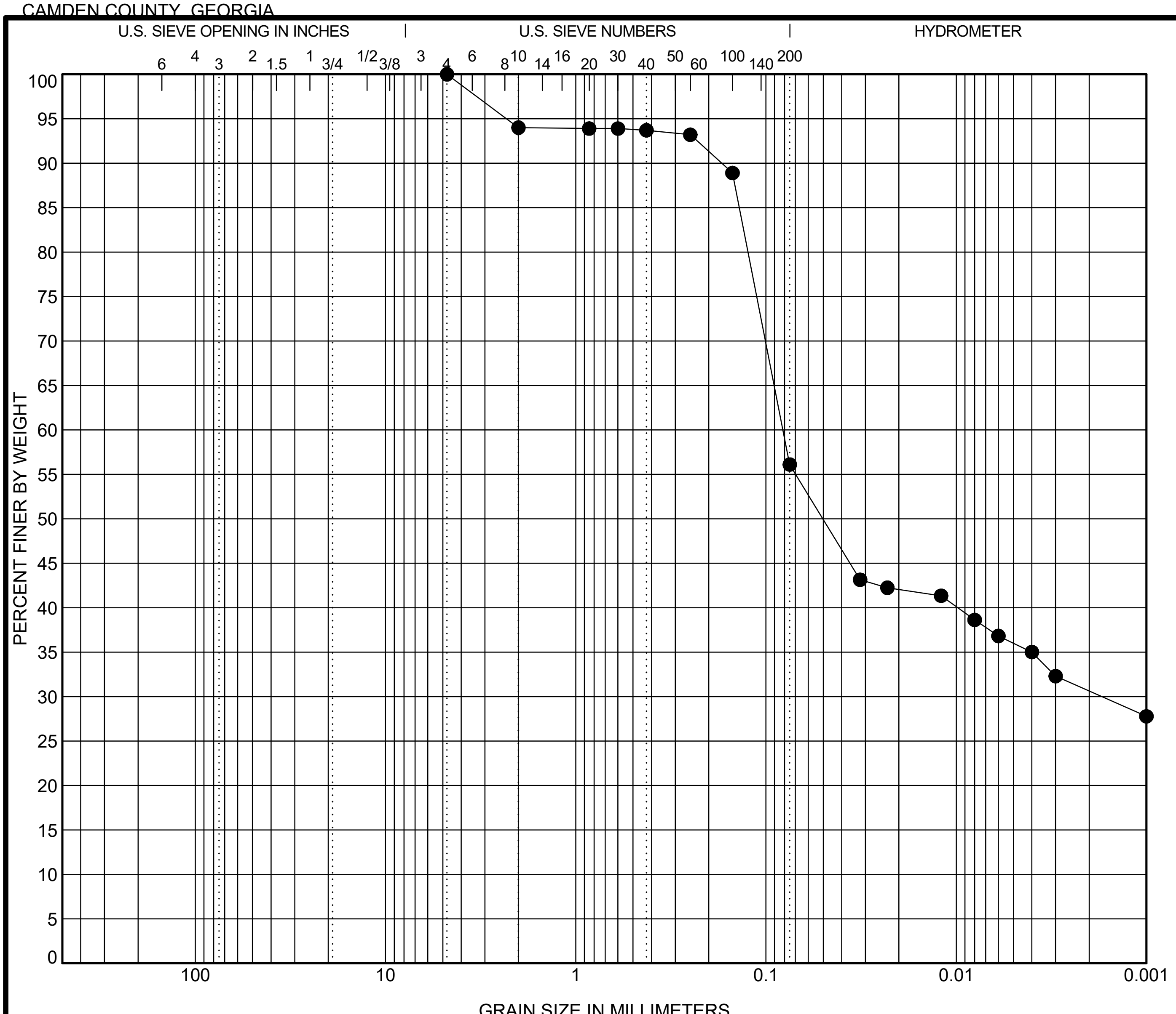
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-13	0.5 ft	**SILT, 10Y 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]**	**97**	**49**	**48**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.081**	**0.002**		**0.0**	**43.9**	**25.5**	**30.6**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	56.1	88.9	93.2	93.7	93.9	93.9	94.0	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/20/19	SRH	EC

Schnabel ENGINEERING

GRADATION CURVE

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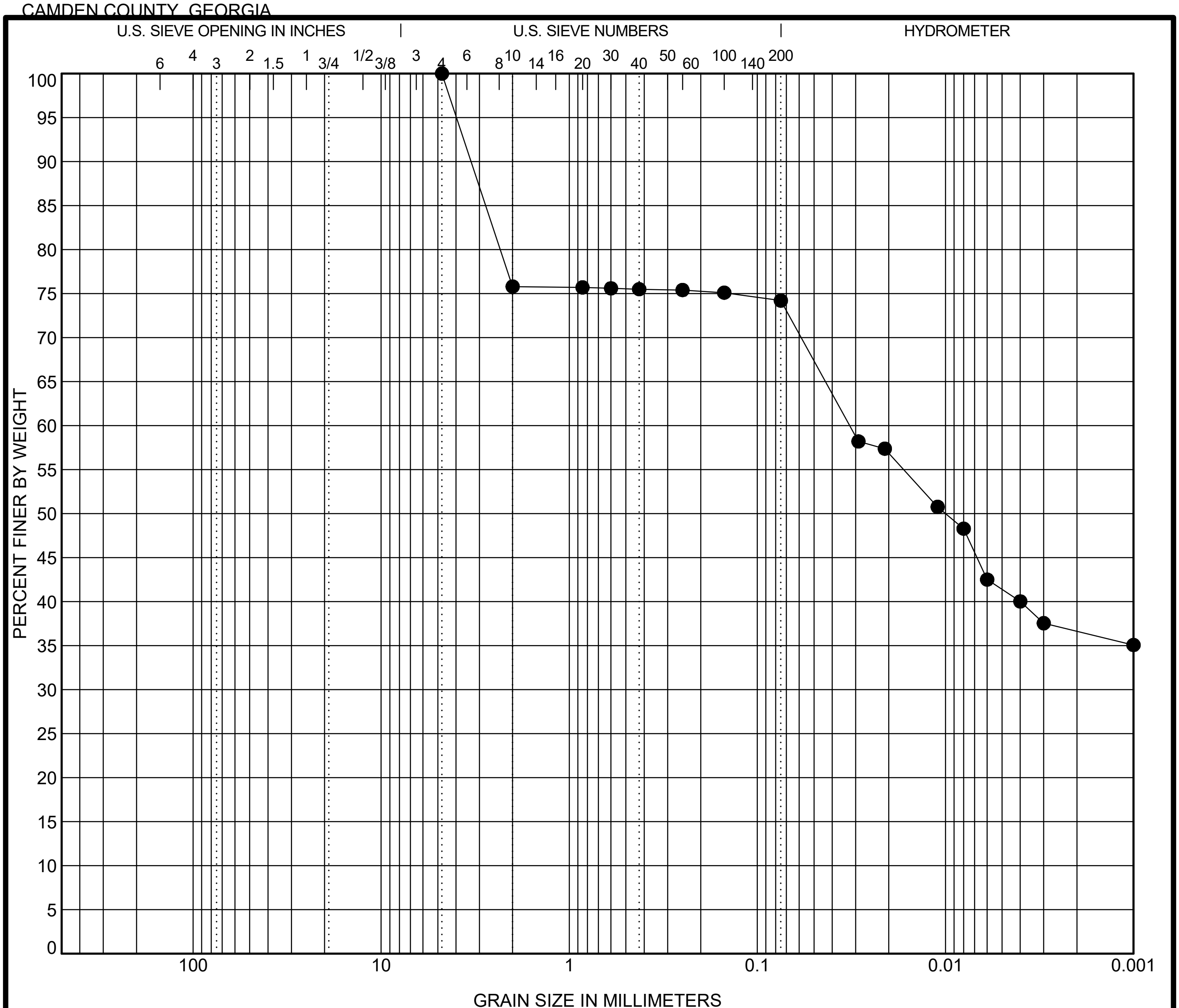
SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-13	1.5 ft	**SILT, 10Y 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]**	**97**	**52**	**45**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.032**			**0.0**	**25.8**	**37.6**	**36.6**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	74.2	75.1	75.4	75.5	75.6	75.7	75.8	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/20/19	SRH	EC

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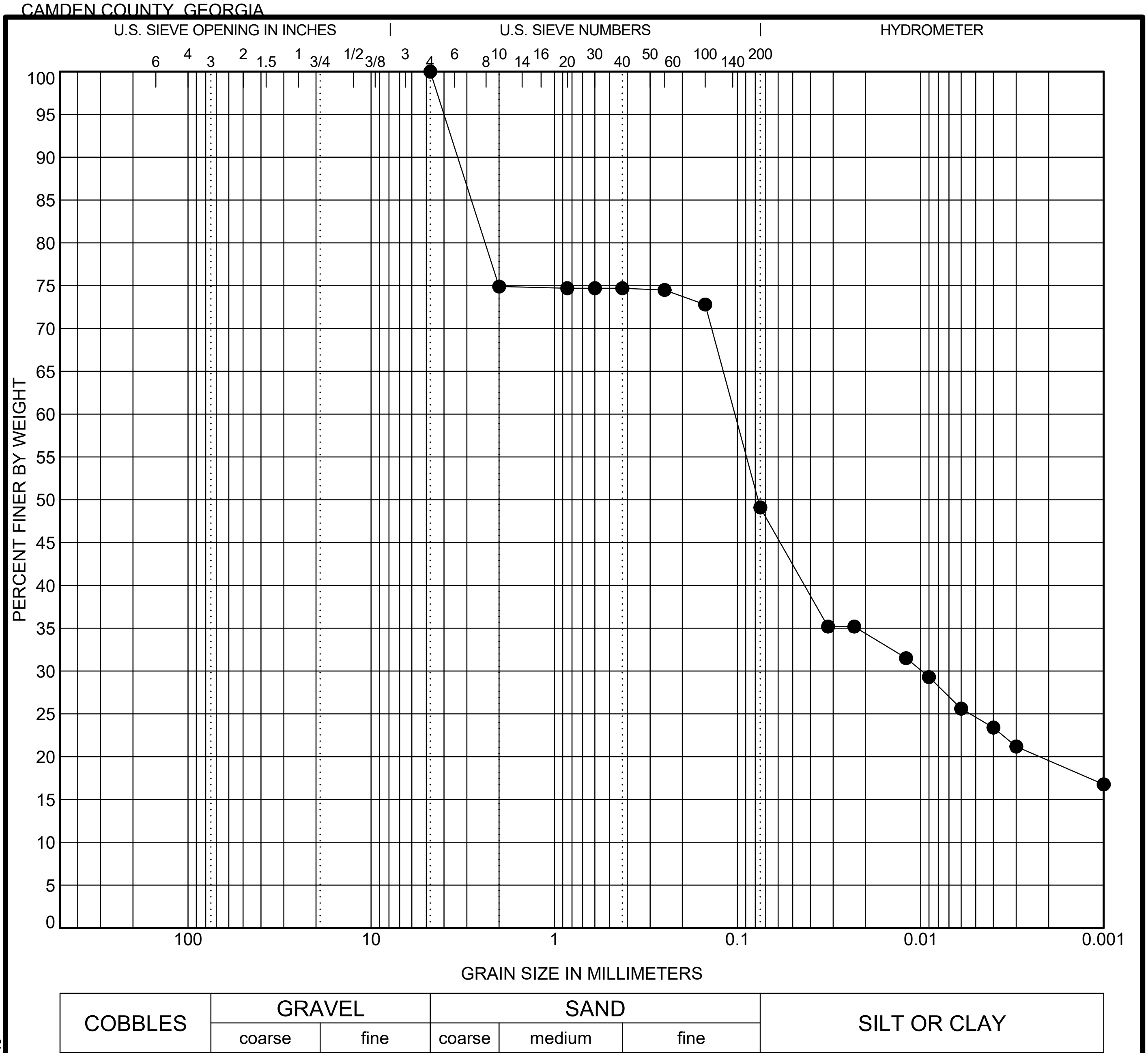
Contract: 19C19018.00

SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-13	3.0 ft	**SAND, silty, 5Y 3/1 very dark gray (SM) Estimated shell content < 1.0%]**	**93**	**46**	**47**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.103**	**0.01**		**0.0**	**50.9**	**29.5**	**19.6**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	49.1	72.8	74.5	74.7	74.7	74.7	74.9	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/20/19	SRH	EC

Schnabel ENGINEERING

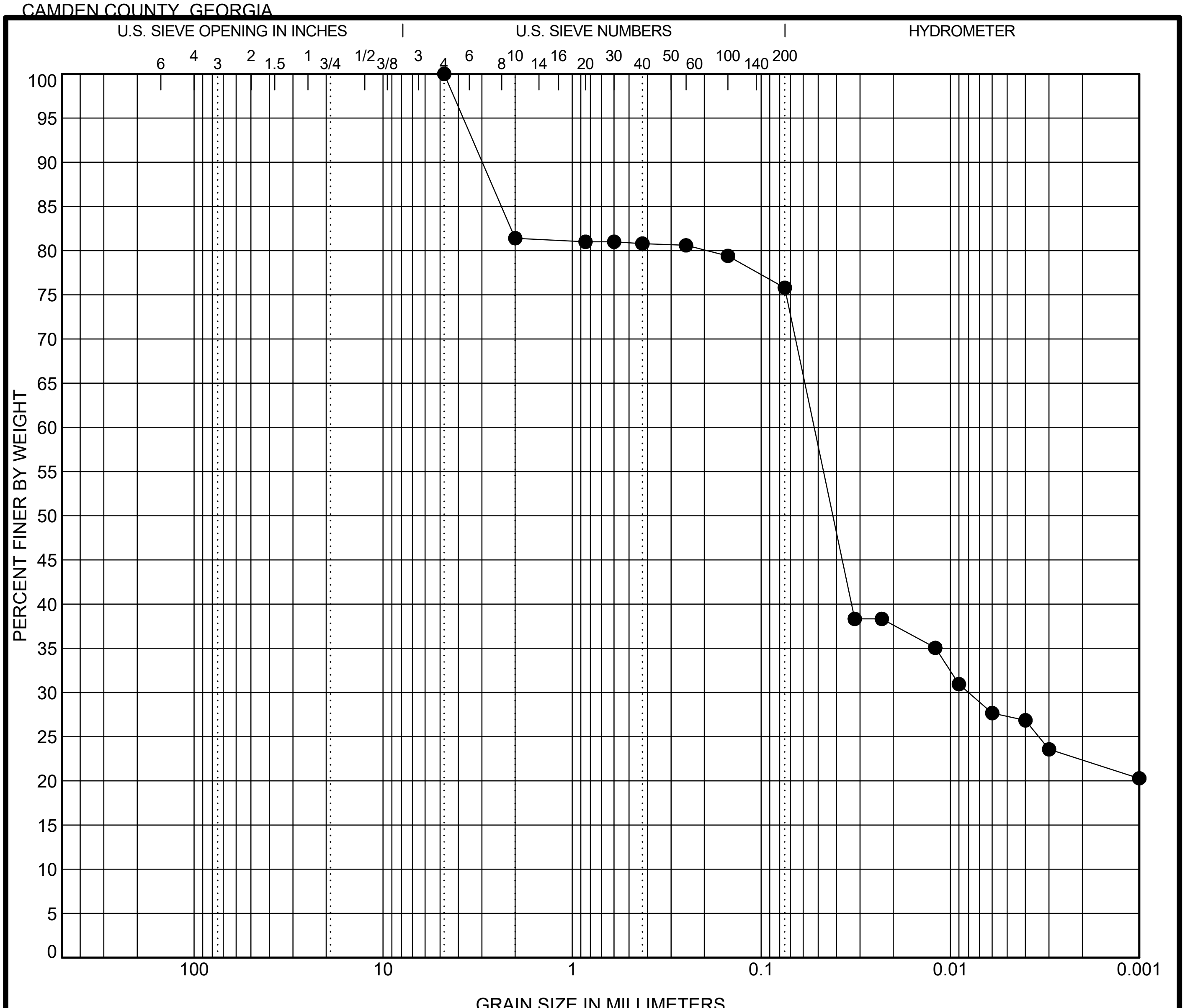
GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00

SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

Specimen		Sample Description	LL	PL	PI	Test Method
● VB-KBIC19-14	0.5 ft	**SILT, 5Y 2.5/1 black (MH) [Estimated shell content < 1.0%]**	**98**	**46**	**52**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.052**	**0.008**		**0.0**	**24.2**	**53.4**	**22.4**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	75.8	79.4	80.6	80.8	81.0	81.0	81.4	100.0

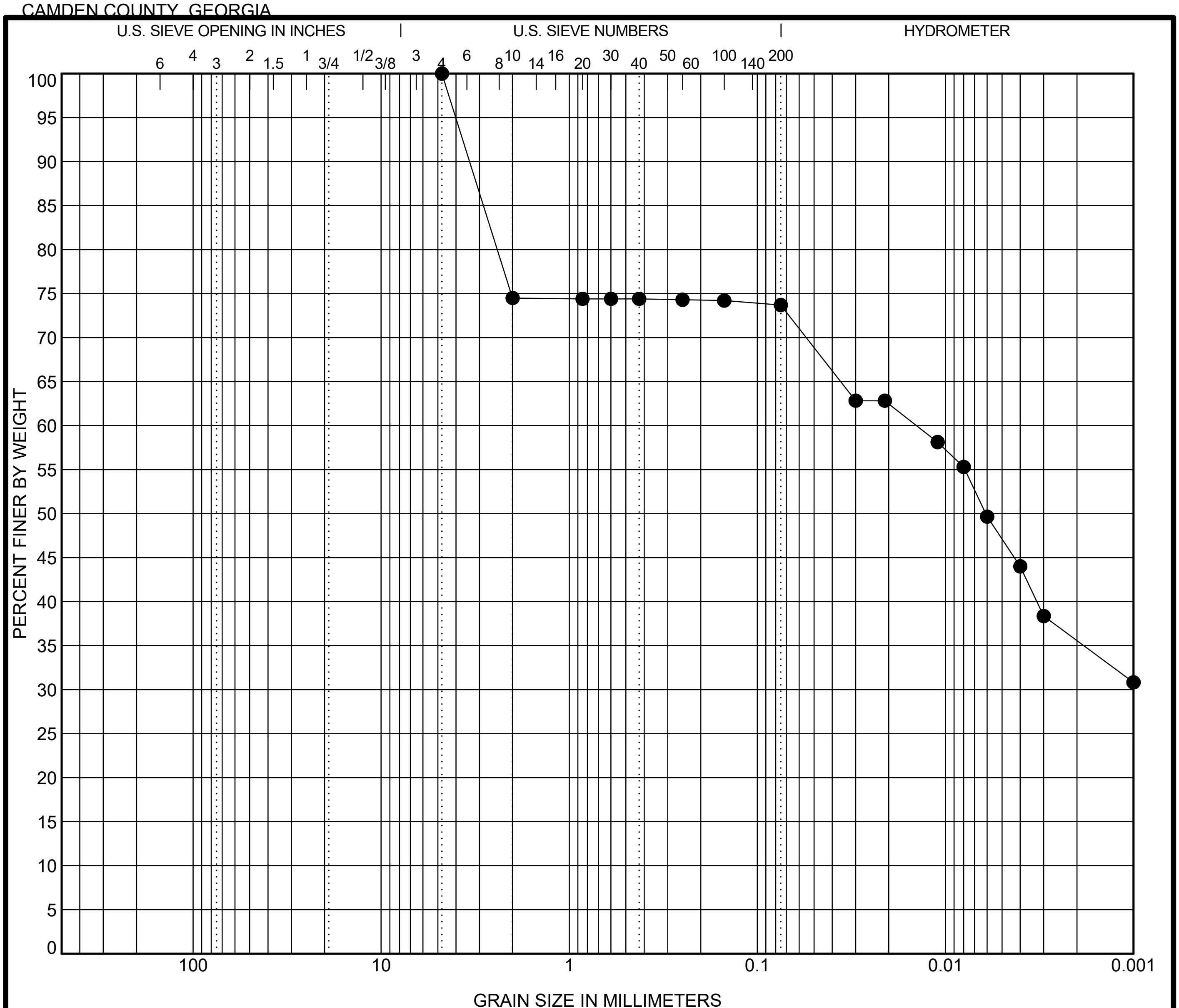
Tested By	Tested Date	Reviewed by	Calc by
CAB	6/20/19	SRH	EC

Schnabel ENGINEERING

GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-14	1.5 ft	**SILT, 5Y 2.5/1 black (MH) [Estimated shell content < 1.0%]**	**105**	**55**	**50**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.014**			**0.0**	**26.3**	**38.1**	**35.6**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	73.7	74.2	74.3	74.4	74.4	74.4	74.5	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/20/19	SRH	EC

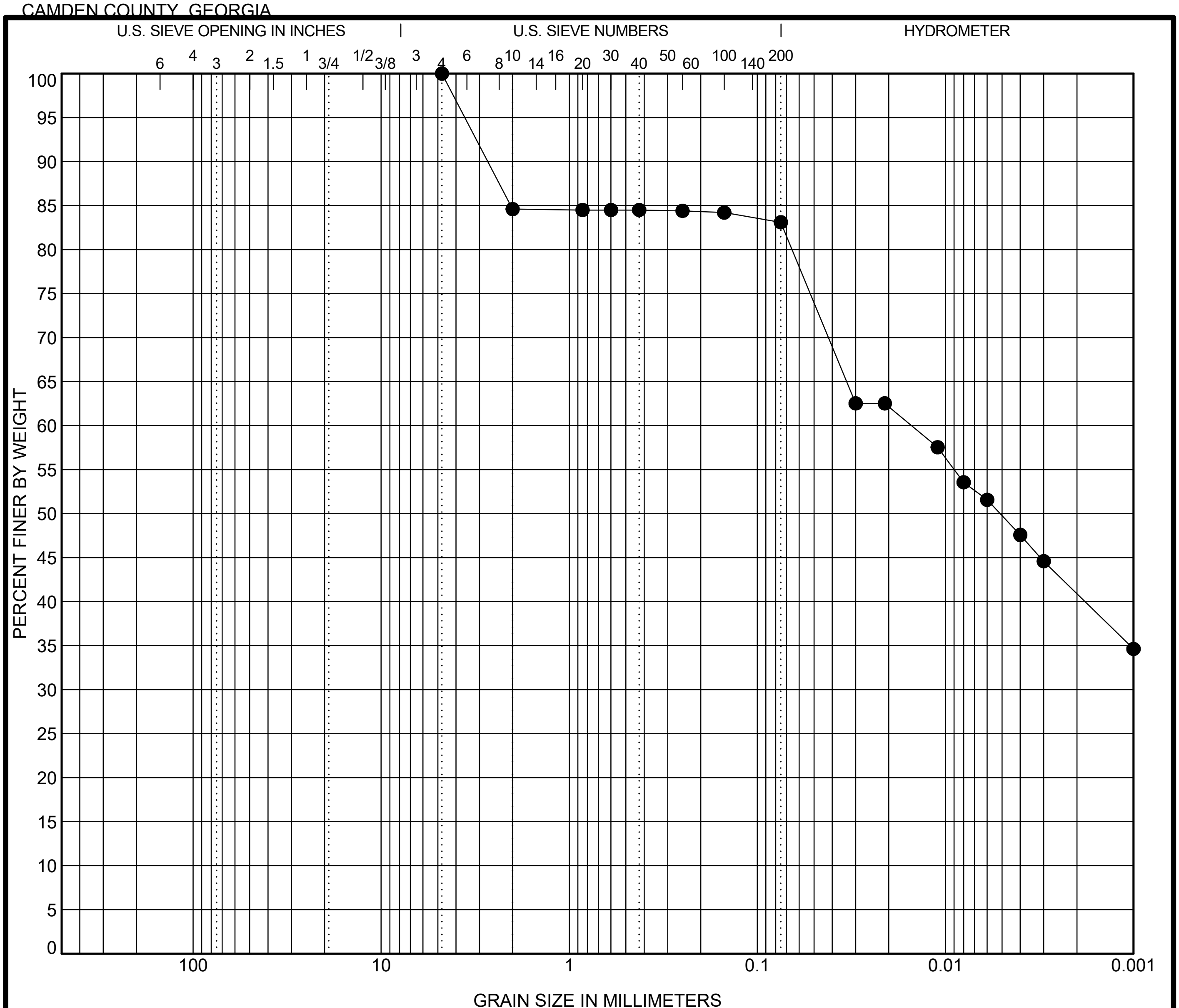
Schnabel ENGINEERING

GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-14	3.0 ft	**SILT, 5Y 3/2 dark olive gray (MH) [Estimated shell content < 1.0%]**	**107**	**52**	**55**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.015**			**0.0**	**16.9**	**42.2**	**40.9**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	83.1	84.2	84.4	84.5	84.5	84.5	84.6	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/20/19	SRH	EC

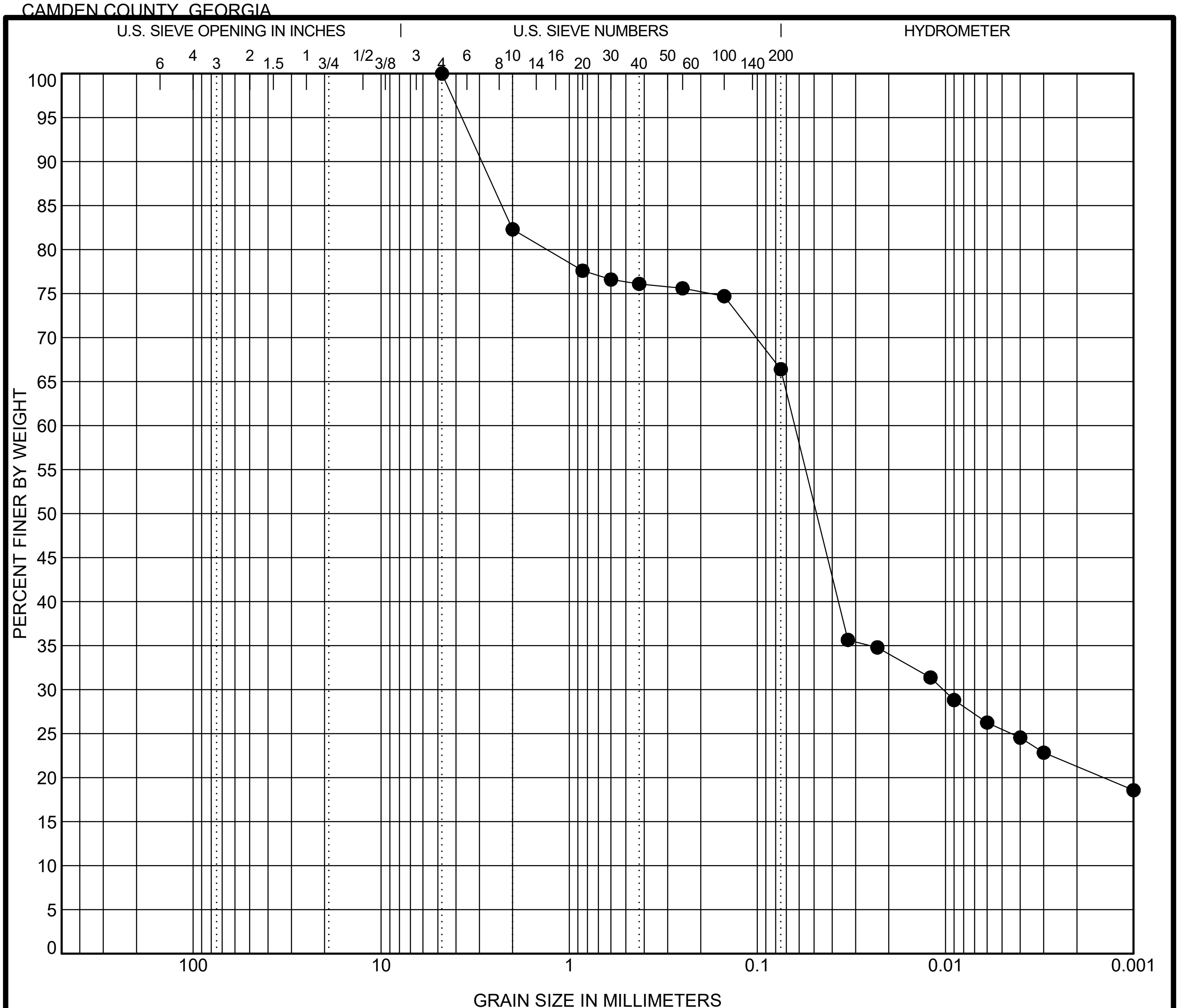
Schnabel ENGINEERING

GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00

SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-15	0.5 ft	**SILT, 10Y 3/1 very dark greenish gray (MH) [Estimated shell content < 1.0%]**	**81**	**37**	**44**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.063**	**0.01**		**0.0**	**33.6**	**45.1**	**21.3**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	66.4	74.7	75.6	76.1	76.6	77.6	82.3	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/25/19	SRH	EC

Schnabel ENGINEERING

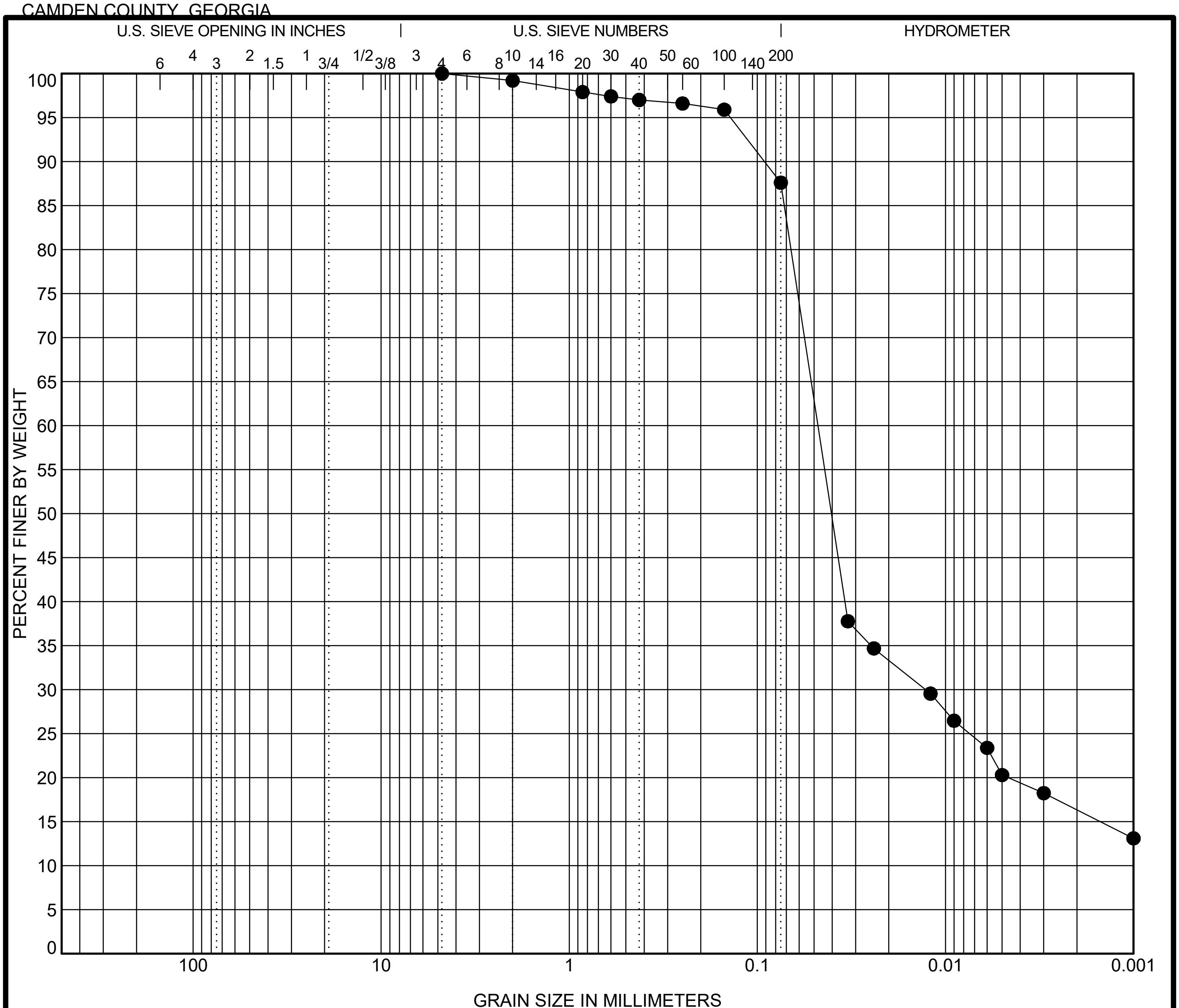
GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00

SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-15	1.5 ft	**SILT, 10Y 3/1 very dark greenish gray (MH) [Estimated shell content < 1.0%]**	**104**	**57**	**47**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.048**	**0.013**		**0.0**	**12.4**	**71.3**	**16.3**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	87.6	95.9	96.6	97.0	97.4	97.9	99.2	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/25/19	SRH	EC

Schnabel ENGINEERING

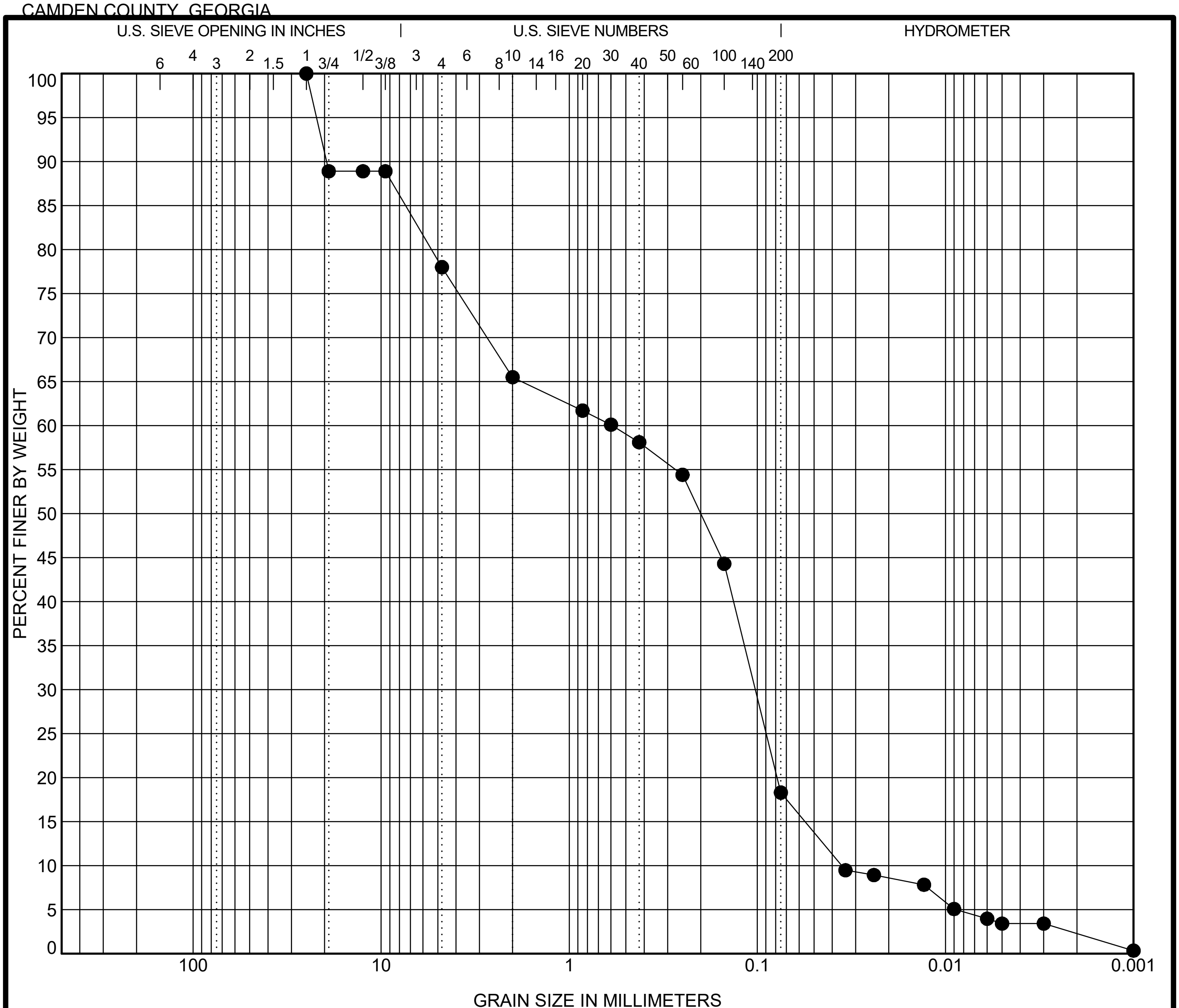
GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00

SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-16	0.3 ft	**SAND, silty, clayey, 10Y 3/1 very dark greenish gray (SC-SM) [Estimated shell content < 22.0%]**	**28**	**21**	**7**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**25**	**0.59**	**0.102**	**0.036**	**22.0**	**59.7**	**16.0**	**2.3**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4	3/8"	1/2"	3/4"	1.0"
% Finer	18.3	44.3	54.4	58.1	60.1	61.7	65.5	78.0	88.9	88.9	88.9	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/19/19	SRH	EC

Schnabel ENGINEERING

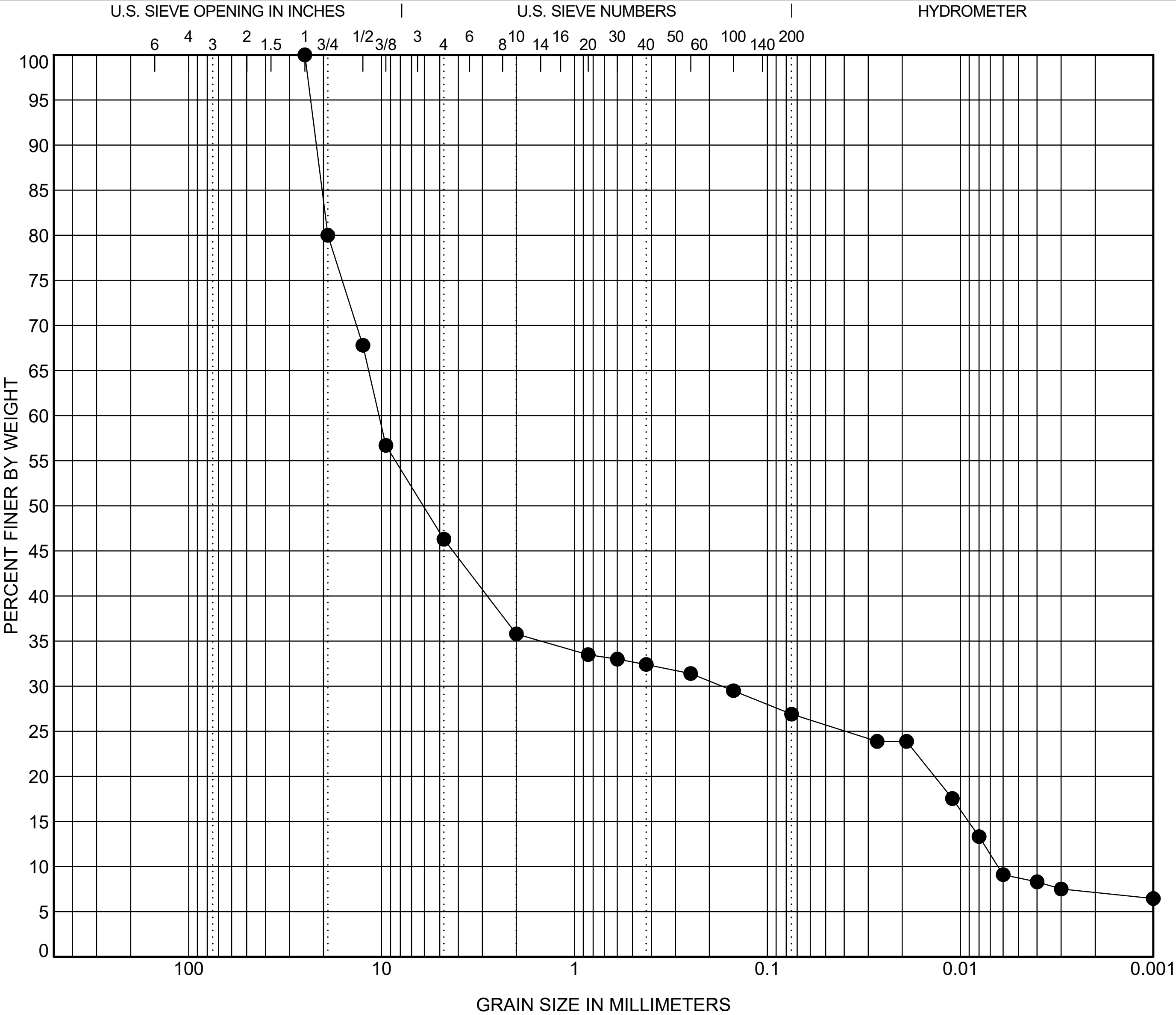
GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00

SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen		Sample Description	LL	PL	PI	Test Method
●	VB-KBIC19-16	1.2 ft	**SHELL, 5Y 4/1 dark gray (GM) [Estimated shell content < 54.0%]**	**22**	**19**	**3**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**25**	**10.308**	**0.172**	**0.006**	**53.7**	**19.4**	**19.8**	**7.1**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4	3/8"	1/2"	3/4"	1.0"
% Finer	26.9	29.5	31.4	32.4	33.0	33.5	35.8	46.3	56.7	67.8	80.0	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/21/19	SRH	EC

Schnabel ENGINEERING

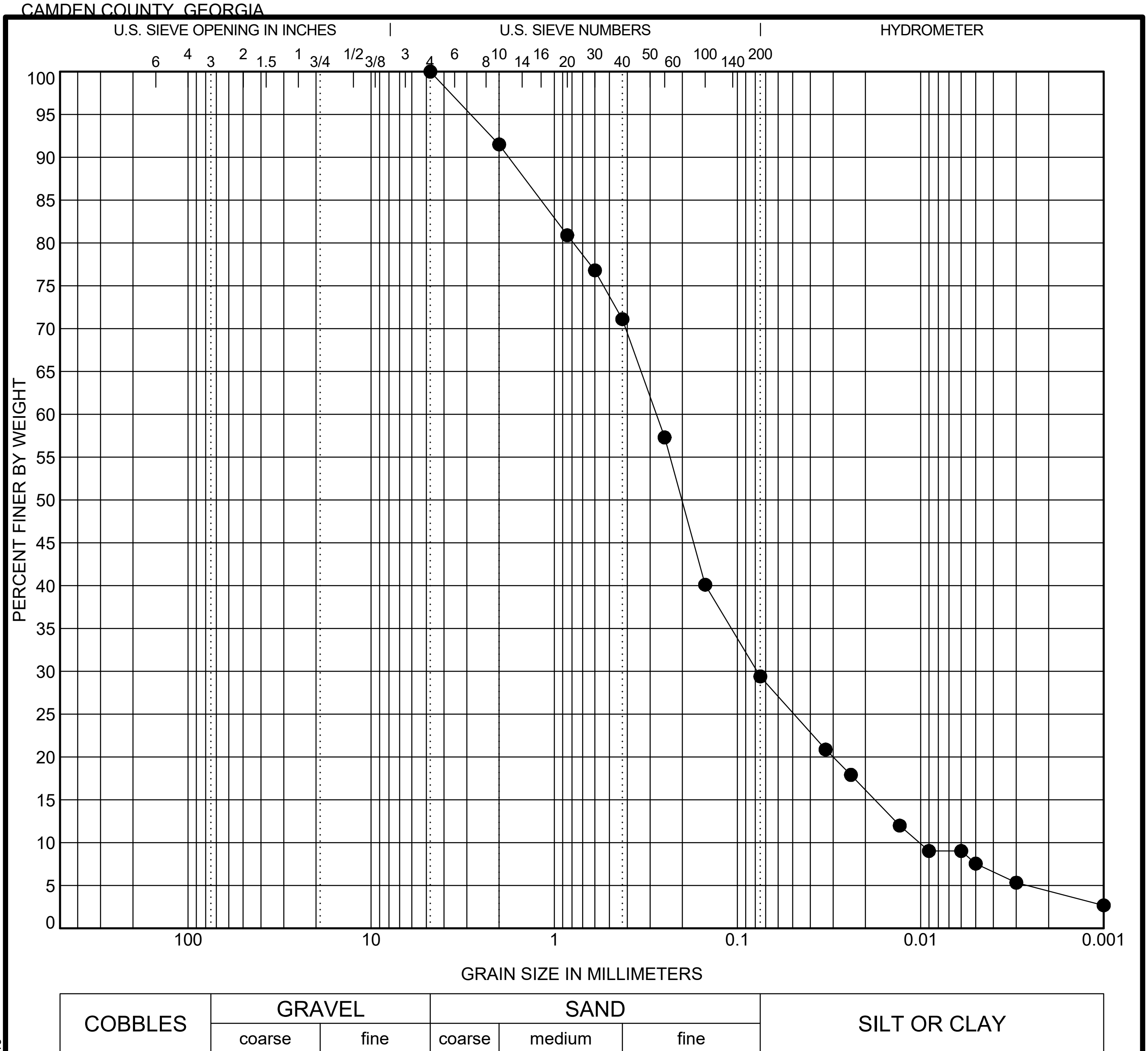
GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00

SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA



Specimen		Sample Description	LL	PL	PI	Test Method
● VB-KBIC19-16	2.5 ft	**SAND, silty, 2.5Y 7/1 light gray (SM) [Estimated shell content < 1.0%]**	**NP**	**NP**	**NP**	Atterberg ASTM D4318

Test Method	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
ASTM D7928	**4.75**	**0.277**	**0.078**	**0.01**	**0.0**	**70.6**	**25.1**	**4.3**

Percent Finer

Sieve Size	200	100	60	40	30	20	10	4
% Finer	29.4	40.1	57.3	71.1	76.8	80.9	91.5	100.0

Tested By	Tested Date	Reviewed by	Calc by
CAB	6/19/19	SRH	EC

Schnabel ENGINEERING

GRADATION CURVE

Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

Contract: 19C19018.00

SIEVE 1/SHEET 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

amec foster wheeler

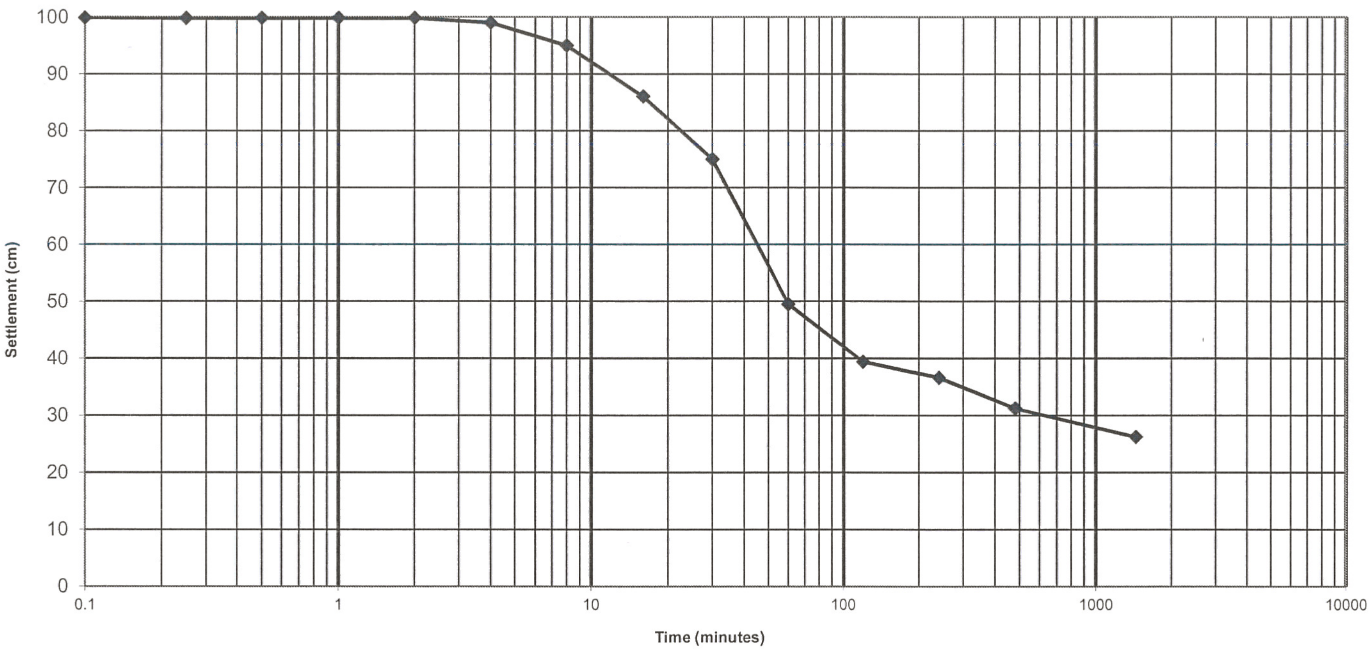
REPORT OF SETTLING RATE TESTING

PROJECT: King's Bay Inner Channel PROJECT NO: 6738-16-5486.10

CLIENT: MAE DATE TESTED: 9/14/16

BORING NO: VB-KBIC16-02 DEPTH: 4.0'-4.2'

SAMPLE NO: 1 CONCENTRATION: 100 g/L



Sedimentation Rate Test COE Method

TIME (min)	INTERFACE (cm)	TIME (min)	INTERFACE (cm)
0.1	99.9	16	86.0
0.25	99.8	30	75.0
0.5	99.8	60	49.5
1	99.8	120	39.4
2	99.8	240	36.6
4	99.0	480	31.2
8	95.0	1440	26.2

Final Concentration: 381.68 g/L

Respectfully Submitted

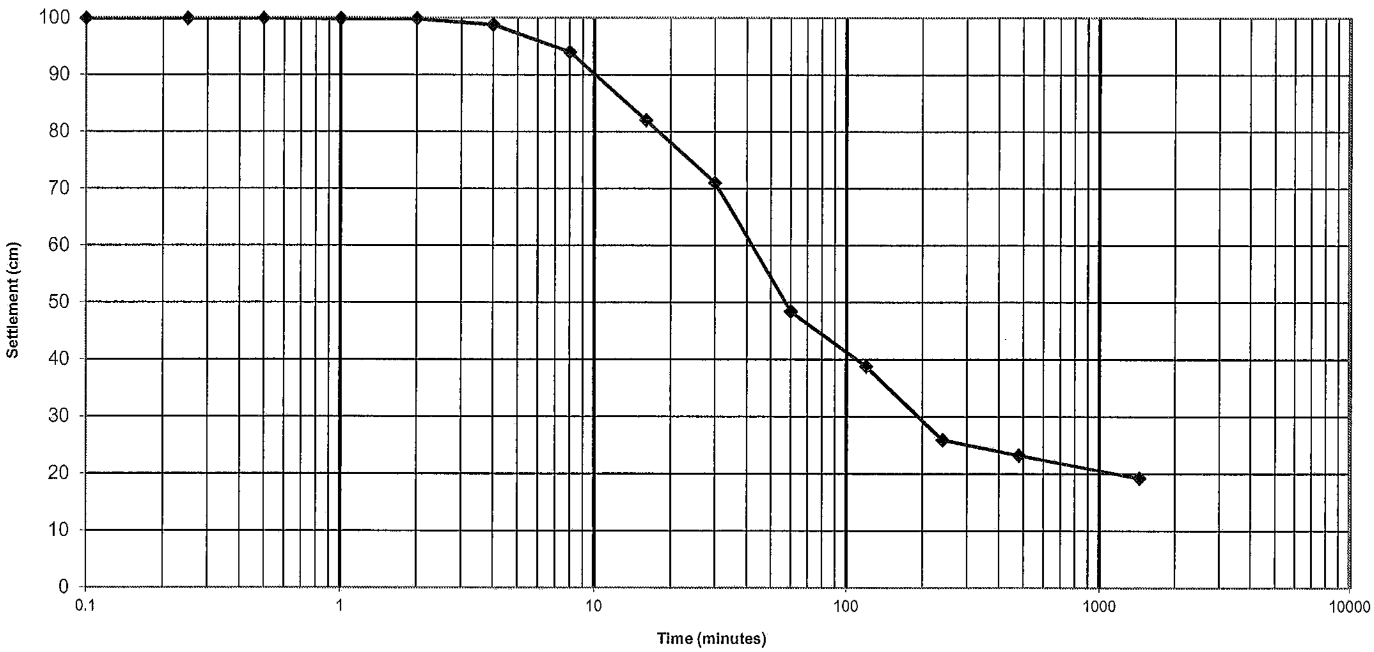
Corey Chascin 9/16/16

Corey T. Chascin, E. I.

amec foster wheeler

REPORT OF SETTLING RATE TESTING

PROJECT:	King's Bay Inner Channel	PROJECT NO:	6738-16-5486.10
CLIENT:	MAE	DATE TESTED:	9/14/16
BORING NO:	VB-KBIC16-03	DEPTH:	3.1'-3.3'
SAMPLE NO:	2	CONCENTRATION:	100 g/L



Sedimentation Rate Test COE Method

TIME (min)	INTERFACE (cm)	TIME (min)	INTERFACE (cm)
0.1	99.9	16	82.0
0.25	99.9	30	71.0
0.5	99.9	60	48.4
1	99.8	120	38.8
2	99.8	240	25.9
4	98.8	480	23.2
8	94.0	1440	19.2

Final Concentration: 520.83 g/L

Respectfully Submitted

Corey Chascin 9/28/16

Corey T. Chascin, E. I.

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

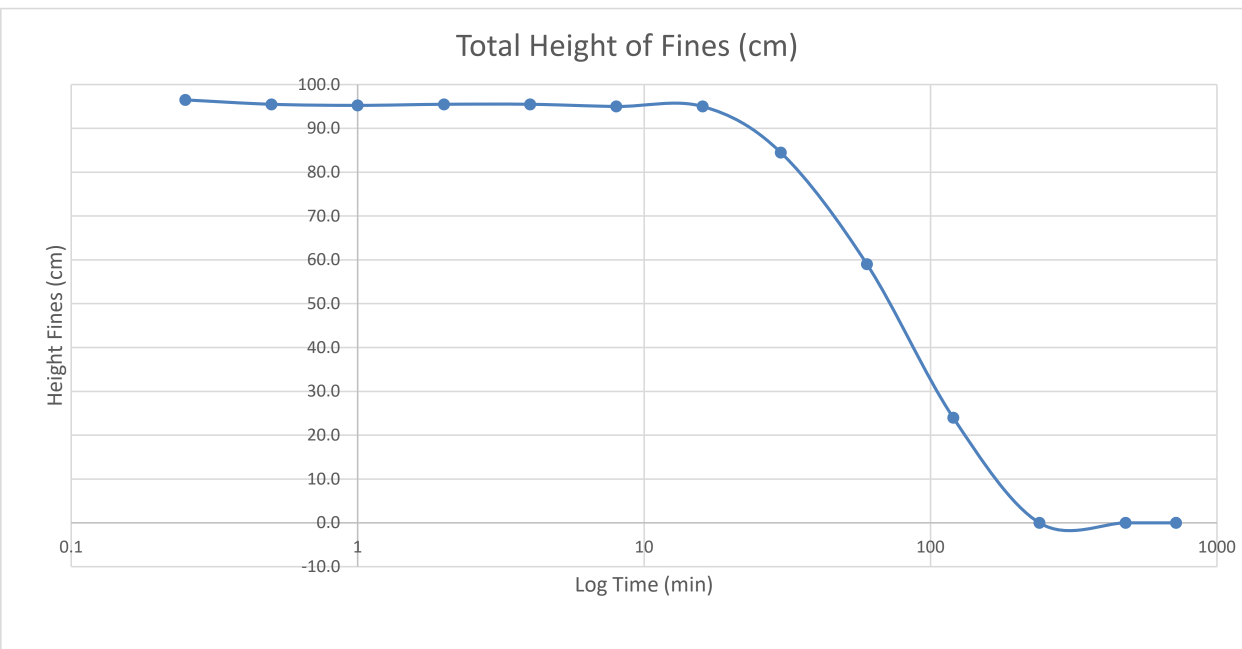
Sample No. : KBIC-19-3
Depth (ft) : 0.2-.07

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
10:27	0.25		100.0	3.5	96.5
10:27	0.5		100.0	4.5	95.5
10:28	1		100.0	4.8	95.3
10:29	2		100.0	4.5	95.5
10:31	4		100.0	4.5	95.5
10:35	8		100.0	5.0	95.0
10:43	16		100.0	5.0	95.0
10:57	30	0.5	90.0	5.5	84.5
11:27	60	1.0	65.0	6.0	59.0
12:27	120	2.0	30.0	6.0	24.0
14:27	240	4.0	6.5	6.5	0.0
18:27	480	8.0	6.5	6.5	0.0
22:27	720	12.0	6.5	6.5	n/a
10:27	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	6.5

Final Concentration (g/l)	769

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left({}^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/9/2019

Tested By KRM
Date 7/10/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

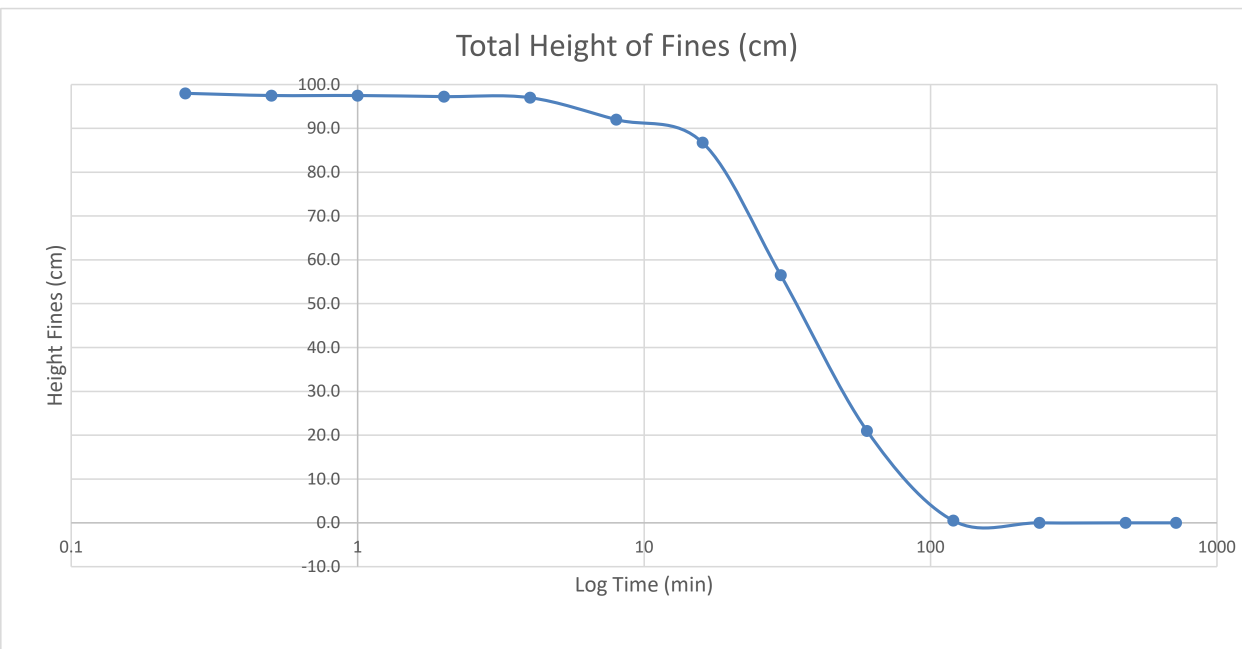
Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

Sample No. : KBIC-19-3
Depth (ft) : 1.7-2.2

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
10:30	0.25		100.0	2.0	98.0
10:30	0.5		100.0	2.5	97.5
10:31	1		100.0	2.5	97.5
10:32	2		100.0	2.8	97.3
10:34	4		100.0	3.0	97.0
10:38	8		95.0	3.0	92.0
10:46	16		90.0	3.3	86.8
11:00	30	0.5	60.0	3.5	56.5
11:30	60	1.0	25.0	4.0	21.0
12:30	120	2.0	5.0	4.5	0.5
14:30	240	4.0	5.0	5.0	0.0
18:30	480	8.0	5.0	5.0	0.0
22:30	720	12.0	n/a	n/a	n/a
10:30	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	5.0
Final Concentration (g/l)	1000

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left({}^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/9/2019

Tested By KRM
Date 7/10/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

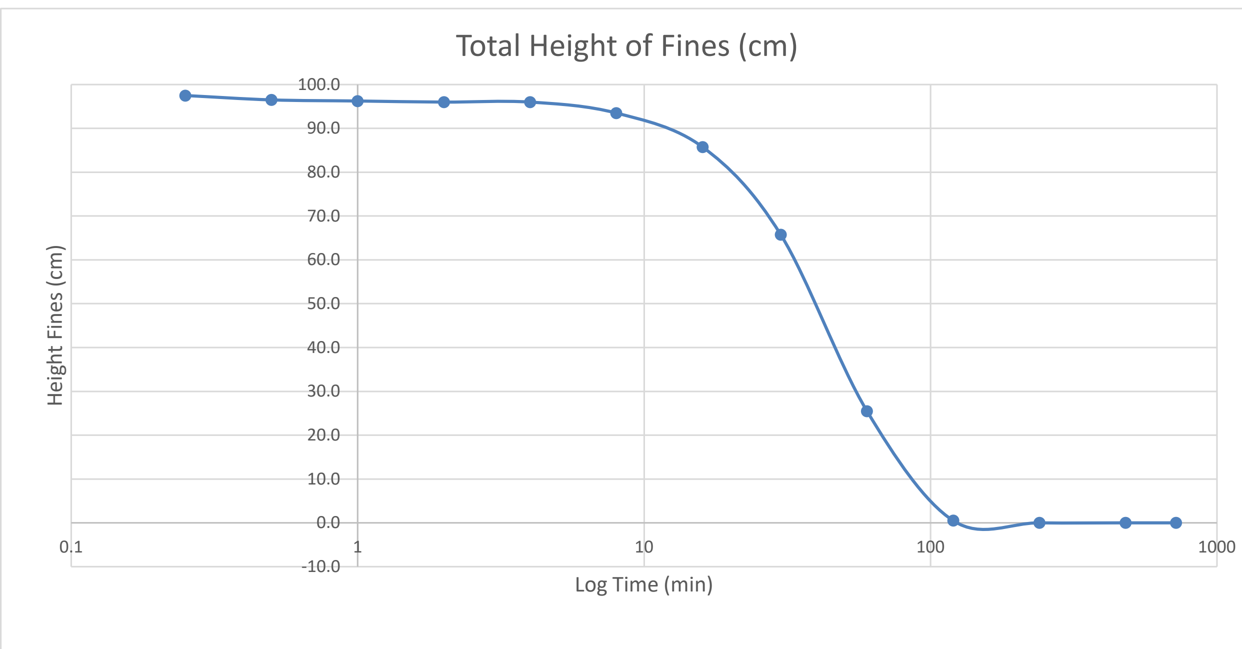
Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00
Sample No. : KBIC-19-3
Depth (ft) : 2.8-3.3

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
10:33	0.25		100.0	2.5	97.5
10:33	0.5		100.0	3.5	96.5
10:34	1		100.0	3.8	96.3
10:35	2		100.0	4.0	96.0
10:37	4		100.0	4.0	96.0
10:41	8		97.5	4.0	93.5
10:49	16		90.0	4.3	85.8
11:03	30	0.5	70.0	4.3	65.8
11:33	60	1.0	30.0	4.5	25.5
12:33	120	2.0	5.0	4.5	0.5
14:33	240	4.0	5.0	5.0	0.0
18:33	480	8.0	5.0	5.0	0.0
22:33	720	12.0	n/a	n/a	n/a
10:33	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	5.0

Final Concentration (g/l)	1000

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(g/_l\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/9/2019

Tested By KRM
Date 7/10/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

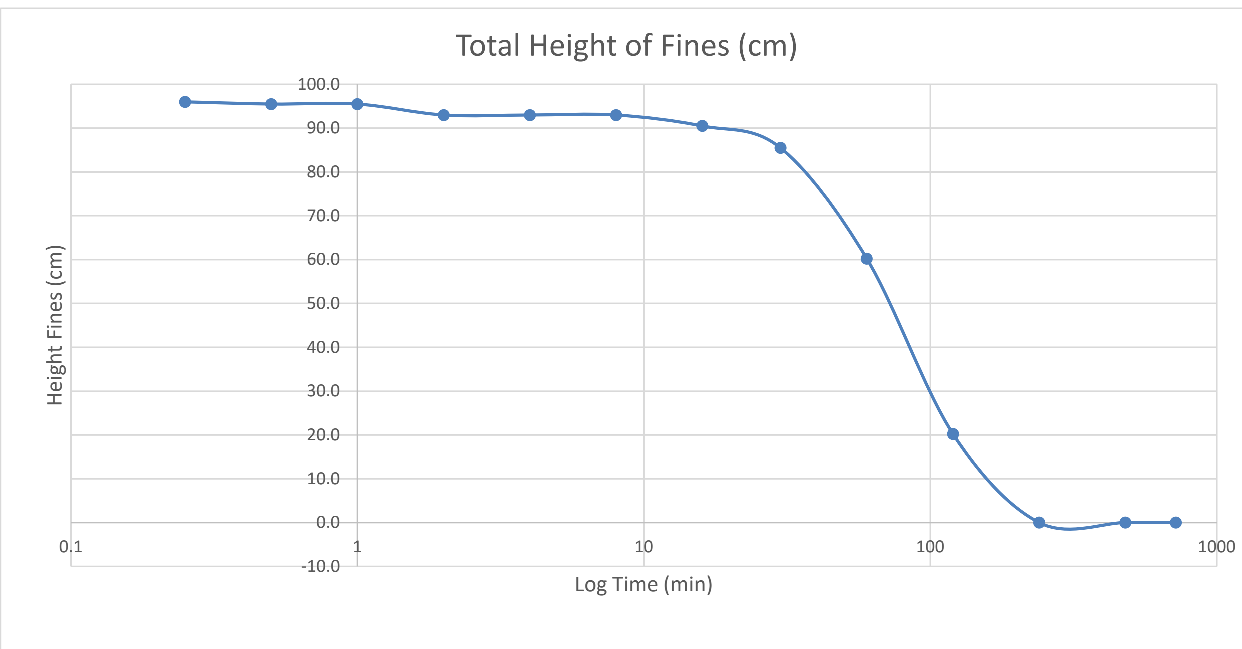
Sample No. : KBIC-19-4
Depth (ft) : 0.8-1.3

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
10:36	0.25		100.0	4.0	96.0
10:36	0.5		100.0	4.5	95.5
10:37	1		100.0	4.5	95.5
10:38	2		97.5	4.5	93.0
10:40	4		97.5	4.5	93.0
10:44	8		97.5	4.5	93.0
10:52	16		95.0	4.5	90.5
11:06	30	0.5	90.0	4.5	85.5
11:36	60	1.0	65.0	4.8	60.3
12:36	120	2.0	25.0	4.8	20.3
14:36	240	4.0	5.0	5.0	0.0
18:36	480	8.0	5.0	5.0	0.0
22:36	720	12.0	5.0	5.0	n/a
10:36	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	5.0

Final Concentration (g/l)	1000

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left({}^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/9/2019

Tested By KRM
Date 7/10/2019

Calc. By KRM
Date 7/11/2019

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Date 7/12/2019

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Schnabel ENGINEERING

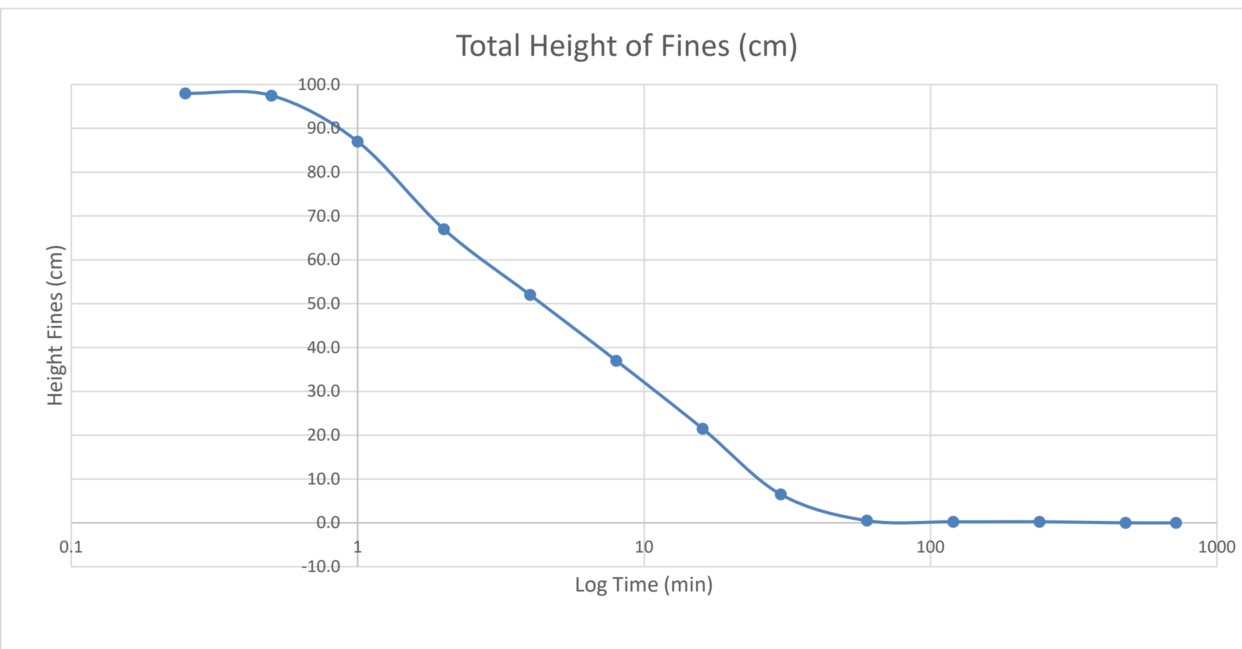
Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00
Sample No. : KBIC-19-4
Depth (ft) : 1.3-1.7

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
10:22	0.25		100.0	2.0	98.0
10:22	0.5		100.0	2.5	97.5
10:23	1		90.0	3.0	87.0
10:24	2		70.0	3.0	67.0
10:26	4		55.0	3.0	52.0
10:30	8		40.0	3.0	37.0
10:38	16		25.0	3.5	21.5
10:52	30	0.5	10.0	3.5	6.5
11:22	60	1.0	4.0	3.5	0.5
12:22	120	2.0	4.0	3.8	0.3
14:22	240	4.0	4.0	3.8	0.3
18:22	480	8.0	4.0	4.0	0.0
22:22	720	12.0	n/a	n/a	n/a
10:22	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	4.0
Final Concentration (g/l)	1250

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\left(g/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/9/2019

Tested By KRM
Date 7/10/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

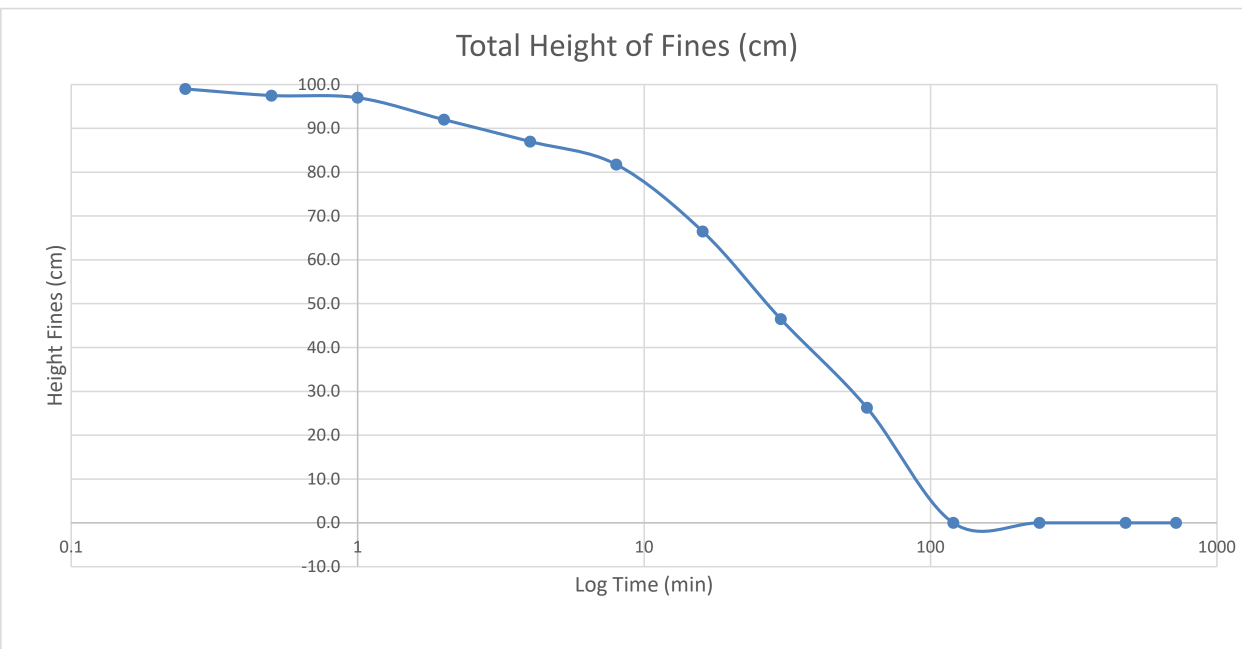
Sample No. : KBIC-19-4
Depth (ft) : 1.7-2.2

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
10:34	0.25		100.0	1.0	99.0
10:34	0.5		100.0	2.5	97.5
10:35	1		100.0	3.0	97.0
10:36	2		95.0	3.0	92.0
10:38	4		90.0	3.0	87.0
10:42	8		85.0	3.3	81.8
10:50	16		70.0	3.5	66.5
11:04	30	0.5	50.0	3.5	46.5
11:34	60	1.0	30.0	3.8	26.3
12:34	120	2.0	4.0	4.0	0.0
14:34	240	4.0	4.0	4.0	0.0
18:34	480	8.0	4.0	4.0	0.0
22:34	720	12.0	n/a	n/a	n/a
10:34	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	4.0

Final Concentration (g/l)	1250

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left({}^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



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NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Schnabel ENGINEERING

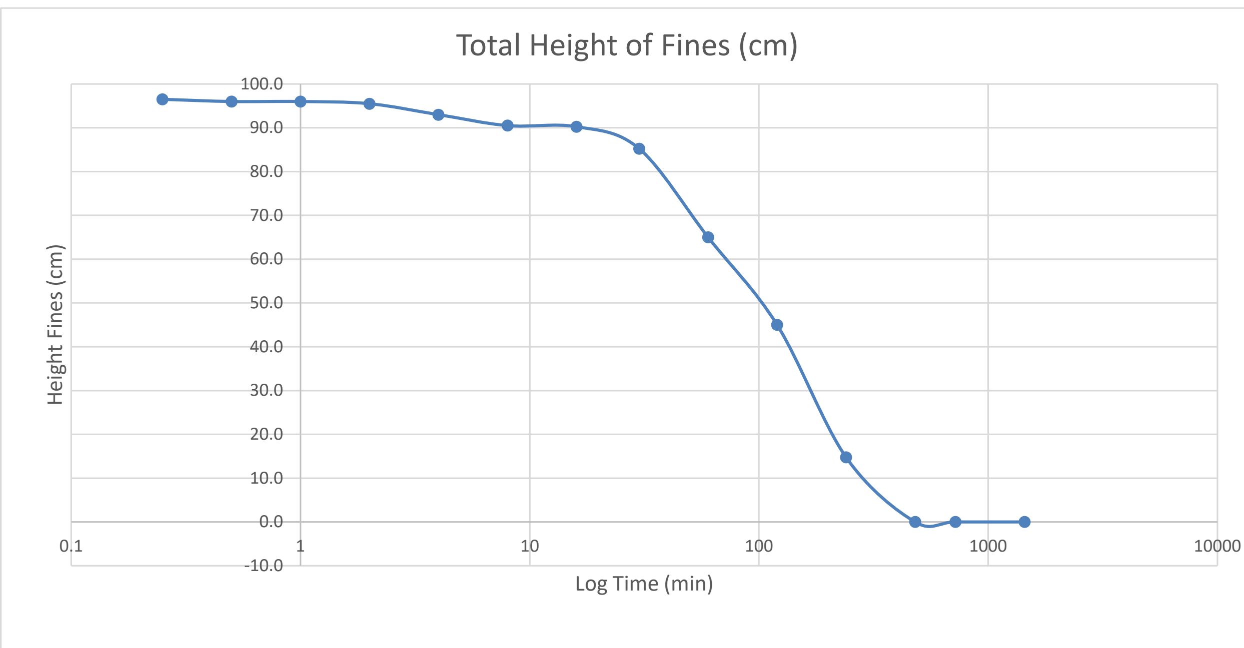
Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00
Sample No. : KBIC-19-4
Depth (ft) : 2.5-3.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
10:38	0.25		100.0	3.5	96.5
10:38	0.5		100.0	4.0	96.0
10:39	1		100.0	4.0	96.0
10:40	2		100.0	4.5	95.5
10:42	4		97.5	4.5	93.0
10:46	8		95.0	4.5	90.5
10:54	16		95.0	4.8	90.3
11:08	30	0.5	90.0	4.8	85.3
11:38	60	1.0	70.0	5.0	65.0
12:38	120	2.0	50.0	5.0	45.0
14:38	240	4.0	20.0	5.3	14.8
18:38	480	8.0	5.5	5.5	0.0
22:38	720	12.0	5.5	5.5	0.0
10:38	1440	24.0	5.5	5.5	0.0

Final Sediment Height (cm)	5.5
Final Concentration (g/l)	909

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(\frac{g}{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



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MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

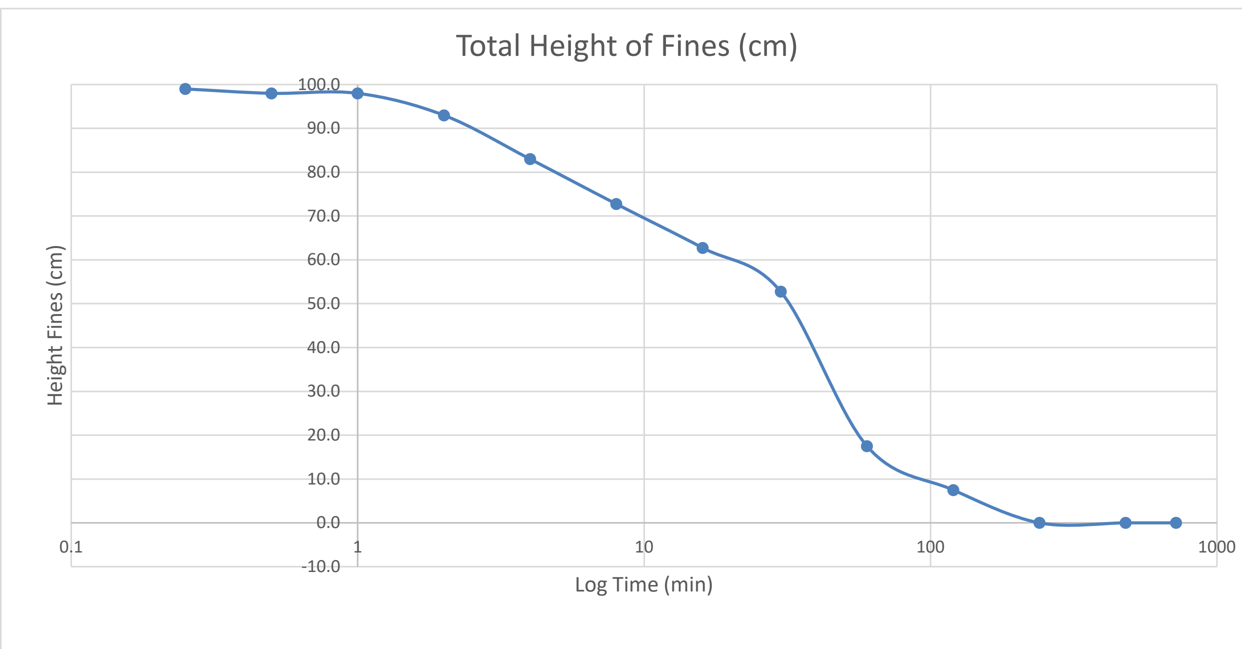
Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00
Sample No. : KBIC-19-5
Depth (ft) : 0.8-1.3

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
10:40	0.25		100.0	1.0	99.0
10:40	0.5		100.0	2.0	98.0
10:41	1		100.0	2.0	98.0
10:42	2		95.0	2.0	93.0
10:44	4		85.0	2.0	83.0
10:48	8		75.0	2.3	72.8
10:56	16		65.0	2.3	62.8
11:10	30	0.5	55.0	2.3	52.8
11:40	60	1.0	20.0	2.5	17.5
12:40	120	2.0	10.0	2.5	7.5
14:40	240	4.0	2.5	2.5	0.0
18:40	480	8.0	2.5	2.5	0.0
22:40	720	12.0	n/a	n/a	n/a
10:40	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	2.5

Final Concentration (g/l)	2000

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(\frac{g}{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



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Date 7/10/2019

Calc. By KRM
Date 7/11/2019

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Date 7/12/2019

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

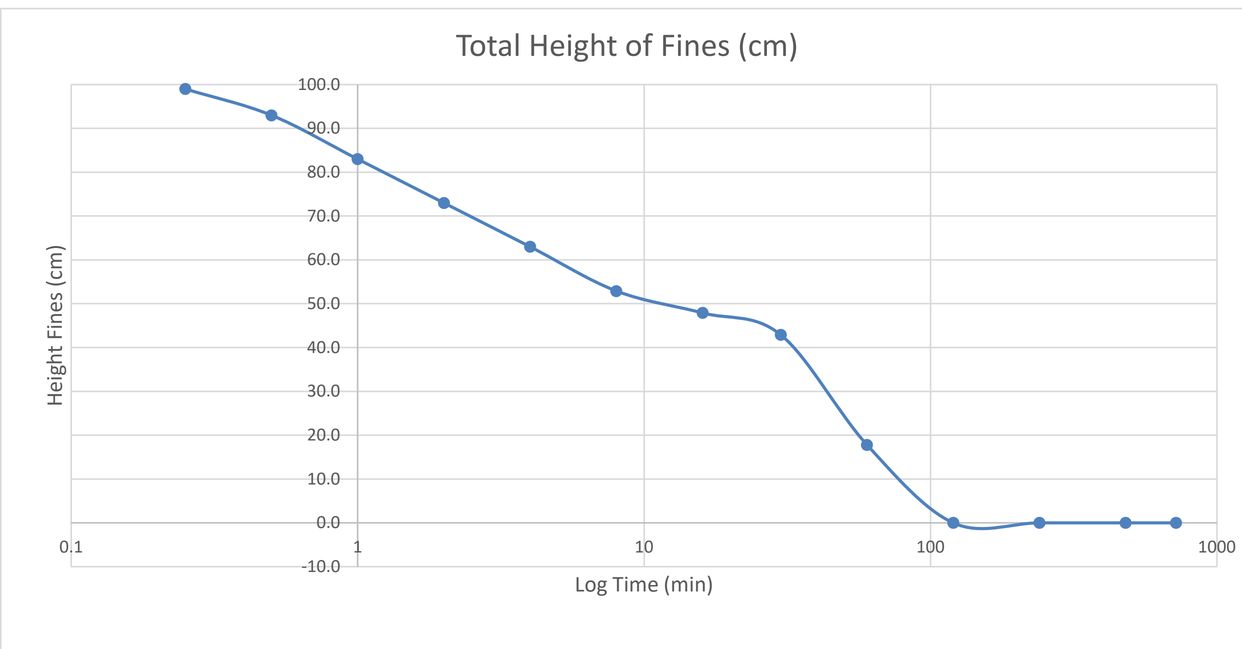
Sample No. : KBIC-19-5
Depth (ft) : 2.5-3.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
10:55	0.25		100.0	1.0	99.0
10:55	0.5		95.0	2.0	93.0
10:56	1		85.0	2.0	83.0
10:57	2		75.0	2.0	73.0
10:59	4		65.0	2.0	63.0
11:03	8		55.0	2.1	52.9
11:11	16		50.0	2.1	47.9
11:25	30	0.5	45.0	2.1	42.9
11:55	60	1.0	20.0	2.2	17.8
12:55	120	2.0	2.2	2.2	0.0
14:55	240	4.0	2.2	2.2	0.0
18:55	480	8.0	2.2	2.2	0.0
22:55	720	12.0	n/a	n/a	n/a
10:55	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	2.2

Final Concentration (g/l)	2273

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left({}^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE	Tested By KRM	Calc. By KRM	Reviewed SRH
Date 7/9/2019	Date 7/10/2019	Date 7/11/2019	Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA

Project No. : 19C19018.00

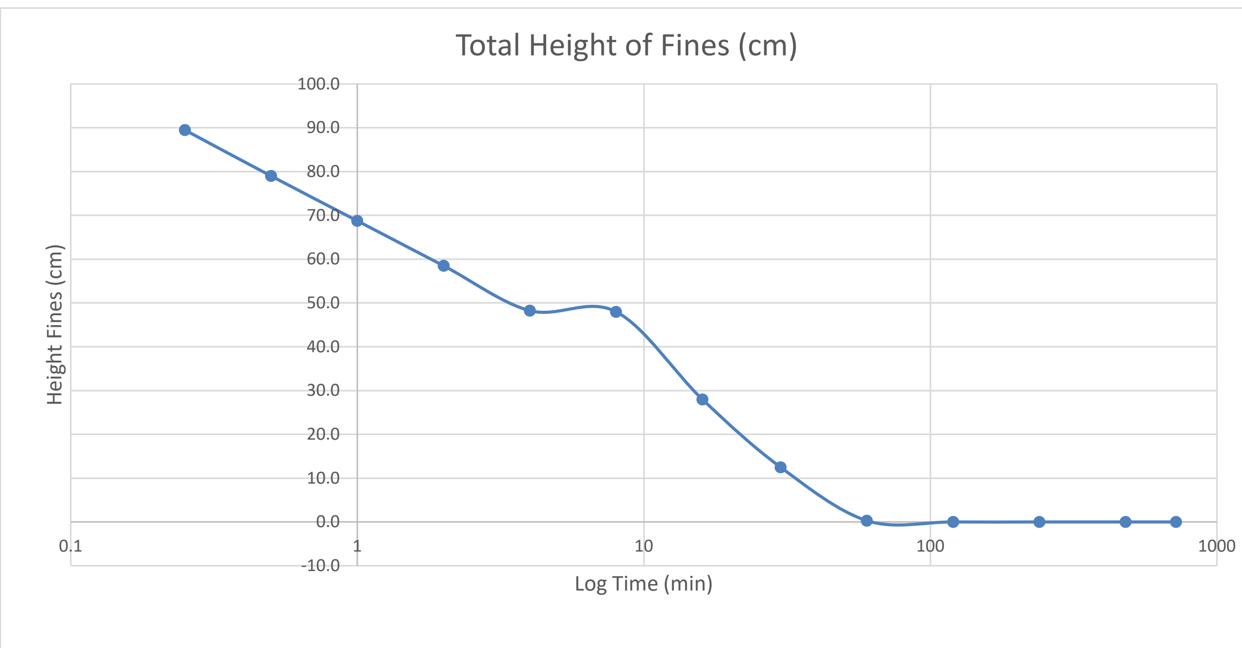
Sample No. : KBIC-19-6

Depth (ft) : 0.2-0.7

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
11:15	0.25		90.0	0.5	89.5
11:15	0.5		80.0	1.0	79.0
11:16	1		70.0	1.3	68.8
11:17	2		60.0	1.5	58.5
11:19	4		50.0	1.8	48.3
11:23	8		50.0	2.0	48.0
11:31	16		30.0	2.0	28.0
11:45	30	0.5	15.0	2.5	12.5
12:15	60	1.0	3.0	2.7	0.3
13:15	120	2.0	2.7	2.7	0.0
15:15	240	4.0	2.7	2.7	0.0
19:15	480	8.0	2.7	2.7	0.0
23:15	720	12.0	n/a	n/a	n/a
11:15	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	2.7
Final Concentration (g/l)	1852

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(g/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/9/2019

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Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

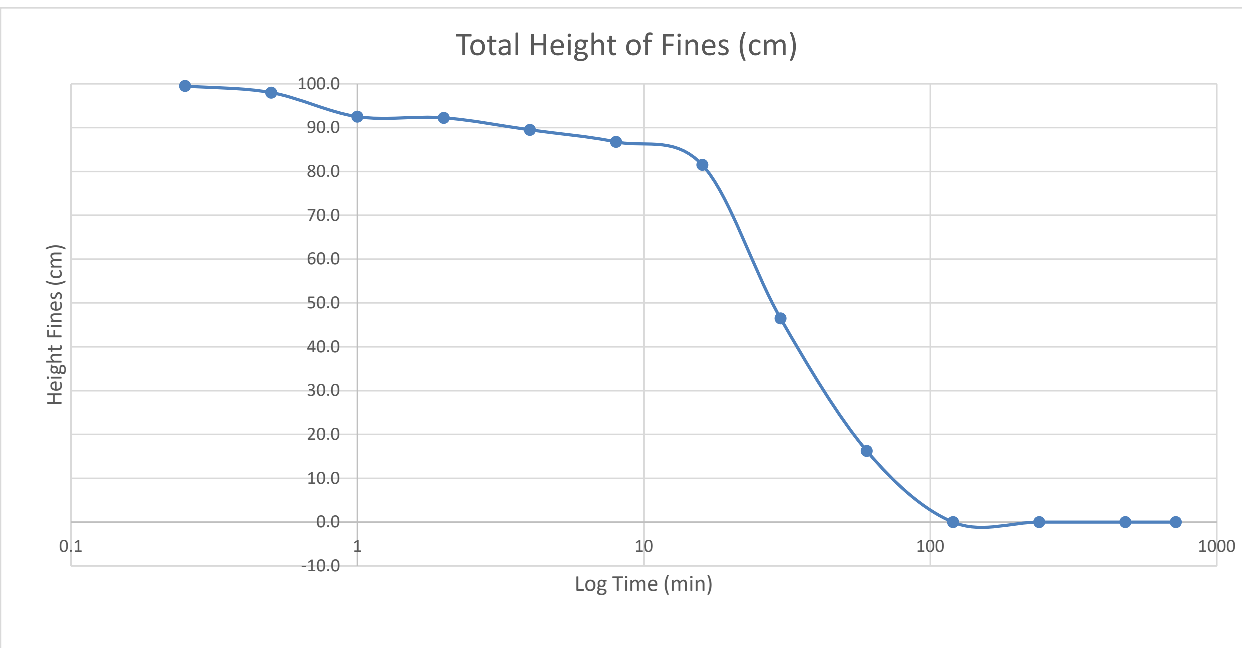
Sample No. : KBIC-19-6
Depth (ft) : 1.8-2.3

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
13:30	0.25		100.0	0.5	99.5
13:30	0.5		100.0	2.0	98.0
13:31	1		95.0	2.5	92.5
13:32	2		95.0	2.8	92.3
13:34	4		92.5	3.0	89.5
13:38	8		90.0	3.3	86.8
13:46	16		85.0	3.5	81.5
14:00	30	0.5	50.0	3.5	46.5
14:30	60	1.0	20.0	3.8	16.3
15:30	120	2.0	4.0	4.0	0.0
17:30	240	4.0	4.0	4.0	0.0
21:30	480	8.0	4.0	4.0	0.0
1:30	720	12.0	n/a	n/a	n/a
13:30	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	4.0

Final Concentration (g/l)	1250

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\left(\frac{g}{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



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NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

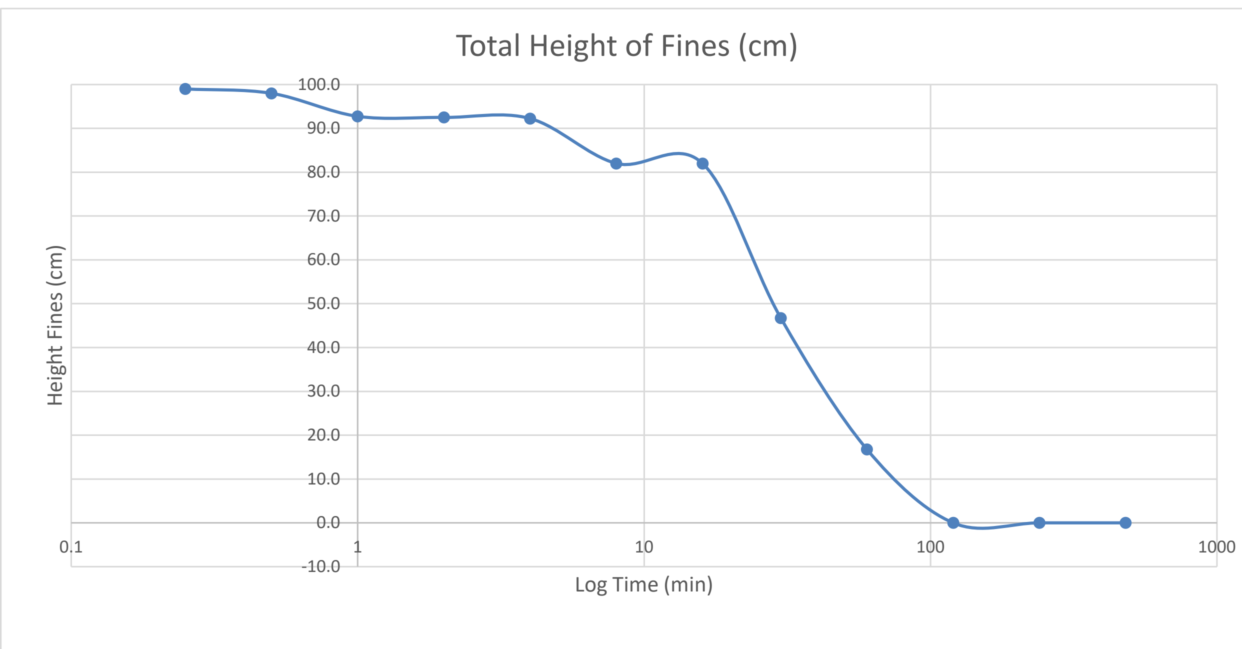
Sample No. : KBIC-19-6
Depth (ft) : 2.5-3.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
13:35	0.25		100.0	1.0	99.0
13:35	0.5		100.0	2.0	98.0
13:36	1		95.0	2.3	92.8
13:37	2		95.0	2.5	92.5
13:39	4		95.0	2.8	92.3
13:43	8		85.0	3.0	82.0
13:51	16		85.0	3.0	82.0
14:05	30	0.5	50.0	3.3	46.8
14:35	60	1.0	20.0	3.3	16.8
15:35	120	2.0	3.5	3.5	0.0
17:35	240	4.0	3.5	3.5	0.0
21:35	480	8.0	3.5	3.5	0.0
1:35	720	12.0	n/a	n/a	n/a
13:35	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	3.5

Final Concentration (g/l)	1429

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(g/_l\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
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Date 7/10/2019

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MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

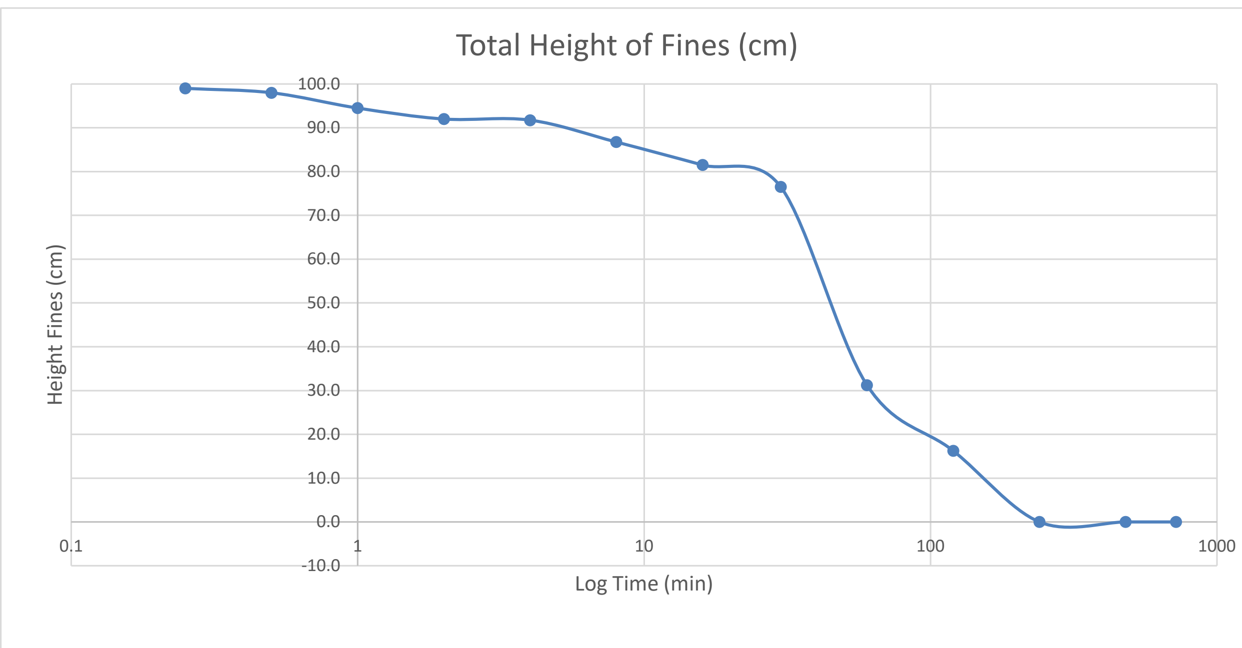
Sample No. : KBIC-19-6
Depth (ft) : 3.8-4.3

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
13:40	0.25		100.0	1.0	99.0
13:40	0.5		100.0	2.0	98.0
13:41	1		97.5	3.0	94.5
13:42	2		95.0	3.0	92.0
13:44	4		95.0	3.3	91.8
13:48	8		90.0	3.3	86.8
13:56	16		85.0	3.5	81.5
14:10	30	0.5	80.0	3.5	76.5
14:40	60	1.0	35.0	3.8	31.3
15:40	120	2.0	20.0	3.8	16.3
17:40	240	4.0	4.0	4.0	0.0
21:40	480	8.0	4.0	4.0	0.0
1:40	720	12.0	4.0	4.0	0.0
13:40	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	4.0

Final Concentration (g/l)	1250

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left({}^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



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Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

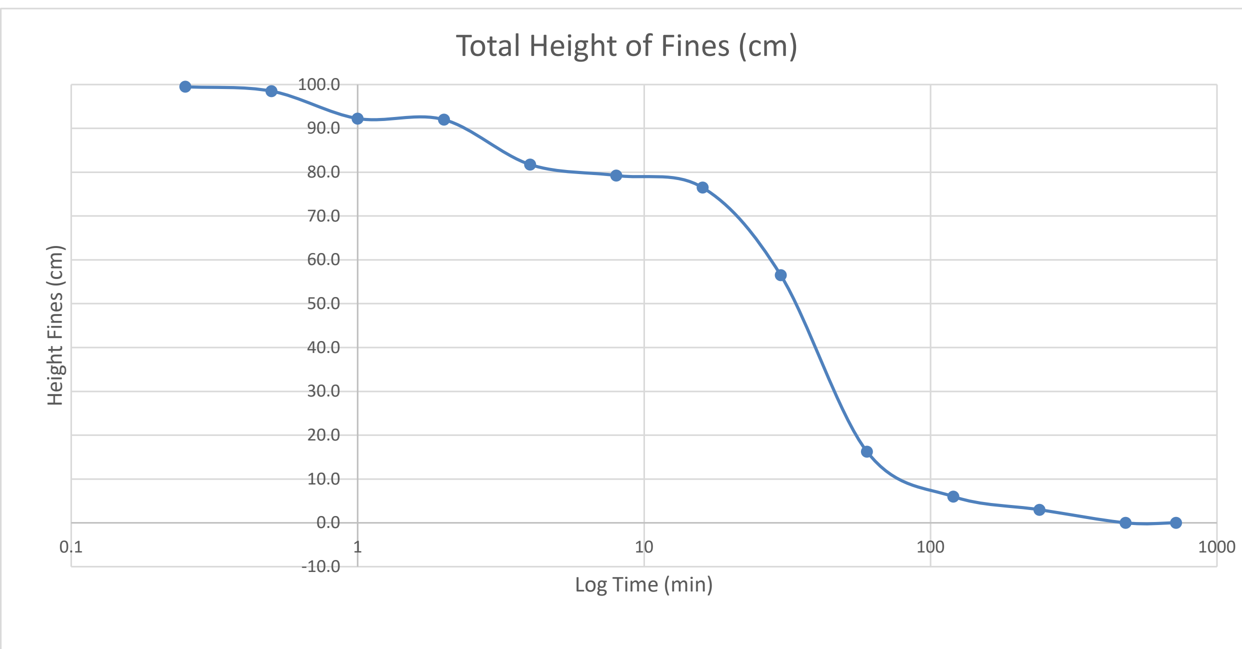
Sample No. : KBIC-19-7
Depth (ft) : 0.2-0.7

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
13:44	0.25		100.0	0.5	99.5
13:44	0.5		100.0	1.5	98.5
13:45	1		95.0	2.8	92.3
13:46	2		95.0	3.0	92.0
13:48	4		85.0	3.3	81.8
13:52	8		82.5	3.3	79.3
14:00	16		80.0	3.5	76.5
14:14	30	0.5	60.0	3.5	56.5
14:44	60	1.0	20.0	3.8	16.3
15:44	120	2.0	10.0	4.0	6.0
17:44	240	4.0	7.0	4.0	3.0
21:44	480	8.0	4.0	4.0	0.0
1:44	720	12.0	4.0	4.0	0.0
13:44	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	4.0

Final Concentration (g/l)	1250

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left({}^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



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Soil Mechanics Laboratory
Sedimentation Rate Worksheet

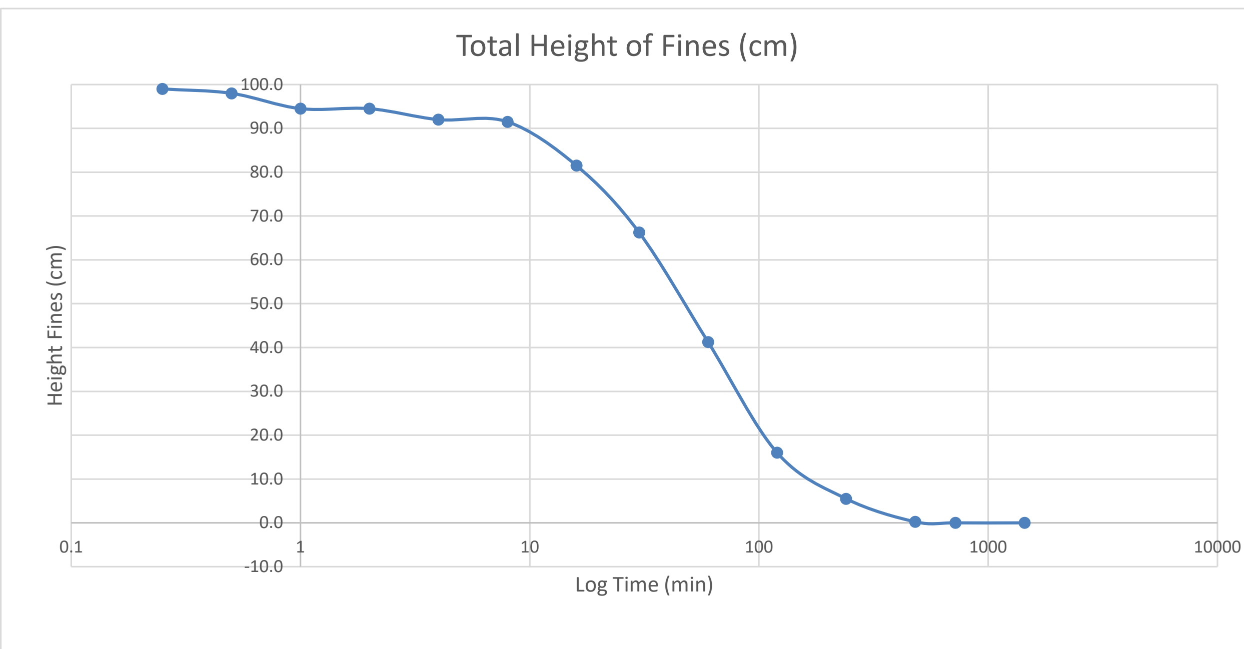
Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00
Sample No. : KBIC-19-7
Depth (ft) : 1.8-2.3

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
13:46	0.25		100.0	1.0	99.0
13:46	0.5		100.0	2.0	98.0
13:47	1		97.5	3.0	94.5
13:48	2		97.5	3.0	94.5
13:50	4		95.0	3.0	92.0
13:54	8		95.0	3.5	91.5
14:02	16		85.0	3.5	81.5
14:16	30	0.5	70.0	3.8	66.3
14:46	60	1.0	45.0	3.8	41.3
15:46	120	2.0	20.0	4.0	16.0
17:46	240	4.0	10.0	4.5	5.5
21:46	480	8.0	5.0	4.8	0.3
1:46	720	12.0	5.0	5.0	0.0
13:46	1440	24.0	5.0	5.0	0.0

Final Sediment Height (cm)	5.0

Final Concentration (g/l)	1000

$$\mathit{Final\ Concentration} = \frac{\mathit{Initial\ Dispersion\ Concentration}\left({}^{g}/_{l}\right) * 100\ cm}{\mathit{Final\ Height}\ (cm)}$$



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Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

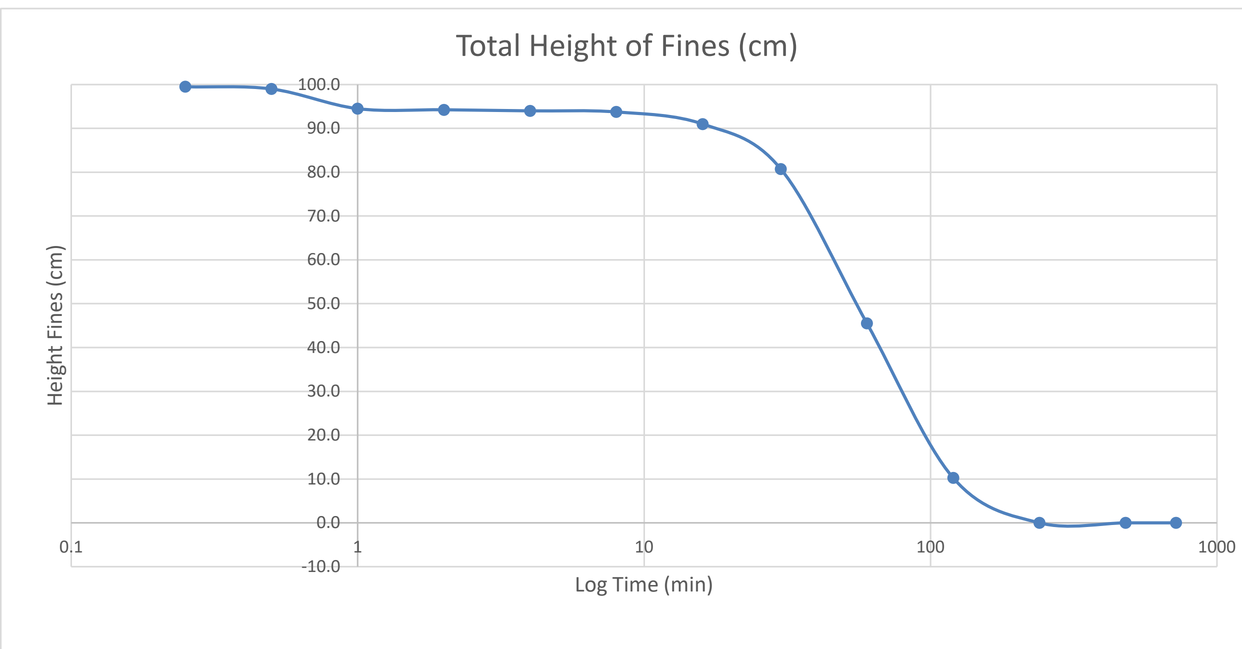
Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00
Sample No. : KBIC-19-7
Depth (ft) : 2.5-3.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
13:49	0.25		100.0	0.5	99.5
13:49	0.5		100.0	1.0	99.0
13:50	1		97.5	3.0	94.5
13:51	2		97.5	3.3	94.3
13:53	4		97.5	3.5	94.0
13:57	8		97.5	3.8	93.8
14:05	16		95.0	4.0	91.0
14:19	30	0.5	85.0	4.3	80.8
14:49	60	1.0	50.0	4.5	45.5
15:49	120	2.0	15.0	4.8	10.3
17:49	240	4.0	5.0	5.0	0.0
21:49	480	8.0	5.0	5.0	0.0
1:49	720	12.0	n/a	n/a	n/a
13:49	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	5.0

Final Concentration (g/l)	1000

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(\frac{g}{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By	MDE	Tested By	KRM	Calc. By	KRM	Reviewed	SRH
Date	7/9/2019	Date	7/10/2019	Date	7/11/2019	Date	7/12/2019

Schnabel ENGINEERING

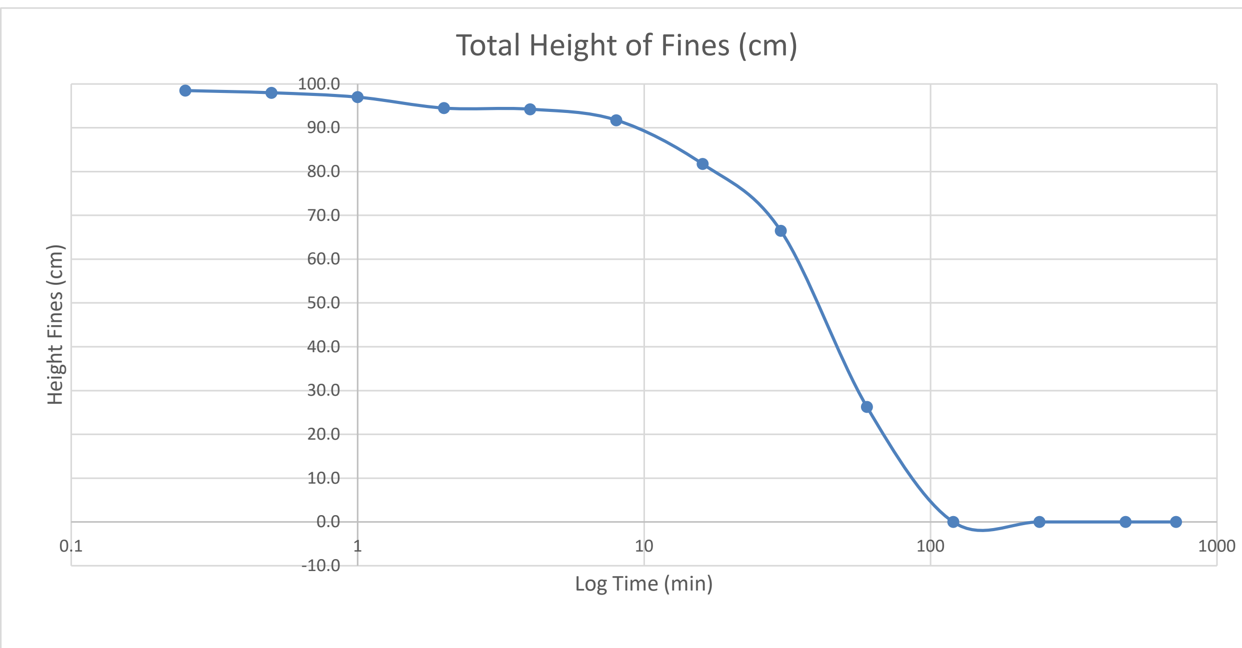
Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00
Sample No. : KBIC-19-8
Depth (ft) : 0.5-1.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
10:42	0.25		100.0	1.5	98.5
10:42	0.5		100.0	2.0	98.0
10:43	1		100.0	3.0	97.0
10:44	2		97.5	3.0	94.5
10:46	4		97.5	3.3	94.3
10:50	8		95.0	3.3	91.8
10:58	16		85.0	3.3	81.8
11:12	30	0.5	70.0	3.5	66.5
11:42	60	1.0	30.0	3.8	26.3
12:42	120	2.0	4.0	4.0	0.0
14:42	240	4.0	4.0	4.0	0.0
18:42	480	8.0	4.0	4.0	0.0
22:42	720	12.0	n/a	n/a	n/a
10:42	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	4.0
Final Concentration (g/l)	1250

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left({}^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

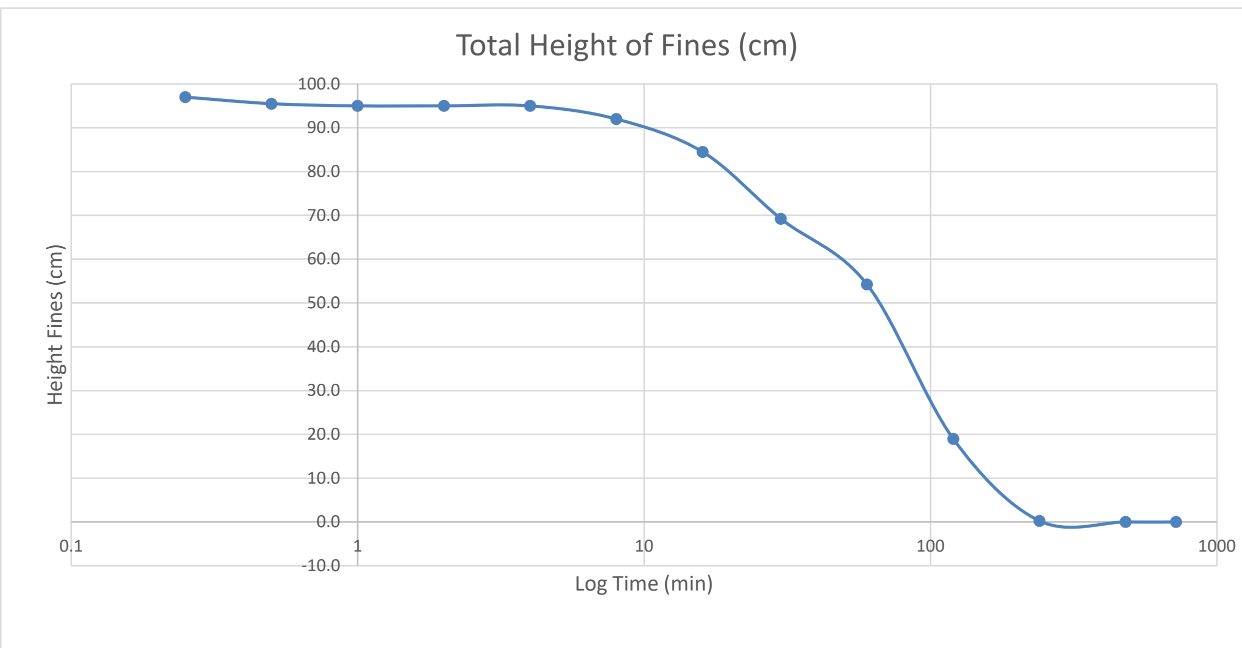
Sample No. : KBIC-19-8
Depth (ft) : 1.5-2.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
10:44	0.25		100.0	3.0	97.0
10:44	0.5		100.0	4.5	95.5
10:45	1		100.0	5.0	95.0
10:46	2		100.0	5.0	95.0
10:48	4		100.0	5.0	95.0
10:52	8		97.5	5.5	92.0
11:00	16		90.0	5.5	84.5
11:14	30	0.5	75.0	5.8	69.3
11:44	60	1.0	60.0	5.8	54.3
12:44	120	2.0	25.0	6.0	19.0
14:44	240	4.0	6.5	6.3	0.3
18:44	480	8.0	6.5	6.5	0.0
22:44	720	12.0	6.5	6.5	0.0
10:44	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	6.5

Final Concentration (g/l)	769

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(g/_l\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

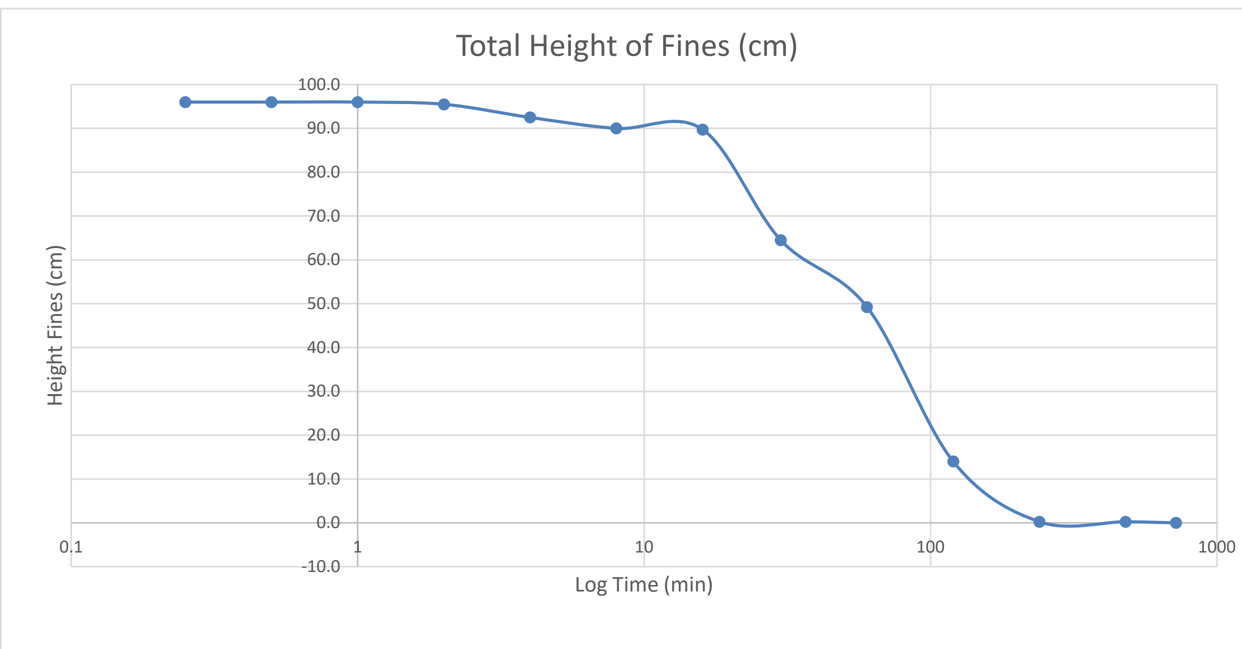
Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

Sample No. : KBIC-19-8
Depth (ft) : 3.0-3.5

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
10:47	0.25		100.0	4.0	96.0
10:47	0.5		100.0	4.0	96.0
10:48	1		100.0	4.0	96.0
10:49	2		100.0	4.5	95.5
10:51	4		97.5	5.0	92.5
10:55	8		95.0	5.0	90.0
11:03	16		95.0	5.3	89.8
11:17	30	0.5	70.0	5.5	64.5
11:47	60	1.0	55.0	5.8	49.3
12:47	120	2.0	20.0	6.0	14.0
14:47	240	4.0	6.5	6.3	0.3
18:47	480	8.0	6.5	6.3	0.3
22:47	720	12.0	6.5	6.5	0.0
10:47	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	6.5
Final Concentration (g/l)	769

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(g/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

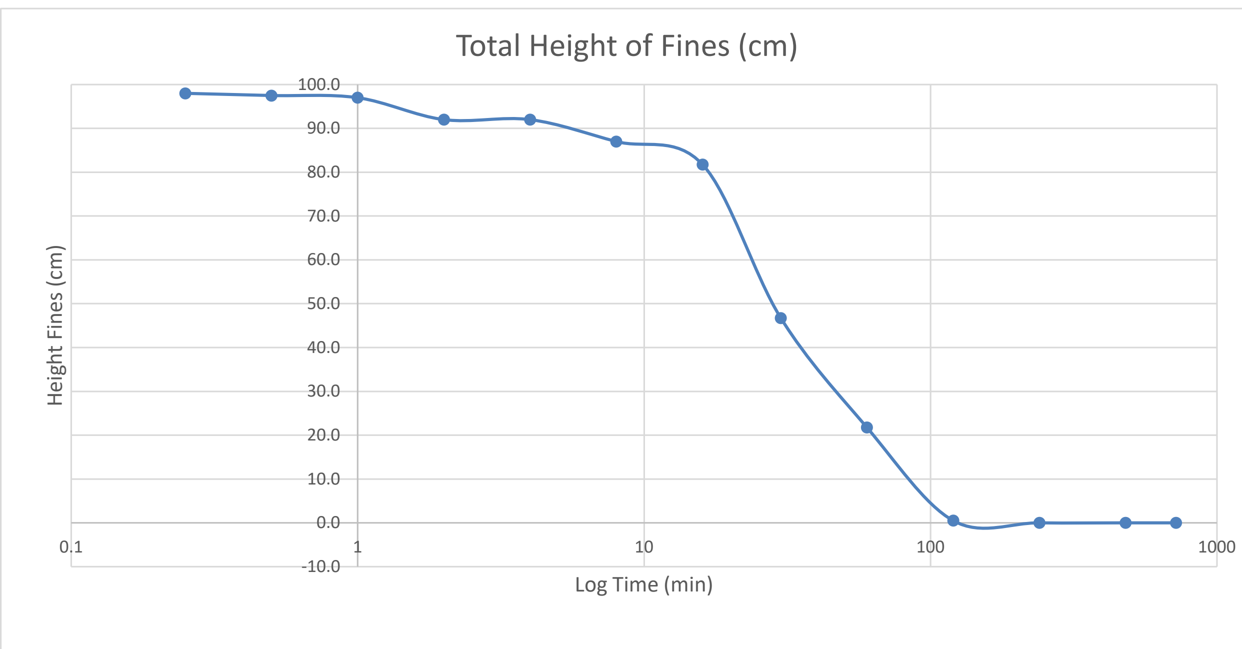
Sample No. : KBIC-19-9
Depth (ft) : 0.5-1.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
12:51	0.25		100.0	2.0	98.0
12:51	0.5		100.0	2.5	97.5
12:52	1		100.0	3.0	97.0
12:53	2		95.0	3.0	92.0
12:55	4		95.0	3.0	92.0
12:59	8		90.0	3.0	87.0
13:07	16		85.0	3.3	81.8
13:21	30	0.5	50.0	3.3	46.8
13:51	60	1.0	25.0	3.3	21.8
14:51	120	2.0	4.0	3.5	0.5
16:51	240	4.0	4.0	4.0	0.0
20:51	480	8.0	4.0	4.0	0.0
0:51	720	12.0	n/a	n/a	n/a
12:51	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	4.0

Final Concentration (g/l)	1250

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(\frac{g}{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

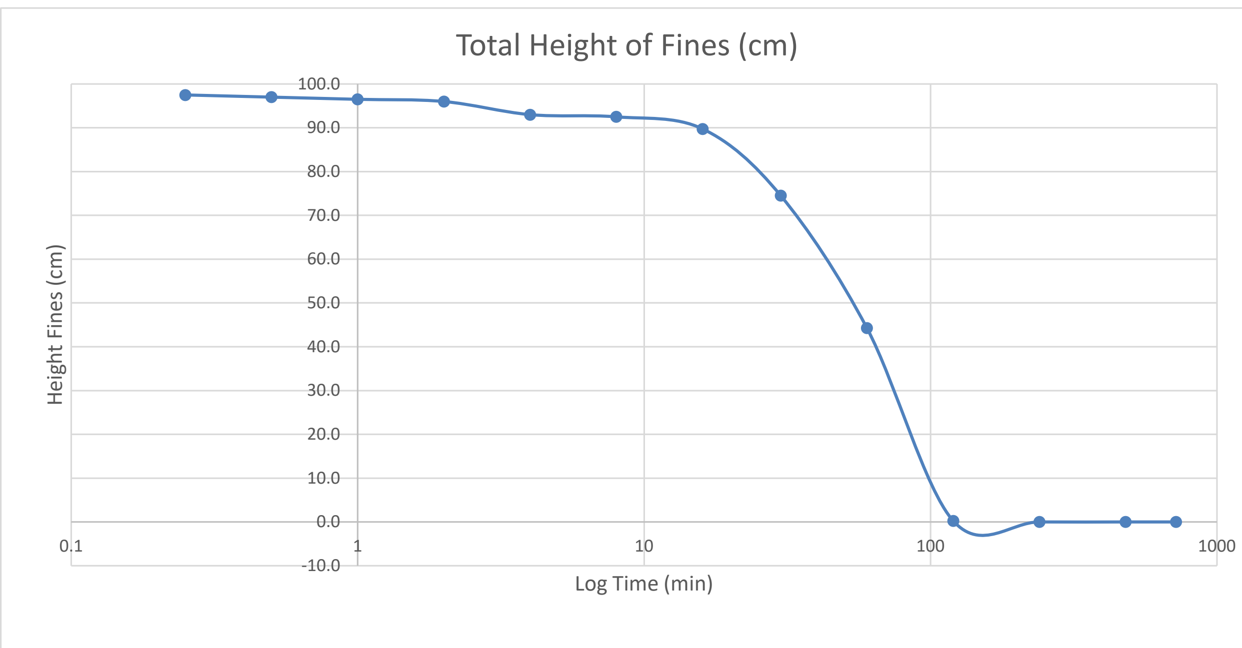
Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00
Sample No. : KBIC-19-9
Depth (ft) : 1.5-2.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
12:53	0.25		100.0	2.5	97.5
12:53	0.5		100.0	3.0	97.0
12:54	1		100.0	3.5	96.5
12:55	2		100.0	4.0	96.0
12:57	4		97.5	4.5	93.0
13:01	8		97.5	5.0	92.5
13:09	16		95.0	5.3	89.8
13:23	30	0.5	80.0	5.5	74.5
13:53	60	1.0	50.0	5.8	44.3
14:53	120	2.0	6.0	5.8	0.3
16:53	240	4.0	6.0	6.0	0.0
20:53	480	8.0	6.0	6.0	0.0
0:53	720	12.0	n/a	n/a	n/a
12:53	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	6.0

Final Concentration (g/l)	833

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(g/_l\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE	Tested By KRM	Calc. By KRM	Reviewed SRH
Date 7/10/2019	Date 7/11/2019	Date 7/11/2019	Date 7/12/2019

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

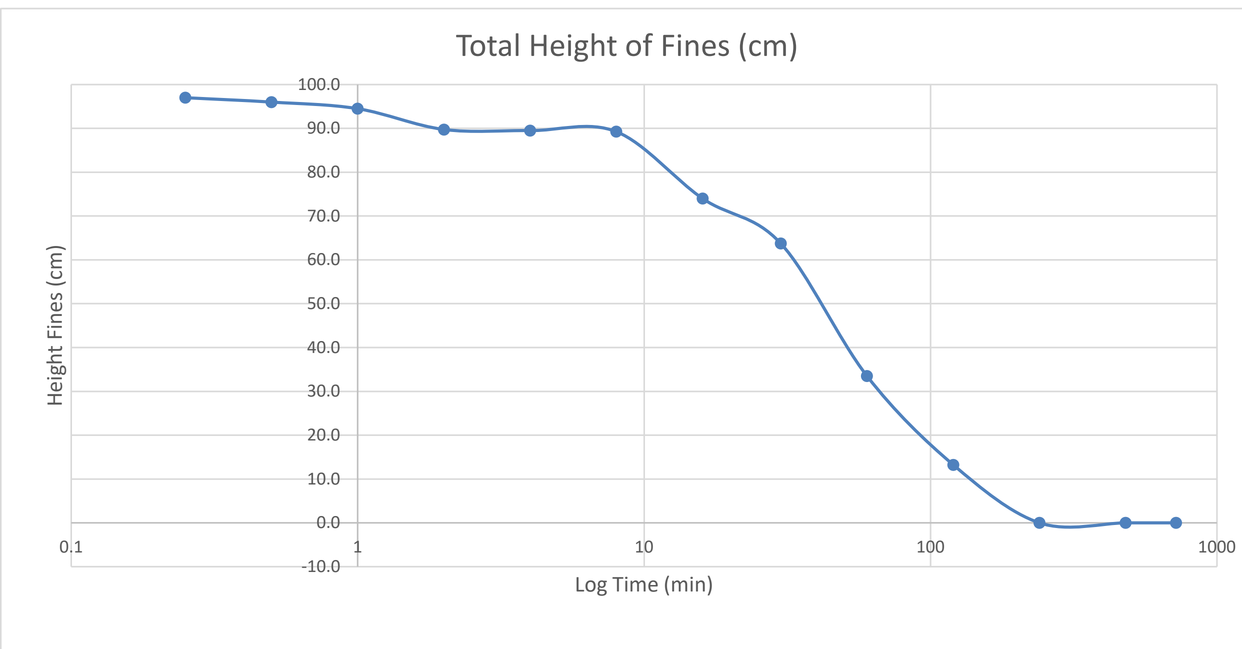
Sample No. : KBIC-19-9
Depth (ft) : 3.0-3.5

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
14:00	0.25		100.0	3.0	97.0
14:00	0.5		100.0	4.0	96.0
14:01	1		100.0	5.5	94.5
14:02	2		95.0	5.3	89.8
14:04	4		95.0	5.5	89.5
14:08	8		95.0	5.8	89.3
14:16	16		80.0	6.0	74.0
14:30	30	0.5	70.0	6.3	63.8
15:00	60	1.0	40.0	6.5	33.5
16:00	120	2.0	20.0	6.8	13.3
18:00	240	4.0	7.0	7.0	0.0
22:00	480	8.0	7.0	7.0	0.0
2:00	720	12.0	n/a	n/a	n/a
14:00	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	7.0

Final Concentration (g/l)	714

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left({}^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA

Project No. : 19C19018.00

Sample No. : KBIC-19-10

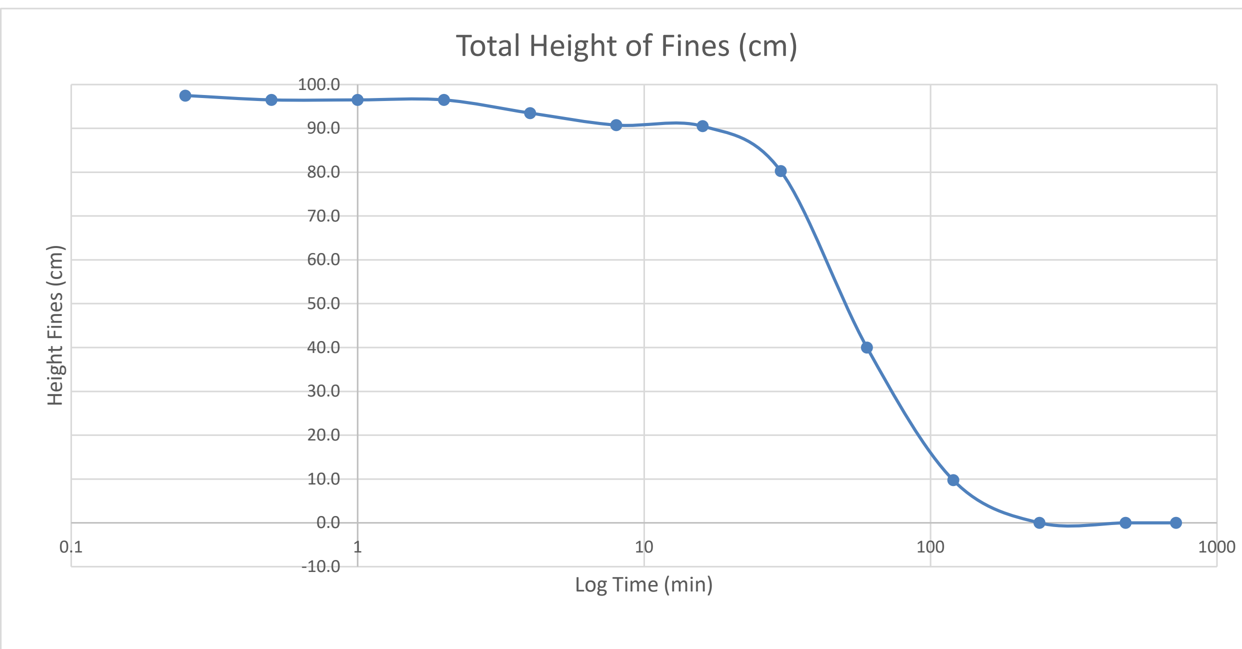
Depth (ft) : 0.5-1.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
12:55	0.25		100.0	2.5	97.5
12:55	0.5		100.0	3.5	96.5
12:56	1		100.0	3.5	96.5
12:57	2		100.0	3.5	96.5
12:59	4		97.5	4.0	93.5
13:03	8		95.0	4.3	90.8
13:11	16		95.0	4.5	90.5
13:25	30	0.5	85.0	4.8	80.3
13:55	60	1.0	45.0	5.0	40.0
14:55	120	2.0	15.0	5.3	9.8
16:55	240	4.0	5.5	5.5	0.0
20:55	480	8.0	5.5	5.5	0.0
0:55	720	12.0	n/a	n/a	n/a
12:55	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	5.5

Final Concentration (g/l)	909

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(g/_l\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE	Tested By KRM	Calc. By KRM	Reviewed SRH
Date 7/10/2019	Date 7/11/2019	Date 7/11/2019	Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

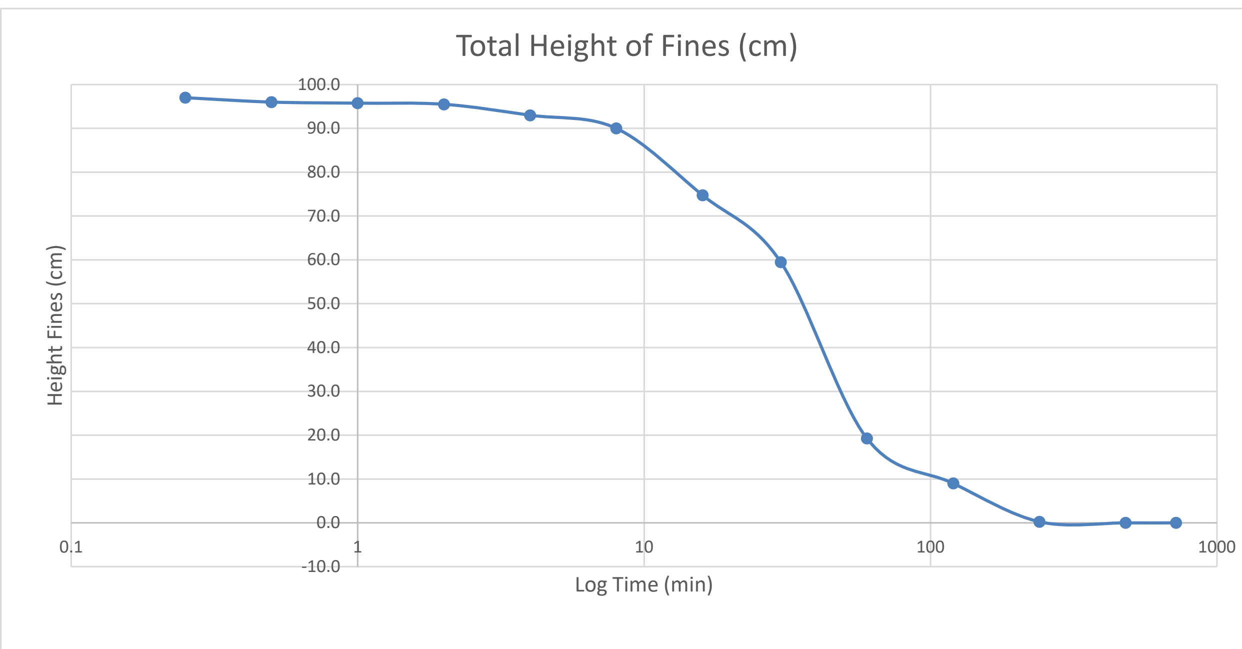
Sample No. : KBIC-19-10
Depth (ft) : 1.5-2.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
14:03	0.25		100.0	3.0	97.0
14:03	0.5		100.0	4.0	96.0
14:04	1		100.0	4.3	95.8
14:05	2		100.0	4.5	95.5
14:07	4		97.5	4.5	93.0
14:11	8		95.0	5.0	90.0
14:19	16		80.0	5.3	74.8
14:33	30	0.5	65.0	5.5	59.5
15:03	60	1.0	25.0	5.8	19.3
16:03	120	2.0	15.0	6.0	9.0
18:03	240	4.0	6.5	6.3	0.3
22:03	480	8.0	6.5	6.5	0.0
2:03	720	12.0	6.5	6.5	0.0
14:03	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	6.5

Final Concentration (g/l)	769

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(g/_l\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

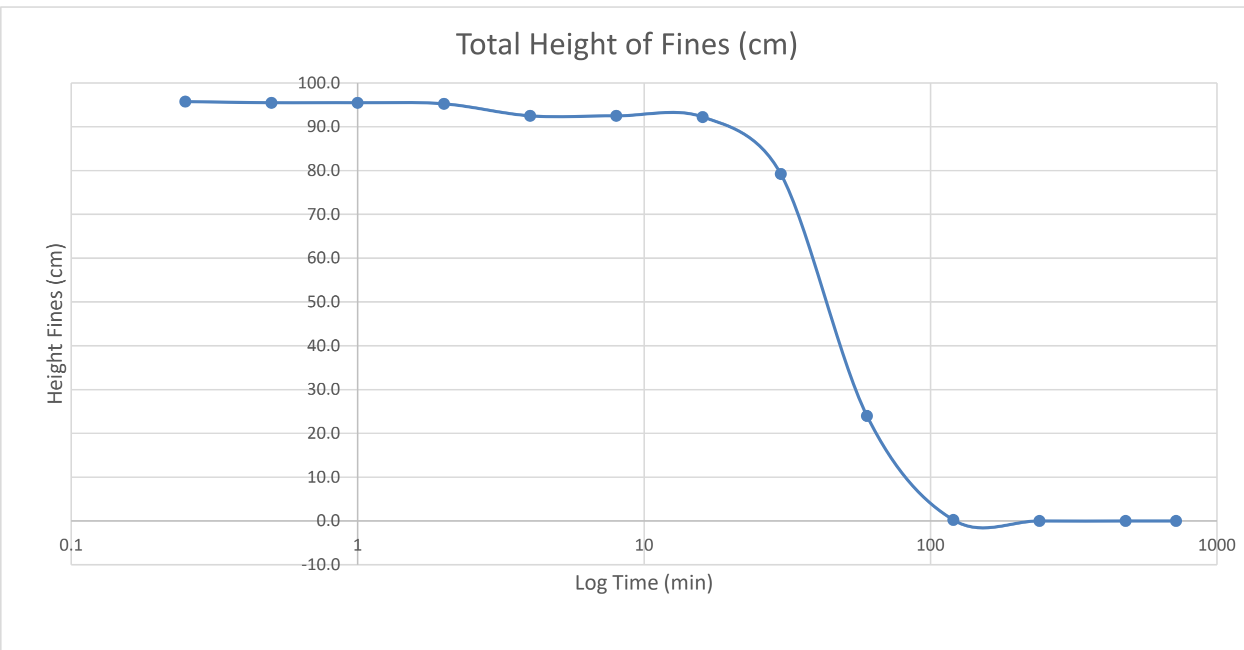
Sample No. : KBIC-19-10
Depth (ft) : 3.0-3.5

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
12:57	0.25		100.0	4.3	95.8
12:57	0.5		100.0	4.5	95.5
12:58	1		100.0	4.5	95.5
12:59	2		100.0	4.8	95.3
13:01	4		97.5	5.0	92.5
13:05	8		97.5	5.0	92.5
13:13	16		97.5	5.3	92.3
13:27	30	0.5	85.0	5.8	79.3
13:57	60	1.0	30.0	6.0	24.0
14:57	120	2.0	6.5	6.3	0.3
16:57	240	4.0	6.5	6.5	0.0
20:57	480	8.0	6.5	6.5	0.0
0:57	720	12.0	n/a	n/a	n/a
12:57	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	6.5

Final Concentration (g/l)	769

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\left(g/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

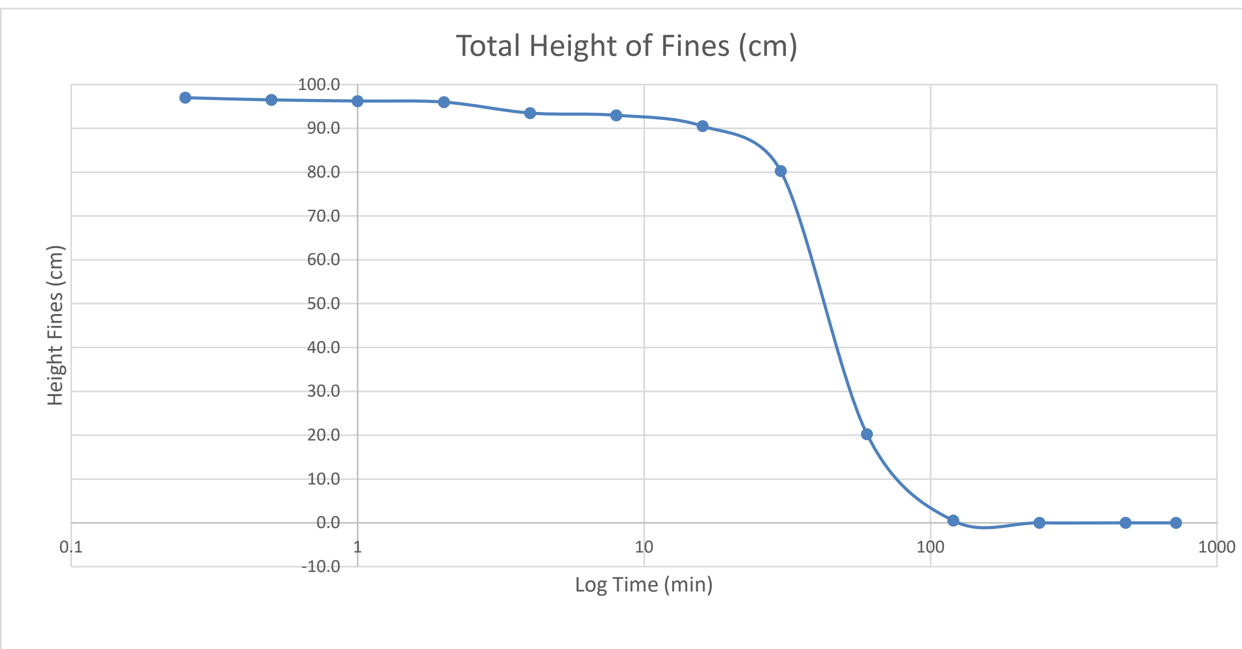
Sample No. : KBIC-19-11
Depth (ft) : 0.5-1.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
12:59	0.25		100.0	3.0	97.0
12:59	0.5		100.0	3.5	96.5
13:00	1		100.0	3.8	96.3
13:01	2		100.0	4.0	96.0
13:03	4		97.5	4.0	93.5
13:07	8		97.5	4.5	93.0
13:15	16		95.0	4.5	90.5
13:29	30	0.5	85.0	4.8	80.3
13:59	60	1.0	25.0	4.8	20.3
14:59	120	2.0	5.5	5.0	0.5
16:59	240	4.0	5.5	5.5	0.0
20:59	480	8.0	5.5	5.5	0.0
0:59	720	12.0	n/a	n/a	n/a
12:59	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	5.5

Final Concentration (g/l)	909

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\left({}^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

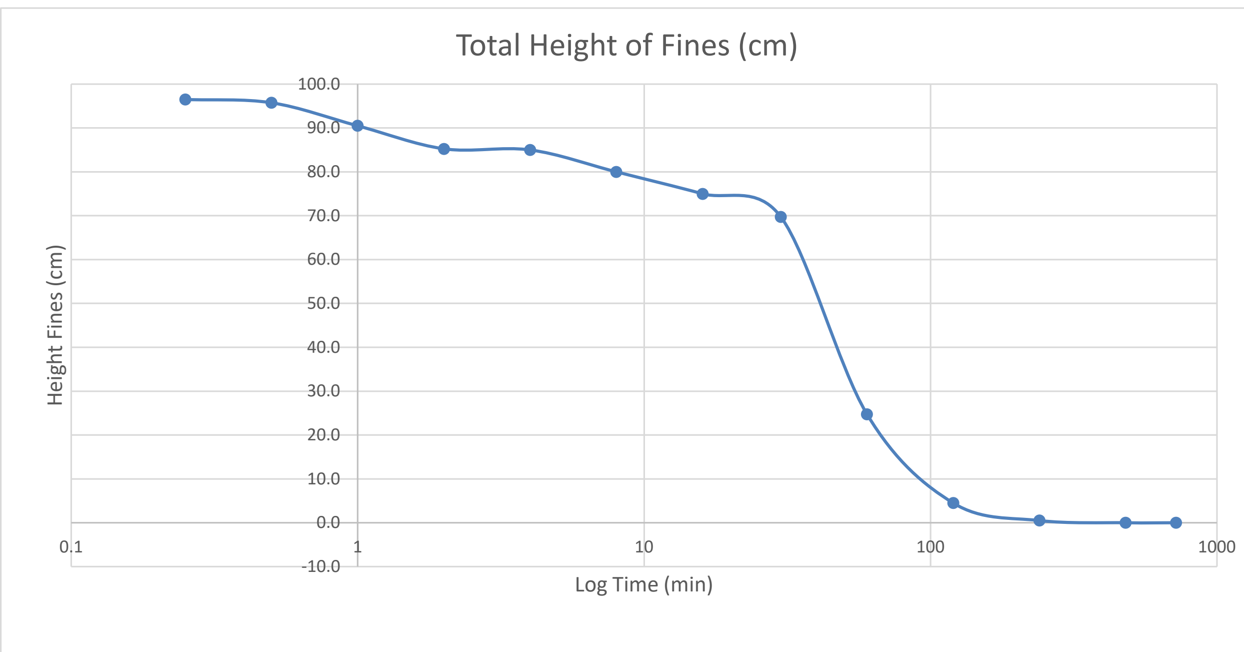
Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

Sample No. : KBIC-19-11
Depth (ft) : 1.5-2.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
7:58	0.25		100.0	3.5	96.5
7:58	0.5		100.0	4.3	95.8
7:59	1		95.0	4.5	90.5
8:00	2		90.0	4.8	85.3
8:02	4		90.0	5.0	85.0
8:06	8		85.0	5.0	80.0
8:14	16		80.0	5.0	75.0
8:28	30	0.5	75.0	5.3	69.8
8:58	60	1.0	30.0	5.3	24.8
9:58	120	2.0	10.0	5.5	4.5
11:58	240	4.0	6.0	5.5	0.5
15:58	480	8.0	6.0	6.0	0.0
19:58	720	12.0	6.0	6.0	0.0
7:58	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	6.0
Final Concentration (g/l)	833

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left({}^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

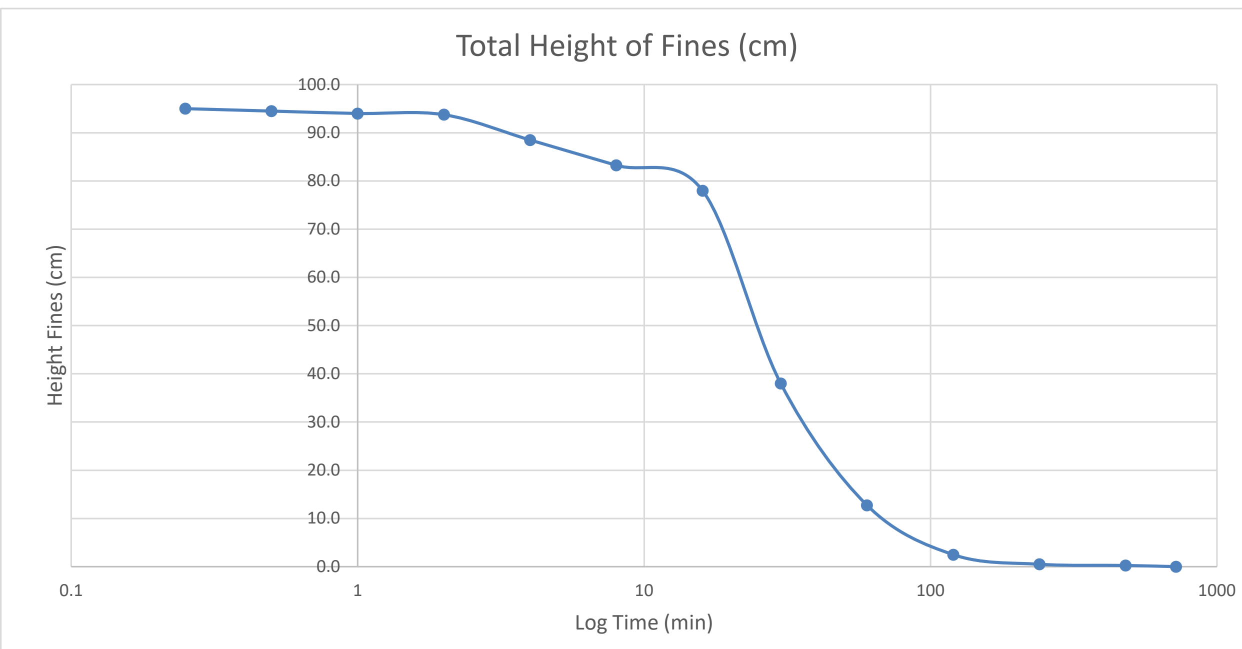
Sample No. : KBIC-19-11
Depth (ft) : 3.0-3.5

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
14:07	0.25		100.0	5.0	95.0
14:07	0.5		100.0	5.5	94.5
14:08	1		100.0	6.0	94.0
14:09	2		100.0	6.3	93.8
14:11	4		95.0	6.5	88.5
14:15	8		90.0	6.8	83.3
14:23	16		85.0	7.0	78.0
14:37	30	0.5	45.0	7.0	38.0
15:07	60	1.0	20.0	7.3	12.8
16:07	120	2.0	10.0	7.5	2.5
18:07	240	4.0	8.5	8.0	0.5
22:07	480	8.0	8.5	8.3	0.3
2:07	720	12.0	8.5	8.5	0.0
14:07	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	8.5

Final Concentration (g/l)	588

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(\frac{g}{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

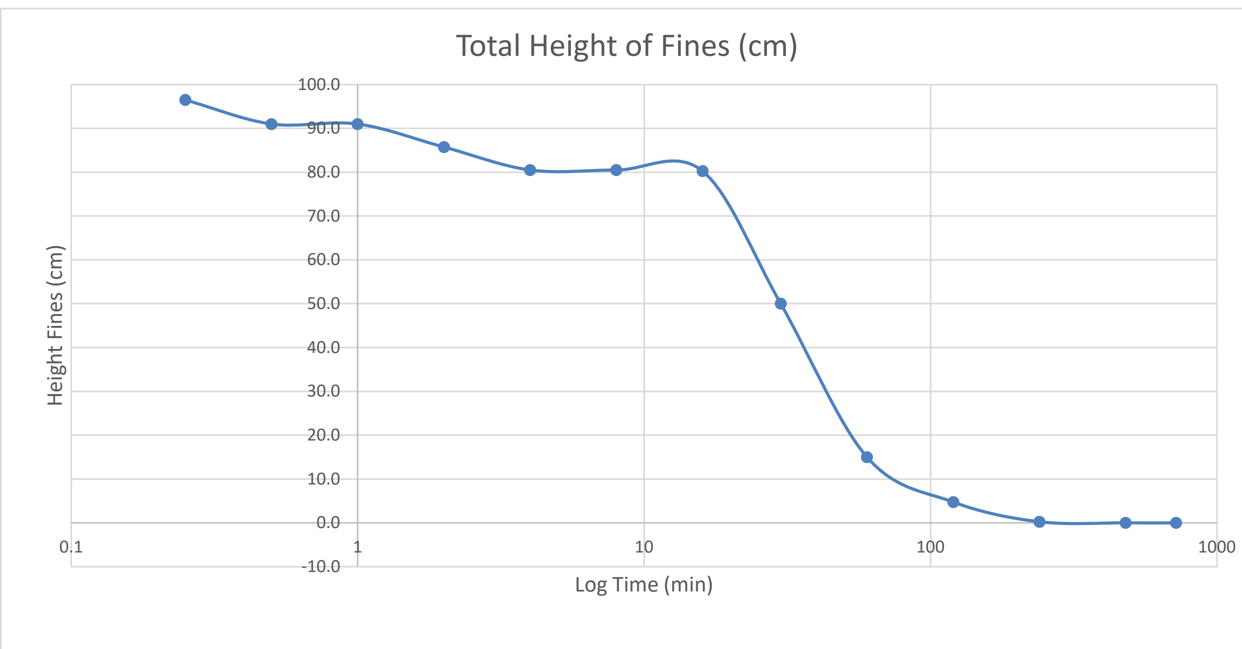
Sample No. : KBIC-19-12
Depth (ft) : 0.5-1.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
8:01	0.25		100.0	3.5	96.5
8:01	0.5		95.0	4.0	91.0
8:02	1		95.0	4.0	91.0
8:03	2		90.0	4.3	85.8
8:05	4		85.0	4.5	80.5
8:09	8		85.0	4.5	80.5
8:17	16		85.0	4.8	80.3
8:31	30	0.5	55.0	5.0	50.0
9:01	60	1.0	20.0	5.0	15.0
10:01	120	2.0	10.0	5.3	4.8
12:01	240	4.0	5.5	5.3	0.3
16:01	480	8.0	5.5	5.5	0.0
20:01	720	12.0	5.5	5.5	0.0
8:01	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	5.5

Final Concentration (g/l)	909

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(g/_l\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

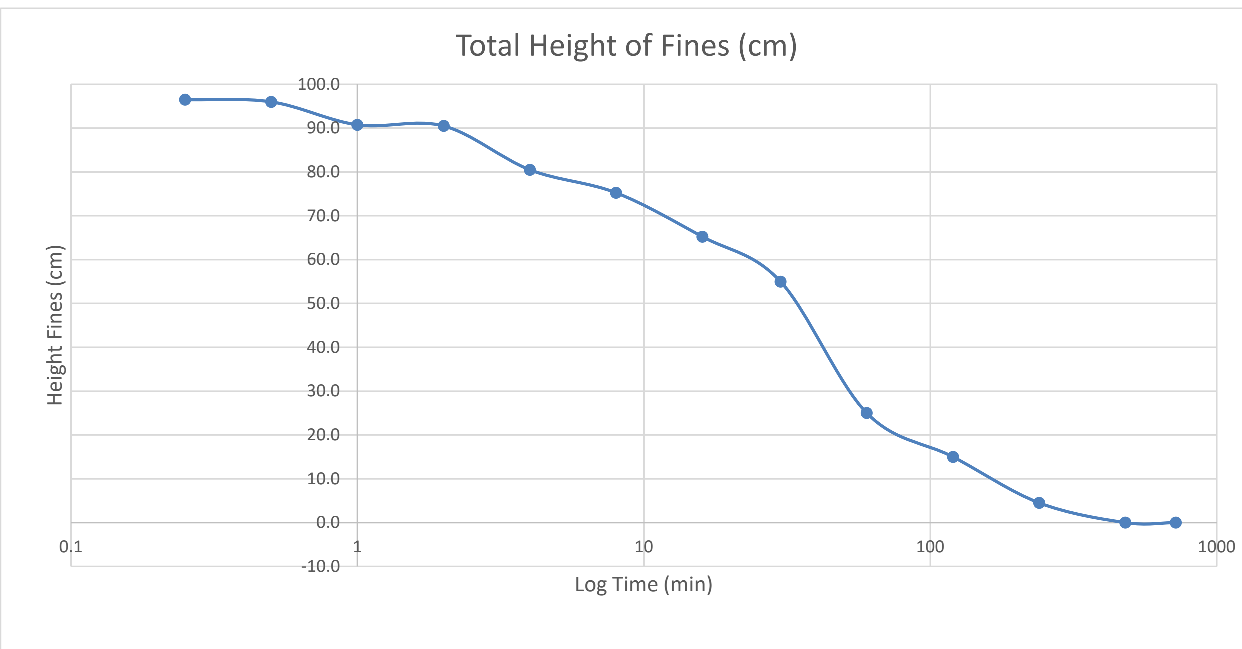
Sample No. : KBIC-19-12
Depth (ft) : 1.5-2.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
8:03	0.25		100.0	3.5	96.5
8:03	0.5		100.0	4.0	96.0
8:04	1		95.0	4.3	90.8
8:05	2		95.0	4.5	90.5
8:07	4		85.0	4.5	80.5
8:11	8		80.0	4.8	75.3
8:19	16		70.0	4.8	65.3
8:33	30	0.5	60.0	5.0	55.0
9:03	60	1.0	30.0	5.0	25.0
10:03	120	2.0	20.0	5.0	15.0
12:03	240	4.0	10.0	5.5	4.5
16:03	480	8.0	5.5	5.5	0.0
20:03	720	12.0	5.5	5.5	0.0
8:03	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	5.5

Final Concentration (g/l)	909

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left({}^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

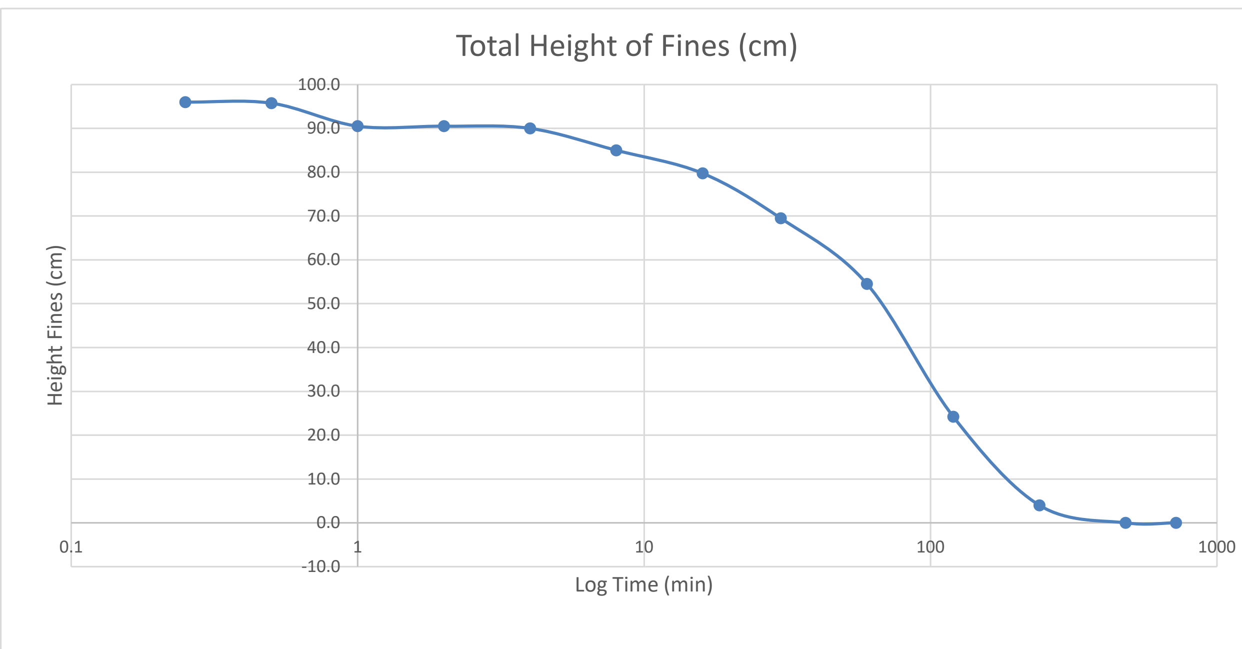
Sample No. : KBIC-19-12
Depth (ft) : 2.5-3.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
8:05	0.25		100.0	4.0	96.0
8:05	0.5		100.0	4.3	95.8
8:06	1		95.0	4.5	90.5
8:07	2		95.0	4.5	90.5
8:09	4		95.0	5.0	90.0
8:13	8		90.0	5.0	85.0
8:21	16		85.0	5.3	79.8
8:35	30	0.5	75.0	5.5	69.5
9:05	60	1.0	60.0	5.5	54.5
10:05	120	2.0	30.0	5.8	24.3
12:05	240	4.0	10.0	6.0	4.0
16:05	480	8.0	6.5	6.5	0.0
20:05	720	12.0	6.5	6.5	0.0
8:05	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	6.5

Final Concentration (g/l)	769

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(g/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

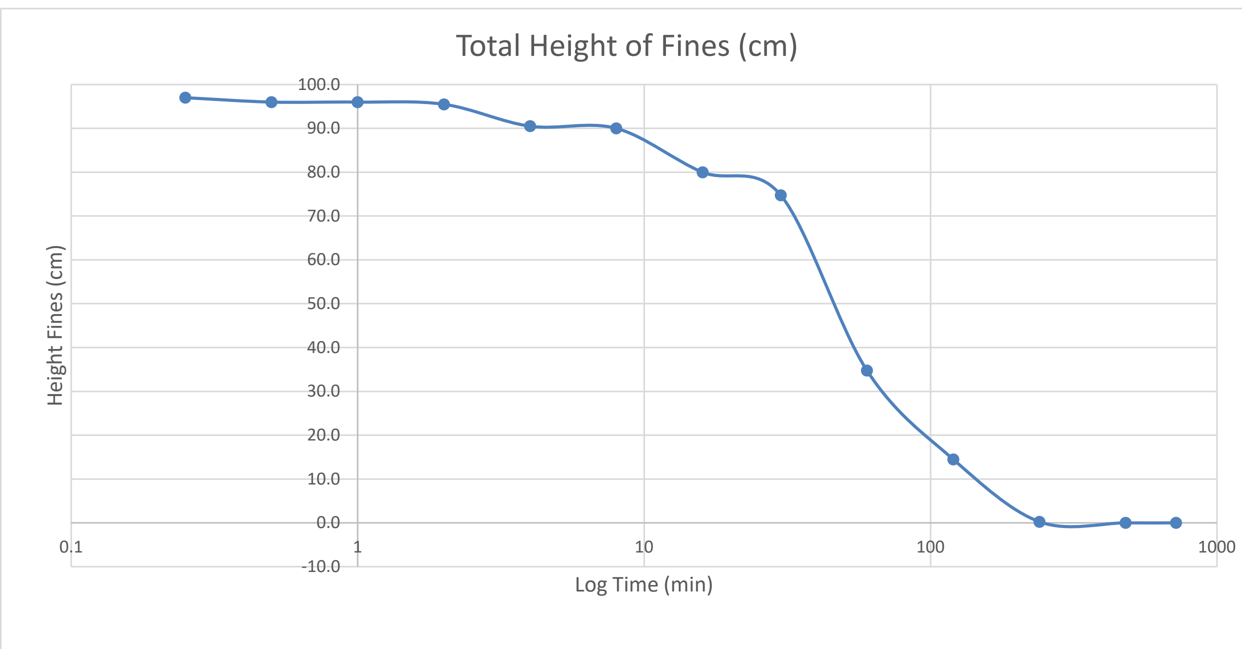
Sample No. : KBIC-19-13
Depth (ft) : 0.5-1.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
8:09	0.25		100.0	3.0	97.0
8:09	0.5		100.0	4.0	96.0
8:10	1		100.0	4.0	96.0
8:11	2		100.0	4.5	95.5
8:13	4		95.0	4.5	90.5
8:17	8		95.0	5.0	90.0
8:25	16		85.0	5.0	80.0
8:39	30	0.5	80.0	5.3	74.8
9:09	60	1.0	40.0	5.3	34.8
10:09	120	2.0	20.0	5.5	14.5
12:09	240	4.0	6.0	5.8	0.3
16:09	480	8.0	6.0	6.0	0.0
20:09	720	12.0	6.0	6.0	0.0
8:09	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	6.0

Final Concentration (g/l)	833

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left({}^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

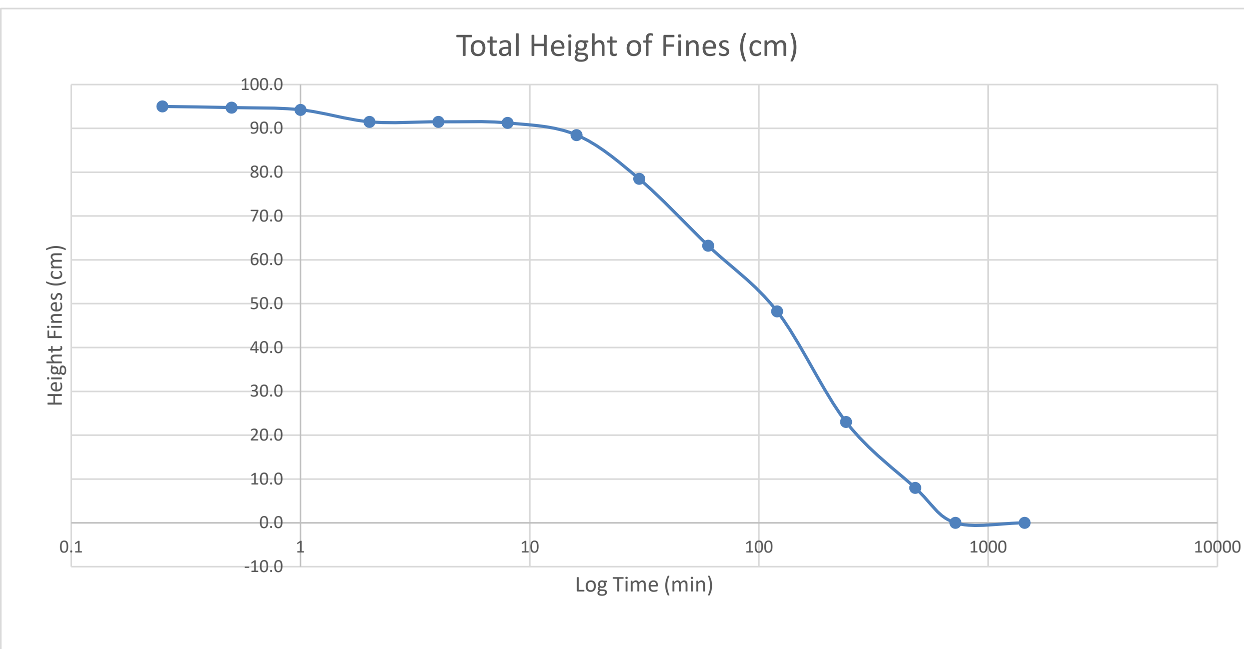
Sample No. : KBIC-19-13
Depth (ft) : 1.5-2.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
8:12	0.25		100.0	5.0	95.0
8:12	0.5		100.0	5.3	94.8
8:13	1		100.0	5.8	94.3
8:14	2		97.5	6.0	91.5
8:16	4		97.5	6.0	91.5
8:20	8		97.5	6.3	91.3
8:28	16		95.0	6.5	88.5
8:42	30	0.5	85.0	6.5	78.5
9:12	60	1.0	70.0	6.8	63.3
10:12	120	2.0	55.0	6.8	48.3
12:12	240	4.0	30.0	7.0	23.0
16:12	480	8.0	15.0	7.0	8.0
20:12	720	12.0	8.0	8.0	0.0
8:12	1440	24.0	8.0	8.0	n/a

Final Sediment Height (cm)	15.0

Final Concentration (g/l)	333

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(g/l\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

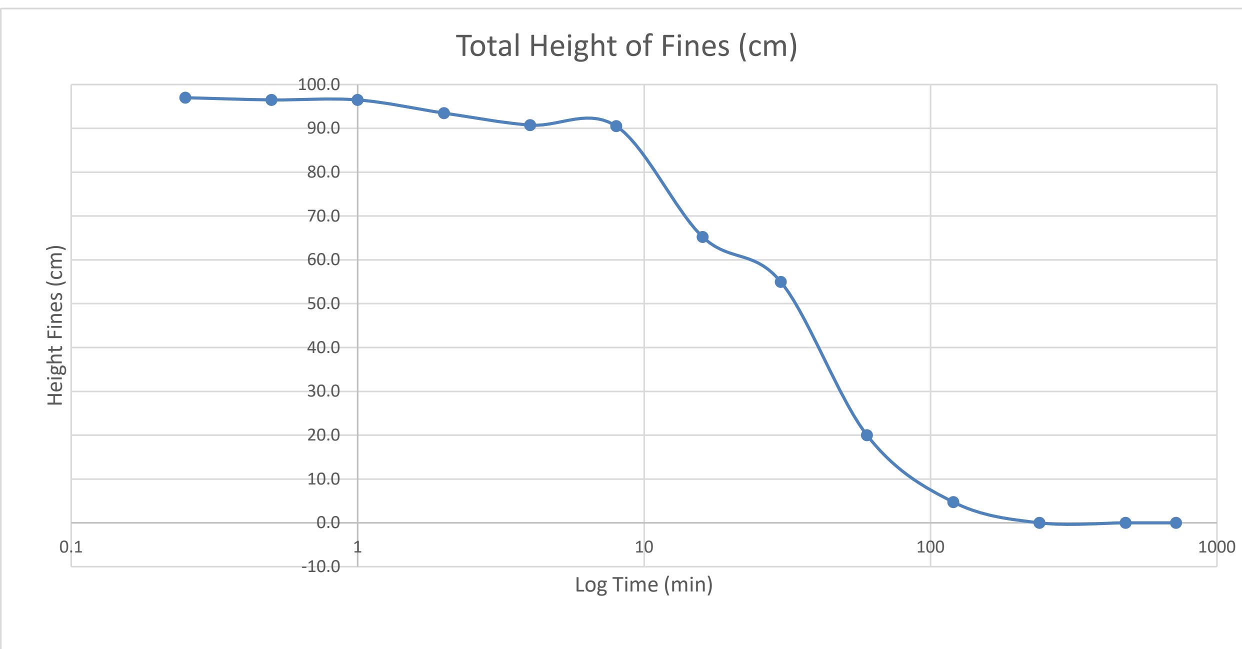
Sample No. : KBIC-19-13
Depth (ft) : 3.0-3.5

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
8:15	0.25		100.0	3.0	97.0
8:15	0.5		100.0	3.5	96.5
8:16	1		100.0	3.5	96.5
8:17	2		97.5	4.0	93.5
8:19	4		95.0	4.3	90.8
8:23	8		95.0	4.5	90.5
8:31	16		70.0	4.8	65.3
8:45	30	0.5	60.0	5.0	55.0
9:15	60	1.0	25.0	5.0	20.0
10:15	120	2.0	10.0	5.3	4.8
12:15	240	4.0	5.5	5.5	0.0
16:15	480	8.0	5.5	5.5	0.0
20:15	720	12.0	n/a	n/a	n/a
8:15	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	5.5

Final Concentration (g/l)	909

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(g/_l\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

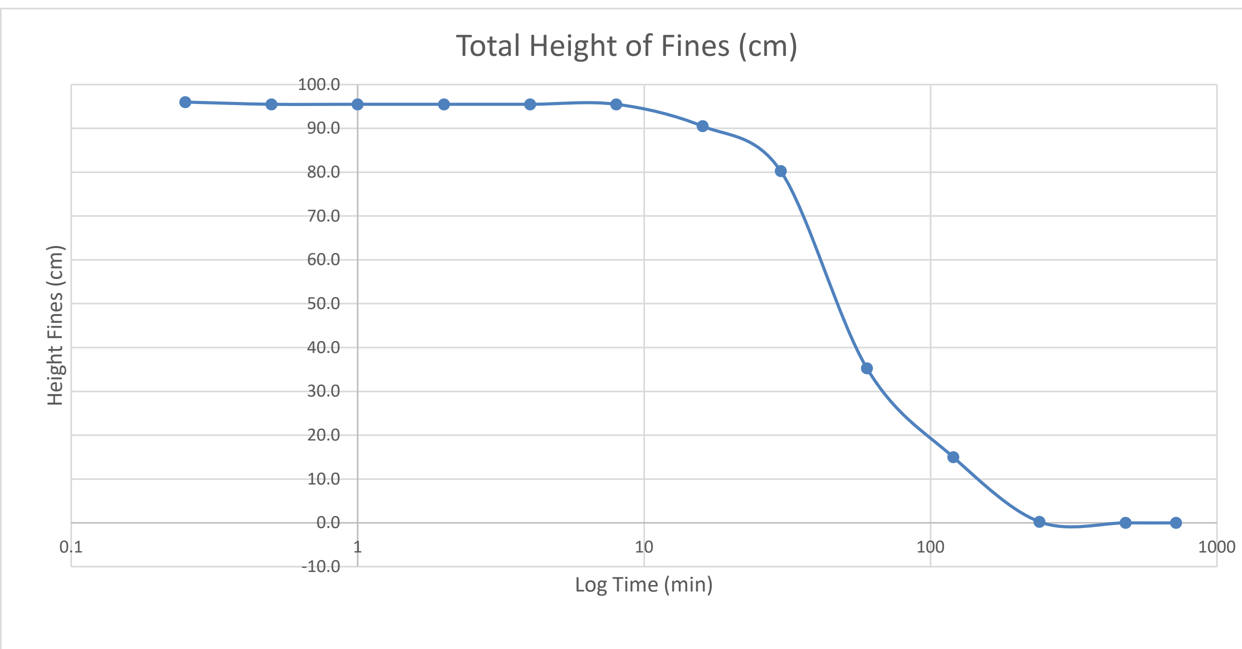
Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

Sample No. : KBIC-19-14
Depth (ft) : 0.5-1.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
13:01	0.25		100.0	4.0	96.0
13:01	0.5		100.0	4.5	95.5
13:02	1		100.0	4.5	95.5
13:03	2		100.0	4.5	95.5
13:05	4		100.0	4.5	95.5
13:09	8		100.0	4.5	95.5
13:17	16		95.0	4.5	90.5
13:31	30	0.5	85.0	4.8	80.3
14:01	60	1.0	40.0	4.8	35.3
15:01	120	2.0	20.0	5.0	15.0
17:01	240	4.0	5.5	5.3	0.3
21:01	480	8.0	5.5	5.5	0.0
1:01	720	12.0	n/a	n/a	n/a
13:01	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	5.5
Final Concentration (g/l)	909

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left({}^{g}/{}_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

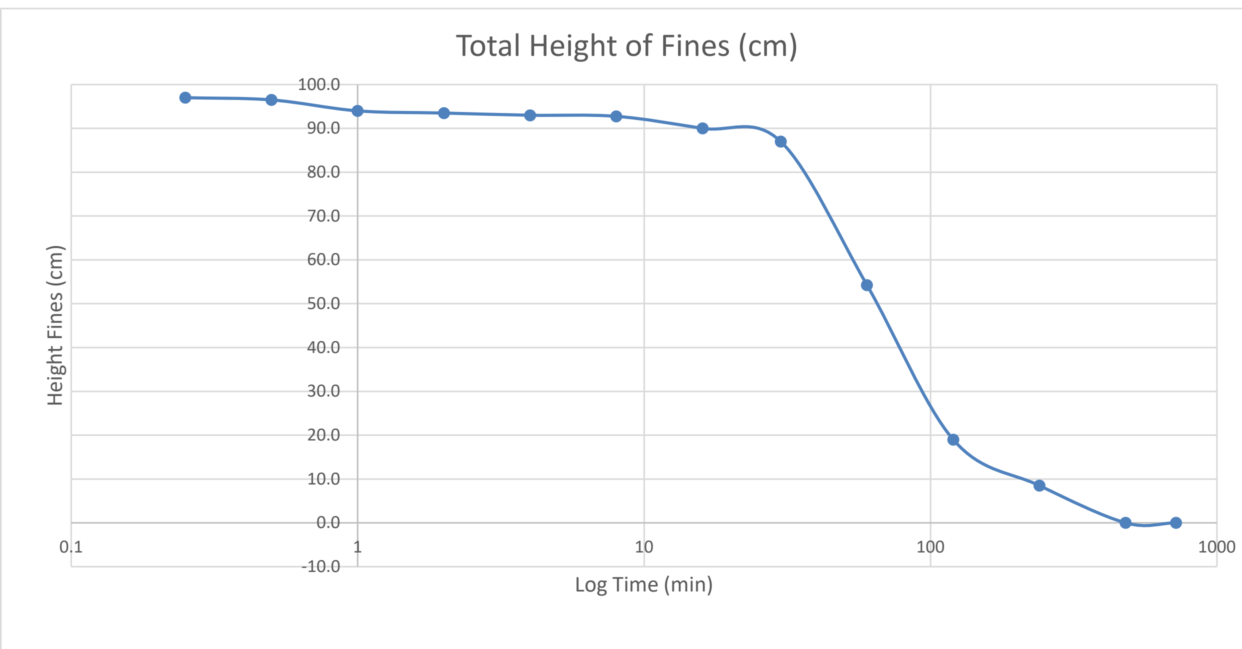
Sample No. : KBIC-19-14
Depth (ft) : 1.5-2.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
8:19	0.25		100.0	3.0	97.0
8:19	0.5		100.0	3.5	96.5
8:20	1		97.5	3.5	94.0
8:21	2		97.5	4.0	93.5
8:23	4		97.5	4.5	93.0
8:27	8		97.5	4.8	92.8
8:35	16		95.0	5.0	90.0
8:49	30	0.5	92.5	5.5	87.0
9:19	60	1.0	60.0	5.8	54.3
10:19	120	2.0	25.0	6.0	19.0
12:19	240	4.0	15.0	6.5	8.5
16:19	480	8.0	7.0	7.0	0.0
20:19	720	12.0	7.0	7.0	0.0
8:19	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	7.0

Final Concentration (g/l)	714

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left({}^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

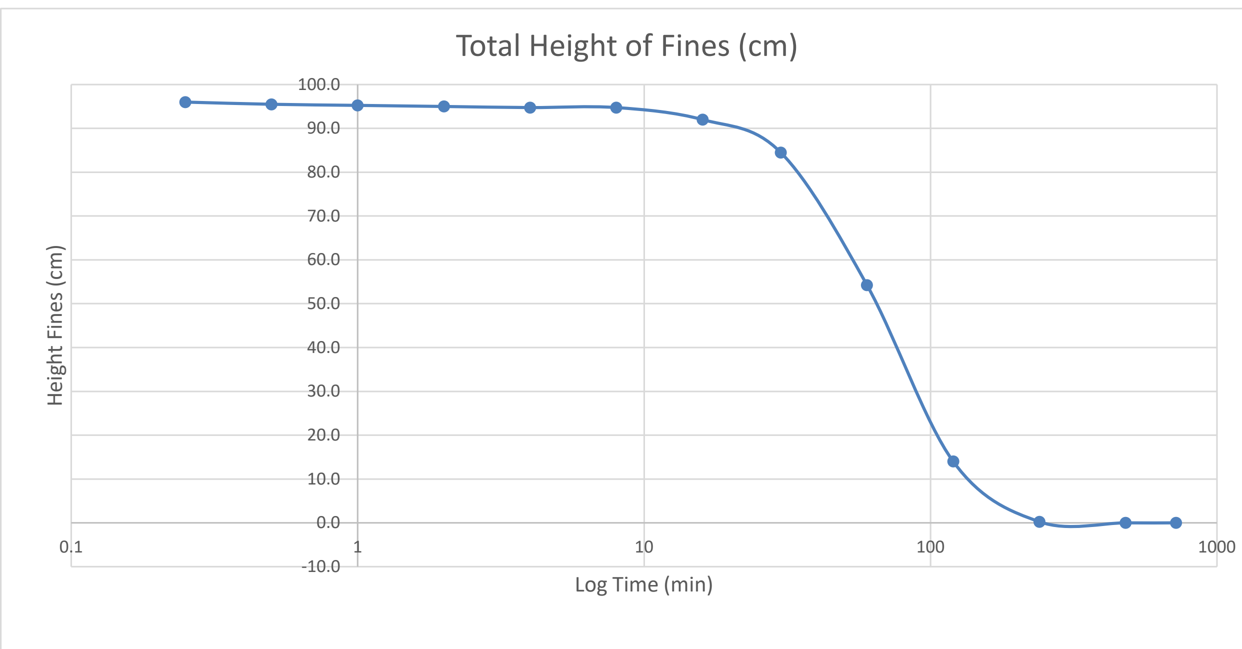
Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

Sample No. : KBIC-19-14
Depth (ft) : 3.0-3.5

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
13:03	0.25		100.0	4.0	96.0
13:03	0.5		100.0	4.5	95.5
13:04	1		100.0	4.8	95.3
13:05	2		100.0	5.0	95.0
13:07	4		100.0	5.3	94.8
13:11	8		100.0	5.3	94.8
13:19	16		97.5	5.5	92.0
13:33	30	0.5	90.0	5.5	84.5
14:03	60	1.0	60.0	5.8	54.3
15:03	120	2.0	20.0	6.0	14.0
17:03	240	4.0	6.5	6.3	0.3
21:03	480	8.0	6.5	6.5	0.0
1:03	720	12.0	n/a	n/a	n/a
13:03	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	6.5
Final Concentration (g/l)	769

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

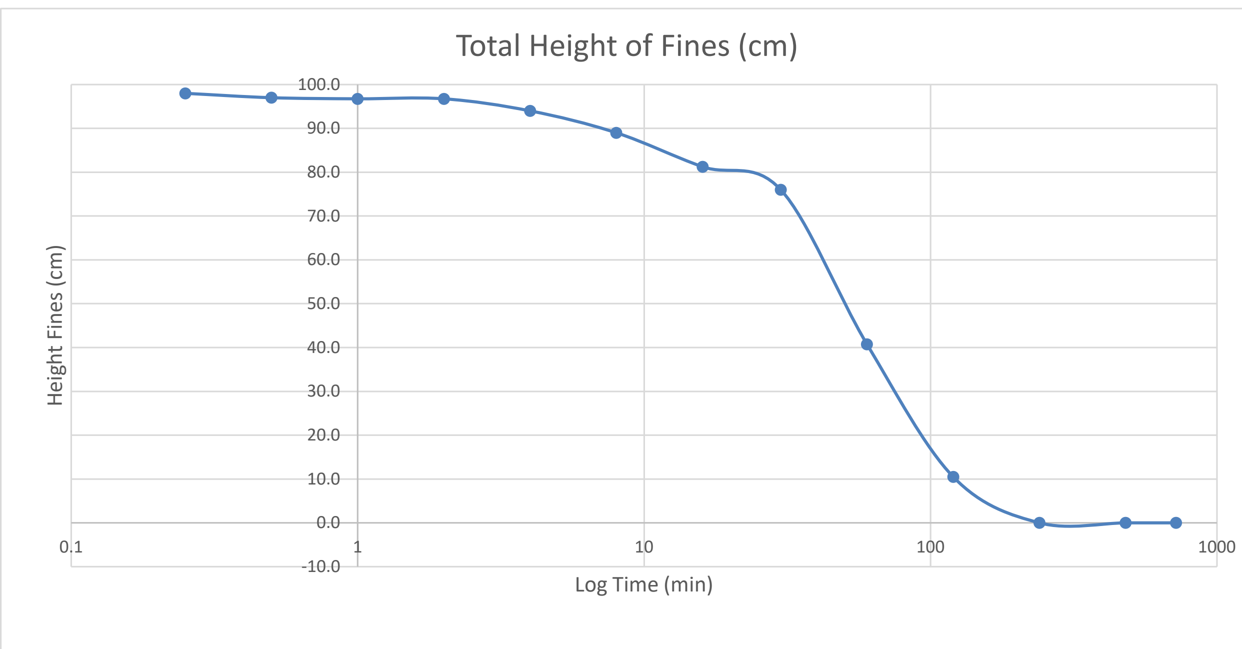
Sample No. : KBIC-19-15
Depth (ft) : 0.5-1.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
8:24	0.25		100.0	2.0	98.0
8:24	0.5		100.0	3.0	97.0
8:25	1		100.0	3.3	96.8
8:26	2		100.0	3.3	96.8
8:28	4		97.5	3.5	94.0
8:32	8		92.5	3.5	89.0
8:40	16		85.0	3.8	81.3
8:54	30	0.5	80.0	4.0	76.0
9:24	60	1.0	45.0	4.3	40.8
10:24	120	2.0	15.0	4.5	10.5
12:24	240	4.0	5.0	5.0	0.0
16:24	480	8.0	5.0	5.0	0.0
20:24	720	12.0	n/a	n/a	n/a
8:24	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	5.0

Final Concentration (g/l)	1000

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(g/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE	Tested By KRM	Calc. By KRM	Reviewed SRH
Date 7/10/2019	Date 7/11/2019	Date 7/11/2019	Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

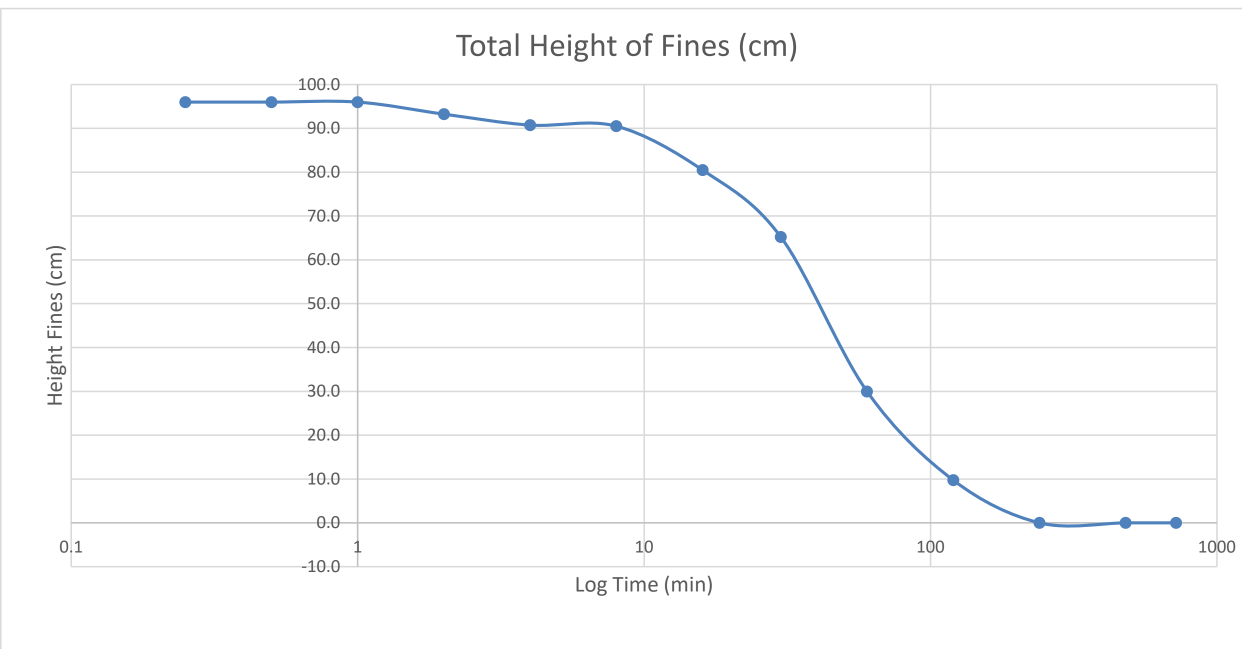
Sample No. : KBIC-19-15
Depth (ft) : 1.5-2.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
8:27	0.25		100.0	4.0	96.0
8:27	0.5		100.0	4.0	96.0
8:28	1		100.0	4.0	96.0
8:29	2		97.5	4.3	93.3
8:31	4		95.0	4.3	90.8
8:35	8		95.0	4.5	90.5
8:43	16		85.0	4.5	80.5
8:57	30	0.5	70.0	4.8	65.3
9:27	60	1.0	35.0	5.0	30.0
10:27	120	2.0	15.0	5.3	9.8
12:27	240	4.0	5.5	5.5	0.0
16:27	480	8.0	5.5	5.5	0.0
20:27	720	12.0	n/a	n/a	n/a
8:27	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	5.5

Final Concentration (g/l)	909

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

NAVAL SUBMARINE BASE KINGS BAY
MAINTENANCE DREDGING
CAMDEN COUNTY, GEORGIA

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

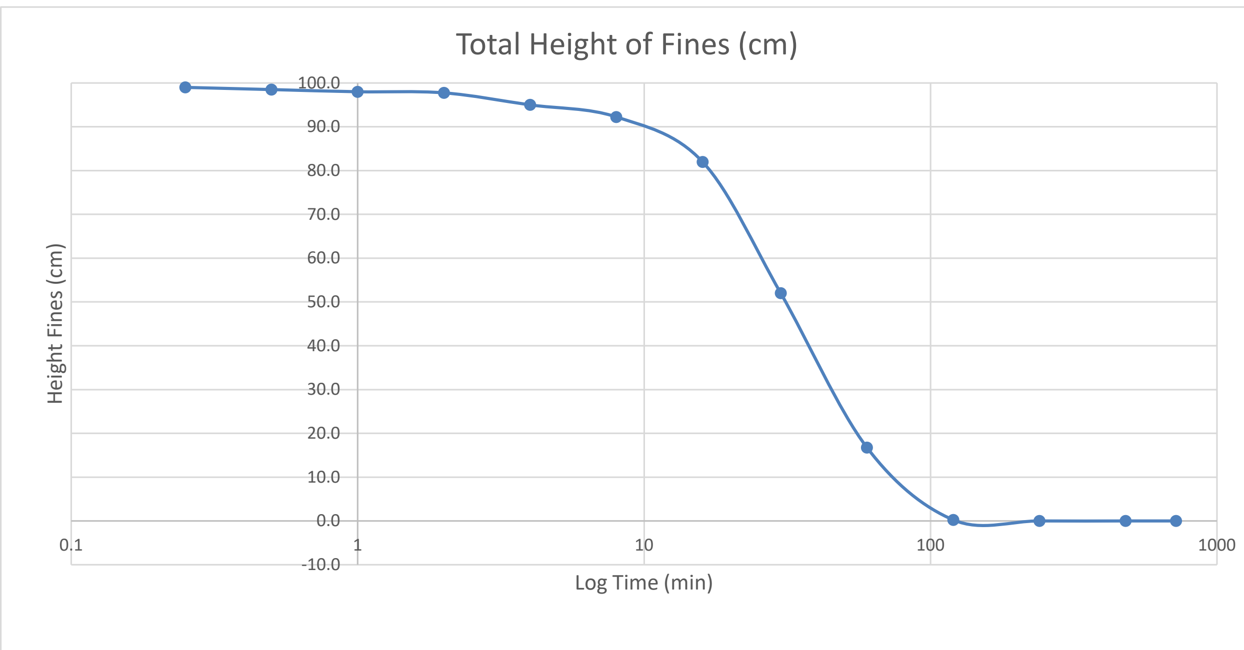
Sample No. : KBIC-19-16
Depth (ft) : 0.3-0.8

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
13:06	0.25		100.0	1.0	99.0
13:06	0.5		100.0	1.5	98.5
13:07	1		100.0	2.0	98.0
13:08	2		100.0	2.3	97.8
13:10	4		97.5	2.5	95.0
13:14	8		95.0	2.8	92.3
13:22	16		85.0	3.0	82.0
13:36	30	0.5	55.0	3.0	52.0
14:06	60	1.0	20.0	3.3	16.8
15:06	120	2.0	3.5	3.3	0.3
17:06	240	4.0	3.5	3.5	0.0
21:06	480	8.0	3.5	3.5	0.0
1:06	720	12.0	n/a	n/a	n/a
13:06	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	3.5

Final Concentration (g/l)	1429

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left({}^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00

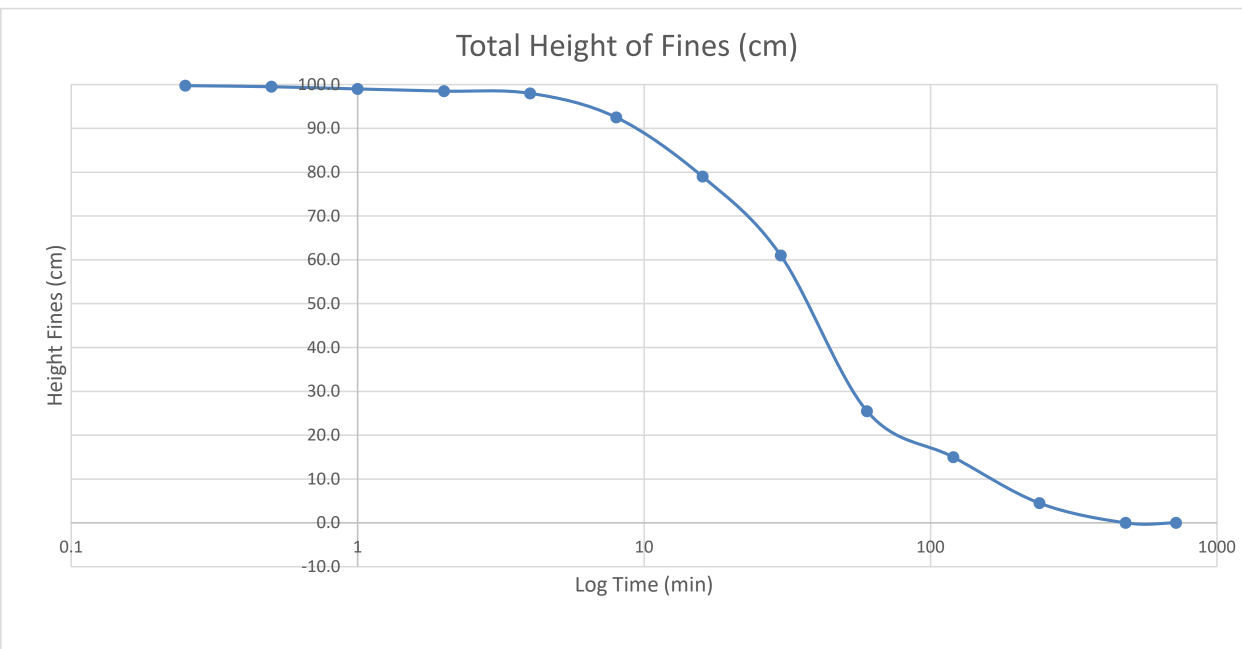
Sample No. : KBIC-19-16
Depth (ft) : 1.2-1.7

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
8:31	0.25		100.0	0.25	99.8
8:31	0.5		100.0	0.5	99.5
8:32	1		100.0	1.0	99.0
8:33	2		100.0	1.5	98.5
8:35	4		100.0	2.0	98.0
8:39	8		95.0	2.5	92.5
8:47	16		82.5	3.5	79.0
9:01	30	0.5	65.0	4.0	61.0
9:31	60	1.0	30.0	4.5	25.5
10:31	120	2.0	20.0	5.0	15.0
12:31	240	4.0	10.0	5.5	4.5
16:31	480	8.0	6.0	6.0	0.0
20:31	720	12.0	6.0	6.0	0.0
8:31	1440	24.0	n/a	n/a	n/a

Final Sediment Height (cm)	6.0

Final Concentration (g/l)	833

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left({}^{g}/_{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE
Date 7/10/2019

Tested By KRM
Date 7/11/2019

Calc. By KRM
Date 7/11/2019

Reviewed SRH
Date 7/12/2019

Schnabel ENGINEERING

Soil Mechanics Laboratory
Sedimentation Rate Worksheet

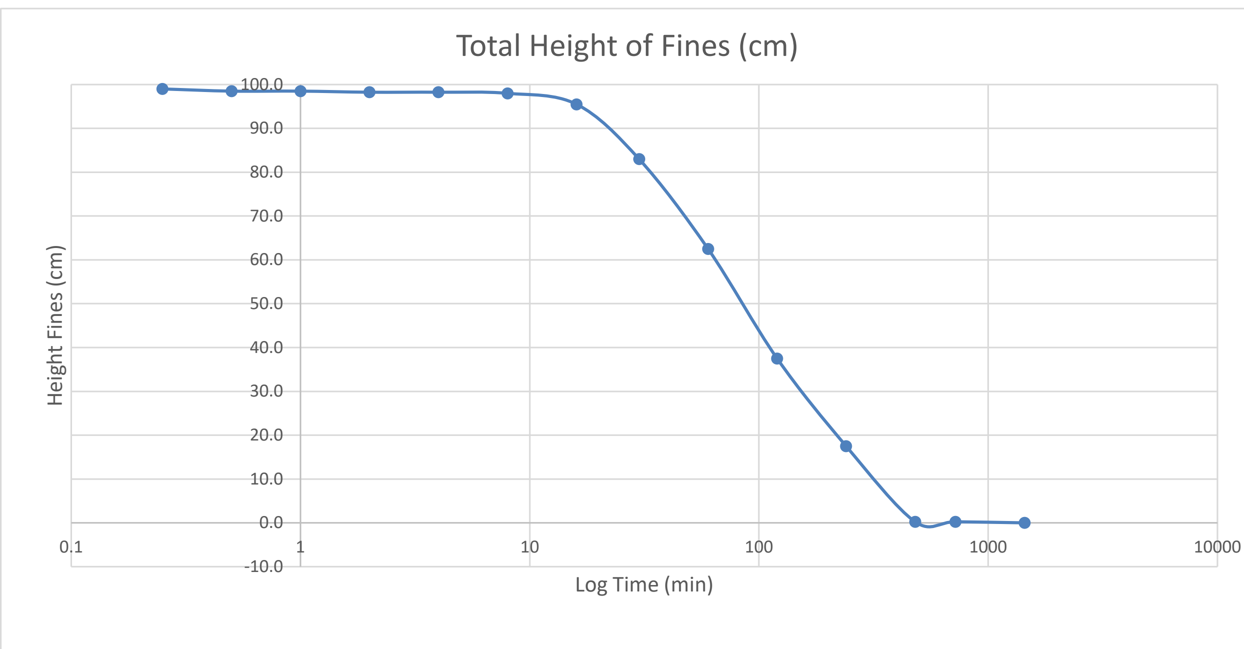
Project Name : P-617 Transit PP, Kings Bay GA
Project No. : 19C19018.00
Sample No. : KBIC-19-16
Depth (ft) : 2.5-3.0

Start Time	Elapsed Time Reading (min)	Elapsed Time Readings (Hours)	Top Reading Suspended Sediment (cm)	Height Settled Sediment (cm)	Total Height of Suspended Sediment (cm)
13:08	0.25		100.0	1.0	99.0
13:08	0.5		100.0	1.5	98.5
13:09	1		100.0	1.5	98.5
13:10	2		100.0	1.8	98.3
13:12	4		100.0	1.8	98.3
13:16	8		100.0	2.0	98.0
13:24	16		97.5	2.0	95.5
13:38	30	0.5	85.0	2.0	83.0
14:08	60	1.0	65.0	2.5	62.5
15:08	120	2.0	40.0	2.5	37.5
17:08	240	4.0	20.0	2.5	17.5
21:08	480	8.0	3.0	2.8	0.3
1:08	720	12.0	3.0	2.8	0.3
13:08	1440	24.0	3.0	3.0	0.0

Final Sediment Height (cm)	3.0

Final Concentration (g/l)	1667

$$Final\ Concentration = \frac{Initial\ Dispersion\ Concentration\ \left(\frac{g}{l}\right) * 100\ cm}{Final\ Height\ (cm)}$$



Set Up By MDE	Tested By KRM	Calc. By KRM	Reviewed SRH
Date 7/10/2019	Date 7/11/2019	Date 7/11/2019	Date 7/12/2019

Summary Of Laboratory Tests

Appendix
Sheet 1 of 7
Project Number: 19C19018.00

Boring No.	Sample Depth ft / Elevation ft	Sample Type	Description of Soil Specimen	Natural Moisture (%)	Liquid Limit	Plastic Limit	Plasticity Index	% Retained No. 4 Sieve	% Passing No. 200 Sieve	Percent Sand	Percent Silt	Percent Clay	Specific Gravity	Organic Content (%)
VB-KBIC19-1B	0.0 - 0.5	Tube	SAND, silty, 2.5Y 3/1 very dark gray (SM) [Estimated shell content < 1.0%]	43.6	NP	NP	NP	0.0	12.6	87.4	12.2	0.4	2.57	1.6
VB-KBIC19-1B	1.5 - 2.0	Tube	SAND, silty, 2.5Y 3/1 very dark gray (SM) [Estimated shell content < 2.0%]	71.3	NP	NP	NP	1.6	20.9	77.5	19.0	1.9	2.85	0.7
VB-KBIC19-1B	2.9 - 3.4	Tube	SAND, silty, 10Y 3/1 very dark greenish gray (SM) [Estimated shell content < 1.0%]	55.6	NP	NP	NP	0.2	19.6	80.2	18.3	1.3	2.86	1.3
VB-KBIC19-2B	0.2 - 0.7	Tube	SAND, poorly-graded with silt, 2.5Y 3/1 very dark gray (SP-SM) [Estimated shell content < 1.0%]	29.8	NP	NP	NP	0.0	8.4	91.6	8.1	0.3	2.65	0.7
VB-KBIC19-2B	2.0 - 2.5	Tube	SAND, silty, 2.5Y 3/1 very dark gray (SM) [Estimated shell content < 6.0%]	28.4	NP	NP	NP	5.6	12.6	81.8	12.3	0.3	2.64	1.6
VB-KBIC19-2B	2.8 - 3.3	Tube	SAND, silty, 2.5Y 3/1 very dark gray (SM) [Estimated shell content < 1.0%]	62.4	36	25	11	0.6	26.2	73.2	22.9	3.3	2.83	1.7
VB-KBIC19-3	0.2 - 0.7	Tube	SAND, silty, 10Y 2.5/1 greenish black (SM) [Estimated shell content < 1.0%]	199.0	75	41	34	0.0	65.3	34.7	41.1	24.2	2.49	7

Notes:
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2. Soil classifications are in general accordance with ASTM D2487(as applicable), based on testing indicated and visual classification.
3. Key to abbreviations: NP=Non-Plastic; -- indicates no test performed.
4. ASTM Methods: Natural Moisture D2216, Atterberg D4318, Gradation D6913, Hydrometer D7928, Specific Gravity D854, Organic Content D2974

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Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

DYNAMIC LAB SUMMARY 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

Summary Of Laboratory Tests

Appendix
Sheet 2 of 7
Project Number: 19C19018.00

Boring No.	Sample Depth ft / Elevation ft	Sample Type	Description of Soil Specimen	Natural Moisture (%)	Liquid Limit	Plastic Limit	Plasticity Index	% Retained No. 4 Sieve	% Passing No. 200 Sieve	Percent Sand	Percent Silt	Percent Clay	Specific Gravity	Organic Content (%)
VB-KBIC19-3	1.7 - 2.2	Tube	SAND, 10Y 2.5/1 greenish black (SM) [Estimated shell content < 1.0%]	173.0	62	32	30	0.0	43.5	56.5	27.5	16.0	2.85	3.7
VB-KBIC19-3	2.8 - 3.3	Tube	SAND, silty, 10Y 2.5/1 greenish black (SM) [Estimated shell content < 1.0%]	163.0	NP	NP	NP	0.0	45.8	54.2	29.4	16.4	2.69	5.7
VB-KBIC19-4	0.8 - 1.3	Tube	SAND, silty, N 2.5/ black (SM) [Estimated shell content < 1.0%]	116.9	80	45	35	0.0	38.6	61.4	19.6	19.0	2.53	6
VB-KBIC19-4	1.3 - 1.7	Tube	SAND, silty, N 2.5/ black (SM) [Estimated shell content <3.0%]	103.2	93	47	46	2.7	26.4	70.9	16.7	9.7	2.72	2.9
VB-KBIC19-4	1.7 - 2.2	Tube	SAND, silty, 2.5Y 3/1 very dark gray (SM) [Estimated shell content <9.0%]	68.6	92	55	37	9.0	16.8	74.2	12.3	4.5	2.70	2.8
VB-KBIC19-4	2.5 - 3.0	Tube	SILT, 5Y 2.5/2 black (MH) [Estimated shell content < 1.0%]	180.8	94	52	42	0.0	57.9	42.1	31.3	26.6	2.78	7.5
VB-KBIC19-5	0.8 - 1.3	Tube	SAND, poorly-graded with silt, 5Y 6/1 gray (SP-SM) [Estimated shell content < 1.0%]	48.9	NP	NP	NP	0.0	10.9	89.1	10.0	0.9	2.59	1.5

Notes:
1. Soil tests in general accordance with ASTM standards.
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3. Key to abbreviations: NP=Non-Plastic; -- indicates no test performed.
4. ASTM Methods: Natural Moisture D2216, Atterberg D4318, Gradation D6913, Hydrometer D7928, Specific Gravity D854, Organic Content D2974

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Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

DYNAMIC LAB SUMMARY 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

Summary Of Laboratory Tests

Appendix
Sheet 3 of 7
Project Number: 19C19018.00

Boring No.	Sample Depth ft / Elevation ft	Sample Type	Description of Soil Specimen	Natural Moisture (%)	Liquid Limit	Plastic Limit	Plasticity Index	% Retained No. 4 Sieve	% Passing No. 200 Sieve	Percent Sand	Percent Silt	Percent Clay	Specific Gravity	Organic Content (%)
VB-KBIC19-5	2.5 - 3.0	Tube	SAND, poorly-graded with silt, 2.5Y 4/1 dark gray (SP-SM) [Estimated shell content < 1.0%]	25.9	NP	NP	NP	0.0	5.8	94.2	2.7	3.1	2.60	1
VB-KBIC19-6	0.2 - 0.7	Tube	SAND, poorly-graded, 2.5YR 4/1 dark reddish gray (SP) [Estimated shell content < 1.0%]	26.1	NP	NP	NP	0.0	4.7	95.3	4.6	0.1	2.59	0.6
VB-KBIC19-6	1.8 - 2.3	Tube	SAND, silty, 2.5YR 3/1 dark reddish gray (SM) [Estimated shell content < 1.0%]	64.7	NP	NP	NP	0.0	16.8	83.2	14.3	2.5	2.86	3.5
VB-KBIC19-6	2.5 - 3.0	Tube	SAND, poorly-graded with silt, 10Y 2.5/1 greenish black (SP-SM) [Estimated shell content <21.0%]	70.9	NP	NP	NP	1.8	7.0	91.2	6.5	0.5	2.68	2.4
VB-KBIC19-6	3.8 - 4.3	Tube	SAND, silty, 5Y 3/1 very dark gray (SM) [Estimated shell content < 1.0%]	40.4	NP	NP	NP	0.0	13.5	86.5	12.6	0.9	2.80	2.2
VB-KBIC19-7	0.2 - 0.7	Tube	SAND, silty, 10Y 4/1 dark greenish gray (SM) [Estimated shell content < 1.0%]	31.8	NP	NP	NP	0.0	14.0	86.0	12.6	1.4	2.86	2.5
VB-KBIC19-7	1.8 - 2.3	Tube	SILT, 5Y 3/1 very dark gray (MH) [Estimated shell content < 1.0%]	99.7	67	33	34	0.0	52.4	47.6	32.2	20.2	2.56	4.9

Notes:
1. Soil tests in general accordance with ASTM standards.
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3. Key to abbreviations: NP=Non-Plastic; -- indicates no test performed.
4. ASTM Methods: Natural Moisture D2216, Atterberg D4318, Gradation D6913, Hydrometer D7928, Specific Gravity D854, Organic Content D2974

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Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

DYNAMIC LAB SUMMARY 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

Summary Of Laboratory Tests

Appendix
Sheet 4 of 7
Project Number: 19C19018.00

Boring No.	Sample Depth ft / Elevation ft	Sample Type	Description of Soil Specimen	Natural Moisture (%)	Liquid Limit	Plastic Limit	Plasticity Index	% Retained No. 4 Sieve	% Passing No. 200 Sieve	Percent Sand	Percent Silt	Percent Clay	Specific Gravity	Organic Content (%)
VB-KBIC19-7	2.5 - 3.0	Tube	SAND, silty, 5Y 3/1 very dark gray (SM) [Estimated shell content < 1.0%]	94.1	39	25	14	0.0	36.4	63.6	27.4	9.0	2.74	3.9
VB-KBIC19-8	0.5 - 1.0	Tube	SILT, 10Y 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	280.5	87	58	29	0.0	95.9	4.1	78.4	17.5	2.47	11.2
VB-KBIC19-8	1.5 - 2.0	Tube	SILT, 10Y 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	287.2	90	56	34	0.0	74.6	25.4	61.3	13.3	2.83	9.8
VB-KBIC19-8	3.0 - 3.5	Tube	SILT, 10Y 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	259.3	97	55	42	0.0	80.2	19.8	65.4	14.8	2.84	12
VB-KBIC19-9	0.5 - 1.0	Tube	SILT, 10Y 3/1 very dark greenish gray (MH) [Estimated shell content < 1.0%]	273.9	84	49	35	0.0	93.0	7.0	77.7	15.3	2.54	8.4
VB-KBIC19-9	1.5 - 2.0	Tube	SILT, 10Y 3/1 very dark greenish gray (MH) [Estimated shell content < 1.0%]	268.7	85	49	36	0.0	90.0	10.0	47.1	42.9	2.78	9.5
VB-KBIC19-9	3.0 - 3.5	Tube	SILT, 10Y 3/1 very dark greenish gray (MH) [Estimated shell content < 1.0%]	263.3	78	47	31	0.0	70.3	29.7	69.7	0.6	2.75	9.4

Notes:
1. Soil tests in general accordance with ASTM standards.
2. Soil classifications are in general accordance with ASTM D2487(as applicable), based on testing indicated and visual classification.
3. Key to abbreviations: NP=Non-Plastic; -- indicates no test performed.
4. ASTM Methods: Natural Moisture D2216, Atterberg D4318, Gradation D6913, Hydrometer D7928, Specific Gravity D854, Organic Content D2974

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Project: P617 Transit Program
Sediment Transportation Modeling
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DYNAMIC LAB SUMMARY 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

Summary Of Laboratory Tests

Appendix
Sheet 5 of 7
Project Number: 19C19018.00

Boring No.	Sample Depth ft / Elevation ft	Sample Type	Description of Soil Specimen	Natural Moisture (%)	Liquid Limit	Plastic Limit	Plasticity Index	% Retained No. 4 Sieve	% Passing No. 200 Sieve	Percent Sand	Percent Silt	Percent Clay	Specific Gravity	Organic Content (%)
VB-KBIC19-10	0.5 - 1.0	Tube	SILT, 2.5Y 3/1 very dark gray (MH) [Estimated shell content < 1.0%]	266.9	105	55	50	0.0	79.7	20.3	79.1	0.6	2.49	8.7
VB-KBIC19-10	1.5 - 2.0	Tube	SILT, 2.5Y 3/1 very dark gray (MH) [Estimated shell content < 4.0%]	238.5	81	49	32	4.0	54.8	41.2	27.8	27.0	2.80	8.1
VB-KBIC19-10	3.0 - 3.5	Tube	SILT, 2.5Y 3/1 very dark gray (MH) [Estimated shell content < 1.0%]	246.8	101	50	51	0.0	62.2	37.8	56.2	6.0	2.74	10.8
VB-KBIC19-11	0.5 - 1.0	Tube	SILT, 5GY 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	327.9	108	58	50	0.0	80.5	19.5	49.2	31.3	2.51	12.3
VB-KBIC19-11	1.5 - 2.0	Tube	SILT, 5GY 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	296.7	110	59	51	0.0	71.0	29.0	37.1	33.9	2.65	14.8
VB-KBIC19-11	3.0 - 3.5	Tube	SILT, 5GY 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	253.5	112	66	46	0.0	82.0	18.0	81.2	0.8	2.76	12.7
VB-KBIC19-12	0.5 - 1.0	Tube	SILT, 5G 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	260.4	102	55	47	0.0	54.6	45.4	27.8	26.8	2.47	13.8

Notes:
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3. Key to abbreviations: NP=Non-Plastic; -- indicates no test performed.
4. ASTM Methods: Natural Moisture D2216, Atterberg D4318, Gradation D6913, Hydrometer D7928, Specific Gravity D854, Organic Content D2974

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Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

DYNAMIC LAB SUMMARY 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

Summary Of Laboratory Tests

Appendix
Sheet 6 of 7
Project Number: 19C19018.00

Boring No.	Sample Depth ft / Elevation ft	Sample Type	Description of Soil Specimen	Natural Moisture (%)	Liquid Limit	Plastic Limit	Plasticity Index	% Retained No. 4 Sieve	% Passing No. 200 Sieve	Percent Sand	Percent Silt	Percent Clay	Specific Gravity	Organic Content (%)
VB-KBIC19-12	1.5 - 2.0	Tube	SILT, 5G 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	264.7	101	54	47	0.0	90.5	9.5	49.0	41.5	2.77	13
VB-KBIC19-12	2.5 - 3.0	Tube	SILT, 5G 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	207.8	79	39	40	0.0	69.6	30.4	41.3	28.3	2.77	8
VB-KBIC19-13	0.5 - 1.0	Tube	SILT, 10Y 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	150.7	97	49	48	0.0	56.1	43.9	25.5	30.6	2.53	5.8
VB-KBIC19-13	1.5 - 2.0	Tube	SILT, 10Y 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	274.1	97	52	45	0.0	74.2	25.8	37.6	36.6	2.56	11
VB-KBIC19-13	3.0 - 3.5	Tube	SAND, silty, 5Y 3/1 very dark gray (SM) Estimated shell content < 1.0%]	158.7	93	46	47	0.0	49.1	50.9	29.5	19.6	2.67	8.6
VB-KBIC19-14	0.5 - 1.0	Tube	SILT, 5Y 2.5/1 black (MH) [Estimated shell content < 1.0%]	231.4	98	46	52	0.0	75.8	24.2	53.4	22.4	2.54	7.6
VB-KBIC19-14	1.5 - 2.0	Tube	SILT, 5Y 2.5/1 black (MH) [Estimated shell content < 1.0%]	241.7	105	55	50	0.0	73.7	26.3	38.1	35.6	2.66	10.9

Notes:
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Project: P617 Transit Program
Sediment Transportation Modeling
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DYNAMIC LAB SUMMARY 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

Summary Of Laboratory Tests

Appendix
Sheet 7 of 7
Project Number: 19C19018.00

Boring No.	Sample Depth ft / Elevation ft	Sample Type	Description of Soil Specimen	Natural Moisture (%)	Liquid Limit	Plastic Limit	Plasticity Index	% Retained No. 4 Sieve	% Passing No. 200 Sieve	Percent Sand	Percent Silt	Percent Clay	Specific Gravity	Organic Content (%)
VB-KBIC19-14	3.0 - 3.5	Tube	SILT, 5Y 3/2 dark olive gray (MH) [Estimated shell content < 1.0%]	242.1	107	52	55	0.0	83.1	16.9	42.2	40.9	2.64	13.1
VB-KBIC19-15	0.5 - 1.0	Tube	SILT, 10Y 3/1 very dark greenish gray (MH) [Estimated shell content < 1.0%]	250.4	81	37	44	0.0	66.4	33.6	45.1	21.3	2.51	9.3
VB-KBIC19-15	1.5 - 2.0	Tube	SILT, 10Y 3/1 very dark greenish gray (MH) [Estimated shell content < 1.0%]	234.0	104	57	47	0.0	87.6	12.4	71.3	16.3	2.74	9.2
VB-KBIC19-16	0.3 - 0.8	Tube	SAND, silty, clayey, 10Y 3/1 very dark greenish gray (SC-SM) [Estimated shell content < 22.0%]	57.5	28	21	7	22.0	18.3	59.7	16.0	2.3	2.58	0.9
VB-KBIC19-16	1.2 - 1.7	Tube	SHELL, 5Y 4/1 dark gray (GM) [Estimated shell content < 54.0%]	34.8	22	19	3	53.7	26.9	19.4	19.8	7.1	2.70	0.8
VB-KBIC19-16	2.5 - 3.0	Tube	SAND, silty, 2.5Y 7/1 light gray (SM) [Estimated shell content < 1.0%]	28.3	NP	NP	NP	0.0	29.4	70.6	25.1	4.3	2.67	0.5

Notes:
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Project: P617 Transit Program
Sediment Transportation Modeling
Kings Bay Inner Channel, Camden Co., GA

DYNAMIC LAB SUMMARY 19C19018.00 SEI LAB SH COE CLASS.GPJ TEST TEMPLATE.GDT 8/30/19

Summary Of Laboratory Tests

Appendix
Sheet 1 of 6
Project Number: 19C19018.00

Boring No.	Sample Depth ft / Elevation ft	Sample Type	Description of Soil Specimen	Bulk Density (lb/ft^3)	Natural Moisture (%)	Dry Density (lb/ft^3)	Porosity	Total Suspended Soilds-Final Sediment Height (cm)	Total Suspended Solids-Final Concentration (g/L)	Sedimentation Rate: Coarse-fraction Concentration (ppm)	Sedimentation Rate: Fine-fraction Concentration (ppm)
VB-KBIC19-1B	0.0 - 0.5	Tube	SAND, silty, 2.5Y 3/1 very dark gray (SM) [Estimated shell content < 1.0%]	100.4	60.7	62.5	0.61	4.5	1111	4.8	202.3
VB-KBIC19-1B	1.5 - 2.0	Tube	SAND, silty, 2.5Y 3/1 very dark gray (SM) [Estimated shell content < 2.0%]	71.1	71.3	41.5	–	4.0	1250		
VB-KBIC19-1B	2.9 - 3.4	Tube	SAND, silty, 10Y 3/1 very dark greenish gray (SM) [Estimated shell content < 1.0%]	69.8	55.6	44.8	–	4.0	1250		
VB-KBIC19-2B	0.2 - 0.7	Tube	SAND, poorly-graded with silt, 2.5Y 3/1 very dark gray (SP-SM) [Estimated shell content < 1.0%]	121.1	27.1	95.3	0.42	3.4	1471	3.6	333.9
VB-KBIC19-2B	2.0 - 2.5	Tube	SAND, silty, 2.5Y 3/1 very dark gray (SM) [Estimated shell content < 6.0%]	80.4	28.4	51.6	–	3.0	1667		
VB-KBIC19-2B	2.8 - 3.3	Tube	SAND, silty, 2.5Y 3/1 very dark gray (SM) [Estimated shell content < 1.0%]	74.7	62.4	45.8	–	4.0	1250		
VB-KBIC19-3	0.2 - 0.7	Tube	SAND, silty, 10Y 2.5/1 greenish black (SM) [Estimated shell content < 1.0%]	77.5	197.2	26.1	0.83	6.5	769	5.4	391.8
VB-KBIC19-3	1.7 - 2.2	Tube	SAND, 10Y 2.5/1 greenish black (SM) [Estimated shell content < 1.0%]	65.3	173.0	23.9	–	5.0	1000		
VB-KBIC19-3	2.8 - 3.3	Tube	SAND, silty, 10Y 2.5/1 greenish black (SM) [Estimated shell content < 1.0%]	66.4	163.0	25.3	–	5.0	1000		

Notes:
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3. Key to abbreviations: -- indicates no test performed.
4. ASTM Methods: Natural Moisture D2216, Bulk Density D7263, Total Suspended Solids (TSS) D3977

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Summary Of Laboratory Tests

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Project Number: 19C19018.00

Boring No.	Sample Depth ft / Elevation ft	Sample Type	Description of Soil Specimen	Bulk Density (lb/ft^3)	Natural Moisture (%)	Dry Density (lb/ft^3)	Porosity	Total Suspended Soilds Final Sediment Height (cm)	Total Suspended Soilds Final Concentration (g/L)	Sedimentation Rate: Coarse-fraction Concentration (ppm)	Sedimentation Rate: Fine-fraction Concentration (ppm)
VB-KBIC19-4	0.8 - 1.3	Tube	SAND, silty, N 2.5/ black (SM) [Estimated shell content < 1.0%]	90.5	87.4	48.3	0.69	5.0	1000	3.3	429.9
VB-KBIC19-4	1.3 - 1.7	Tube	SAND, silty, N 2.5/ black (SM) [Estimated shell content <3.0%]	74.4	103.2	36.6	–	4.0	1250		
VB-KBIC19-4	1.7 - 2.2	Tube	SAND, silty, 2.5Y 3/1 very dark gray (SM) [Estimated shell content <9.0%]	76.6	68.6	45.4	–	4.0	1250		
VB-KBIC19-4	2.5 - 3.0	Tube	SILT, 5Y 2.5/2 black (MH) [Estimated shell content < 1.0%]	69.4	186.8	24.7	–	5.5	909		
VB-KBIC19-5	0.8 - 1.3	Tube	SAND, poorly-graded with silt, 5Y 6/1 gray (SP-SM) [Estimated shell content < 1.0%]	85.4	48.9	57.3	–	2.5	2000	4.9	325.2
VB-KBIC19-5	2.5 - 3.0	Tube	SAND, poorly-graded with silt, 2.5Y 4/1 dark gray (SP-SM) [Estimated shell content < 1.0%]	122.0	22.4	99.7	0.39	2.2	2273		
VB-KBIC19-6	0.2 - 0.7	Tube	SAND, poorly-graded, 2.5YR 4/1 dark reddish gray (SP) [Estimated shell content < 1.0%]	117.3	24.4	94.3	0.42	2.7	1852	10.0	264.2
VB-KBIC19-6	1.8 - 2.3	Tube	SAND, silty, 2.5YR 3/1 dark reddish gray (SM) [Estimated shell content < 1.0%]	81.2	64.7	49.3	–	4.0	1250		
VB-KBIC19-6	2.5 - 3.0	Tube	SAND, poorly-graded with silt, 10Y 2.5/1 greenish black (SP-SM) [Estimated shell content <21.0%]	80.5	70.9	47.1	–	3.5	1429		
VB-KBIC19-6	3.8 - 4.3	Tube	SAND, silty, 5Y 3/1 very dark gray (SM) [Estimated shell content < 1.0%]	77.9	40.4	55.5	–	4.0	1250		

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Summary Of Laboratory Tests

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Project Number: 19C19018.00

Boring No.	Sample Depth ft / Elevation ft	Sample Type	Description of Soil Specimen	Bulk Density (lb/ft^3)	Natural Moisture (%)	Dry Density (lb/ft^3)	Porosity	Total Suspended Solids Final Sediment Height (cm)	Total Suspended Solids Final Concentration (g/L)	Sedimentation Rate: Coarse-fraction Concentration (ppm)	Sedimentation Rate: Fine-fraction Concentration (ppm)
VB-KBIC19-7	0.2 - 0.7	Tube	SAND, silty, 10Y 4/1 dark greenish gray (SM) [Estimated shell content < 1.0%]	71.6	31.8	54.8	–	4.0	1250	9.8	246.6
VB-KBIC19-7	1.8 - 2.3	Tube	SILT, 5Y 3/1 very dark gray (MH) [Estimated shell content < 1.0%]	84.3	124.0	37.6	0.76	5.0	1000		
VB-KBIC19-7	2.5 - 3.0	Tube	SAND, silty, 5Y 3/1 very dark gray (SM) [Estimated shell content < 1.0%]	73.1	94.1	37.7	–	5.0	1000		
VB-KBIC19-8	0.5 - 1.0	Tube	SILT, 10Y 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	71.6	347.6	16.0	0.90	4.0	1250	31.4	616.8
VB-KBIC19-8	1.5 - 2.0	Tube	SILT, 10Y 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	57.3	287.2	14.8	–	6.5	769		
VB-KBIC19-8	3.0 - 3.5	Tube	SILT, 10Y 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	59.4	259.3	16.5	–	6.5	769		
VB-KBIC19-9	0.5 - 1.0	Tube	SILT, 10Y 3/1 very dark greenish gray (MH) [Estimated shell content < 1.0%]	73.9	275.2	19.7	0.88	4.0	1250	18.5	274.0
VB-KBIC19-9	1.5 - 2.0	Tube	SILT, 10Y 3/1 very dark greenish gray (MH) [Estimated shell content < 1.0%]	60.6	268.7	16.4	–	6.0	833		
VB-KBIC19-9	3.0 - 3.5	Tube	SILT, 10Y 3/1 very dark greenish gray (MH) [Estimated shell content < 1.0%]	62.4	263.3	17.2	–	7.0	714		

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Summary Of Laboratory Tests

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Project Number: 19C19018.00

Boring No.	Sample Depth ft / Elevation ft	Sample Type	Description of Soil Specimen	Bulk Density (lb/ft^3)	Natural Moisture (%)	Dry Density (lb/ft^3)	Porosity	Total Suspended Solids Final Sediment Height (cm)	Total Suspended Solids Final Concentration (g/L)	Sedimentation Rate: Coarse-fraction Concentration (ppm)	Sedimentation Rate: Fine-fraction Concentration (ppm)
VB-KBIC19-10	0.5 - 1.0	Tube	SILT, 2.5Y 3/1 very dark gray (MH) [Estimated shell content < 1.0%]	72.2	254.1	20.4	0.87	5.5	909		
VB-KBIC19-10	1.5 - 2.0	Tube	SILT, 2.5Y 3/1 very dark gray (MH) [Estimated shell content < 4.0%]	64.4	238.5	19.0	–	6.5	769	5.8	161.4
VB-KBIC19-10	3.0 - 3.5	Tube	SILT, 2.5Y 3/1 very dark gray (MH) [Estimated shell content < 1.0%]	59.8	246.8	17.2	–	6.5	769		
VB-KBIC19-11	0.5 - 1.0	Tube	SILT, 5GY 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	71.7	356.5	15.7	0.90	5.5	909		
VB-KBIC19-11	1.5 - 2.0	Tube	SILT, 5GY 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	60.3	296.7	15.2	–	6.0	833	4.6	165.2
VB-KBIC19-11	3.0 - 3.5	Tube	SILT, 5GY 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	63.0	253.5	17.8	–	8.5	588		
VB-KBIC19-12	0.5 - 1.0	Tube	SILT, 5G 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	70.7	280.1	18.6	0.88	5.5	909		
VB-KBIC19-12	1.5 - 2.0	Tube	SILT, 5G 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	65.7	264.7	18.0	–	5.5	909	13.3	327.1
VB-KBIC19-12	2.5 - 3.0	Tube	SILT, 5G 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	64.5	207.8	21.0	–	6.5	769		

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Summary Of Laboratory Tests

Boring No.	Sample Depth ft / Elevation ft	Sample Type	Description of Soil Specimen	Bulk Density (lb/ft^3)	Natural Moisture (%)	Dry Density (lb/ft^3)	Porosity	Total Suspended Solids Final Sediment Height (cm)	Total Suspended Solids Final Concentration (g/L)	Sedimentation Rate: Coarse-fraction Concentration (ppm)	Sedimentation Rate: Fine-fraction Concentration (ppm)
VB-KBIC19-13	0.5 - 1.0	Tube	SILT, 10Y 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	70.9	312.1	17.2	0.89	6.0	833		
VB-KBIC19-13	1.5 - 2.0	Tube	SILT, 10Y 2.5/1 greenish black (MH) [Estimated shell content < 1.0%]	61.3	274.1	16.4	–	15.0	333	3.9	217.7
VB-KBIC19-13	3.0 - 3.5	Tube	SAND, silty, 5Y 3/1 very dark gray (SM) Estimated shell content < 1.0%]	62.2	158.7	24.1	–	5.5	909		
VB-KBIC19-14	0.5 - 1.0	Tube	SILT, 5Y 2.5/1 black (MH) [Estimated shell content < 1.0%]	76.7	224.9	23.6	0.85	5.5	909		
VB-KBIC19-14	1.5 - 2.0	Tube	SILT, 5Y 2.5/1 black (MH) [Estimated shell content < 1.0%]	64.1	241.7	18.8	–	7.0	714	6.3	148.6
VB-KBIC19-14	3.0 - 3.5	Tube	SILT, 5Y 3/2 dark olive gray (MH) [Estimated shell content < 1.0%]	68.2	242.1	19.9	–	6.5	769		
VB-KBIC19-15	0.5 - 1.0	Tube	SILT, 10Y 3/1 very dark greenish gray (MH) [Estimated shell content < 1.0%]	72.9	296.0	18.4	0.88	5.0	1000	6.3	337.3
VB-KBIC19-15	1.5 - 2.0	Tube	SILT, 10Y 3/1 very dark greenish gray (MH) [Estimated shell content < 1.0%]	62.1	234.0	18.6	–	5.5	909		

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Summary Of Laboratory Tests

Appendix
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Boring No.	Sample Depth ft / Elevation ft	Sample Type	Description of Soil Specimen	Bulk Density (lb/ft^3)	Natural Moisture (%)	Dry Density (lb/ft^3)	Porosity	Total Suspended Soilds Final Sediment Height (cm)	Total Suspended Soilds Final Concentration (g/L)	Sedimentation Rate: Coarse-fraction Concentration (ppm)	Sedimentation Rate: Fine-fraction Concentration (ppm)
VB-KBIC19-16	0.3 - 0.8	Tube	SAND, silty, clayey, 10Y 3/1 very dark greenish gray (SC-SM) [Estimated shell content < 22.0%]	82.8	57.5	52.6	–	3.5	1429	1.5	61.6
VB-KBIC19-16	1.2 - 1.7	Tube	GRAVEL, silty, 5Y 4/1 dark gray (GM) [Estimated shell content < 54.0%]	119.3	31.8	90.5	0.46	6.0	833		
VB-KBIC19-16	2.5 - 3.0	Tube	SAND, silty, 2.5Y 7/1 light gray (SM) [Estimated shell content < 1.0%]	86.1	28.3	67.1	–	3.0	1667		

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